Signals and Systems

Detailed Solutions • 1987–2026

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

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Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Signals and Systems (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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Q1.24.1 | MSQ | 2024 | 2M | Ans: A,C,D | ● ○ ○

Given the discrete-time signal:

$$x[n] = A\cos\left[\Omega n + \frac{\pi}{3}\right]$$

For the signal to be periodic, the condition is:

$$\frac{\Omega}{2\pi} = \frac{m}{N}$$

where m and N are integers, and N is the fundamental period.

Given $N = 40$, we have:

$$\frac{\Omega}{2\pi} = \frac{m}{40}$$

$$\Omega = m \cdot \frac{2\pi}{40} = m \cdot \frac{\pi}{20} = \pi \cdot m(0.05)$$

$$\Omega = 0.05\pi \cdot m$$

By taking different integer values for m, we can find the possible values for the radian frequency Ω:

- For $m = 1$, $\Omega = 0.05\pi$
- For $m = 2$, $\Omega = 0.10\pi$
- For $m = 3$, $\Omega = 0.15\pi$
- For $m = 4$, $\Omega = 0.20\pi$
- For $m = 5$, $\Omega = 0.25\pi$
- For $m = 6$, $\Omega = 0.30\pi$
- For $m = 7$, $\Omega = 0.35\pi$
- For $m = 8$, $\Omega = 0.40\pi$
- For $m = 9$, $\Omega = 0.45\pi$

From the calculated values, we can determine the correct options.

$$\boxed{\text{A-}0.15\pi,\ \text{C-}0.3\pi,\ \text{D-}0.45\pi}$$

Key Points

- A discrete-time sinusoidal signal $x[n] = A\cos(\Omega n + \theta)$ is periodic if and only if its frequency Ω is a rational multiple of 2π.
- The condition for periodicity is $\frac{\Omega}{2\pi} = \frac{m}{N}$, where m is an integer and N is the fundamental period.

Mistakes to be avoided

- Forgetting that m can be any integer, leading to multiple possible frequencies for a given period N.
- Assuming all discrete-time sinusoids are periodic. This is only true for continuous-time sinusoids.

Q1.19.1 | NAT | 2019 | 1M | Ans: 12 | ● ○ ○

Given the continuous-time signal:

$$f(t) = 1 + 2\cos(\pi t) + 3\sin\left(\frac{2\pi}{3}t\right) + 4\cos\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)$$

The individual radian frequencies of the sinusoidal components are:

$$\omega_1 = \pi$$
$$\omega_2 = \frac{2\pi}{3}$$
$$\omega_3 = \frac{\pi}{2}$$

The fundamental frequency ω_0 is the Greatest Common Divisor (GCD) of the individual frequencies:

$$\omega_0 = \text{GCD}\left(\pi, \frac{2\pi}{3}, \frac{\pi}{2}\right) = \frac{\pi}{6}$$

The fundamental period is:

$$N = \frac{2\pi}{\omega_0} = \frac{2\pi}{(\pi/6)} = 12$$

$$\boxed{12}$$

Key Points

- For a signal $x(t) = x_1(t) + x_2(t) + \cdots + x_n(t)$, the fundamental frequency is $\omega_0 = \text{GCD}(\omega_1, \omega_2, \ldots, \omega_n)$.
- To find the GCD of fractions: $\text{GCD}\left(\frac{a}{b}, \frac{c}{d}\right) = \frac{\text{GCD}(a,c)}{\text{LCM}(b,d)}$.
- Alternatively, the fundamental period T_0 can be found by taking the Least Common Multiple (LCM) of the individual periods: $T_0 = \text{LCM}(T_1, T_2, \ldots, T_n)$.

Mistakes to be avoided

- Calculating the LCM of the frequencies instead of the GCD. Remember: Fundamental Period $\to$ LCM, Fundamental Frequency $\to$ GCD.
- Getting confused by the constant term (1). A constant term is a DC signal ($\omega = 0$) and does not dictate the fundamental period of the varying components.

Q1.15.1 NAT 2015 1M Ans: 2 ● ○ ○

From the given waveform, the signal $x(t)$ is periodic. Let's analyze one complete cycle to determine its fundamental period T and equation.
Observing the waveform, one cycle starts at $t = -1$ and ends at $t = 2$. The fundamental period is:

$$T = 2 - (-1) = 3$$

Over this period $t \in [-1, 2]$, the signal can be piecewise defined as:

$$x(t) = \begin{cases} -3t & \text{for } -1 \le t \le 1 \\ 0 & \text{for } 1 < t \le 2 \end{cases}$$

The average power of a periodic signal $x(t)$ is given by the integral of its squared magnitude over one period, divided by the period:

$$P_x = \frac{1}{T}\int_{\langle T \rangle} |x(t)|^2 \, dt$$
$$P_x = \frac{1}{3}\int_{-1}^{2} |x(t)|^2 \, dt$$

$$P_x = \frac{1}{3}\left[\int_{-1}^{1}(-3t)^2\,dt + \int_{1}^{2}(0)^2\,dt\right]$$

$$P_x = \frac{1}{3}\int_{-1}^{1} 9t^2\,dt$$

$$P_x = \frac{9}{3}\left[\frac{t^3}{3}\right]_{-1}^{1}$$

$$P_x = 3\left(\frac{1^3}{3} - \frac{(-1)^3}{3}\right) = 3\left(\frac{1}{3}+\frac{1}{3}\right) = 3\left(\frac{2}{3}\right) = 2$$

The problem defines a new signal $g(t) = x\left(\frac{t-1}{2}\right)$. This operation involves both time shifting (by 1) and time scaling (by a factor of $1/2$).

The average power of a signal remains invariant under both time shifting and time scaling. Therefore, the power of $g(t)$ is identical to the power of $x(t)$.

$$P_g = P_x = 2$$

$$\boxed{2}$$

Key Points

- The average power of a periodic signal $x(t)$ with period T is $P = \frac{1}{T}\int_T |x(t)|^2\,dt$.
- **Time Shifting Invariance:** The power of $x(t \pm t_0)$ is equal to the power of $x(t)$. Shifting merely delays or advances the waveform without changing the area under its squared curve over a period.
- **Time Scaling Invariance:** The power of $x(at)$ is equal to the power of $x(t)$. While scaling compresses or expands the signal in time (altering its period to T/a), the average power computed over the new period remains mathematically identical.

Q1.14.1 MCQ | 2014 | 1M | Ans: D | ● ○ ○

Given the discrete-time signal:

$$x[n] = \sin(\pi^2 n)$$

The fundamental angular frequency is:

$$\omega_0 = \pi^2$$

For a discrete-time sinusoidal signal to be periodic, its period N must be an integer, which requires:

$$N = \frac{2\pi}{\omega_0}\cdot m$$

where m is the smallest integer that converts $\frac{2\pi}{\omega_0}$ into an integer value.

Substituting $\omega_0 = \pi^2$ into the equation:

$$N = \frac{2\pi}{\pi^2}\cdot m = \frac{2}{\pi}\cdot m$$

So, there exists no such integer value of m which could make N an integer. Therefore, the signal (or system) is not periodic.

$$\boxed{\text{(D) Not periodic}}$$

Key Points

- A discrete-time sinusoid $x[n] = A\sin(\omega_0 n + \phi)$ is periodic if and only if $\frac{\omega_0}{2\pi}$ is a rational number $= \frac{m}{N}$, where m and N are integers.

- If the ratio $\frac{\omega_0}{2\pi}$ is irrational (as in this case, $\frac{\pi}{2}$), the discrete-time signal is aperiodic.

Mistakes to be avoided

- Assuming that all sinusoidal functions are periodic. While true in continuous-time, discrete-time sinusoids are only periodic if their frequency is a rational multiple of 2π.

Q1.13.1 NAT 2013 1M Ans: A ● ○ ○

Given the individual angular frequencies:

$$\omega_1 = 100$$
$$\omega_2 = 300$$
$$\omega_3 = 500$$

The fundamental angular frequency ω is the Highest Common Factor (H.C.F.) of the individual frequencies:

$$\text{H.C.F. of } (\omega_1, \omega_2 \text{ and } \omega_3) = \text{H.C.F.}(100, 300, 500)$$

$$\omega = 100\,\text{rad/sec}$$

$$\boxed{(A)100}$$

Key Points

- For a composite signal consisting of a sum of continuous-time sinusoids, the fundamental angular frequency ω_0 is the Greatest Common Divisor (GCD) or Highest Common Factor (HCF) of the individual frequencies: $\omega_0 = \text{HCF}(\omega_1, \omega_2, \ldots, \omega_n)$.
- Conversely, the fundamental period T_0 is the Least Common Multiple (LCM) of the individual fundamental periods: $T_0 = \text{LCM}(T_1, T_2, \ldots, T_n)$.

Mistakes to be avoided

- Not checking if the signals are continuous-time vs. discrete-time, as discrete-time signals have an additional rational multiple requirement for periodicity.

Q1.09.1 MCQ 2009 1M Ans: B ● ○ ○

Given the function:

$$f(t) = \sin^2 t + \cos 2t$$

To determine the frequency components, we first express the function in terms of linear sinusoidal functions without powers. We use the standard trigonometric identity:

$$\sin^2 t = \frac{1 - \cos 2t}{2} = 0.5 - 0.5\cos 2t$$

Substituting this back into the original equation for $f(t)$:

$$f(t) = (0.5 - 0.5\cos 2t) + \cos 2t$$

$$f(t) = 0.5 + 0.5\cos 2t$$

Now, we analyze the frequency components of this simplified expression: 1. DC Component (0.5): A constant term corresponds to an angular frequency $\omega = 0\,\text{rad/s}$, which means the linear frequency is:

$$f_1 = 0\,\text{Hz}$$

2. AC Component ($0.5 \cos 2t$): Comparing this to the standard form $A\cos(\omega t)$, the angular frequency is $\omega = 2$ rad/s. The corresponding linear frequency f is:

$$f_2 = \frac{\omega}{2\pi} = \frac{2}{2\pi} = \frac{1}{\pi}\,\text{Hz}$$

Therefore, the function $f(t)$ has frequency components at 0 and $1/\pi$ Hz.

$$\boxed{\text{(B) f has frequency components at 0 and } 1/\pi \text{ Hz}}$$

Key Points

- Non-linear terms (like $\sin^2 t$) must be expanded using trigonometric identities (e.g., $\sin^2\theta = \frac{1-\cos 2\theta}{2}$) to correctly identify the frequency components of a signal.
- The relationship between angular frequency (ω in rad/s) and linear frequency (f in Hz) is $\omega = 2\pi f$.
- A constant (DC) value in a time-domain signal always corresponds to a frequency of 0 Hz.

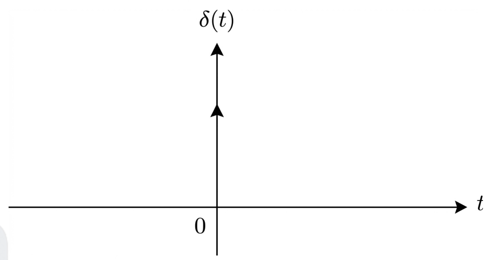
Q1.06.1 MCQ 2006 1M Ans: D ● ○ ○

The Dirac delta function, written as $\delta(x)$, It is used to model very sudden changes, like a sharp spike at a single point. This is why it is also called the impulse symbol or Dirac's delta.

This function is defined as zero everywhere except at $x = 0$, where it becomes infinitely large, but in such a way that the total area under the curve is equal to 1. That means:

- $\delta(x) = 0$ when $x \neq 0$
- $\delta(x) = \infty$ at $x = 0$
- The total area under $\delta(x)$ from just before to just after 0 is 1:

$$\int_b^c \delta(x)\,dx = 1, \quad \text{if 0 is between } b \text{ and } c$$



$$\boxed{(D)}$$

Key Points

- The Dirac delta function (or unit impulse) $\delta(x)$ represents a signal of infinite amplitude, zero duration/width, and unit area.
- The fundamental definition relies on its integral over the entire domain: $\int_{-\infty}^{\infty} \delta(x)\,dx = 1$.

Q1.05.1 NAT 2005 1M Ans: A ● ○ ○

Given the signal:

$$s(t) = 8\cos\left(\frac{\pi}{2} - 20\pi t\right) + 4\sin(15\pi t)$$

Using the trigonometric identity $\cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$, we can simplify the first term:

$$s(t) = 8\sin(20\pi t) + 4\sin(15\pi t)$$

The average power of a sum of sinusoids with distinct frequencies is the sum of the powers of the individual sinusoids. The power of a single sinusoid with amplitude A is $\frac{A^2}{2}$.

$$P = \frac{8^2}{2} + \frac{4^2}{2}$$

$$P = 32 + 8 = 40$$

$$\boxed{(A)\ \ 40}$$

Key Points

- The average power of a continuous-time sinusoidal signal $x(t) = A\cos(\omega t + \theta)$ or $A\sin(\omega t + \theta)$ is $P = \frac{A^2}{2}$.
- For a signal $s(t) = \sum_{k=1}^{N} A_k \sin(\omega_k t + \theta_k)$, if all frequencies ω_k are distinct (orthogonal), the total average power is the sum of individual powers: $P = \sum_{k=1}^{N} \frac{A_k^2}{2}$.
- Trigonometric conversion rule: $\cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$.

Mistakes to be avoided

- Applying the superposition of power ($P = P_1 + P_2$) without first ensuring the frequency components are distinct (orthogonal). If components share the same frequency, they must be combined into a single equivalent phasor before calculating power.

Q1.05.2 MCQ 2005 1M Ans: A ● ○ ○

Any signal $x(t)$ can be decomposed into its even and odd parts. For the unit step signal $u(t)$, the decomposition is performed as follows:

$$\text{Even part} = \frac{u(t) + u(-t)}{2}$$

$$\text{Odd part} = \frac{u(t) - u(-t)}{2}$$

- The even part $u_e(t)$ results in a constant value:

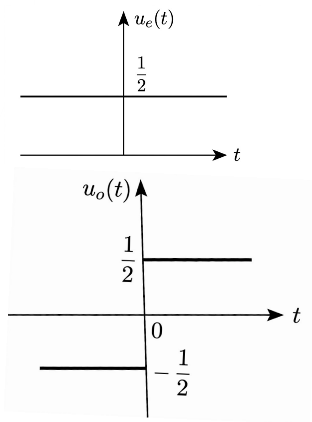
$$u_e(t) = \frac{1}{2}$$

- The odd part $u_o(t)$ is given by:

$$u_o(t) = \frac{\text{sgn}(t)}{2}$$

Which can be represented as:

$$u_o(t) = \begin{cases} \frac{1}{2}, & t > 0 \\ -\frac{1}{2}, & t < 0 \end{cases}$$



$$\boxed{(A)\ \ u_e(t) = \frac{1}{2}, u_o(t) = \frac{\text{sgn}(t)}{2} = \frac{x(t)}{2}}$$

Key Points

- A signal is even if $x(t) = x(-t)$ and odd if $x(t) = -x(-t)$.
- The even component is calculated as $x_e(t) = \frac{x(t)+x(-t)}{2}$.
- The odd component is calculated as $x_o(t) = \frac{x(t)-x(-t)}{2}$.
- For a unit step $u(t)$, the sum $u(t) + u(-t) = 1$ for all $t \neq 0$.

Q1.01.1 | MCQ | 2001 | 1M | Ans: B | ● ○ ○

The energy E of a continuous-time signal $f(t)$ is defined by the integral:

$$E = \int_{-\infty}^{\infty} |f(t)|^2 \, dt$$

We are required to find the energy of the time-scaled signal $f(2t)$. Let E_{scaled} be the energy of this new signal:

$$E_{scaled} = \int_{-\infty}^{\infty} |f(2t)|^2 \, dt$$

To solve this, we can use the substitution method. Let $\tau = 2t$. Differentiating both sides with respect to t gives $d\tau = 2\,dt$, which means $dt = \frac{d\tau}{2}$. The limits of integration remain unchanged (as $t \to \pm\infty$, $\tau \to \pm\infty$).
Substitute τ and dt back into the energy equation:

$$E_{scaled} = \int_{-\infty}^{\infty} |f(\tau)|^2 \frac{d\tau}{2}$$

$$E_{scaled} = \frac{1}{2} \int_{-\infty}^{\infty} |f(\tau)|^2 \, d\tau$$

Recognizing that the integral term is exactly the definition of the original energy E:

$$E_{scaled} = \frac{1}{2}E = \frac{E}{2}$$

This corresponds to option B.

$$\boxed{\text{(B) E/2}}$$

Key Points

- **Time Scaling Property of Signal Energy:** If a continuous-time signal $x(t)$ has energy E, then the time-scaled signal $x(at)$ has an energy equal to $\frac{E}{|a|}$.
- **Physical Interpretation:** Time compression ($|a| > 1$, e.g., $a = 2$) speeds up the signal, effectively halving its duration and therefore halving its total energy. Time expansion ($|a| < 1$) stretches the signal, increasing its total energy.

Q1.01.2 | MCQ | 2001 | 1M | Ans: A | ● ○ ○

To solve the given integral, we apply the sampling property of the Dirac delta function $\delta(t)$.

$$\int_{-\infty}^{\infty} \delta(t) \cos\left(\frac{3t}{2}\right) dt$$

According to the sampling property, $\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$. In this case, $t_0 = 0$ and $f(t) = \cos\left(\frac{3t}{2}\right)$.

$$\int_{-\infty}^{\infty} \delta(t) \cos\left(\frac{3t}{2}\right) dt = f(0)$$

$$f(0) = \cos\left(\frac{3 \times 0}{2}\right) = \cos 0 = 1$$

$$\boxed{(A)\ 1}$$

Key Points

- The Dirac delta function $\delta(t)$ is defined such that it is zero everywhere except at $t = 0$, where it is infinite, and its total area is 1.
- Sampling Property: $\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$, provided $f(t)$ is continuous at $t = t_0$.

Mistakes to be avoided

- Ensure that the limits of integration include the point where the delta function is impulse-active ($t = 0$ in this case).

Q1.01.3 | MCQ | 2001 | 1M | Ans: B | ● ○ ○

Consider the energy E of a continuous-time signal $f(t)$ defined as:

$$E = \int_{-\infty}^{\infty} f(t)^2 dt$$

To find the energy E' of the time-scaled signal $f(2t)$, we evaluate the following integral:

$$E' = \int_{-\infty}^{\infty} f(2t)^2 dt$$

Applying the method of substitution, let $p = 2t$. Then, the differential is $dp = 2dt$, which gives $dt = \frac{dp}{2}$. The limits of integration remain $-\infty$ to ∞.

$$E' = \int_{-\infty}^{\infty} f(p)^2 \frac{dp}{2}$$

Factoring out the constant:

$$E' = \frac{1}{2}\int_{-\infty}^{\infty} f(p)^2 dp$$

Since the integral $\int_{-\infty}^{\infty} f(p)^2 dp$ is equivalent to the original energy E, we have:

$$\boxed{(B)\ E' = \frac{E}{2}}$$

Key Points

- The energy of a signal $x(t)$ is defined as $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$.
- Time Scaling Property: If a signal $x(t)$ has energy E, then the time-scaled signal $x(at)$ has energy $\frac{E}{|a|}$.
- Time scaling compresses or expands the signal on the time axis, which directly affects the area under the squared magnitude of the signal.

Mistakes to be avoided

- Do not confuse time scaling with amplitude scaling; amplitude scaling $A \cdot f(t)$ results in energy A^2E.

Q1.95.1 | MCQ | 1995 | 1M | Ans: A | ● ○ ○

The RMS (Root Mean Square) value of a periodic signal $v(t)$ with period T is calculated using the formula:

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\, dt}$$

Based on the problem description, the waveform over one complete period T is defined as: $v(t) = +V$ for a duration of T_1 $v(t) = -V$ for a duration of T_2 where $T_1 + T_2 = T$.

We evaluate the integral by splitting it into the two defined durations:

$$V_{rms} = \sqrt{\frac{1}{T}\left(\int_0^{T_1} (+V)^2\, dt + \int_{T_1}^{T} (-V)^2\, dt\right)}$$

Squaring the amplitudes:

$$V_{rms} = \sqrt{\frac{1}{T}\left(\int_0^{T_1} V^2\, dt + \int_{T_1}^{T} V^2\, dt\right)}$$

Evaluating the definite integrals:

$$V_{rms} = \sqrt{\frac{1}{T}\left(V^2[t]_0^{T_1} + V^2[t]_{T_1}^{T}\right)}$$

$$V_{rms} = \sqrt{\frac{1}{T}\left(V^2(T_1 - 0) + V^2(T - T_1)\right)}$$

Substitute $T - T_1 = T_2$:

$$V_{rms} = \sqrt{\frac{1}{T}\left(V^2T_1 + V^2T_2\right)}$$

$$V_{rms} = \sqrt{\frac{V^2}{T}(T_1 + T_2)}$$

Since the total period $T = T_1 + T_2$, we substitute T back into the equation:

$$V_{rms} = \sqrt{\frac{V^2}{T}(T)}$$

$$V_{rms} = \sqrt{V^2} = V$$

$$\boxed{\text{(A) V}}$$

Key Points

- The RMS value measures the equivalent heating power of a signal, calculated as the square root of the mean of the squared signal over one period.
- If the absolute magnitude of a signal is constant (e.g., it only switches between $+V$ and $-V$), its squared waveform is a constant flat DC line at V^2. Consequently, its RMS value is simply the magnitude V, regardless of the widths T_1 and T_2.

Q1.92.1 MSQ | 1992 | 2M | Ans: A,B,D | ● ○ ○

Signal (a):

$$s(t) = \cos(2t) + \cos(3t) + \cos(5t)$$

This signal is periodic as the ratio of any two frequencies $= m/n$ where m and n are integers. Note that period is 2π and fundamental frequency is 1 rad/sec.

Signal (b):

$$s(t) = e^{j8\pi t}$$

This signal is periodic with $\omega = 8\pi$.

Signal (c):

$$s(t) = e^{-7t}\sin(10\pi t)$$

Here, the term e^{-7t} is not periodic, therefore the entire signal is not periodic.

Signal (d):

$$s(t) = \cos(2t)\cos(4t)$$

Using the trigonometric identity $\cos(A)\cos(B) = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$:

$$= \frac{1}{2}[\cos(6t) + \cos(2t)]$$

This is periodic with fundamental frequency = 2 rad/sec.

$\therefore$ Signals (a), (b) & (d) are periodic.

$$\boxed{\text{Signals (a), (b) \& (d) are periodic.}}$$

Key Points

- A continuous-time signal consisting of a sum of sinusoids is periodic if and only if the ratio of any two component frequencies is a rational number (m/n).
- The fundamental frequency of the sum of periodic signals is the greatest common divisor (GCD) of the individual fundamental frequencies.
- Signals modulated by a decaying exponential component (e.g., e^{-at}) are strictly non-periodic.

DEBUG INFO

Chapter I

Basics of Signals

Total Questions : 14

MCQ : 8

MSQ : 2

NAT : 4

PrepFusion

SOLUTIONS

II

Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Q2.25.1 MCQ 2025 1M Ans: D

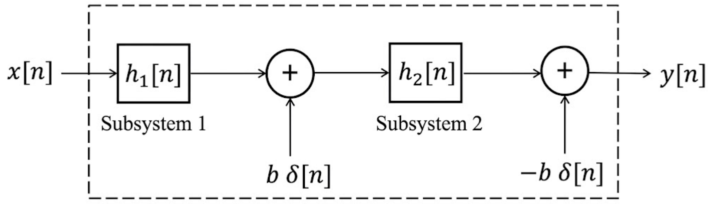


Fig. Solution Q2.25.1

The output from Subsystem 1 is given by $y_1[n] = x[n] * h_1[n]$.
According to the block diagram, the intermediate signal before Subsystem 2 is $\{y_1[n] + b\delta[n]\}$. Therefore, the output of Subsystem 2 is:

$$y_2[n] = \{y_1[n] + b\delta[n]\} * h_2[n]$$

The overall system output $y[n]$ is obtained by subtracting $b\delta[n]$ from $y_2[n]$:

$$y[n] = \{x[n] * h_1[n] + b\delta[n]\} * h_2[n] - b\delta[n]$$

Expanding the convolution:

$$y(n) = \underbrace{x[n] * h_1[n] * h_2[n]}_{\text{This is LTI}} + \underbrace{bh_2[n] - b\delta[n]}_{\text{N.L. terms}}$$

Given that $h_2[n] \neq \delta[n]$, the term $bh_2[n] - b\delta[n]$ does not vanish. Since $b > 0$, this constant additive term (independent of input $x[n]$) makes the system non-linear.
Furthermore, if we consider a case like $h_2[n] = nu[n]$, then:

$$y[n] \text{ is a variable function of time} \Rightarrow \text{Time variant}$$

Thus, the system is non-linear and time variant.

$$\boxed{\text{D) Non linear and Time Variant}}$$

Key Points

- A system is Linear if it satisfies the properties of superposition and homogeneity. An additive term independent of the input (zero-input response) violating $y[n] = 0$ for $x[n] = 0$ makes it non-linear.
- A system is Time-Invariant if a shift in the input $x[n-k]$ results in an identical shift in the output $y[n-k]$.
- Convolution property: $x[n] * \delta[n] = x[n]$.

Mistakes to be avoided

- Do not assume the system is LTI just because the internal subsystems are LTI; the addition of external impulses $b\delta[n]$ can change the system characteristics.

Q2.23.1 MCQ 2023 1M Ans: B

The given system relationship is:

$$y(t) = x(e^t)$$

To check for causality, we examine the output at a specific time. For example, at $t = 0$:

$$y(0) = x(e^0) = x(1)$$

Since the present value of the output ($t = 0$) depends on a future value of the input ($t = 1$), the system is non-causal.
To check for time-invariance, we compare the response to a delayed input with a delayed version of the original output. Let t_0 be a non-zero delay.

Step 1: Delay the input Let $y_1(t)$ be the output when the input is $x(t - t_0)$:

$$y_1(t) = x(e^t - t_0) \quad \ldots (i)$$

Step 2: Delay the output Let $y_2(t)$ be the original output delayed by t_0:

$$y_2(t) = y(t - t_0) = x(e^{t-t_0}) \quad \ldots (ii)$$

Comparing equations (i) and (ii), we see that:

$$x(e^t - t_0) \neq x(e^{t-t_0})$$

Since $y_1(t) \neq y_2(t)$, the system is time variant.
Combining these properties, the system is non-causal and time varying.

$$\boxed{\text{B) Non-causal and time varying}}$$

Key Points

- **Causality:** A system is causal if the output at any time t depends only on values of the input at the present time and/or in the past. If it depends on future values, it is non-causal.
- **Time-Invariance:** A system is time-invariant if a time shift in the input signal results in an identical time shift in the output signal. Mathematically, $\mathcal{T}\{x(t - t_0)\} = y(t - t_0)$.

Mistakes to be avoided

- When checking for time-invariance, ensure you replace every t in the input argument with $(t - t_0)$ for the delayed output, but only subtract t_0 from the existing argument for the delayed input.
- For causality, testing only positive values of t might be misleading; always check $t = 0$ or negative values if the function involves exponents or squares.

Q2.22.1 MCQ 2022 2M Ans: C,D ● ○ ○

Given the system mappings:

$$\sin(t) \xrightarrow{S_1} \sin(-t)$$
$$\sin(t) \xrightarrow{S_2} \sin(t + 1)$$
$$\sin(t) \xrightarrow{S_3} \sin(2t)$$
$$\sin(t) \xrightarrow{S_4} \sin^2(t)$$

For an LTI system, the input & output frequency must be the same with amplitude & phase change. S_1 & S_2 can be LTI.

For S_3:

$$x(t) = \sin t \longrightarrow y(t) = \sin 2t$$

I/P frequency = 1 rad/sec O/P frequency = 2 rad/sec $\longrightarrow$ Definitely not LTI

For S_4:

$$x(t) = \sin t \longrightarrow y(t) = \sin^2 t = \frac{1 - \cos 2t}{2}$$

$x(t)$ is input frequency = 1 rad/sec $y(t)$ is output frequency $\Rightarrow \omega = 0$ & 2 rad/sec $\longrightarrow$ Definitely not LTI

S_3 and S_4 are definitely not LTI.

$$\boxed{S_3 \text{ and } S_4}$$

Key Points

- An LTI system cannot generate new frequency components. It can only modify the amplitude and phase of the existing input frequencies.
- If the input is $x(t) = A\sin(\omega_0 t)$, the output of an LTI system must be of the form $y(t) = B\sin(\omega_0 t + \phi)$.

Mistakes to be avoided

- Assuming non-linear operations like squaring a signal (e.g., $\sin^2 t$) or time-variant operations like time-scaling (e.g., $\sin 2t$) are LTI. These operations produce new frequencies (like DC and harmonics), violating the core properties of LTI systems.

Q2.21.1 NAT | 2021 | 2M | Ans: 0 | ● ○ ○

The impulse response $h(n)$ can be determined from the unit step response $s(n)$ using the relation:

$$h(n) = s(n) - s(n-1)$$

Given the unit step response $s(n) = 2\delta(n+1) + \delta(n) + \delta(n-1)$, we substitute this into the equation:

$$h(n) = 2\delta(n+1) + \delta(n) + \delta(n-1) - [2\delta(n) + \delta(n-1) + \delta(n-1-1)]$$

$$h(n) = 2\delta(n+1) + \delta(n) + \delta(n-1) - 2\delta(n) - \delta(n-1) - \delta(n-2)$$

$$h(n) = 2\delta(n+1) - \delta(n) - \delta(n-2)$$

For the input $x_1(n) = (1/2)^n u(n)$, the output $y_1(n)$ is given by the convolution of the input and the impulse response:

$$y_1(n) = x_1(n) * h(n)$$

$$y_1(n) = \left(\frac{1}{2}\right)^n u(n) * [2\delta(n+1) - \delta(n) - \delta(n-2)]$$

$$y_1(n) = \left(\frac{1}{2}\right)^n u(n) * 2\delta(n+1) - \left(\frac{1}{2}\right)^n u(n) * \delta(n) - \left(\frac{1}{2}\right)^n u(n) * \delta(n-2)$$

Applying the shifting property of convolution with an impulse function, $x(n) * \delta(n-k) = x(n-k)$:

$$y_1(n) = 2\left(\frac{1}{2}\right)^{n+1} u(n+1) - \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{2}\right)^{n-2} u(n-2)$$

To find the value at $n = 0$, we evaluate the expression:

$$y_1(n)|_{n=0} = 2\left(\frac{1}{2}\right)^1 u(1) - \left(\frac{1}{2}\right)^0 u(0) - \left(\frac{1}{2}\right)^{-2} u(-2)$$

Since $u(1) = 1$, $u(0) = 1$, and $u(-2) = 0$:

$$= 2\left(\frac{1}{2}\right) - 1 - 0$$

$$= 1 - 1$$

$$y_1(0) = 0$$

$$\boxed{0}$$

Key Points

- The impulse response $h(n)$ of a discrete-time LTI system is the first difference of its unit step response $s(n)$: $h(n) = s(n) - s(n-1)$.
- Shifting property of convolution: Convolving any discrete signal $x(n)$ with a shifted impulse function $\delta(n-k)$ results in the shifted signal $x(n-k)$.
- The unit step sequence $u(n)$ is defined as 1 for $n \geq 0$ and 0 for $n < 0$. Therefore, causal signals evaluated at negative indices (like $u(-2)$) are zero.

Q2.20.1 MCQ 2020 1M Ans: A ● ○ ○

It is given that

$$y(n) = \max_{-\infty \leq k \leq n} |x(k)|$$

We need to obtain its unit impulse response.

If the input is a unit impulse, i.e., $x(k) = \delta(k)$, then the output is the impulse response $y(n) = h(n)$.

So replacing $x(k)$ by $\delta(k)$ and $y(n)$ by $h(n)$ in the above equation we have:

$$h(n) = \max_{-\infty \leq k \leq n} |\delta(k)|$$

Evaluating for $n = -2$:

$$h(-2) = \max_{-\infty \leq k \leq -2} |\delta(k)|$$

$$= \max[|\delta(-\infty)|, \ldots, |\delta(-3)|, |\delta(-2)|]$$

$$= \max[|0|, |0|, |0|, \ldots, |0|] = 0$$

From this we can say if $n \leq -1$ then $h(n) = 0$.

If $n \geq 0$ then $h(n) = 1$ for all n. For example, evaluating at $n = 5$:

$$h(5) = \max_{-\infty \leq k \leq 5} |\delta(k)|$$

$$= \max[\ldots, |\delta(-11)|, |\delta(0)|, |\delta(1)|, |\delta(2)|, |\delta(3)|, |\delta(4)|, |\delta(5)|]$$

$$= \max[\ldots, |0|, |1|, |0|, |0|, |0|, |0|, |0|] = \max[0, 1, 0, 0, 0] = 1$$

Overall:

$$h(n) = 0; \quad n < 0$$

$$h(n) = 1; \quad n \geq 0$$

$$\Rightarrow h(n) = u(n) \text{ i.e., unit step signal}$$

Hence the correct answer is option (A).

$\boxed{\text{A) unit step signal u[n].}}$

Key Points

- The impulse response $h(n)$ of a system is the output produced when the input is a discrete-time unit impulse $\delta(n)$.
- The unit impulse sequence $\delta(n)$ is defined as 1 at $n = 0$ and 0 for all $n \neq 0$.
- The unit step sequence $u(n)$ is defined as 1 for $n \geq 0$ and 0 for $n < 0$.

Q2.18.1 MCQ | 2018 | 1M | Ans: C | ● ○ ○

If we observe all the 4 options carefully, we can notice option C (i.e., $y = au + b$, $b \neq 0$) is a non-linear system due to the addition of non-zero constant b, since it does not satisfy homogeneity and additivity as follows:

$$[au_1 + b] + [au_2 + b] \neq a(u_1 + u_2) + b$$

$$a(ku) + b \neq k[au + b]$$

Option A is a linear differential equation with constant co-efficients, so it is linear.
Integrator system is generally linear, so option B represents a linear system.
$y = au$, is a perfect linear system.

$$\boxed{\text{C) } y = au + b, b \neq 0}$$

Key Points

- A system is considered linear if and only if it satisfies the principle of superposition, which consists of two properties: additivity and homogeneity.
- **Additivity:** The response to $x_1(t) + x_2(t)$ must be $y_1(t) + y_2(t)$.
- **Homogeneity (Scaling):** The response to $k \cdot x(t)$ must be $k \cdot y(t)$.
- Systems modeled by equations of the form $y = mx + c$ (where $c \neq 0$) are non-linear because the constant c prevents the system from satisfying either the additivity or homogeneity conditions.

Mistakes to be avoided

- **Confusing geometric linearity with system linearity:** A straight-line equation like $y = mx + c$ is often called a "linear equation" in basic algebra, but in signals and systems, it represents an *affine* system. It is non-linear unless $c = 0$.

Q2.17.1 MCQ | 2017 | 1M | Ans: B | ● ○ ○

Given the system:

$$y(t) = \int_{t-T}^{t} x(u)du$$

Check for Linearity: The system is linear, as it satisfies both the superposition and scaling properties (integration is a linear operation).
Check for Time Invariance: Let the response to a delayed input $x(t-\tau)$ be $y_1(t)$:

$$y_1(t) = \int_{t-T}^{t} x(u-\tau)du$$

Let $u - \tau = \lambda$ Then, $du = d\lambda$ Adjusting the limits: When $u = t - T \implies \lambda = t - T - \tau$ When $u = t \implies \lambda = t - \tau$

$$y_1(t) = \int_{t-T-\tau}^{t-\tau} x(\lambda)d\lambda \quad \ldots\ldots(1)$$

Now, find the delayed output $y(t-\tau)$ by replacing t with $t-\tau$ in the original equation:

$$y(t-\tau) = \int_{t-\tau-T}^{t-\tau} x(u)du \quad \ldots\ldots(2)$$

From equations (1) and (2), we can observe that:

$$y_1(t) = y(t-\tau)$$

Since the response to the delayed input is equal to the delayed response, the system is **Time Invariant**.
Therefore, the system is linear and time invariant.

$$\boxed{\text{B) linear and time invariant}}$$

Key Points

- **Linearity:** A system is linear if it obeys the principle of superposition, which requires both additivity and homogeneity (scaling). Integration inherently satisfies these properties.
- **Time Invariance:** A system is time-invariant if a time shift in the input signal results in an identical time shift in the output signal. The test is to show that $y(x(t-\tau)) = y(t-\tau)$.
- A system that computes the definite integral of an input over a sliding window of constant width T is a classic example of a Linear Time-Invariant (LTI) system.

Mistakes to be avoided

- **Forgetting to change integration limits:** When evaluating the delayed input response $y_1(t)$ using the substitution $u - \tau = \lambda$, a very common error is failing to apply the substitution to the upper and lower limits of the integral.
- **Incorrectly shifting the dummy variable:** When calculating the delayed output $y(t-\tau)$, only the independent variable t in the integration limits should be replaced with $t - \tau$. The dummy variable of integration (u) must remain unchanged.

Q2.17.2 MCQ 2017 1M Ans: A ● ○ ○

The given system is

$$y(n) = \begin{cases} n\,x(n), & 0 \le n \le 10 \\ x(n) - x(n-1), & \text{otherwise} \end{cases}$$

→ Present output depends on present input and past input, so it is a causal system
→ For a bounded input, bounded output yields, so it is a stable system

$\boxed{\text{A) It is causal and stable}}$

Key Points

- **Causality:** A discrete-time system is causal if its output $y(n)$ at any time n depends only on present and/or past values of the input (e.g., $x(n)$, $x(n-1)$). It must not depend on future values (e.g., $x(n+1)$).
- **BIBO Stability:** A system is Bounded-Input Bounded-Output (BIBO) stable if every bounded input sequence (i.e., $|x(n)| \le M_x < \infty$) produces a bounded output sequence (i.e., $|y(n)| \le M_y < \infty$).

Mistakes to be avoided

- **Misjudging the $n\,x(n)$ term:** It is tempting to declare the system unstable because multiplying an input by n generally leads to unbounded growth as $n \to \infty$. However, pay close attention to the piecewise condition: n is restricted to a finite interval ($0 \le n \le 10$). The maximum scaling factor is 10, thus keeping the output bounded.
- **Confusing causality with memory:** The term $x(n-1)$ indicates the system has memory (it is dynamic), but because it does not look forward in time, it remains perfectly causal.

Q2.17.3 NAT 2017 2M Ans: 7 ● ○ ○

Given that,

$$h_1(t) = 2\delta(t+2) - 3\delta(t+1)$$

$$h_2(t) = \delta(t-2)$$

Overall impulse response is,

$$h(t) = h_1(t) + h_2(t)$$

$$= 2\delta(t+2) - 3\delta(t+1) + \delta(t-2)$$

If input, $x(t) = u(t)$, then the output will be,

$$y(t) = x(t) * h(t)$$
$$= u(t) * [2\delta(t+2) - 3\delta(t+1) + \delta(t-2)]$$
$$= 2u(t+2) - 3u(t+1) + u(t-2)$$

By plotting $y(t)$, we get,

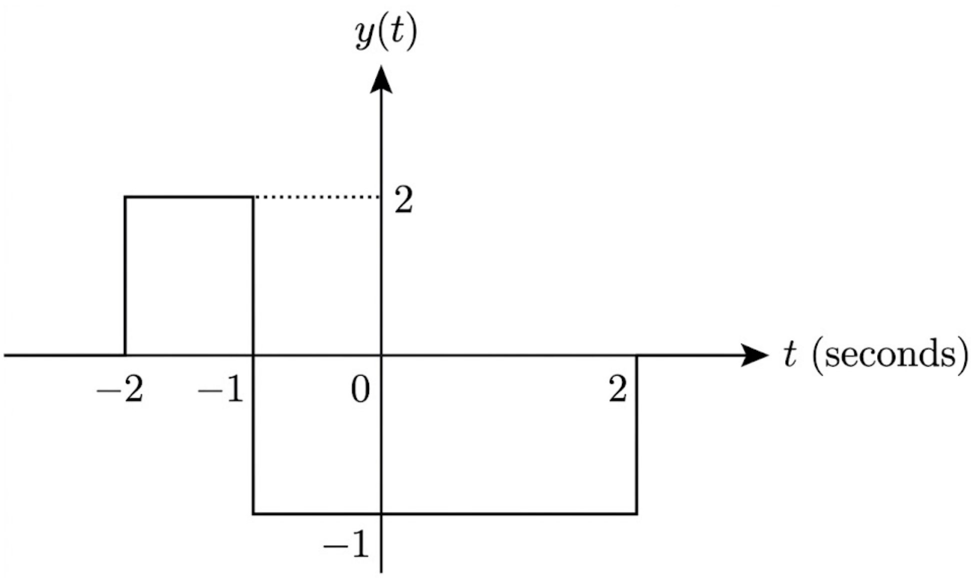


Fig. Solution Q2.17.3

The energy of $y(t)$ can be given as,

$$E_y = \int_{-\infty}^{\infty} y^2(t)dt$$

From the plot, we can see the signal is non-zero only in two intervals: Amplitude is 2 for $-2 < t < -1$ (Duration = 1) Amplitude is -1 for $-1 < t < 2$ (Duration = 3)

Expanding the integral:

$$E_y = \int_{-2}^{-1} (2)^2 dt + \int_{-1}^{2} (-1)^2 dt$$
$$= 4[t]_{-2}^{-1} + 1[t]_{-1}^{2}$$
$$= 4(-1-(-2)) + 1(2-(-1))$$
$$= 4(1) + 1(3)$$
$$= 4 + 3 = 7$$

$\boxed{7}$

Key Points

- **Parallel Systems:** The equivalent impulse response of two LTI systems connected in parallel is the sum of their individual impulse responses: $h(t) = h_1(t) + h_2(t)$.
- **Convolution with Impulse:** Convolving a signal with a shifted impulse simply shifts the signal: $x(t) * \delta(t-t_0) = x(t-t_0)$.

- **Signal Energy:** The total energy of a real-valued continuous-time signal $y(t)$ is defined as $E = \int_{-\infty}^{\infty} y^2(t)dt$. For piecewise constant signals, this simplifies to the sum of (amplitude)^2 × (duration) for each rectangular segment.

Mistakes to be avoided

- **Incorrect interval durations:** When evaluating piecewise integrals from a plot, always carefully check the t-axis bounds. For example, the interval from −1 to 2 has a length of 3, not 1 or 2.
- **Sign errors in energy calculation:** Energy is the integral of the *squared* magnitude of the signal. A common pitfall is treating negative amplitudes as negative energy areas. The square of a negative amplitude is always positive (e.g., $(-1)^2 = 1$), meaning all non-zero segments *add* to the total energy.
- **Convolution limits:** Don't forget that a step function $u(t+2)$ turns on at $t = -2$. Ensure the plot lines up exactly with the shifted $u(t)$ sequence.

Q2.17.4 NAT 2017 2M Ans: 31 ● ○ ○

Given the discrete-time sequences:

$$x[n] = \{1, 2, 1\}$$

$$h[n] = \{1, a, b\}$$

$$y[n] = \{A, 3, 4, B, C\}$$

Using the tabular method for linear convolution, we set up a matrix multiplying the elements of x and h:

$x \backslash h$	1	a	b
1	1	a	b
2	2	$2a$	$2b$
1	1	a	b

By summing the anti-diagonals, we obtain the individual elements of the output sequence $y[n]$:

$$y[0] = 1$$

$$y[1] = 2 + a$$

$$y[2] = 1 + 2a + b$$

$$y[3] = a + 2b$$

$$y[4] = b$$

From the given sequence $y[n]$, we know that $y[1] = 3$ and $y[2] = 4$.

$$\text{Given, } y[1] = 2 + a = 3 \implies a = 1$$

$$\text{Given, } y[2] = 1 + 2a + b = 4 \implies 1 + 2(1) + b = 4 \implies b = 1$$

Now, substituting $a = 1$ and $b = 1$ into the expressions for the remaining terms:

$$\text{So, } y[3] = a + 2b = 1 + 2(1) = 3$$

$$\text{and } y[4] = b = 1$$

Finally, we evaluate the required expression:

$$10y[3] + y[4] = 10 \times 3 + 1 = 31$$

$$\boxed{31}$$

Key Points

- **Length of Convolution:** For two discrete-time signals of finite lengths L and M, their linear convolution results in a sequence of length $N = L + M - 1$. Here, both sequences have a length of 3, so the output length is $3 + 3 - 1 = 5$.
- **Tabular Method:** A highly efficient and organized graphical method to compute linear convolution. The sequences form the axes of a grid, their cross-products fill the cells, and summing along the anti-diagonals directly yields the convolution sum $y[n] = \sum x[k]h[n-k]$.

Mistakes to be avoided

- **Misaligning Anti-Diagonals:** The most common error in the tabular method is summing elements across incorrect diagonal paths. Always ensure you are summing purely bottom-left to top-right diagonals to represent the constant (n) index in the convolution sum.
- **Cascading Arithmetic Errors:** Because the system of equations is solved sequentially (finding a first, then using it to find b), a simple sign or addition error early on will invalidate the entire rest of the sequence calculation. Double-check the first substitution step.

Q2.15.1

MCQ | 2015 | 1M | Ans: B | ● ○ ○

For a Linear Time-Invariant (LTI) system, the relationship between different standard responses is governed by the derivative/integral properties of the input signals.
Given that the unit impulse function $\delta(t)$ is the derivative of the unit step function $u(t)$, the impulse response $h(t)$ is the derivative of the step response $y(t)$.

$$IR = \frac{d}{dt}(SR)$$

Step response,

$$y(t) = \int_{-\infty}^{t} h(\tau) \cdot d\tau$$

Impulse response,

$$h(t) = \frac{d}{dt} y(t)$$

B) Differentiating the unit step response

Key Points

- The impulse response $h(t)$ is the derivative of the step response $s(t)$.
- The step response $s(t)$ is the derivative of the ramp response $r(t)$.
- Therefore, the impulse response can also be obtained by double-differentiating the ramp response.

Q2.15.2

MCQ | 2015 | 1M | Ans: D | ● ○ ○

The given expression involves the convolution of a signal with a shifted and scaled impulse function.
To solve this, we utilize the property of the Dirac delta function under time scaling:

$$\delta(at + b) = \frac{1}{|a|}\delta\left(t + \frac{b}{a}\right)$$

For the given impulse $\delta(-t - t_0)$, where $a = -1$:

$$\delta(-(t + t_0)) = \frac{1}{|-1|}\delta(t + t_0) = \delta(t + t_0)$$

Now, applying the convolution property with a shifted impulse, $x(t) * \delta(t - t_s) = x(t - t_s)$:

$$x(-t) * \delta(-t - t_0) = x(-t) * \delta(t + t_0)$$

By replacing t with $t + t_0$ in the signal $x(-t)$, we get:

$$x(-(t + t_0)) = x(-t - t_0)$$

$$\boxed{\text{D) } x(-t - t_0)}$$

Key Points

- **Even Property of Impulse:** The Dirac delta function is an even function, $\delta(t) = \delta(-t)$.
- **Convolution with Impulse:** Convolving any signal $f(t)$ with $\delta(t - t_0)$ results in the signal shifted to the location of the impulse: $f(t) * \delta(t - t_0) = f(t - t_0)$.

Mistakes to be avoided

- **Incorrect Time Shifting:** Avoid confusing $x(-(t + t_0))$ with $x(-t + t_0)$. When substituting $t \to t + t_0$ in $x(-t)$, the negative sign must distribute over the entire bracket.
- **Impulse Scaling:** Remember that $\delta(-t)$ is identical to $\delta(t)$, but this does not mean the shift t_0 inside the argument changes its sign relative to t unless the scaling property is applied carefully.

Q2.15.3 | NAT | 2015 | 2M | Ans: 1.5 | ● ○ ○

The energy of a discrete-time sequence $x[n]$ is defined as the sum of the squared magnitudes of its samples.

Energy of $x_1[n] = \sum_{n=-\infty}^{\infty} |x_1[n]|^2$

$$= \sum_{n=0}^{\infty} \alpha^2 \left(\frac{1}{4}\right)^n = \alpha^2 \cdot \frac{1}{1 - \frac{1}{4}} = \alpha^2 \cdot \frac{4}{3}$$

Energy of $x_2[n] = \sum_{n=-\infty}^{\infty} |x_2[n]|^2$

$$= 1.5 + 1.5 = 3$$

Given that both sequences have the same energy:

$$\alpha^2 \frac{4}{3} = 3$$

$$\alpha^2 = \frac{9}{4}$$

$$\alpha = 1.5$$

$$\boxed{1.5}$$

Key Points

- The energy E of a discrete signal $x[n]$ is calculated using $E = \sum_{n=-\infty}^{\infty} |x[n]|^2$.
- For an infinite geometric series $\sum_{n=0}^{\infty} ar^n$, the sum is $\frac{a}{1-r}$ provided $|r| < 1$.

Mistakes to be avoided

- When calculating energy, ensure you square the entire expression, including the scaling constant α.
- Check the limits of the summation; for sequences involving $u[n]$, the lower limit typically starts at $n = 0$.

Q2.14.1

NAT | 2014 | 2M | Ans: 10 | ● ○ ○

Method 1

The discrete-time convolution of a signal $x[n]$ with itself is given by the summation:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]x[n-k]$$

Given the signal definition:

$$x[n] = \begin{cases} n & \text{for } 0 \le n \le 10 \\ 0 & \text{otherwise} \end{cases}$$

To find $y[4]$, we evaluate the sum for $n = 4$. Note that $x[k]$ is non-zero only for $0 \le k \le 10$, and $x[4-k]$ is non-zero only when $0 \le 4-k \le 10$, which implies $k \le 4$. Thus, the effective summation limits are from $k = 0$ to $k = 4$:

$$y[4] = \sum_{k=0}^{4} x[k]x[4-k]$$

$$y[4] = x[0]x[4] + x[1]x[3] + x[2]x[2] + x[3]x[1] + x[4]x[0]$$

Substituting the values $x[0] = 0, x[1] = 1, x[2] = 2, x[3] = 3, x[4] = 4$:

$$y[4] = (0 \times 4) + (1 \times 3) + (2 \times 2) + (3 \times 1) + (4 \times 0)$$

$$y[4] = 0 + 3 + 4 + 3 + 0$$

$$y[4] = 10$$

Method 2

Using the Matrix Method for convolution, we can represent $y = x * h$ (where $h = x$) as a matrix multiplication. To find the output up to index $n = 4$, we only need the input samples up to index 4. The convolution matrix formulation is:

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \end{bmatrix} = \begin{bmatrix} x[0] & 0 & 0 & 0 & 0 \\ x[1] & x[0] & 0 & 0 & 0 \\ x[2] & x[1] & x[0] & 0 & 0 \\ x[3] & x[2] & x[1] & x[0] & 0 \\ x[4] & x[3] & x[2] & x[1] & x[0] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \\ x[4] \end{bmatrix}$$

Substituting the sequence values $\{0, 1, 2, 3, 4\}$:

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ 4 & 3 & 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Extracting the 5th row to calculate $y[4]$:

$$y[4] = (4 \times 0) + (3 \times 1) + (2 \times 2) + (1 \times 3) + (0 \times 4)$$

$$y[4] = 0 + 3 + 4 + 3 + 0 = 10$$

$$\boxed{10}$$

Key Points

- The discrete-time convolution of two causal signals $x[n]$ and $h[n]$ evaluated at a specific index N only requires the signal samples from $n = 0$ to $n = N$.
- $y[n] = \sum_{k=0}^{n} x[k]h[n-k]$ for causal systems.
- The Matrix method uses a Toeplitz matrix formed by shifted versions of the input sequence multiplied by the impulse response vector, providing a structured approach for finite-length sequences.

Mistakes to be avoided

- Ensure the limits of summation are correctly identified by checking where both $x[k]$ and $x[n-k]$ are simultaneously non-zero.
- Do not forget to include the $k = 0$ term; even if $x[0] = 0$ in this specific case, it is part of the formal definition.

Q2.13.1 MCQ 2013 1M Ans: C ● ○ ○

The overall impulse response $h(t)$ of the cascade system is given by:

$$h(t) = h_1(t) * h_2(t)$$

$$\boxed{\text{C) Convolution of } h_1(t) \text{ and } h_2(t)}$$

Key Points

- When two Linear Time-Invariant (LTI) systems are connected in cascade (series), the overall impulse response is the convolution of their individual impulse responses.
- Due to the commutative property of convolution, the order in which LTI systems are cascaded does not affect the overall impulse response ($h_1(t) * h_2(t) = h_2(t) * h_1(t)$).

Mistakes to be avoided

- **Addition vs. Convolution:** Do not add the impulse responses. Addition, $h(t) = h_1(t) + h_2(t)$, is the overall response for systems connected in parallel, not cascade.
- **Multiplication Error:** Do not multiply the signals in the time domain. Time-domain multiplication $h_1(t) \cdot h_2(t)$ corresponds to convolution in the frequency domain, which represents a completely different system operation.

Q2.13.2 MCQ 2013 2M Ans: B ● ○ ○

The step response $s(t)$ of a continuous-time LTI system is obtained by integrating its impulse response $h(t)$:

$$s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$$

Given the impulse response:

$$h(t) = \delta(t-1) + \delta(t-3)$$

Integrating $h(t)$ yields the step response. Since the integral of a shifted impulse function $\delta(t - t_0)$ is a shifted unit step function $u(t - t_0)$, we have:

$$s(t) = u(t-1) + u(t-3)$$

To find the value of the step response at $t = 2$, we substitute $t = 2$ into the equation:

$$s(2) = u(2-1) + u(2-3)$$

$$s(2) = u(1) + u(-1)$$

By the definition of the unit step function, $u(t) = 1$ for $t > 0$ and $u(t) = 0$ for $t < 0$:

$$u(1) = 1$$

$$u(-1) = 0$$

Substituting these values back:

$$s(2) = 1 + 0 = 1$$

$$\boxed{\text{B) } 1}$$

Key Points

- The relationship between the step response $s(t)$ and impulse response $h(t)$ is given by $s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$.
- The integral of a delayed Dirac delta function is a delayed unit step function: $\int_{-\infty}^{t} \delta(\tau - t_0)\, d\tau = u(t - t_0)$.
- The unit step function $u(t)$ has a value of 1 for all positive time $t > 0$ and 0 for all negative time $t < 0$.

Q2.12.1 MCQ 2012 2M Ans: D ● ○ ○

To determine the properties of the system, we need to test for time-invariance and BIBO (Bounded-Input Bounded-Output) stability separately.

1. Time-Invariance Check: A system is time-invariant if delaying the input by t_0 delays the output by exactly the same amount t_0. Let the response to an input $x(t)$ be $y(t)$:

$$y(t) = \int_{-\infty}^{t} x(\tau) \cos(3\tau) d\tau$$

Now, let's delay the input by t_0, giving a new input $x_1(t) = x(t - t_0)$. The corresponding output $y_1(t)$ is:

$$y_1(t) = \int_{-\infty}^{t} x_1(\tau) \cos(3\tau) d\tau = \int_{-\infty}^{t} x(\tau - t_0) \cos(3\tau) d\tau$$

To compare this with a delayed output, let's perform a change of variables in the integral: let $u = \tau - t_0$, so $d\tau = du$, and the upper limit becomes $t - t_0$:

$$y_1(t) = \int_{-\infty}^{t-t_0} x(u) \cos(3(u + t_0)) du$$

Next, we determine the actual delayed output $y(t - t_0)$ by simply replacing t with $t - t_0$ in the original system equation:

$$y(t - t_0) = \int_{-\infty}^{t-t_0} x(\tau) \cos(3\tau) d\tau$$

Comparing the two results, we see that $y_1(t) \neq y(t-t_0)$ because of the presence of the $\cos(3(u+t_0))$ term instead of $\cos(3\tau)$. Thus, the system is **not time-invariant** (it is time-varying).

2. Stability Check: A system is BIBO stable if every bounded input results in a bounded output. To prove a system is unstable, we only need to find one bounded input that produces an unbounded output. Let's choose a bounded input signal: $x(t) = \cos(3t)u(t)$. Note that $|x(t)| \leq 1$ for all time, making it bounded. Substituting this input into the system equation for $t > 0$:

$$y(t) = \int_{0}^{t} \cos(3\tau) \cos(3\tau) d\tau = \int_{0}^{t} \cos^2(3\tau) d\tau$$

Using the trigonometric identity $\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$:

$$y(t) = \int_{0}^{t} \frac{1 + \cos(6\tau)}{2} d\tau = \left[\frac{\tau}{2} + \frac{\sin(6\tau)}{12}\right]_0^t$$

$$y(t) = \frac{t}{2} + \frac{\sin(6t)}{12}$$

As $t \to \infty$, the term $\frac{t}{2} \to \infty$. Thus, the output $y(t)$ grows unbounded even though the input was strictly bounded. Therefore, the system is **not stable**.

$$\boxed{\text{D) Not time-invariant and not stable}}$$

Key Points

- **Time-Invariance:** A system $y(t) = \mathcal{T}\{x(t)\}$ is time-invariant if $\mathcal{T}\{x(t-t_0)\} = y(t-t_0)$. The presence of an explicit function of time (like $\cos(3t)$ or t) directly multiplying the input variable inside the integral generally indicates a time-varying system.
- **BIBO Stability:** A system is BIBO stable if $|y(t)| \le M_y < \infty$ for every possible input where $|x(t)| \le M_x < \infty$. A pure integration operation without a decaying mechanism is inherently prone to instability for certain inputs.

Mistakes to be avoided

- **Incorrect shift application:** When checking time-invariance, carefully evaluate $y_1(t)$ by substituting $x(t-t_0)$ into the integral without altering the other explicit t or τ variables, and then separately evaluate $y(t-t_0)$ by replacing the upper bound.
- **Assuming integral boundedness:** Do not assume an integral of a bounded function is bounded over an infinite interval. For example, integrating a constant yields a ramp function, which is unbounded.

Q2.12.2 MCQ 2012 1M Ans: A ● ○ ○

Method 1

$$y[n] = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k g(n-k)$$

$$y[0] = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k g(-k) = 1$$

$$\implies \left(\frac{1}{2}\right)^0 g(0) = 1$$

$$\implies g(0) = 1$$

Since $g(n)$ is Causal sequence

$$g(-1), g(-2), \ldots\ldots\ldots\ldots = 0$$

$$y[1] = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k g[1-k]$$

$$\implies \left(\frac{1}{2}\right)^0 g[1] + \left(\frac{1}{2}\right)^1 g(0) = \frac{1}{2}$$

$$g[1] = 0$$

Method 2

Using the Matrix Method for the convolution of causal sequences, the relationship up to index $n = 1$ can be expressed as:

$$\begin{bmatrix} y[0] \\ y[1] \end{bmatrix} = \begin{bmatrix} h[0] & 0 \\ h[1] & h[0] \end{bmatrix} \begin{bmatrix} g[0] \\ g[1] \end{bmatrix}$$

Substitute the known values into the matrix equation:

$$\begin{bmatrix} 1 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix} \begin{bmatrix} g[0] \\ g[1] \end{bmatrix}$$

Expanding the matrix multiplication row by row: From the first row, we get:

$$1 = 1 \cdot g[0] + 0 \cdot g[1] \implies g[0] = 1$$

From the second row, we get:

$$\frac{1}{2} = \frac{1}{2} \cdot g[0] + 1 \cdot g[1]$$

Substitute $g[0] = 1$ into this equation:

$$\frac{1}{2} = \frac{1}{2}(1) + g[1] \implies g[1] = 0$$

$$\boxed{\text{A) 0}}$$

Key Points

- A causal sequence is defined as a sequence that is zero for all negative indices: $g[n] = 0$ for $n < 0$.
- When performing discrete convolution $y[n] = \sum h[k]g[n-k]$ with a causal signal $g[n]$, the term $g[n-k]$ forces the non-zero terms to satisfy $n - k \geq 0$, which means $k \leq n$.
- This limits the upper bound of the summation to n (i.e., k goes from 0 to n), simplifying the evaluation for specific points like $n = 0$ and $n = 1$.

Mistakes to be avoided

- For causal sequences, the convolution sum simplifies strictly to $y[n] = \sum_{k=0}^{n} h[k]x[n-k]$, meaning the output at time n depends only on inputs and impulse responses up to time n.
- The matrix formulation converts the convolution operation into a system of linear equations, which is highly effective for discrete deconvolution (finding an unknown input sequence when the output and impulse response are known).

Q2.11.1 MCQ 2011 1M Ans: B ● ○ ○

$$h(n) = 2^n u(n-2)$$

For causal system $h(n) = 0$ for $n < 0$

Hence given system is causal.

For stability:

$\sum_{n=2}^{\infty} 2^n = \infty$, so given system is not stable.

$$\boxed{\text{B) causal but not stable}}$$

Key Points

- **Causality:** A discrete-time LTI system is causal if its impulse response satisfies $h(n) = 0$ for all $n < 0$. The presence of $u(n-2)$ ensures $h(n)$ is zero for $n < 2$, fulfilling this condition.
- **Stability:** A discrete-time LTI system is bounded-input bounded-output (BIBO) stable only if its impulse response is

absolutely summable: $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$. An exponentially growing function like 2^n will diverge, making the system unstable.

Mistakes to be avoided

- Do not assume a system is non-causal just because of a time-shifted step function. A delay $(n-2)$ keeps the non-zero values in the positive time domain $(n \geq 2)$.
- Be careful to evaluate the summation bounds correctly. The unit step $u(n-2)$ changes the lower limit of the summation from $-\infty$ to 2.

Q2.10.1 MCQ 2010 1M Ans: C ● ○ ○

When two discrete-time Linear Time-Invariant (LTI) systems are connected in cascade, their overall equivalent impulse response $h[n]$ is the convolution of their individual impulse responses.
Given the individual impulse responses:

$$h_1[n] = \delta[n-1]$$

$$h_2[n] = \delta[n-2]$$

The overall impulse response is:

$$h[n] = h_1[n] * h_2[n]$$

$$h[n] = \delta[n-1] * \delta[n-2]$$

Using the property of the discrete-time impulse function, convolving two shifted impulses results in an impulse shifted by the sum of their individual delays:

$$\delta[n-n_1] * \delta[n-n_2] = \delta[n-(n_1+n_2)]$$

Substituting $n_1 = 1$ and $n_2 = 2$:

$$h[n] = \delta[n-(1+2)]$$

$$h[n] = \delta[n-3]$$

$$\boxed{\text{C) } \delta[n-3]}$$

Key Points

- **Cascaded LTI Systems:** The overall impulse response is the convolution of the individual impulse responses, $h[n] = h_1[n] * h_2[n]$.
- **Shifting Property of Convolution:** Convolving any signal with a delayed impulse shifts the signal by that exact delay: $x[n] * \delta[n-n_0] = x[n-n_0]$.
- **Convolution of Impulses:** As a direct consequence, $\delta[n-n_1] * \delta[n-n_2] = \delta[n-(n_1+n_2)]$.

Mistakes to be avoided

- **Parallel vs. Cascade:** Do not add the impulse responses to get $\delta[n-1] + \delta[n-2]$. Addition represents systems connected in parallel, not in cascade.
- **Time-Domain Multiplication:** Do not simply multiply the sequences to get $\delta[n-1]\delta[n-2]$. Multiplication in the time domain corresponds to convolution in the frequency domain, which does not represent the equivalent behavior of cascaded systems.

Q2.09.1 MCQ 2009 2M Ans: B ● ○ ○

Here $h(t) \neq 0$ for $t < 0$. Thus system is non causal. Again any bounded input $x(t)$ gives bounded output $y(t)$. Thus it is BIBO stable.

Here we can conclude that option (B) is correct. Hence (B) is correct answer.

$$\boxed{\text{B) BIBO, LTI}}$$

Key Points

- **Causality:** An LTI system is causal if and only if its impulse response satisfies $h(t) = 0$ for all $t < 0$. If $h(t) \neq 0$ for any $t < 0$, the system depends on future inputs and is therefore non-causal.
- **BIBO Stability:** A system is Bounded-Input Bounded-Output (BIBO) stable if every bounded input sequence (or signal) produces a bounded output sequence.

Q2.08.1 MCQ 2008 1M Ans: C ● ○ ○

A continuous-time system is causal if its output $y(t)$ at any given time t depends only on the present and past values of the input $x(t)$. It must not depend on future values of the input (i.e., $x(t + \tau)$ where $\tau > 0$).

Let us analyze the dependence of $y(t)$ on $x(t)$ for each option:

- **Option (A):**
$$y(t) = x(t - 2) + x(t + 4)$$
The term $x(t + 4)$ represents a future input (e.g., at $t = 0$, the output $y(0)$ depends on $x(4)$). Thus, the system is non-causal.

- **Option (B):**
$$y(t) = (t - 4)x(t + 1)$$
The term $x(t + 1)$ depends on a future input. Thus, the system is non-causal.

- **Option (C):**
$$y(t) = (t + 4)x(t - 1)$$
The term $x(t - 1)$ depends only on a past input for any given t. The coefficient $(t + 4)$ is a time-varying gain and does not affect the causality of the system. Thus, the system is causal.

- **Option (D):**
$$y(t) = (t + 5)x(t + 5)$$
The term $x(t + 5)$ represents a future input. Thus, the system is non-causal.

$$\boxed{\text{C) } y(t) = (t + 4)x(t - 1)}$$

Key Points

- A system is **causal** if the output at any time t_0 depends only on inputs $x(t)$ for $t \leq t_0$.
- Time-varying coefficients (like $t + 4$ or $t - 4$) make a system time-varying but do not determine causality. Causality is solely dependent on the time shift of the input signal $x(\cdot)$.

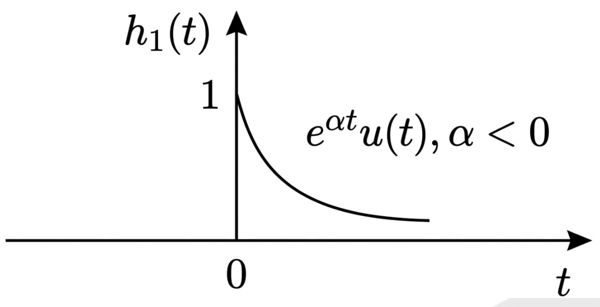
Mistakes to be avoided

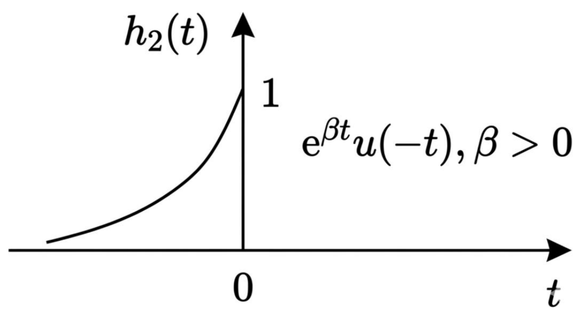
- Confusing time-variance with non-causality. The presence of t as an external multiplier has no bearing on whether the system requires future inputs.
- Misinterpreting the sign inside the function argument. For $\tau > 0$, $x(t - \tau)$ represents a delay (past, causal) while

$x(t+\tau)$ represents an advance (future, non-causal).

Q2.08.2 MCQ 2008 1M Ans: D

For the given LTI system, the impulse response is $h(t) = h_1(t) + h_2(t)$. Where $h_1(t) = e^{\alpha t}u(t)$ is shown in Fig. 1 for $\alpha < 0$ and $h_2(t) = e^{\beta t}u(-t)$ is shown in Fig. 2 for $\beta > 0$.





For stability of LTI system

$$I = \int_{t=-\infty}^{\infty} |h(t)|\, dt < \infty$$

$$I_1 = \int_{t=0}^{\infty} e^{\alpha t}\, dt = \frac{1}{|\alpha|} = \text{constant}(< \infty)$$

$$I_2 = \int_{t=-\infty}^{0} e^{\beta t}\, dt = \frac{1}{\beta} = \text{constant}(< \infty)$$

$$I = I_1 + I_2 = \text{constant}(< \infty)$$

$$\boxed{\text{D) } \alpha \text{ is negative and } \beta \text{ is positive}}$$

Key Points

- An LTI system is Bounded-Input Bounded-Output (BIBO) stable if and only if its impulse response $h(t)$ is absolutely integrable: $\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$.

Q2.08.3 MCQ 2008 2M Ans: A

To determine the correct mapping, we need to test each relation for the properties of linearity and time-invariance.

Relation R1: $y(t) = t^2 x(t)$

- **Linearity:** Let $x(t) = ax_1(t) + bx_2(t)$.

$$y(t) = t^2(ax_1(t) + bx_2(t)) = at^2x_1(t) + bt^2x_2(t) = ay_1(t) + by_2(t)$$

  Since it satisfies the superposition principle, it is **Linear**.

- **Time-Invariance:** Let the input be delayed by t_0, forming $x_1(t) = x(t - t_0)$. The response is $y_1(t) = t^2x(t - t_0)$. Now, delay the original output by t_0: $y(t-t_0) = (t-t_0)^2x(t-t_0)$. Since $y_1(t) \neq y(t-t_0)$, the system is **NOT time-invariant**.

Conclusion for R1: Linear but NOT time-invariant (**P1**).

Relation R2: $y(t) = t|x(t)|$

- **Linearity:** The absolute value operator is non-linear. For a constant $a < 0$, the response to $ax(t)$ is $t|ax(t)| = t|a||x(t)| \neq at|x(t)|$. Thus, it is **NOT linear**.

- **Time-Invariance:** Due to the explicit time multiplier t, applying the same shift test as in R1 will yield $y_1(t) = t|x(t-t_0)|$ and $y(t-t_0) = (t-t_0)|x(t-t_0)|$. It is **NOT time-invariant**.

Conclusion for R2: Neither linear nor time-invariant (Does not match P1, P2, or P3).

Relation R3: $y(t) = |x(t)|$

- **Linearity:** Because of the absolute value, $|ax(t)| = |a||x(t)| \neq a|x(t)|$ for $a < 0$. It is **NOT linear**.
- **Time-Invariance:** Let the input be delayed by t_0: $y_1(t) = |x(t-t_0)|$. Delaying the output by t_0 gives $y(t-t_0) = |x(t-t_0)|$. Since $y_1(t) = y(t-t_0)$, the system is **Time-invariant**.

Conclusion for R3: Time-invariant but NOT linear (**P2**).

Relation R4: $y(t) = x(t-5)$

- **Linearity:** This represents a pure time delay. $ax_1(t-5) + bx_2(t-5) = ay_1(t) + by_2(t)$. It is **Linear**.
- **Time-Invariance:** Delaying the input by t_0 yields $y_1(t) = x((t-t_0)-5) = x(t-t_0-5)$. Delaying the output yields $y(t-t_0) = x((t-t_0)-5)$. Since they are equal, it is **Time-invariant**.

Conclusion for R4: Linear and time-invariant (**P3**).

Matching the properties to the relations: P1 $\longrightarrow$ R1 P2 $\longrightarrow$ R3 P3 $\longrightarrow$ R4

$$\boxed{\text{A) (P1, R1), (P2, R3), (P3, R4)}}$$

Key Points

- **Linearity Check:** A system is linear if it obeys superposition: $\mathcal{H}\{ax_1(t) + bx_2(t)\} = a\mathcal{H}\{x_1(t)\} + b\mathcal{H}\{x_2(t)\}$. Operations like absolute value $|x(t)|$, squaring $x^2(t)$, or adding a non-zero constant violate linearity.
- **Time-Invariance Check:** A system is time-invariant if $\mathcal{H}\{x(t-t_0)\} = y(t-t_0)$. Multiplying the input signal by explicit functions of time (like t or t^2) or time-scaling (like $x(2t)$) violates time-invariance.
- **Pure Delay:** A pure time-shift system $y(t) = x(t-t_d)$ is always an LTI (Linear Time-Invariant) system.

Mistakes to be avoided

- **Ignoring negative scalars in linearity tests:** When checking absolute values or even-power functions, students often forget that linearity requires scaling to hold for all real (or complex) numbers, including negative ones. $|-1 \cdot x(t)| \neq -1 \cdot |x(t)|$.
- **Improper shift application:** When checking for time-invariance, you must find the response to a delayed input $x_1(t) = x(t-t_0)$. Do not mistakenly replace the independent variable t outside the function argument $x(\dots)$. For $y(t) = t^2x(t)$, the delayed input response is $t^2x(t-t_0)$, *not* $(t-t_0)^2x(t-t_0)$.

Q2.06.1 | MCQ | 2006 | 2M | Ans: C | ● ○ ○

To determine the properties of the system, we analyze linearity, stability, and invertibility separately.

1. Linearity Check: Let the responses to inputs $x_1[n]$ and $x_2[n]$ be $y_1[n]$ and $y_2[n]$, respectively:

$$y_1[n] = \left(\sin\frac{5}{6}\pi n\right)x_1[n]$$

$$y_2[n] = \left(\sin\frac{5}{6}\pi n\right)x_2[n]$$

For a linear combination of inputs $x_3[n] = ax_1[n] + bx_2[n]$, the output is:

$$y_3[n] = \left(\sin\frac{5}{6}\pi n\right)(ax_1[n] + bx_2[n])$$

$$y_3[n] = a\left(\sin\frac{5}{6}\pi n\right)x_1[n] + b\left(\sin\frac{5}{6}\pi n\right)x_2[n]$$

$$y_3[n] = ay_1[n] + by_2[n]$$

Since the system satisfies the superposition principle, it is **linear**.

2. Stability Check: A system is Bounded-Input Bounded-Output (BIBO) stable if every bounded input results in a bounded output. Assume the input is bounded such that $|x[n]| \leq M_x < \infty$ for all n.

$$|y[n]| = \left|\sin\frac{5}{6}\pi n\right||x[n]|$$

Since the sine function is bounded by 1 ($\left|\sin\frac{5}{6}\pi n\right| \leq 1$), we get:

$$|y[n]| \leq 1 \cdot M_x = M_x < \infty$$

Thus, the output is always bounded for a bounded input. The system is **stable**.

3. Invertibility Check: A system is invertible if distinct inputs produce distinct outputs, meaning we can uniquely recover $x[n]$ from $y[n]$. From the given relation, recovering $x[n]$ requires division:

$$x[n] = \frac{y[n]}{\sin\left(\frac{5}{6}\pi n\right)}$$

However, at $n = 0$, $\sin(0) = 0$. This gives $y[0] = 0 \cdot x[0] = 0$. Because any value of $x[0]$ yields $y[0] = 0$, we cannot uniquely determine $x[0]$ from $y[0]$. The system loses information at $n = 0$ (and other integer multiples like $n = 6$), making it **non-invertible**.

$\boxed{\text{C) linear, stable and non-invertible}}$

Key Points

- **Linearity:** A system is linear if it obeys superposition: $\mathcal{T}\{ax_1[n]+bx_2[n]\} = a\mathcal{T}\{x_1[n]\}+b\mathcal{T}\{x_2[n]\}$. Multiplying the input by a time-varying function (such as a sine wave) preserves linearity.
- **Stability:** A system is BIBO stable if $|y[n]| \leq M_y < \infty$ whenever $|x[n]| \leq M_x < \infty$. Multiplying by a bounded sequence preserves stability.
- **Invertibility:** A system is invertible if it has a one-to-one mapping between the input and the output. If the multiplier evaluates to exactly zero at any discrete time index n, the mapping is no longer one-to-one, and the system is non-invertible.

Mistakes to be avoided

- **Confusing time-varying with non-linear:** The presence of n in the coefficient $\sin(\frac{5}{6}\pi n)$ makes the system time-varying, but it is still linear because the operation on $x[n]$ is a simple scalar multiplication at each instant.
- **Ignoring specific indices for invertibility:** Do not assume that because a function is non-zero for most values of n, it is invertible. Always check for zero-crossings at integer values of n.

Q2.05.1 | MCQ | 2005 | 1M | Ans: B | ● ○ ○

For a causal system, the impulse response $h(t)$ must satisfy the following condition:

$$h(t) = 0 \text{ for } t < 0$$

By analyzing the given figures:

- **Figure (a):** Shows non-zero values for $t < 0$. Thus, it is non-causal.

- **Figure (b):** The signal starts at $t > 0$ and is zero for all $t < 0$. Thus, it is **causal**.
- **Figure (c):** The signal starts before $t = 0$ (negative time). Thus, it is non-causal.
- **Figure (d):** The signal exists entirely for $t < 0$. Thus, it is anti-causal (non-causal).

$\boxed{\text{B}}$

Key Points

- **Causality in Time-Domain:** A continuous-time LTI system is causal if and only if its impulse response $h(t)$ is zero for all negative time ($t < 0$).
- **Physical Interpretation:** In a causal system, the output at any time depends only on the present and past values of the input; it cannot anticipate future inputs.

Mistakes to be avoided

- **Visual Misinterpretation:** Ensure you check the horizontal axis (t-axis) carefully. Any part of the graph appearing to the left of the vertical $h(t)$ axis represents $t < 0$.
- **Confusing Causal with Stable:** A system can be causal but unstable (e.g., $e^t u(t)$). Causality only refers to the timing of the response, not its magnitude over time.

Q2.04.1 MCQ | 2004 | 1M | Ans: A | ● ○ ○

To determine the stability and causality of the system, we analyze the given impulse response:

$$h[n] = u[n+3] + u[n-2] - 2u[n-7]$$

1. Stability Check: A discrete-time LTI system is BIBO stable if its impulse response is absolutely summable:

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

Substituting the given $h[k]$:

$$\sum_{k=-\infty}^{\infty} h(k) = \sum_{k=-3}^{\infty} u(k+3) + \sum_{k=2}^{\infty} u(k-2) - 2\sum_{k=7}^{\infty} u(k-7)$$

The terms cancel out for higher values of k. Specifically, the three unit steps overlap such that for $k \geq 7$, the sum is $1 + 1 - 2(1) = 0$. Thus, the summation is finite:

$$= \sum_{k=-3}^{6} 1 + \sum_{k=2}^{6} 1$$

$$= 10 + 5 = 15 < \infty$$

Since the sum is finite, the system is **stable**.

2. Causality Check: A system is causal if $h[n] = 0$ for all $n < 0$. In this case, the term $u[n+3]$ is non-zero starting from $n = -3$. Since $h[-3], h[-2], h[-1] \neq 0$, the response depends on future values of the input signal. Therefore, the system is **not causal**.

$\boxed{\text{A) stable but not causal}}$

Key Points

- **BIBO Stability:** For discrete LTI systems, stability requires $\sum |h[n]| < \infty$. If the impulse response is of finite duration or decays sufficiently, the system is stable.
- **Causality:** A system is causal if the impulse response $h[n]$ is zero for all $n < 0$. Any term with a positive time shift, like $u[n+k]$ where $k > 0$, violates causality.

Mistakes to be avoided

- **Infinite Summation Error:** Do not assume the system is unstable just because it contains unit step functions. If the steps are subtracted in a way that limits the response duration, the sum can be finite.
- **Causality Misconception:** Remember that $u[n-2]$ is causal, but $u[n+3]$ is non-causal. The presence of even one non-causal term makes the entire system non-causal.

Q2.03.1 | MCQ | 2003 | 2M | Ans: A | ● ○ ○

To analyze the properties of the given discrete-time system, we examine linearity, time-invariance, causality, and stability based on the input-output relationship:

$$y(n) = \begin{cases} x(n), & n \geq 1 \\ 0, & n = 0 \\ x(n+1), & n \leq -1 \end{cases}$$

1. Linearity (P): Each branch of the relationship is a linear operation on the input $x(n)$. There are no non-linear terms like $x^2(n)$, $|x(n)|$, or constant offsets. Therefore, the system satisfies the principle of superposition and is **linear**.

2. Time-Invariance (Q): The system's behavior changes depending on the specific time index n (it has different rules for $n \geq 1$, $n = 0$, and $n \leq -1$). For example, a shift in the input will not result in an identical shift in the output because the "zero-out" rule at $n = 0$ is fixed to that specific coordinate. Thus, the system is **time-varying**.

3. Causality (R): A system is causal if the output $y(n)$ depends only on present or past values of the input. For the branch $n \leq -1$:

$$\text{If } n = -1, \quad y(-1) = x(-1+1) = x(0)$$

Here, $x(0)$ is a future value relative to the time index $n = -1$. Since the output depends on a future value of the input, the system is **non-causal**.

4. Stability (S): A system is BIBO stable if a bounded input $|x(n)| \leq M_x < \infty$ produces a bounded output. Since $y(n)$ is either equal to an input value or zero, $|y(n)|$ will never exceed M_x. Therefore, the system is **stable**.

Based on the analysis, the system is linear (P) and stable (S), but not time-invariant (Q) or causal (R).

$$\boxed{\text{A) P, S but not Q, R}}$$

Key Points

- **Linearity:** Check if the transformation $\mathcal{T}\{ax_1(n) + bx_2(n)\} = a\mathcal{T}\{x_1(n)\} + b\mathcal{T}\{x_2(n)\}$.
- **Causality:** A system is non-causal if $y(n)$ depends on any $x(n+k)$ where $k > 0$.
- **Stability:** For BIBO stability, the output must remain bounded if the input is bounded.

Mistakes to be avoided

- **Misidentifying Causality:** Don't just look at one branch. Even if the $n \geq 1$ branch is causal, the $n \leq -1$ branch makes the entire system non-causal.
- **Assuming Time-Invariance:** Systems defined by different equations over different time intervals (piecewise) are almost always time-varying.

Q2.02.1 MCQ 2002 1M Ans: C ● ○ ○

The convolution of a signal with a shifted impulse function results in the signal being shifted by the same amount. This is a direct application of the shifting property of the Dirac delta function.

The shifting property states:

$$f(t) * \delta(t - t_0) = f(t - t_0)$$

In this problem, we are given the expression:

$$x(t + 5) * \delta(t - 7)$$

Treating $x(t + 5)$ as our function $f(t)$ and applying a delay of $t_0 = 7$, we substitute t with $(t - 7)$:

$$x(t + 5) * \delta(t - 7) = x((t - 7) + 5)$$

$$x(t + 5) * \delta(t - 7) = x(t - 2)$$

$$\boxed{\text{C) } x(t - 2)}$$

Key Points

- **Shifting Property of Convolution:** Convolving any continuous-time signal with a delayed unit impulse shifts the original signal by that specific delay: $x(t) * \delta(t - t_0) = x(t - t_0)$.

Mistakes to be avoided

- **Adding Instead of Subtracting:** Carelessly combining the shifts as $x(t + 5 + 7) = x(t + 12)$. When applying the shift $t \rightarrow (t - t_0)$, you must substitute $(t - 7)$ exactly where t is in the argument.

Q2.01.1 MCQ 2001 2M Ans: D ● ○ ○

$$h_1(t) \neq 0 \text{ for } t < 0$$

Therefore, is non causal

$$h_2(t) = u(t) = 0 \text{ for } t < 0$$

$$\int_{-\infty}^{\infty} n_2(t)dt = \int_{-\infty}^{\infty} u(t)dt = \int_{0}^{\infty} dt = \infty$$

Therefore s_2 is unstable and causal

$$h_3(t) = \frac{u(t)}{t + 1}$$

at $t = -1$

$$h_3(t) = \infty, \quad h_3(t) = 0 \text{ for } t < 0$$

Therefore s_3 is unstable and causal

$$h_4(t) = e^{-3t}u(t)$$

s_4 is time invariant, causal and stable

$$\boxed{\text{D) S4}}$$

Key Points

- **Causality:** A system is causal if its impulse response $h(t) = 0$ for all $t < 0$.
- **Stability:** A continuous-time LTI system is Bounded-Input Bounded-Output (BIBO) stable if its impulse response is absolutely integrable: $\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$.

Mistakes to be avoided

- **Ignoring the limits imposed by $u(t)$:** The unit step function $u(t)$ enforces causality (making the signal zero for $t < 0$). When checking for stability, this changes the lower limit of the integral from $-\infty$ to 0.
- **Assuming a causal signal is stable:** As seen with s_2 and s_3, a signal can be zero for $t < 0$ (causal) but still have an integral that diverges to infinity (unstable).

Q2.00.1 MCQ 2000 2M Ans: B ● ○ ○

To determine the properties of the system $y(t) = tx(t)$, we evaluate it for linearity and time-invariance.

1. Linearity Check Let $x_1(t)$ and $x_2(t)$ be two inputs, producing outputs $y_1(t) = tx_1(t)$ and $y_2(t) = tx_2(t)$ respectively. For a linear combination of inputs $x_3(t) = ax_1(t) + bx_2(t)$, the output is:

$$y_3(t) = tx_3(t) = t[ax_1(t) + bx_2(t)]$$

$$y_3(t) = a[tx_1(t)] + b[tx_2(t)] = ay_1(t) + by_2(t)$$

Since the system satisfies the superposition principle, it is linear.

2. Time-Invariance Check Let the input be delayed by t_0, so the new input is $x_1(t) = x(t - t_0)$. The output due to this delayed input is:

$$y_1(t) = tx_1(t) = tx(t - t_0)$$

Now, delaying the original output $y(t)$ by t_0 yields:

$$y(t - t_0) = (t - t_0)x(t - t_0)$$

Comparing the two expressions, we observe:

$$y_1(t) \neq y(t - t_0)$$

Since a time shift in the input does not result in an identical time shift in the output, the system is time-varying.

Therefore, the system is linear and time-varying.

$\boxed{\text{B) Linear and time varying}}$

Key Points

- A system is **linear** if it satisfies additivity and homogeneity (the superposition principle).
- A system is **time-invariant** if an output $y_1(t)$ due to a delayed input $x(t - t_0)$ equals the delayed output $y(t - t_0)$.
- Multiplying a signal by the time variable t introduces a time-varying property.

Q2.95.1 MCQ 1995 1M Ans: A ● ○ ○

Note that here $u(t)$ is not a unit step function. It is any arbitrary input.

For LTI system:

$$y(t) = u(t) * h(t)$$

By definition of the convolution integral:

$$y(t) = \int_{\tau=-\infty}^{\infty} h(\tau)u(t - \tau)d\tau$$

or, using the commutative property:

$$= \int_{\tau=-\infty}^{\infty} h(t-\tau)u(\tau)d\tau$$

If the system is **causal**, the impulse response must satisfy $h(t) = 0$ for $t < 0$. This means $h(\tau) = 0$ for $\tau < 0$. Therefore, the lower limit of the integral becomes 0:

$$= \int_{\tau=0}^{\infty} h(\tau)u(t-\tau)d\tau$$

If, in addition to the system being causal, the **input** $u(t)$ **is also causal**, then $u(t) = 0$ for $t < 0$. In the integral $u(t-\tau)$, this means $u(t-\tau) = 0$ when $t - \tau < 0$, which implies $\tau > t$. Therefore, the upper limit of the integral is bounded by t:

$$y(t) = \int_{\tau=0}^{t} h(\tau)u(t-\tau)d\tau$$

Key Points

- **Convolution Integral:** The fundamental relationship for continuous-time LTI systems is $y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$.
- **Causal System limit:** A causal system restricts the lower limit to 0 because the system cannot respond before the impulse occurs ($h(t) = 0$ for $t < 0$).
- **Causal Input limit:** A causal input restricts the upper limit to t because future values of the input ($x(t-\tau)$ for $\tau > t$) are zero.

Mistakes to be avoided

- **Notation confusion:** Always clarify the meaning of symbols like $u(t)$ in a specific context. Usually, $u(t)$ is the unit step, but here it represents an arbitrary input signal (often denoted as $x(t)$).
- **Incorrect Limits:** Do not automatically use limits from 0 to t unless you are certain both the system and the input signal are causal.

Q2.94.1 NAT 1994 2M Ans: A-1, B-2, C-3 ● ○ ○

Let's analyze each given differential equation to determine its type.

Equation (A):

$$a_1\frac{d^2y}{dx^2} + a_2y\frac{dy}{dx} + a_3y = a_4$$

The presence of the term $a_2y\frac{dy}{dx}$ involves the product of the dependent variable y and its derivative $\frac{dy}{dx}$. This property makes the differential equation non-linear. Thus, (A) matches with (1) Non-linear differential equation.

Equation (B):

$$a_1\frac{d^3y}{dx^3} + a_2y = a_3$$

The dependent variable y and its derivatives appear only with a degree of 1, and there are no cross-products between them. Assuming a_1, a_2, and a_3 are constants, this represents a linear equation with constant coefficients. Thus, (B) matches with (2) Linear differential equation with constant coefficients.

Equation (C):

$$a_1\frac{d^2y}{dx^2} + a_2x\frac{dy}{dx} + a_3x^2y = 0$$

This equation is linear because y and its derivatives are of the first degree. The right-hand side is equal to 0, which signifies that the equation is homogeneous. Although the coefficients are functions of x, it satisfies the condition of being linear and homogeneous. Thus, (C) matches with (3) Linear homogeneous differential equation.

$$\boxed{\mathrm{A} \to 1, \mathrm{B} \to 2, \mathrm{C} \to 3}$$

Key Points

- A differential equation is **linear** if the dependent variable and all its derivatives appear only to the first power and are not multiplied together.
- A differential equation is **non-linear** if it contains products of the dependent variable and its derivatives (e.g., $y\frac{dy}{dx}$), or if the dependent variable/derivatives are raised to a power other than 1.
- A linear differential equation is **homogeneous** if the term independent of the dependent variable and its derivatives is zero (i.e., the RHS is 0).

Q2.91.1 MCQ 1991 2M Ans: C ● ○ ○

A system is defined as causal if its output at any given time depends only on the present and past values of the input, and not on future inputs.

In practical terms, this means a causal system cannot produce a response before an excitation is applied. Mathematically, if an input excitation is applied at $t = T$, it implies that the input is zero for $t < T$. For the system to be causal, its corresponding response must also be zero for all times before the input is applied, i.e., $t < T$.

The problem states that the excitation is applied at $t = T$ and the response is zero for $-\infty < t < T$. This perfectly satisfies the fundamental condition for causality.

Therefore, the system is a causal system.

$$\boxed{\text{(C) Causal system}}$$

Key Points

- **Causal System:** A system whose response to an input does not begin before the input itself is applied. If an input $x(t) = 0$ for $t < t_0$, then the output $y(t) = 0$ for $t < t_0$.
- **Non-causal System:** A system whose output depends on future inputs, meaning it can theoretically produce a response before the input is applied.
- All physical, real-time systems are inherently causal because they cannot anticipate future events.

Q2.90.1 MCQ 1990 2M Ans: 0.5 ● ○ ○

Method 1

The output $y(t)$ of a linear time-invariant (LTI) system is determined by the convolution of the input excitation $x(t)$ and the impulse response $h(t)$:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

To find the output at $t = 2$ sec, we evaluate the integral at $t = 2$:

$$y(2) = \int_{-\infty}^{\infty} x(\tau)h(2-\tau)\,d\tau$$

Based on Fig. a, the impulse response is a rectangular pulse defined as:

$$h(t) = \begin{cases} 1, & 0 \le t \le 6 \\ 0, & \text{otherwise} \end{cases}$$

Thus, the shifted and time-reversed impulse response $h(2-\tau) = 1$ when:

$$0 \le 2 - \tau \le 6$$

$$-2 \le -\tau \le 4$$

$$-4 \leq \tau \leq 2$$

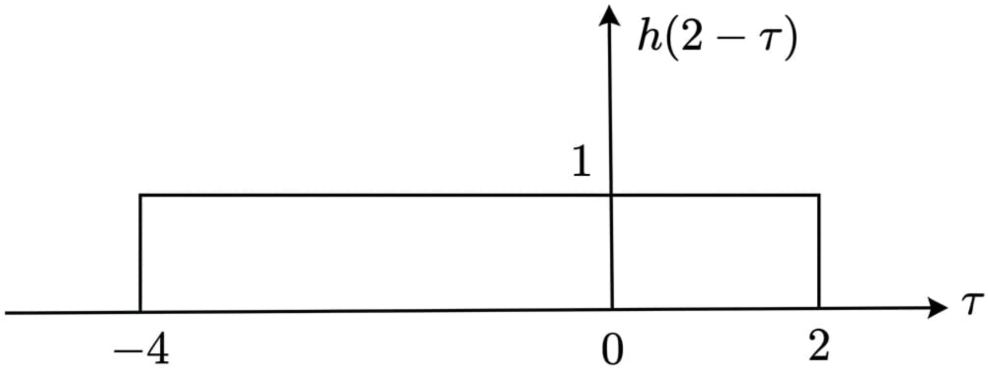


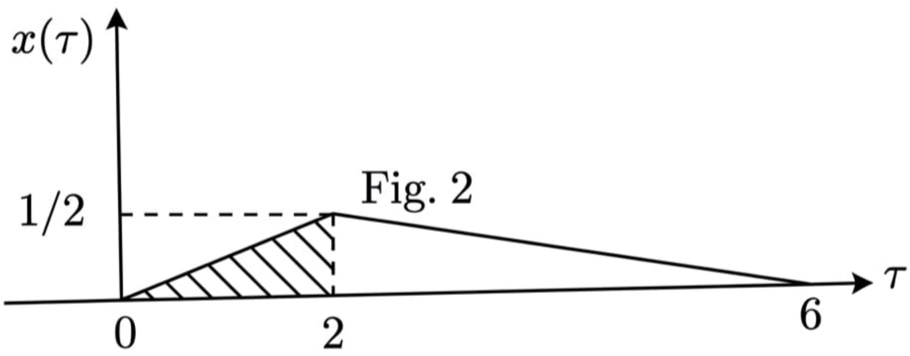
Fig. 1

Since the excitation function $x(\tau) = 0$ for $\tau < 0$ (from Fig. b), the effective non-zero overlap between $x(\tau)$ and $h(2-\tau)$ occurs only in the interval $0 \leq \tau \leq 2$. The convolution integral simplifies to:

$$y(2) = \int_0^2 x(\tau)(1)\, d\tau$$

This integral represents the area under the curve of the excitation function $x(t)$ from $t = 0$ to $t = 2$ sec.

From Fig. b, the geometric shape of $x(t)$ in the interval $[0, 2]$ is a right-angled triangle with a base of 2 and a height of $1/2$.



$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$y(2) = \frac{1}{2} \times 2 \times \frac{1}{2} = 0.5$$

Method 2

$$h(t) = u(t) - u(t-6)$$

$$x(t) = \frac{1}{4}r(t) - \frac{3}{8}r(t-2) + \frac{1}{8}r(t-6)$$

$$y(t) = x(t) * h(t)$$

$$= \frac{1}{4}r(t) * u(t) - \frac{3}{8}r(t-2) * u(t) + \frac{1}{8}r(t-6) * u(t)$$

$$-\frac{1}{4}r(t) * u(t-6) + \frac{3}{8}r(t-2) * u(t-6) - \frac{1}{8}r(t-6)u(t-6)$$

Evaluating at $t = 2$:

$$y(2) = \frac{1}{4}r(2) * u(2) - \frac{3}{8}r(2-2) * u(2) + \frac{1}{8}r(2-6) * u(2)$$

$$-\frac{1}{4}r(2) * u(2-6) + \frac{3}{8}r(2-2) * u(2-6) - \frac{1}{8}r(2-6)u(2-6)$$

Noting that $r(t) = 0$ for $t \leq 0$ and $u(t) = 0$ for $t < 0$, all terms evaluate to zero except the first term:

$$= \frac{1}{4} \times 2 = \frac{1}{2}$$

$$\boxed{\text{B) } 0.5}$$

Key Points

- The output of an LTI system is the convolution integral of the input signal and the impulse response: $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$.
- Convolution with a rectangular pulse of unit amplitude mathematically simplifies to finding the area under the input signal's curve over the overlapping time interval.

DEBUG INFO

Chapter II

Basics of Systems

Total Questions : 35

MCQ : 29

MSQ : 0

NAT : 6

SOLUTIONS

III

Continuous Time Fourier Series

Topics

1. Introduction to Fourier Series
2. Exponential Fourier Series
3. Predicting the Nature of a Signal from Fourier Coefficients
4. Power Using Fourier Series Coefficients
5. Properties of Exponential Fourier Series
6. Fourier Series of Periodic Waveforms
7. Trigonometric Fourier Series and Symmetry
8. Polar Form, Dirichlet's Conditions and Gibbs Phenomenon

Q3.26.1 MCQ 2026 2M Ans: A ● ○ ○

Given that,

$$x\left(t-\frac{T}{2}\right) = -x(t)$$

where 'T' is FTP.
Thus, $x(t)$ is half-wave symmetric signal.
For HWS: Even harmonics will be absent i.e. $a_n = 0$ if n is even-integer.

$$\Rightarrow a_{2m} = 0 \text{ where } m = \text{integer}$$

$$\boxed{A) a_{2m} = 0}$$

Key Points

- A periodic signal $x(t)$ with a fundamental time period T possesses Half-Wave Symmetry (HWS) if it satisfies the condition $x(t \pm T/2) = -x(t)$.
- A key property of half-wave symmetric signals is that their Fourier series expansion contains only odd harmonics. All even harmonics evaluate to zero ($a_n = 0$ for all even integers n).

Q3.25.1 MSQ 2025 2M Ans: A,D ● ○ ○

Given,

$$f(t) \underset{\text{FTP: } T_0}{\rightleftharpoons} c_k$$

$$y(t) \underset{\text{FTP: } \frac{T_0}{\alpha}}{\rightleftharpoons} d_k$$

and

$$y(t) = f(\alpha t)$$

By time-scaling property of FS,

$$d_k = c_k \text{ if } \alpha \text{ is positive}$$

Given that,

$$\alpha > 1$$

$\therefore$

$$\boxed{\text{A) } c_k = d_k \text{ for all } k}$$

$$\boxed{\text{D) } y(t) \text{ is periodic with a fundamental period } T_0/\alpha}$$

Key Points

- **Time Scaling Property:** If $f(t)$ is a periodic signal with fundamental period T_0 and Fourier series coefficients c_k, the time-scaled signal $f(\alpha t)$ for $\alpha > 0$ has a fundamental period of T_0/α.
- The Fourier series coefficients of the time-scaled signal remain c_k. The scaling affects the signal's period and fundamental frequency, but not the amplitude or phase of its harmonic components.

Mistakes to be avoided

- **Scaling the Coefficients:** A very common trap is assuming that the Fourier coefficients scale with time (e.g., $d_k = c_k/\alpha$). Time scaling only compresses or expands the signal in time; it does not change the magnitude or phase of the individual harmonic components.
- **Inverting the Period Calculation:** When $\alpha > 1$, the signal is compressed in time, meaning it completes a cycle faster.

Therefore, the new period is shorter (T_0/α), not longer (αT_0).

Q3.21.1 NAT | 2021 | 2M | Ans: 32 | ● ○ ○

1. $x(t)$ is real and even so a_k is also real and even $a_k = a_{-k}$
2. Average of $x(t)$ is 2 i.e., $a_0 = 2$.
3. $x(t) \to a_k = \begin{cases} k, & 1 \le k \le 3 \\ 0, & k > 3 \end{cases}$

$$a_1 = 1, \qquad a_{-1} = 1$$
$$a_2 = 2, \qquad a_{-2} = 2$$
$$a_3 = 3, \qquad a_{-3} = 3$$

4. $T_0 = 6$

Parsval's Power Theorem

$$\frac{1}{T}\int_0^T |x(t)|^2 dt = \sum_{n=-\infty}^{+\infty} |a_k|^2$$

$$P_{x(t)} = \sum_{n=-\infty}^{+\infty} |a_k|^2 = |a_{-3}|^2 + |a_{-2}|^2 + |a_{-1}|^2 + |a_0|^2 + |a_1|^2 + |a_2|^2 + |a_3|^2$$
$$= 2|a_1|^2 + 2|a_2|^2 + 2|a_3|^2 + |a_0|^2$$
$$= 2 \times 1^2 + 2 \times (2)^2 + 2(3)^2 + (2)^2$$
$$= 2 + 8 + 18 + 4$$
$$P_{x(t)} = 32$$

$$\boxed{32}$$

Key Points

- **Fourier Series Symmetry:** For a real and even continuous-time periodic signal $x(t)$, its exponential Fourier series coefficients a_k are also real and even, satisfying $a_k = a_{-k}$.
- **DC Component:** The average value of the signal corresponds to the a_0 coefficient in the exponential Fourier series.
- **Parseval's Power Theorem:** The total average power P of a periodic signal is equal to the sum of the squared magnitudes of its Fourier series coefficients: $P = \sum |a_k|^2$.

Q3.18.1 MCQ | 2018 | 1M | Ans: C | ● ○ ○

Method 1

Only for case study let us consider the periodic waveform $x(t)$ as a square wave.

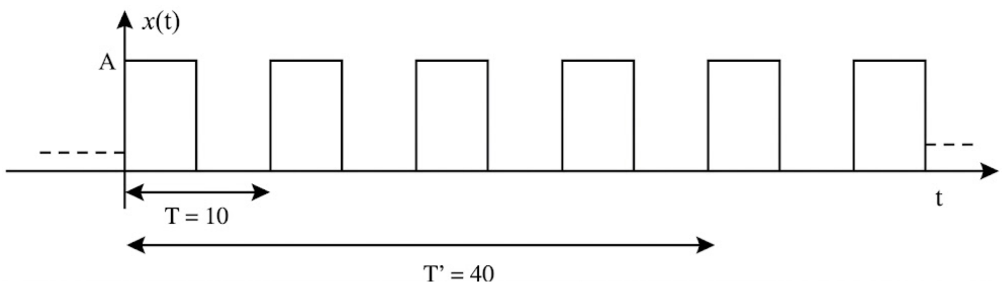


Fig. Solution Q3.18.1

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t}$$

We know a_k is the exponential Fourier series coefficient and is given by

$$a_k = \frac{1}{T}\int_0^T x(t)e^{-jk\omega_o t}dt$$

Since the waveform repeat itself, we can say if we evaluate a_k using the period, as any integer multiple of T_o like $2T_o, 3T_o \dots$ the coefficient will not change, we can notice this in the above integration.
So we can say, if $T = 10$, the E.F.S coefficient is a_k, then if $T = 40$, the E.F.S coefficient is $b_k = a_k$
Since, $a_k = b_k \Rightarrow \sum_{k=-\infty}^{\infty} |a_k| = \sum_{k=-\infty}^{\infty} |b_k| = 16$
For further more clarity, if we find the D.C. component ($k = 0$) of $x(t)$, then if we consider $T = 10, T = 20, T = 40$, etc, the value of a_o remains same.

Method 2

Time scaling property of fourier series is

$$x(t) \leftrightarrow a_k, \omega_0$$

$$x(at) \leftrightarrow a_k, a\omega_0$$

Even if we scale periodic functions no change in the coefficient amplitude

$$\sum_{k=-\infty}^{\infty} |a_k| = \sum_{k=-\infty}^{\infty} |b_k|$$

$$\boxed{\text{C) 16}}$$

Key Points

- **Independence of Integration Interval:** The Fourier series coefficients a_k of a periodic signal are invariant when evaluated over an integer multiple of the fundamental period T_o. Mathematically, integrating over nT_o and dividing by nT_o yields the exact same value as integrating over T_o and dividing by T_o.
- **Time Scaling Property:** Time scaling a periodic signal changes its fundamental frequency ($\omega_0 \to a\omega_0$) but does not alter the values of its Fourier series coefficients (a_k).

Mistakes to be avoided

- **Ignoring the Normalization Factor:** A common pitfall is assuming that because the integration window is 4 times longer ($T = 40$ vs $T = 10$), the coefficients will scale up by a factor of 4. Students must remember the $1/T$ multiplier outside the integral, which perfectly balances the increased integration area.
- **Confusing Total Power/Energy:** While the energy over a longer period increases, the average power (and thus the magnitude of the Fourier coefficients) remains constant for a steady periodic signal regardless of the observation

window, provided the window is an integer multiple of the fundamental period.

Q3.17.1 MCQ 2017 1M Ans: A

Given the first condition:

$$x(t) = -x(-t)$$

This implies that $x(t)$ is an odd signal. The trigonometric Fourier series of an odd signal contains only sine terms. The DC component and all cosine terms evaluate to zero. Therefore, $a_n = 0$ for all n (and $a_0 = 0$).

Given the second condition:

$$x(t) = -x\left(t - \frac{\pi}{\omega_0}\right)$$

We know that the fundamental time period is $T_0 = \frac{2\pi}{\omega_0}$. Thus, $\frac{\pi}{\omega_0} = \frac{T_0}{2}$. Substituting this into the condition gives:

$$x(t) = -x\left(t - \frac{T_0}{2}\right)$$

This is the standard condition for Half-Wave Symmetry (HWS). A signal possessing half-wave symmetry contains only odd harmonics; all even harmonics are zero. Since the signal only has sine terms (b_n), this means $b_n = 0$ for all even integers n.

Combining these observations: a_n are zero for all n, and b_n are zero for n even.

$$\boxed{\text{(A) } a_n \text{ are zero for all } n \text{ and } b_n \text{ are zero for } n \text{ even}}$$

Key Points

- **Odd Symmetry:** If $x(t) = -x(-t)$, the signal is odd. Its Fourier series expansion consists entirely of sine terms ($a_n = 0$ for all n).
- **Half-Wave Symmetry (HWS):** If a signal satisfies $x(t) = -x(t \pm T_0/2)$, it has half-wave symmetry. Such signals contain only odd harmonics (coefficients for even n are zero).

Mistakes to be avoided

- **Misidentifying the Time Shift:** A common mistake is failing to recognize that the shift π/ω_0 corresponds exactly to $T_0/2$. Always relate the given angular frequency ω_0 back to the fundamental period $T_0 = 2\pi/\omega_0$ to check for half-wave symmetry.
- **Confusing Even/Odd Functions with Even/Odd Harmonics:** Remember that an "odd function" (time-domain symmetry) eliminates cosine terms (a_n), whereas "odd harmonics" (frequency-domain property due to HWS) eliminates the even-indexed coefficients ($n = 2, 4, 6 \ldots$).

Q3.17.2 MCQ 2017 2M Ans: B

Given that the fundamental period of $x(t)$ is $T = 1$ s. The fundamental angular frequency of $x(t)$ is:

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{1} = 2\pi \text{ rad/s}$$

Let the complex Fourier series representation of $x(t)$ be denoted as:

$$x(t) \xleftrightarrow{FS} a_k \quad \text{with fundamental frequency } \omega_0$$

According to the time-scaling property of the continuous-time Fourier series, if a periodic signal is compressed or expanded in time by a factor 'a' ($a > 0$), its Fourier series coefficients remain unchanged, but its fundamental frequency scales by the same factor:

$$x(at) \xleftrightarrow{FS} a_k \quad \text{with new fundamental frequency } \omega_0' = a\omega_0$$

For the scaled signal $x(3t)$, we have the scaling factor $a = 3$. Therefore, the complex Fourier series coefficients of $x(3t)$ are $\{a_k\}$. This means **Statement I is true** and Statement II is false.
The fundamental angular frequency of $x(3t)$ evaluates to:

$$\omega_0' = 3 \times 2\pi = 6\pi \text{ rad/s}$$

This means **Statement III is true**.
Since only statements I and III are true, the correct option is (b).

$$\boxed{\text{B) only I and III are true}}$$

Key Points

- **Time Scaling Property:** Time scaling a periodic signal $x(t)$ to $x(at)$ changes the fundamental period to T/a and the fundamental angular frequency to $a\omega_0$.
- The Fourier series coefficients (a_k) themselves are invariant to time scaling. This is because time scaling affects the rate at which the signal cycles, not the relative amplitude or phase of its constituent harmonics.

Mistakes to be avoided

- **Scaling Coefficients Incorrectly:** A frequent error is assuming that a multiplier in the time domain $(x(3t))$ results in a proportional multiplier for the Fourier coefficients $(3a_k)$. Always remember that time scaling only affects the frequency/time axis, not the amplitude axis.

Q3.16.1 MCQ 2016 2M Ans: B ● ○ ○

$$H(j\omega) = e^{j\omega} + e^{-j\omega} = 2\cos\omega$$

For an input $A\cos(\omega_0 t)$ to a system $H(j\omega)$, the steady-state output is given by $A|H(j\omega_0)|\cos(\omega_0 t + \angle H(j\omega_0))$.
If $x(t) = 2\cos\left(\frac{2\pi}{3}t\right)$:

$$\omega_0 = \frac{2\pi}{3}$$
$$H(j\omega_0) = 2\cos\left(\frac{2\pi}{3}\right) = 2\left(\frac{-1}{2}\right) = -1$$
$$y(t) = 2\cos\left(\frac{2\pi}{3}t + 180^\circ\right)$$

If $x(t) = \cos(\pi t)$:

$$\omega_0 = \pi$$
$$H(j\omega_0) = 2\cos(\pi) = -2$$
$$y(t) = 2\cos(\pi t + 180^\circ)$$

Combining the outputs for the total system response:

$$y(t) = 2\cos\left(\frac{2\pi}{3}t + \pi\right) - 2\cos(\pi t + \pi)$$

Finding the fundamental frequency (ω_0):

$$\omega_1 = \frac{2\pi}{3}, \quad \omega_2 = \pi$$
$$T_1 = 3, \quad T_2 = 2$$
$$T_0 = 6$$

$$\omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{3}$$

Expressing $y(t)$ in terms of the fundamental frequency ω_0:

$$y(t) = 2\cos(2\omega_0 t + \pi) - 2\cos(3\omega_0 t + \pi)$$

Expanding into exponential Fourier series form using Euler's formula:

$$y(t) = e^{j(2\omega_0 t+\pi)} + e^{-j(2\omega_0 t+\pi)} - e^{j(3\omega_0 t+\pi)} - e^{-j(3\omega_0 t+\pi)}$$

Using the property $e^{j\pi} = e^{-j\pi} = -1$:

$$y(t) = -e^{j(2\omega_0 t)} - e^{-j(2\omega_0 t)} + e^{j(3\omega_0 t)} + e^{-j(3\omega_0 t)}$$

The coefficient C_3 corresponds to the term $e^{j3\omega_0 t}$. From the equation above:

$$C_3 = 1$$

$$\boxed{\text{B) } 1}$$

Key Points

- The response of an LTI system to a sinusoidal input $A\cos(\omega_0 t)$ modifies the amplitude by $|H(j\omega_0)|$ and shifts the phase by $\angle H(j\omega_0)$.
- The fundamental period of a composite periodic signal is the Least Common Multiple (LCM) of the individual periods: $T_0 = \text{LCM}(T_1, T_2)$.
- Euler's identity relates trigonometric functions to complex exponentials: $2\cos(\theta) = e^{j\theta} + e^{-j\theta}$.
- The k^{th} coefficient C_k in an exponential Fourier series is the multiplier for the $e^{jk\omega_0 t}$ term.

Q3.15.1 MCQ 2015 1M Ans: A ● ○ ○

Method 1

$$a_k = |a_k| e^{j\angle a_k}$$

$$a_{-3} = 3e^{-j\pi} = -3$$

$$a_{-2} = 2e^{-j\pi} = -2$$

$$a_{-1} = 0$$

$$a_1 = 0$$

$$a_2 = 2e^{-j\pi} = -2$$

$$a_3 = 3e^{-j\pi} = -3$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jn\omega_0 t}$$

$$= -3e^{-j3\omega_0 t} - 2e^{-2j\omega_0 t} + 1 - 2e^{2j\omega_0 t} - 2e^{3j\omega_0 t}$$

$$x(t) = 1 - 4\cos(2\omega_0 t) - 6\cos(3\omega_0 t)$$

$$x(t) \in R$$

Method 2

Graphical Solution

Based on the graphical properties of the frequency spectrum: First, observe the magnitude spectrum. Since $|a_k| = |a_{-k}|$, the magnitude spectrum is EVEN. Next, analyze the phase spectrum for positive and negative frequencies:

$$\angle a_k + \angle a_{-k} = -\pi + (-\pi) = -2\pi$$

Since the magnitude spectrum is even and in the phase spectrum the sum of angles at positive and negative frequencies is equal to an even multiple of π $(2n\pi)$, the signal $x(t)$ will be purely real.

$$\boxed{\text{A) } x(t) \in R}$$

Key Points

- A complex number can be represented in exponential form as $z = |z|e^{j\angle z}$.
- The inverse Fourier series synthesizes the time-domain signal: $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$.
- By Euler's identity, pairs of complex exponentials simplify to sinusoidal signals: $e^{j\theta} + e^{-j\theta} = 2\cos(\theta)$.
- If the Fourier coefficients satisfy conjugate symmetry ($a_k = a^*_{-k}$), the resulting signal is purely real ($x(t) \in R$).

Mistakes to be avoided

- Assuming that the phase spectrum must strictly visually appear as an odd function passing through the origin. A phase of $-\pi$ is mathematically equivalent to $+\pi$. Therefore, even if the phase plot visually appears symmetric (even) at $\pm\pi$, it still identically satisfies the odd symmetry condition modulo 2π.

Q3.14.1 NAT 2014 1M Ans: 0.51 ● ○ ○

Method 1

$$X(\omega) = \sum_{K=-\infty}^{\infty} 2\pi a_k \delta(\omega - K\omega_0)$$

$$a_K = \frac{1}{2\pi}\int_0^2 x(t)e^{-jK\omega_0 t}\,dt$$

$$= \frac{1}{2\pi}\left\{\int_0^1 1\cdot e^{-jK\omega_0 t}dt - \int_1^2 1\cdot e^{-jK\omega_0 t}dt\right\}$$

$$= \frac{1}{2\pi}\left[\frac{1}{-jK\omega_0}\left[e^{-jK\omega_0} - 1\right] - \frac{1}{-jK\omega_0}\left[e^{-j2K\omega_0} - e^{-jK\omega_0}\right]\right]$$

$$= \frac{1}{2\pi}\left[\frac{e^{-jK\omega_0/2}}{jK\omega_0}\cdot 2j\sin\left(\frac{K\omega_0}{2}\right) - \frac{1}{jK\omega_0}e^{-j\frac{K\omega_0 3}{2}}\left(2j\sin\left(\frac{K\omega_0}{2}\right)\right)\right]$$

$$= \frac{1}{\pi}\left[\frac{\sin\left(\frac{K\omega_0}{2}\right)}{K\omega_0}\left(e^{-jK\omega_0}\right)\left(e^{j\frac{K\omega_0}{2}} - e^{-j\frac{K\omega_0}{2}}\right)\right]$$

$$= \frac{1}{\pi}\left(\frac{\sin(K\omega_0/2)}{K\omega_0}\right)e^{-jK\omega_0}\times 2j\sin\left(\frac{K\omega_0}{2}\right)$$

$$= \frac{1}{\pi^2}\frac{\left(\sin\left(K\frac{\pi}{2}\right)\right)^2}{K}2j\times\cos(K\pi)$$

7^{th} harmonic power $= (2\pi a_7)^2$
5^{th} harmonic power $= (2\pi a_5)^2$

$$\text{Ratio} = \left(\frac{a_7}{a_5}\right)^2 = \left(\frac{5}{7}\right)^2 = \frac{25}{49} = 0.51$$

Method 2

We know the Fourier Series (FS) of a square wave of amplitude A is given by:

$$x(t) = \sum_{n=1}^{\infty} \frac{4A}{n\pi}(-1)^n \cos(n\omega_0 t) \quad ; \text{if } x(t) \text{ is even}$$

$$x(t) = \sum_{n=1}^{\infty} \frac{4A}{n\pi} \sin(n\omega_0 t) \quad ; \text{if } x(t) \text{ is odd}$$

In both cases, n only takes odd values ($n \in$ odd). Assuming an amplitude $A = 1$, the power for the n-th harmonic is proportional to the square of its Fourier coefficient.

$$\text{Power for 5th harmonic} = \left(\frac{4}{5\pi}\right)^2$$

$$\text{Power for 7th harmonic} = \left(\frac{4}{7\pi}\right)^2$$

Taking the ratio of the power of the 7th harmonic to the 5th harmonic:

$$\text{Ratio} = \frac{\left(\frac{4}{7\pi}\right)^2}{\left(\frac{4}{5\pi}\right)^2} = \left(\frac{5}{7}\right)^2 = \frac{25}{49} \approx 0.51$$

$$\boxed{0.51}$$

Key Points

- The power of the k^{th} harmonic is proportional to the square of the magnitude of its Fourier coefficient, i.e., $|a_k|^2$.
- For a standard periodic square wave, the magnitude of the k^{th} harmonic coefficient a_k is inversely proportional to k ($|a_k| \propto 1/k$).
- Taking the ratio of the power of two harmonics simplifies to the inverse squared ratio of their harmonic indices.

Q3.11.1 MCQ 2011 1M Ans: C ● ○ ○

A periodic signal $f(t)$ can be represented by its trigonometric Fourier series as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

For an even function, the condition $f(t) = f(-t)$ holds true.
The coefficient for the sine terms, b_n, is evaluated over a symmetric period T_0 as:

$$b_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} f(t) \sin(n\omega_0 t)\, dt$$

Since $f(t)$ is an even function and $\sin(n\omega_0 t)$ is an odd function, their product $f(t)\sin(n\omega_0 t)$ results in an odd function. The definite integral of an odd function evaluated over a symmetrical interval $[-T_0/2, T_0/2]$ is always zero.

$$\implies b_n = 0$$

Therefore, the trigonometric Fourier series of an even function consists only of the DC term (a_0) and cosine terms (a_n). It does not contain sine terms.

$$\boxed{\text{C) sine terms}}$$

Key Points

- **Even Symmetry:** If $f(t) = f(-t)$, the Fourier series contains only DC and cosine terms ($b_n = 0$).
- **Odd Symmetry:** If $f(t) = -f(-t)$, the Fourier series contains only sine terms ($a_0 = 0, a_n = 0$).
- **Half-Wave Symmetry:** If $f(t) = -f(t \pm T_0/2)$, the Fourier series contains only odd harmonics (coefficients for even n are zero).

Q3.10.1 MCQ 2010 1M Ans: C ● ○ ○

Since $f(t)$ is an even function, its trigonometric Fourier series contains only cosine terms.

D.C. component,

$$A_0 = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\,dt = \frac{2}{T}\int_0^{T/2} f(t)\,dt$$

$$= \frac{2}{T}\left[\int_0^{T/4} A\,dt + \int_{T/4}^{T/2} (-2A)\,dt\right]$$

$$= \frac{2}{T}\left[\frac{AT}{4} - 2A\left(\frac{T}{2} - \frac{T}{4}\right)\right]$$

$$= \frac{2}{T}\left[-\frac{AT}{4}\right] = -\frac{A}{2}$$

Therefore, the trigonometric Fourier series for the waveform $f(t)$ contains only cosine terms and a negative value for the dc component.

$$\boxed{\text{C) only cosine terms and a negative value for the dc components}}$$

Key Points

- For an even function where $f(t) = f(-t)$, the Fourier series coefficients for sine terms (b_n) evaluate to zero, leaving only the DC and cosine terms.
- The DC component (A_0) is the average value of the waveform over one complete period T.
- The integral of an even function over a symmetric interval $[-T/2, T/2]$ can be simplified to twice the integral over $[0, T/2]$.

Q3.09.1 MCQ 2009 1M Ans: A ● ○ ○

The Fourier series of a real periodic function has only cosine terms if it is even and only sine terms if it is odd. Therefore, statement (P) "cosine terms if it is even" and statement (S) "sine terms if it is odd" are correct.

$$\boxed{\text{A) P and S}}$$

Key Points

- **Even Symmetry:** If $f(t) = f(-t)$, the product $f(t)\sin(n\omega_0 t)$ is an odd function. Its integral over a fundamental period is zero, making all sine coefficients $b_n = 0$. The series contains only DC and cosine terms.
- **Odd Symmetry:** If $f(t) = -f(-t)$, the product $f(t)\cos(n\omega_0 t)$ is an odd function. Its integral over a fundamental period is zero, making the DC component $a_0 = 0$ and all cosine coefficients $a_n = 0$. The series contains only sine terms.

Q3.05.1 MCQ 2005 1M Ans: C ● ○ ○

For a continuous-time signal to have a defined Fourier series representation, a primary condition is that the signal must be periodic over the interval $-\infty < t < \infty$. Let us analyze each given function:
Option (A): $f(t) = 3\sin(25t)$ is a standard sinusoidal signal, which is inherently periodic.

Option (B): $f(t) = 4\cos(20t+3)+2\sin(10t)$ is a sum of two sinusoids. The ratio of their angular frequencies is $\frac{\omega_1}{\omega_2} = \frac{20}{10} = 2$, which is a rational number. Because the ratio is rational, this composite signal is also periodic.

Option (C): $f(t) = \exp(-|t|)\sin(25t)$. The presence of the exponential term $\exp(-|t|)$ acts as an envelope that forces the signal's amplitude to decay toward zero as $t \to \pm\infty$. Consequently, the waveform does not repeat itself over time, making it an aperiodic (non-periodic) signal.

Option (D): $f(t) = 1$ is a constant (DC) signal. Constant signals are considered periodic with an undefined fundamental period and have a trivial Fourier series containing only the DC term.

Since the signal in option (C) is aperiodic, a Fourier series cannot be defined for it.

$$\boxed{\text{C) } \exp(-|t|)\sin(25t)}$$

Key Points

- A fundamental requirement for the existence of a Fourier series is that the function must be periodic, satisfying the condition $f(t) = f(t \pm nT_0)$.
- Signals that are multiplied by decaying exponentials, such as e^{-at} or $e^{-|t|}$, are transient energy signals and are strictly aperiodic.
- While periodic power signals can be expanded into a Fourier series, aperiodic energy signals require the Fourier Transform for frequency domain representation.

Q3.04.1 MCQ 2004 2M Ans: A ● ○ ○

Method 1

The given rectangular pulse train $s(t)$ is periodic signal with period $T_1 = 0.1$ msec and $f_1 = 10$ kHz
It can be expressed in terms of Fourier series as

$$s(t) = \sum c_n e^{j2\pi n f_1 t},$$

Let the signal convolving with it be $g(t)$

$$g(t) = \cos^2(4\pi 10^3 t)$$

$$g(t) = \frac{1}{2} + \frac{1}{2}\cos(2\pi f_2 t),$$

$f_2 = 4\,\text{kHz}$ and $T_2 = 0.25\,\text{msec}$

$$= \frac{1}{2} + \frac{1}{4}e^{j2\pi f_2 t} + \frac{1}{4}e^{-j2\pi f_2 t}$$

From the definition of convolution,

for $n \neq 0$,

$$\left[\frac{1}{2} * c_n e^{j2\pi n f_1 t}\right] = \frac{1}{2}c_n \int_{t=-\infty}^{\infty} e^{j2\pi n f_1 (t-\tau)}\, d\tau$$

$$= \frac{1}{2}c_n e^{j2\pi n f_1 t} \int_{-\infty}^{\infty} e^{-j2\pi n f_1 \tau}\, d\tau \ldots\ldots (1)$$

= 0 as the integral becomes zero over a period, T_1.

for $n = 0$, $\frac{1}{2} * c_0 = K \to \infty$ (d.c. signal)

for all n, $c_n e^{j2\pi n f_1 t} * \frac{1}{4}e^{\pm j2\pi f_2 t} = 0$ in accordance with (1)

$\therefore\ s(t) * g(t)$ is a large DC signal.

Method 2

The fundamental frequency f_0 of the signal $s(t)$ is given by:

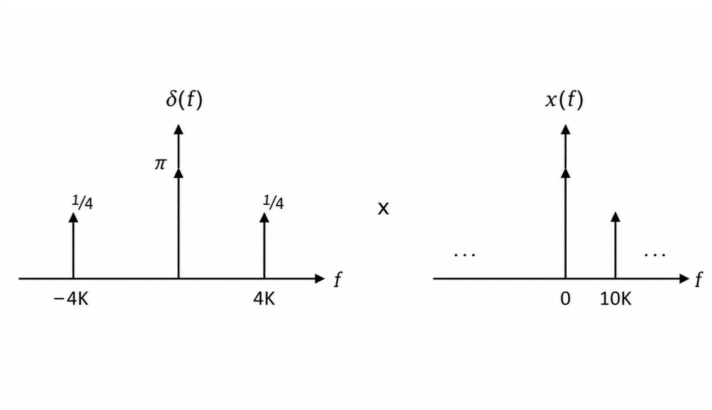
$$f_0 = \frac{1}{0.1\text{ ms}} = 10\,\text{kHz}$$

$\therefore$ The fundamental frequency components will be present at 0, 10 kHz, 20 kHz, ... For the signal term $\cos^2(4\pi \times 10^3 t)$, expanding using the standard trigonometric identity gives:

$$\cos^2(4\pi \times 10^3 t) = \frac{1}{2} + \frac{\cos(8\pi \times 10^3 t)}{2}$$

The frequency spectrum for this term will have impulses at 0 Hz (the DC component) and at $f = \pm 4\,\text{kHz}$. According to the properties of the Fourier Transform:

$$\text{Convolution in time domain} \rightleftharpoons \text{Multiplication in frequency domain}$$



When evaluating the final output, we must multiply the two frequency spectra. The first spectrum has components at $\{0, \pm 4\,\text{kHz}\}$, and the second spectrum $X(f)$ has components at $\{0, 10\,\text{kHz}, 20\,\text{kHz}, \ldots\}$. Since multiplication yields a non-zero result only where both spectra have overlapping components, the only surviving frequency is at 0 Hz. Hence, only the DC component will be present in the final output.

$\boxed{\text{A) DC}}$

Key Points

- Convolution in the time domain corresponds to multiplication in the frequency domain: $x(t) * y(t) \xleftrightarrow{\mathcal{F}} X(f)Y(f)$.
- The convolution of two signals with discrete spectra (or a periodic signal with another signal) yields a non-zero output only at frequencies present in both signals.
- The fundamental frequency of $s(t)$ is 10 kHz, generating harmonics at $0, 10, 20, 30\ldots$ kHz. The signal $g(t)$ has components at 0 kHz (DC) and 4 kHz. Since the only common frequency is 0 kHz, the resulting convolution contains only a DC component.

Q3.03.1 MCQ 2003 1M Ans: D

$$c_{-k} = c_k^*$$

$$c_3 = 3 + j5$$

$$c_{-3} = c_k^* = 3 - j5$$

$$\boxed{\text{D) } 3 - j5}$$

Key Points

- For a real-valued periodic signal, the exponential Fourier series coefficients exhibit conjugate symmetry, mathematically expressed as $c_{-k} = c_k^*$.
- To find the complex conjugate of a number in rectangular form $(a + jb)$, you simply negate the imaginary part to get $(a - jb)$.

Q3.02.1 MCQ 2002 1M Ans: B

$$x(t) = 2\cos\pi t + 7\cos t$$

$$T_1 = \frac{2\pi}{(\pi)} = 2$$

$$T_2 = \frac{2\pi}{1} = 2\pi$$

$$\frac{T_1}{T_2} = \frac{1}{\pi} = \text{irrational}$$

$\therefore$ $x(t)$ is not periodic and does not satisfy Dirichlet condition.

$$\boxed{\text{B) } x(t) = 2\cos\pi t + 7\cos t}$$

Key Points

- For a continuous-time signal consisting of the sum of two or more sinusoids to be periodic, the ratio of their individual fundamental periods (T_1/T_2) must be a rational number.
- Since the ratio is irrational (due to the presence of π), the combined signal $x(t)$ is aperiodic.
- A fundamental requirement for a signal to have a Fourier series expansion is that it must be periodic over an interval. Thus, an aperiodic signal cannot be expanded into a Fourier series.

Q3.00.1 MCQ 2000 2M Ans: C ● ○ ○

W_1 is a periodic square waveform with period T and it is having odd symmetry and also odd harmonic symmetry (or Half-wave symmetry).
W_2 is a periodic triangular waveform with period T and it is having odd symmetry and also odd harmonic symmetry (or Half-wave symmetry).
$\therefore$ Only odd harmonics: nf_0, $n = 1, 3, 5$ etc. of sine terms are present in wave forms W_1 and W_2 in their Fourier series expansion.

Note that waveform, W_2 can be obtained by integrating the waveform, W_1.

If C_n is the exponential FS coefficient of the n^{th} harmonic component,

$$|C_n| \propto \left|\frac{1}{n}\right| = |n^{-1}| \quad \text{for waveform } W_1$$

$$|C_n| \propto \left|\frac{1}{n^2}\right| = |n^{-2}| \quad \text{for waveform } W_2$$

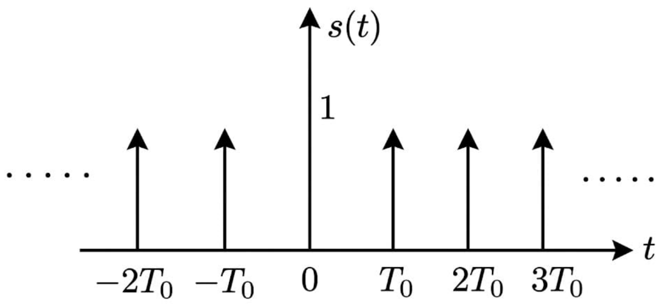
$$\boxed{\text{(c) } |n^{-1}| \text{ and } |n^{-2}|}$$

Key Points

- A waveform with both odd symmetry and odd harmonic (half-wave) symmetry contains only odd harmonic sine terms in its Fourier series expansion.
- The magnitude of the Fourier series coefficients for a square wave decays at a rate proportional to $1/n$.
- Integrating a signal in the time domain corresponds to dividing its Fourier coefficients by $jn\omega_0$, meaning the coefficients of the resulting triangular wave decay at a rate proportional to $1/n^2$.

Q3.99.1 MCQ 1999 2M Ans: D ● ○ ○

The given impulse train, $s(t)$ with strength of each impulse as 1, is a periodic function with period, T_0 as shown in Fig. 1.



In the form of Fourier series

$$s(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}$$

and C_n is given by

$$C_n = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \delta(t) e^{-jn\omega_0 t}\, dt = \left.\frac{1 e^{-jn\omega_0 t}}{T_0}\right|_{t=0}$$

$$C_n = \frac{1}{T_0}, \therefore\ s(t) = \frac{1}{T_0}\sum_{n=-\infty}^{\infty} e^{j\frac{2\pi}{T_0}nt}$$

D) $\frac{1}{T_0}\sum_{n=-\infty}^{\infty}\exp(\frac{j2\pi nt}{T_0})$

Key Points

- The exponential Fourier series coefficients of a unit impulse train with period T_0 are constant and equal to $\frac{1}{T_0}$ for all frequencies.
- **Sifting Property of Impulse Function:** $\int_a^b x(t)\delta(t-t_0)\,dt = x(t_0)$ provided $a < t_0 < b$.

Q3.98.1

MCQ | 1998 | 1M | Ans: C | ● ○ ○

The given periodic signal $x(t)$ is shown in Fig. 1, over one period : $|t| < \frac{T_0}{2}$

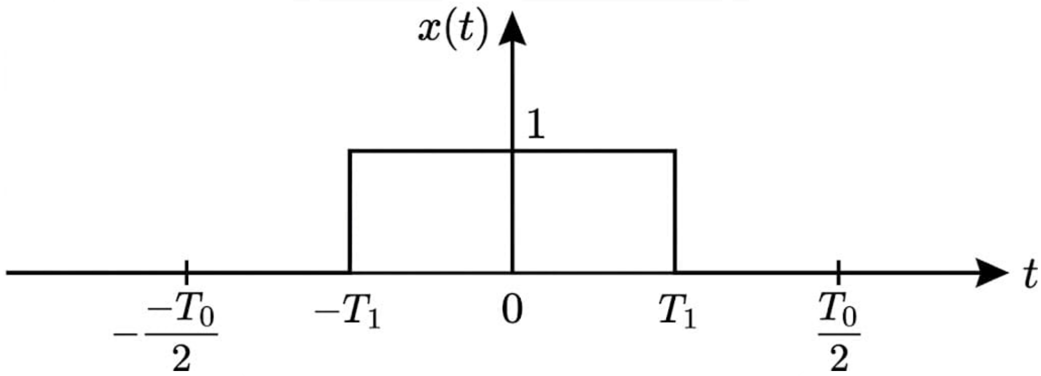


Fig. Solution Q3.98.1

$$a_0 = \text{d.c. component} = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\,dt$$

From the given figure, $x(t) = 1$ for $-T_1 < t < T_1$ and 0 otherwise in the fundamental period. Evaluating the integral:

$$a_0 = \frac{1}{T_0}\int_{-T_1}^{T_1} 1\,dt$$

$$a_0 = \frac{1}{T_0}\Big[t\Big]_{-T_1}^{T_1}$$

$$a_0 = \frac{1}{T_0}(T_1 - (-T_1))$$

$$a_0 = \frac{2T_1}{T_0}$$

C) $\frac{2T_1}{T_0}$

Key Points

- The DC component (or average value) of a continuous-time periodic signal $x(t)$ with fundamental period T_0 is defined as $a_0 = \frac{1}{T_0}\int_{T_0} x(t)\,dt$.
- Geometrically, the DC component is equal to the net area under the waveform over one complete period divided by the fundamental period T_0.

Q3.98.2 MCQ 1998 1M Ans: C ● ○ ○

The trigonometric Fourier series of a general periodic time function $f(t)$ is expressed as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

This general expression fundamentally consists of: 1. A DC component (a_0) 2. Cosine terms ($a_n \cos(n\omega_0 t)$) 3. Sine terms ($b_n \sin(n\omega_0 t)$)
Mathematically, the DC component a_0 can be interpreted as a zero-frequency cosine term, since $\cos(0) = 1$. Therefore, the complete orthogonal basis set for representing any general periodic waveform is composed of **cosine and sine terms**.
Unless a specific symmetry (such as even or odd) is stated, a periodic signal requires both sine and cosine terms for a complete representation.

C) cosine and sine terms

Key Points

- The trigonometric Fourier series utilizes harmonically related sine and cosine functions to decompose and represent periodic signals.

Q3.96.1 MCQ 1996 1M Ans: C ● ○ ○

A periodic function $f(t)$ is even if it satisfies the condition $f(t) = f(-t)$.
The trigonometric Fourier series of a periodic function is given by:

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

The coefficients for the sine terms, b_n, are evaluated over a symmetric period T_0 as:

$$b_n = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} f(t)\sin(n\omega_0 t)\,dt$$

Since $f(t)$ is an even function and $\sin(n\omega_0 t)$ is an odd function, their product $f(t)\sin(n\omega_0 t)$ results in an odd function. The definite integral of an odd function evaluated over a symmetrical interval $[-T_0/2, T_0/2]$ is always zero.

$$\implies b_n = 0$$

Therefore, the trigonometric Fourier series of an even function consists only of the DC term (a_0) and cosine terms (a_n). It does not contain sine terms.

C) sine terms

Q3.94.1 MCQ 1994 1M Ans: D ● ○ ○

A periodic function $f(t)$ is classified as an odd function if it satisfies the condition $f(t) = -f(-t)$.

The trigonometric Fourier series of a periodic function is given by:

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

The Fourier coefficients are evaluated over a symmetric period T_0 as follows:

1. **DC Component (a_0):**

$$a_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t)\, dt$$

Since $f(t)$ is an odd function, its integral over a symmetrical interval $[-T_0/2, T_0/2]$ evaluates to zero. Thus, $a_0 = 0$.

2. **Cosine Coefficients (a_n):**

$$a_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} f(t) \cos(n\omega_0 t)\, dt$$

The product of an odd function $f(t)$ and an even function $\cos(n\omega_0 t)$ yields an odd function. Integrating this odd function over a symmetrical interval results in zero. Thus, $a_n = 0$.

3. **Sine Coefficients (b_n):**

$$b_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} f(t) \sin(n\omega_0 t)\, dt$$

The product of an odd function $f(t)$ and another odd function $\sin(n\omega_0 t)$ yields an even function. The integral of an even function over a symmetric interval is generally non-zero.

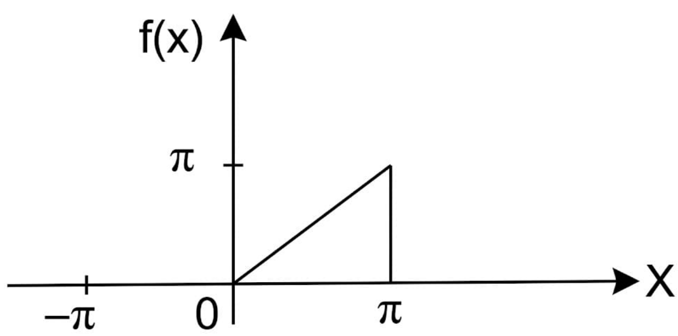
$$\implies b_n \neq 0$$

Therefore, the Fourier series of an odd periodic function contains only sine terms.

$$\boxed{\text{D) sine terms}}$$

Q3.93.1 NAT 1993 2M Ans: $\frac{\pi^2}{8}$ ● ○ ○

Given the Fourier series



$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{1}{\pi n^2} (\cos n\pi - 1) \cos nx - \frac{1}{n} \cos n\pi \sin nx \right].$$

At the point $x = \pi$, the Fourier series converges to the average of the left and right limits:

$$f(\pi) = \frac{\pi + 0}{2} = \frac{\pi}{2}.$$

Substituting $x = \pi$,

$$\frac{\pi}{2} = \frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{1}{\pi n^2}(\cos n\pi - 1)\cos n\pi.$$

Since

$$\sin(n\pi) = 0,$$

all sine terms vanish. Also,

$$\cos n\pi = (-1)^n.$$

Therefore,

$$(\cos n\pi - 1)\cos n\pi = ((-1)^n - 1)(-1)^n = 1 - (-1)^n.$$

Hence,

$$\frac{\pi}{2} = \frac{\pi}{4} + \frac{1}{\pi}\sum_{n=1}^{\infty} \frac{1-(-1)^n}{n^2}.$$

Multiplying throughout by π,

$$\frac{\pi^2}{2} = \frac{\pi^2}{4} + \sum_{n=1}^{\infty} \frac{1-(-1)^n}{n^2}.$$

Thus,

$$\sum_{n=1}^{\infty} \frac{1-(-1)^n}{n^2} = \frac{\pi^2}{4}.$$

Since

$$1-(-1)^n = \begin{cases} 0, & n \text{ even}, \\ 2, & n \text{ odd}, \end{cases}$$

we obtain

$$2\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{4}.$$

Therefore,

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}.$$

$$\boxed{\frac{\pi^2}{8}}$$

Key Points

- At a discontinuity, a Fourier series converges to the average of the left-hand and right-hand limits.
- $\cos(n\pi) = (-1)^n$ and $\sin(n\pi) = 0$.
- The factor $1-(-1)^n$ isolates the odd-indexed terms of the series.

Mistakes to be avoided

- Using $f(\pi) = \pi$ instead of the Fourier convergence value $\frac{\pi}{2}$.
- Missing the factor of 2 when converting the summation to odd terms only.

Q3.87.1 MCQ 1987 2M Ans: C ● ○ ○

A half-wave rectified sinusoidal waveform, $v(t)$ is shown in $Fig.Q3.87.1$

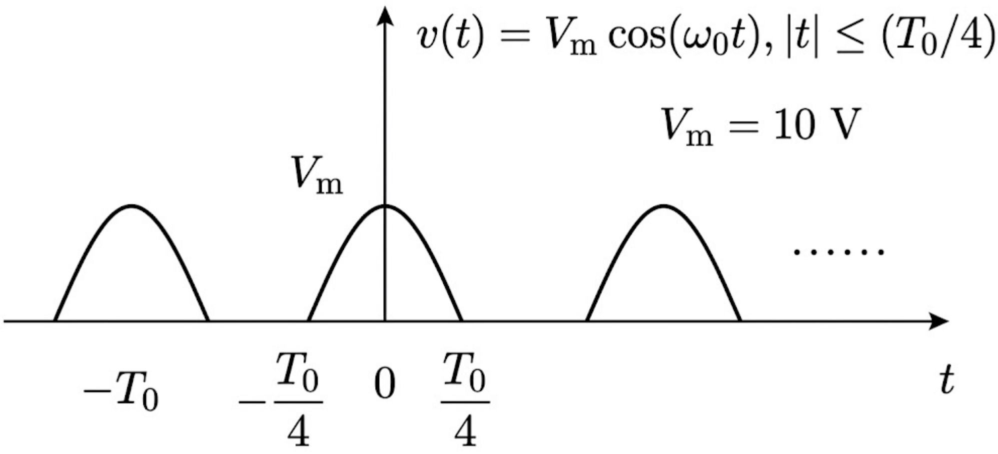


Fig. Solution Q3.87.1

From the figure, we have $v(t) = V_m \cos(\omega_0 t)$ for $|t| \leq (T_0/4)$ and $V_m = 10$ V.
Period = T_0
$f_0 = 1/T_0$ = fundamental frequency = 1st harmonic
$\omega_0 = 2\pi f_0$, $n\omega_0 = n^{\text{th}}$ harmonic
Fourier series of $v(t)$ is given by

$$v(t) = a_0 + a_1 \cos(1\omega_0 t) + a_2 \cos(2\omega_0 t) + \ldots$$

a_0 = Average (or mean or dc) value

$$a_0 = \frac{1}{T_0} \int_{T_0} v(t)\, dt$$

$$= \frac{1}{T_0} \int_{-T_0/4}^{T_0/4} V_m \cos(\omega_0 t)\, dt = \frac{V_m}{\pi}$$

Peak value of the fundamental component

$$a_1 = \frac{2}{T_0} \int_{\langle T_0 \rangle} v(t) \cos(1\omega_0 t)\, dt$$

$$= \frac{2}{T_0} \int_{-T_0/4}^{T_0/4} V_m \cos^2(\omega_0 t)\, dt = \frac{V_m}{2}$$

Remember the answers:

$$a_0 = \frac{V_m}{\pi}, \quad a_1 = \frac{V_m}{2}$$

Substituting $V_m = 10$ V:

$$\therefore a_0 = \frac{10}{\pi}\text{ V}, \quad a_1 = 5\text{ V}$$

$$\boxed{\text{C) } \frac{10}{\pi}\text{ V}, 5\text{ V}}$$

Key Points

- The DC (average) component of a half-wave rectified sinusoidal signal with peak amplitude V_m is always V_m/π.
- The peak amplitude of the fundamental frequency component for a half-wave rectified sinusoid is always $V_m/2$.

DEBUG INFO

Chapter III

Continuous Time Fourier Series

Total Questions : 24

MCQ : 20

MSQ : 1

NAT : 3

SOLUTIONS

IV

Continuous Time Fourier Series

Topics

Q4.26.1 MCQ | 2026 | 1M | Ans: D | ● ○ ○

$x_1(t) = \cos 20\pi t$

On sampling

$$t = nT_s = \frac{n}{f_s} = \frac{n}{40}$$

$$x_1(n) = x_1(nT_s) = \cos\left(20\pi \times \frac{n}{40}\right) = \cos\frac{\pi}{2}n$$

Similarly, for $x_2(t)$

$$x_2(t) = \cos 100\pi t$$

On sampling

$$t = nT_s = \frac{n}{40}$$

$$x_2(n) = x_2(nT_s) = \cos\left(100\pi\frac{n}{40}\right) = \cos\left(\frac{5\pi}{2}n\right)$$

$$= \cos\left[\left(2\pi + \frac{\pi}{2}\right)n\right] = \cos\frac{\pi}{2}n$$

Therefore,

$$x_1(n) = x_2(n)$$

Hence, all of the fourth to seventh samples of $x_1(t)$ are equal to the corresponding samples of $x_2(t)$.

D) All of the fourth to seventh samples of $x_1(t)$ are equal to the corresponding samples of $x_2(t)$.

Key Points

- **Aliasing Concept:** When sampling a continuous-time signal, frequency components that differ by integer multiples of the sampling frequency (f_s) map to the same discrete-time sequence.
- **Discrete Cosine Periodicity:** The sequence $\cos(\omega_0 n)$ is periodic with respect to the digital frequency ω_0 with period 2π. That is, $\cos((\omega_0 + 2\pi m)n) = \cos(\omega_0 n)$ for any integer m.

Q4.25.1 MCQ | 2025 | 1M | Ans: A | ● ○ ○

We are given that $F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$ exists. This means that $f(t)$ is absolutely integrable, i.e.,

$$\int_{-\infty}^{\infty} |f(t)|dt < \infty$$

To find a general bound on $|F(\omega)|$, we apply the triangle inequality for integrals:

$$|F(\omega)| = \left|\int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt\right| \leq \int_{-\infty}^{\infty} \left|f(t)e^{-j\omega t}\right| dt = \int_{-\infty}^{\infty} |f(t)|dt$$

This inequality is always true.

Now check the options:

- (A) $|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)|dt$ – Correct
- (B) Strict inequality – not always true
- (C) Inequality involving $\int f(t)dt$ – this can be 0 for symmetric signals (e.g., odd functions), but $|F(\omega)|$ may be nonzero
- (D) Similar issue – not always true

$$\boxed{\text{A) } |F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)|dt}$$

Key Points

- **Triangle Inequality for Integrals:** The absolute value of an integral is less than or equal to the integral of the absolute value, $\left|\int g(t)dt\right| \le \int |g(t)|dt$.
- **Magnitude of Complex Exponentials:** For any real arguments ω and t, the magnitude $\left|e^{-j\omega t}\right| = 1$.

Q4.25.2 MCQ 2025 2M Ans: A ● ○ ○ CLICK FOR VIDEO SOLUTION

Method 1

We are given that $f(t)$ is bandlimited to $[-\omega_c, \omega_c]$, meaning its Fourier transform $F(j\omega)$ is zero outside this interval.

Step 1: Sampling process

Sampling is done using the impulse train:

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - \tau - nT_s)$$

Multiplying this with $f(t)$ gives:

$$y(t) = f(t) \cdot p(t) = \sum_{n=-\infty}^{\infty} f(\tau + nT_s) \cdot \delta(t - \tau - nT_s)$$

This means we are taking samples of $f(t)$ at times $t = \tau + nT_s$.

Step 2: Filtering with ideal LPF

The sampled signal $y(t)$ is passed through an ideal lowpass filter with impulse response:

$$h(t) = T_s \cdot \text{sinc}\left(\frac{\omega_c t}{\pi}\right)$$

The output of the system is:

$$z(t) = y(t) * h(t) = \sum_{n=-\infty}^{\infty} f(\tau + nT_s) \cdot h(t - \tau - nT_s)$$

Substitute $h(t)$:

$$z(t) = T_s \sum_{n=-\infty}^{\infty} f(\tau + nT_s) \cdot \text{sinc}\left(\frac{\omega_c(t - \tau - nT_s)}{\pi}\right)$$

This is the reconstruction formula for $f(t)$, but it appears to reconstruct $f(t - \tau)$ at first glance.

Step 3: Time-invariance and interpretation

Here's the key point: Even though the samples are taken at $t = \tau + nT_s$, the reconstruction system is linear and time-invariant. A shift in sampling time does not affect the shape of the reconstructed signal, because the sinc interpolation is symmetric and consistent. Hence, the filter effectively undoes the delay, and we recover the original signal $f(t)$.

Step 4: Condition for perfect reconstruction

Perfect reconstruction using sinc interpolation is possible only if there is no aliasing, which happens when:

$$T_s < \frac{\pi}{\omega_c}$$

This is the Nyquist condition for bandlimited signals.

Method 2

The sampled signal in the time domain is given by the multiplication of the input continuous-time signal and the sampling pulse train:

$$y_s(t) = f(t)p(t)$$

The sampling pulse train is a shifted impulse train:

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - 2 - nT_s) = c(t - 2)$$

where $c(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$. Taking the Fourier transform of the unshifted standard impulse train yields:

$$C(\omega) = \frac{2\pi}{T_s} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

Using the time-shifting property of the Fourier transform, the spectrum of the shifted pulse train $p(t)$ is:

$$P(\omega) = e^{-j2\omega} C(\omega) = \frac{2\pi}{T_s} \sum_{n=-\infty}^{\infty} e^{-j2n\omega_s} \delta(\omega - n\omega_s)$$

In the frequency domain, time-domain sampling translates to a convolution operation:

$$Y_s(\omega) = \frac{1}{2\pi}[F(\omega) * P(\omega)]$$

Applying the sifting property of the Dirac delta function ($F(\omega) * \delta(\omega - \omega_0) = F(\omega - \omega_0)$), the sampled spectrum becomes:

$$Y_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{-j2n\omega_s} F(\omega - n\omega_s)$$

This shows that the sampled spectrum consists of infinitely many shifted clones of $F(\omega)$ spaced by ω_s, where each clone is scaled by $\frac{1}{T_s}$ and multiplied by a phase factor $e^{-j2n\omega_s}$. The sampled signal is then passed through an ideal low-pass filter $H(\omega)$ with a cutoff frequency ω_c and a passband gain of T_s. Assuming no aliasing occurs, the filter rejects all higher-order clones ($n \neq 0$) and allows only the baseband component ($n = 0$) to pass:

$$Y(\omega) = H(\omega) \cdot \left(\frac{1}{T_s} e^{-j2(0)\omega_s} F(\omega - 0) \right)$$

$$Y(\omega) = T_s \cdot \left[\frac{1}{T_s} F(\omega) \right] = F(\omega)$$

Taking the inverse Fourier transform, the final output signal is precisely the original input signal:

$$y(t) = f(t)$$

To avoid aliasing and ensure perfect reconstruction, the separation between the baseband and the first shifted clone ($n = 1$) must be sufficiently large such that the first clone does not overlap into the filter's passband. The lowest frequency edge of the first clone is located at $\omega_s - \omega_c$.

$$\omega_s - \omega_c \geq \omega_c \implies \omega_s \geq 2\omega_c$$

Substituting $\omega_s = \frac{2\pi}{T_s}$:

$$\frac{2\pi}{T_s} \geq 2\omega_c \implies T_s \leq \frac{\pi}{\omega_c}$$

$$\boxed{y(t) = f(t), \quad T_s \leq \frac{\pi}{\omega_c}}$$

$$\boxed{\text{A) } f(t) \text{ if } T_s < \pi/\omega_c}$$

Key Points

- **Nyquist-Shannon Sampling Theorem:** A continuous-time signal bandlimited to ω_c can be perfectly reconstructed if the sampling period satisfies $T_s < \pi/\omega_c$.
- **Shifted Sampling:** Sampling a signal with a shifted impulse train $\sum \delta(t-\tau-nT_s)$ does not prevent perfect reconstruction via an ideal lowpass filter, provided the Nyquist rate is satisfied. The ideal LPF interpolates the shifted samples back to the original continuous signal $f(t)$.

Q4.24.1 MCQ 2024 2M Ans: C CLICK FOR VIDEO SOLUTION

Method 1

$$x(t) = 2\cos\left(8\pi t + \frac{\pi}{3}\right), \quad \omega_o = 8\pi$$

$$f_s = 15\,\text{Hz}$$

$$x(t) \xrightarrow{\text{sample}} x_s(t) \xrightarrow{h(t)} x_0(t)$$

$$\Rightarrow \quad h(t) = \frac{\sin 2\pi t}{\pi t}\cos\left(38\pi t - \frac{\pi}{2}\right)$$

Let $p(t) = \frac{\sin 2\pi t}{\pi t}$

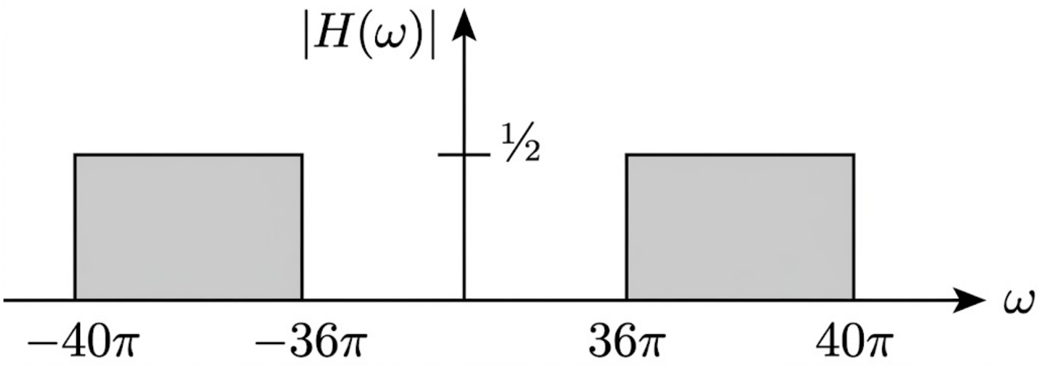


Fig. Solution Q4.24.1

After sampling:

$$X_s(\omega) = f_s \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

Frequency components present in sampler output $n\omega_s \pm \omega_o$

$$\omega_o, \omega_s \pm \omega_o, 2\omega_s \pm \omega_o, \ldots$$

$$8\pi, 30\pi \pm 8\pi, 60\pi \pm 8\pi, \ldots$$

$$8\pi, 22\pi, 38\pi, 52\pi, 68\pi, \ldots \text{(rad/sec)}$$

System will pass '38π' component of input.

$$x_o(t) = 2 \times \frac{f_s}{2}\cos\left[38\pi t + \frac{\pi}{3} - \frac{\pi}{2}\right] = 15\cos\left(38\pi t - \frac{\pi}{6}\right)$$

Method 2

The continuous-time input signal is given as:

$$x(t) = 2\cos\left(8\pi t + \frac{\pi}{3}\right)$$

The fundamental frequency of the input signal is $f_m = 4\,\text{Hz}$. In the frequency domain, the spectrum $X(f)$ consists of impulses at $f = 4\,\text{Hz}$ (with a phase shift of $+\frac{\pi}{3}$) and $f = -4\,\text{Hz}$ (with a phase shift of $-\frac{\pi}{3}$). The signal is sampled at a frequency of $f_s = 15\,\text{Hz}$. The spectrum of the ideally sampled signal, $X_s(f)$, consists of infinitely many shifted clones of $X(f)$, each scaled by the sampling frequency f_s:

$$X_s(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

The positive frequency components in $X_s(f)$ will be located at $nf_s \pm f_m = 15n \pm 4\,\text{Hz}$. For $n = 1$, the spectral clones are situated at $15 - 4 = 11\,\text{Hz}$ and $15 + 4 = 19\,\text{Hz}$. The component at 19 Hz originates from the positive frequency +4 Hz shifted by +15 Hz. Therefore, it retains its original phase of $+\frac{\pi}{3}$ and has a scaled amplitude of $f_s \cdot 1 = 15$. The sampled signal is passed through an LTI filter with the impulse response:

$$h(t) = \frac{\sin(2\pi t)}{\pi t}\cos\left(38\pi t - \frac{\pi}{2}\right)$$

The baseband component of this filter, $\frac{\sin(2\pi t)}{\pi t}$, corresponds to an ideal low-pass filter with a cutoff frequency of 1 Hz and a passband gain of 1. Multiplying this by $\cos(38\pi t - \frac{\pi}{2})$ modulates the filter, shifting its center frequency to $f_0 = 19\,\text{Hz}$. This forms a bandpass filter with a passband of $[18\,\text{Hz}, 20\,\text{Hz}]$. Due to Euler's formula expansion for the cosine term, the filter introduces an amplitude scaling of $\frac{1}{2}$ and a phase shift of $-\frac{\pi}{2}$ at the positive center frequency (+19 Hz). When passing $X_s(f)$ through $H(f)$: Only the 19 Hz clone falls within the filter's passband. The output component at 19 Hz will have its amplitude modified and its phase shifted by the filter:

$$\text{Output Amplitude} = 15 \cdot \frac{1}{2} = \frac{15}{2}$$

$$\text{Output Phase} = \frac{\pi}{3} - \frac{\pi}{2} = -\frac{\pi}{6}$$

Recombining the filtered positive and negative frequency components yields the final time-domain output:

$$x_0(t) = 2\left(\frac{15}{2}\right)\cos\left(2\pi(19)t - \frac{\pi}{6}\right)$$

$$x_0(t) = 15\cos\left(38\pi t - \frac{\pi}{6}\right)$$

Using the trigonometric identity $\cos(\theta - \frac{\pi}{6}) = \sin(\theta + \frac{\pi}{3})$, this can equivalently be written as $15\sin(38\pi t + \frac{\pi}{3})$.

$$\boxed{15\cos\left(38\pi t - \frac{\pi}{6}\right)}$$

$$\boxed{\text{C) } 15\cos(38\pi t - \pi/6)}$$

Key Points

- **Spectrum of Sampled Signal:** When a continuous-time signal with spectrum $X(\omega)$ is sampled at a frequency f_s (where $\omega_s = 2\pi f_s$), the resulting sampled signal spectrum $X_s(\omega)$ consists of periodic replications of $X(\omega)$ shifted by integer multiples of ω_s and scaled by f_s.
- **Filter Characteristics:** The impulse response $h(t) = \frac{\sin(2\pi t)}{\pi t}\cos\left(38\pi t - \frac{\pi}{2}\right)$ represents an ideal bandpass filter. The sinc term defines a baseband bandwidth of 2π, and the cosine term modulates (shifts) this passband to be centered at the carrier frequency $\omega_c = 38\pi$.
- **System Output:** Only the frequency component of the sampled signal that falls within the filter's passband (in this

case, the 38π component) will be passed to the output $x_o(t)$, with appropriate scaling and phase shifts applied.

Q4.24.2 MCQ | 2024 | 2M | Ans: B | ● ○ ○

Length of $x(t)$ is 1 and length of $y(t)$ is 4.

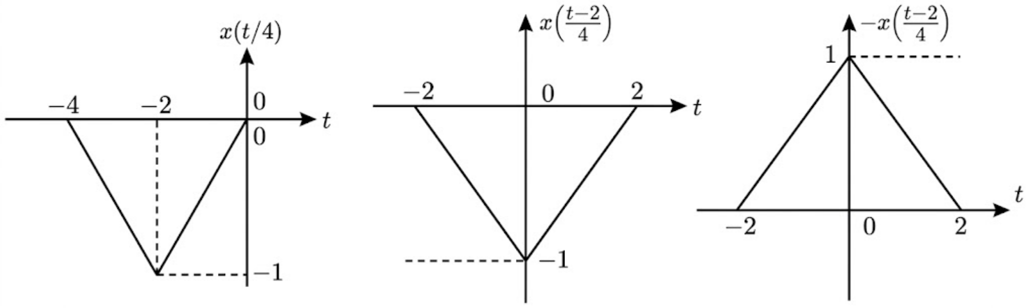


Fig. Solution Q4.24.2

$$\therefore \quad y(t) = -x\left(\frac{t-2}{4}\right)$$

$$y(t) = -x\left(\frac{t}{4} - \frac{2}{4}\right) = -x\left(\frac{t}{4} - \frac{1}{2}\right)$$

On taking Fourier transform, we get

$$x(t) \leftrightarrow X(f)$$

$$x\left(\frac{t}{4} - \frac{1}{2}\right) \leftrightarrow 4X(4f)e^{-j4\pi f}$$

$$-x\left(\frac{t}{4} - \frac{1}{2}\right) \leftrightarrow -4X(4f)e^{-j4\pi f}$$

$$\boxed{B) - 4X(4f)e^{-j4\pi f}}$$

Key Points

- **Time Scaling Property:** $x(at) \leftrightarrow \frac{1}{|a|}X\left(\frac{f}{a}\right)$
- **Time Shifting Property:** $x(t - t_0) \leftrightarrow X(f)e^{-j2\pi f t_0}$
- **Linearity Property:** $ax(t) \leftrightarrow aX(f)$

Q4.23.1 MCQ | 2023 | 1M | Ans: C | ● ○ ○

We know;

$$e^{-at^2}; a > 0 \overset{FT}{\rightleftharpoons} \sqrt{\frac{\pi}{a}} \cdot e^{-\omega^2/4a}$$

Here; $a = 1$

$$\therefore X(\omega) = \boxed{\sqrt{\pi} \cdot e^{-\omega^2/4}}$$

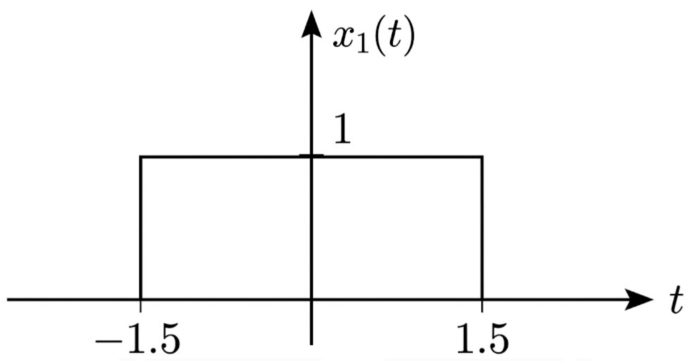
Key Points

- The Fourier transform of a Gaussian signal is also a Gaussian signal.
- Standard formula: $\mathcal{F}\{e^{-at^2}\} = \sqrt{\frac{\pi}{a}} e^{-\omega^2/4a}$ for $a > 0$.

$$\boxed{\text{C) } \sqrt{\pi} e^{-\frac{\omega^2}{4}}}$$

Q4.23.2 NAT 2023 2M Ans: 15 ● ○ ○

$$x_1(t) = u(t + 1.5) - u(t - 1.5)$$



$$\Rightarrow x_1(t) = \text{rect}\left(\frac{t}{3}\right)$$

$$x_1(t) = \text{rect}\left(\frac{t}{3}\right) \overset{FT}{\longleftrightarrow} 3Sa(1.5\omega)$$

Now, $x_2(t) = \delta(t + 3) + \text{rect}\left(\frac{t}{2}\right) + 2\delta(t - 2)$
Taking Fourier transform

$$X_2(\omega) = e^{3j\omega} + 2Sa(\omega) + 2e^{-2j\omega}$$

$$y(t) = x_1(t) * x_2(t)$$

$$Y(\omega) = X_1(\omega) \cdot X_2(\omega)$$

$$Y(\omega) = \int_{-\infty}^{\infty} y(t) \cdot e^{-j\omega t} \cdot dt$$

We know

$$\therefore \quad \int_{-\infty}^{\infty} y(t)dt = Y(0)$$

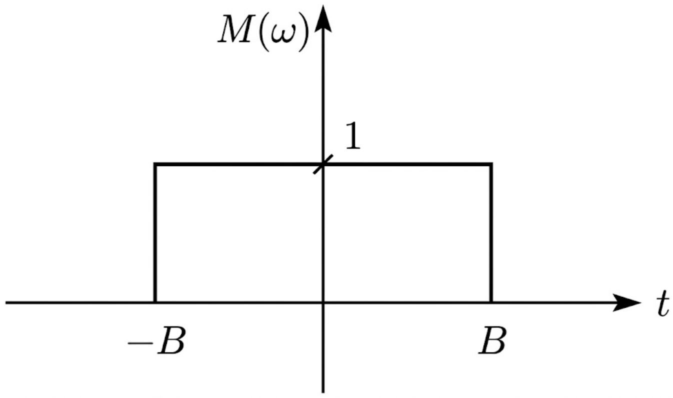
$$\therefore \quad Y(0) = X_1(0) \cdot X_2(0) = 3[1 + 2 + 2] = 15$$

$$\boxed{B)15}$$

Key Points

- **Convolution Property:** Convolution in the time domain corresponds to multiplication in the frequency domain: $x_1(t) * x_2(t) \overset{FT}{\longleftrightarrow} X_1(\omega) \cdot X_2(\omega)$.
- x(t) = $A \cdot \text{rect}\left(\frac{t}{\tau}\right) \rightarrow X(f) = A\tau \cdot \text{sinc}(f\tau))$
- **Area Property:** The area under a signal in the time domain is equal to the value of its Fourier Transform evaluated at $\omega = 0$: $\int_{-\infty}^{\infty} y(t)dt = Y(0)$.

Q4.23.3 MCQ 2023 1M Ans: B ● ○ ○



$$m(t) \rightleftharpoons M(\omega)$$

$$\text{Energy }(E) = \frac{1}{2\pi}\int_{-B}^{B}(1)^2 \cdot d\omega$$

$$E = \frac{B}{\pi}$$

Now, let

$$y(t) = m(t)\cos\omega_0 t$$

$$Y(\omega) = \frac{1}{2}[M(\omega-\omega_0) + M(\omega+\omega_0)]$$

Here;

$$\omega_0 = 10B$$

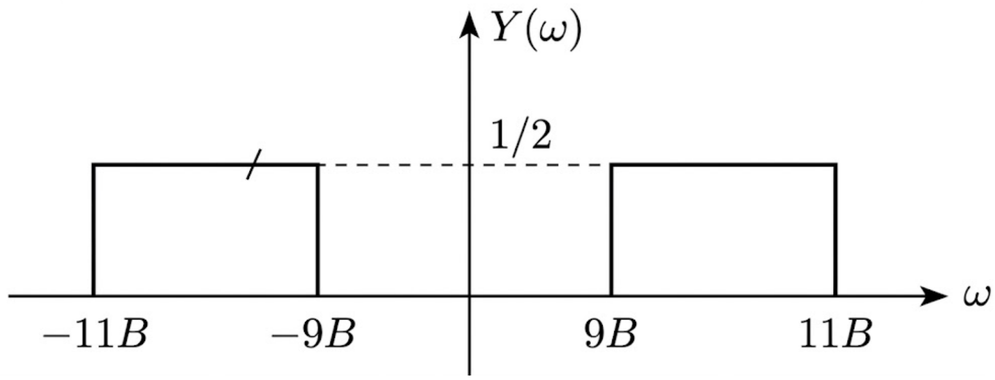


Fig. Solution Q4.23.3

Now,

$$\text{Energy }(E') = \frac{1}{2\pi}\int_{-\infty}^{\infty}|Y(\omega)|^2 \cdot d\omega$$

$$= \frac{1}{2\pi}\left[\int_{-11B}^{-9B}\left(\frac{1}{2}\right)^2 \cdot d\omega + \int_{9B}^{11B}\left(\frac{1}{2}\right)^2 \cdot d\omega\right]$$

$$= \frac{1}{2\pi}\left[\frac{1}{4}(2B) + \frac{1}{4}(2B)\right]$$

$$= \frac{1}{2\pi}[B] = \frac{1}{2}\left(\frac{B}{\pi}\right)$$

$$E' = \frac{E}{2}$$

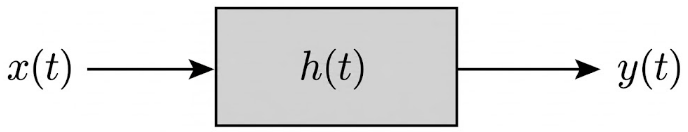
$$\boxed{B)\frac{E}{2}}$$

Key Points

- **Parseval's Theorem**: The energy of a signal can be computed in the frequency domain using $E = \frac{1}{2\pi}\int_{-\infty}^{\infty}|X(\omega)|^2 d\omega$.
- **Modulation Property**: Multiplying a signal by $\cos(\omega_0 t)$ shifts its frequency spectrum to $\pm\omega_0$ and scales its amplitude by $\frac{1}{2}$.

Q4.23.4 MCQ 2023 2M Ans: A ● ○ ○

Given $h(t)$ is Real and Even. When sinusoidal input applied to LTI system having even impulse response, then output will also be sinusoidal.



here,

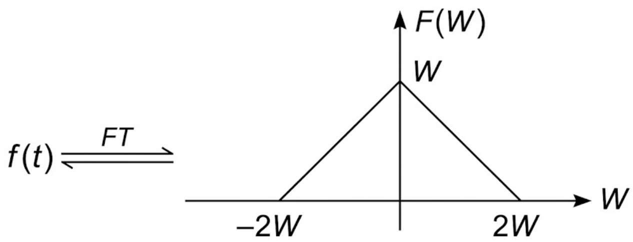
$$y(t) = H(W)|_{W=10.5W} \cdot 10\cos(10.5Wt)$$

let,

$$h(t) = f(t)\cos 10Wt$$

where,

$$f(t) = \pi\left(\frac{\sin Wt}{\pi t}\right)^2$$



Now;

$$H(W) = \frac{1}{2}[F(W + 10W) + F(W - 10W)]$$

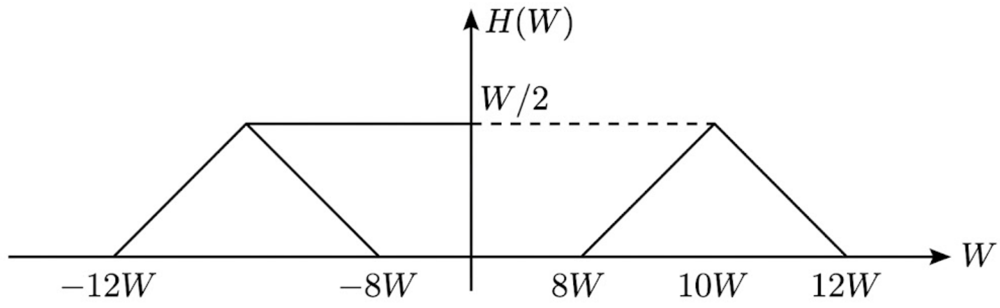


Fig. Solution Q4.23.4

$$\therefore \quad H(W)|_{W=10.5W} = \frac{3}{8}W$$

Hence,

$$y(t) = \left(\frac{3}{8}W\right)(10\cos 10.5Wt)$$

$$\boxed{A)\frac{15}{4}W\cos 10.5Wt}$$

Key Points

- **Eigenfunction Property:** For an LTI system, the response to a sinusoidal input is the input scaled by the magnitude of the system's frequency response evaluated at the input frequency.
- **Modulation Property:** Multiplying a signal by a cosine in the time domain shifts its frequency spectrum: $f(t)\cos(\omega_0 t) \overset{FT}{\longleftrightarrow} \frac{1}{2}[F(\omega+\omega_0)+F(\omega-\omega_0)]$.

Q4.22.1 MCQ 2022 1M Ans: A ● ○ ○

Consider,

$$x(t) = e^{-|t|}$$

by taking Fourier transform,

$$e^{-|t|} \overset{F.T}{\longleftrightarrow} \frac{2}{1+\omega^2}$$

Apply differentiation in frequency,

$$tx(t) \overset{F.T}{\longleftrightarrow} j\frac{d}{d\omega}X(\omega)$$

$$te^{-|t|} \overset{F.T}{\longleftrightarrow} j\left[\frac{d}{d\omega}\left(\frac{2}{1+\omega^2}\right)\right]$$

$$te^{-|t|} \overset{F.T}{\longleftrightarrow} j\left[\frac{0-2(2\omega)}{(1+\omega^2)^2}\right]$$

$$te^{-|t|} \overset{F.T}{\longleftrightarrow} \frac{-4j\omega}{(1+\omega^2)^2}$$

Apply duality property,

$$\frac{-4jt}{(1+t^2)^2} \overset{F.T}{\longleftrightarrow} 2\pi(-\omega)e^{-|-\omega|}$$

$$\frac{t}{(1+t^2)^2} \overset{F.T}{\longleftrightarrow} \frac{-2\pi\omega e^{-|\omega|}}{-4j}$$

$$\frac{t}{(1+t^2)^2} \xleftrightarrow{F.T} \boxed{\frac{\pi}{2j}\omega e^{-|\omega|}}$$

$$\boxed{\text{A) } \frac{\pi}{2j}\omega e^{-|\omega|}}$$

Key Points

- **Standard Pair:** $\mathcal{F}\{e^{-a|t|}\} = \frac{2a}{a^2+\omega^2}$.
- **Differentiation in Frequency:** Multiplication by t in the time domain corresponds to differentiation in the frequency domain: $tx(t) \xleftrightarrow{F.T} j\frac{d}{d\omega}X(\omega)$.
- **Duality Property:** If $x(t) \xleftrightarrow{F.T} X(\omega)$, then $X(t) \xleftrightarrow{F.T} 2\pi x(-\omega)$.

Q4.21.1 MCQ 2021 1M Ans: B ● ○ ○

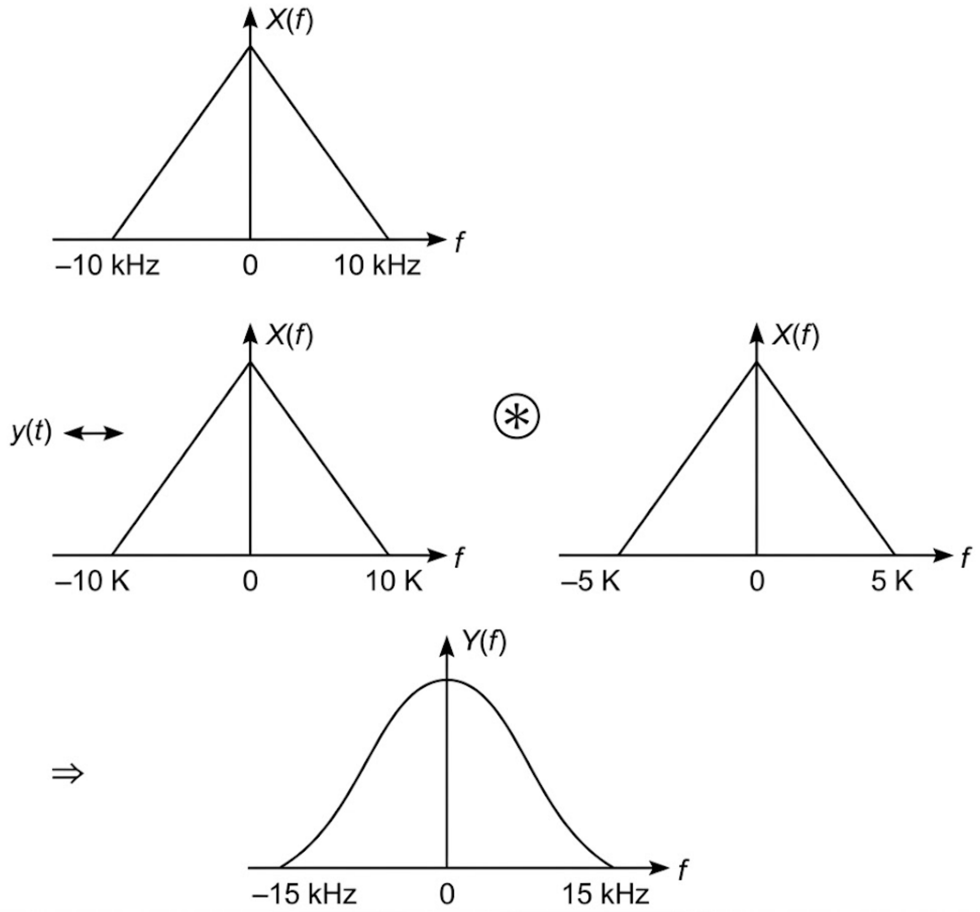


Fig. Solution Q4.21.1

From the convolution in the frequency domain shown in the figure, the maximum frequency of the resulting signal $Y(f)$ is the sum of the maximum frequencies of the convolved signals:

$$f_{\max} = 10\,\text{kHz} + 5\,\text{kHz} = 15\,\text{kHz}$$

$$\text{NR} = 2 \times f_{\max} = 2 \times 15 = 30\,\text{kHz}$$

$$\boxed{\text{B) } 30\,\text{kHz}}$$

Key Points

- **Time Scaling Property:** If $x(t)$ is band-limited to B, then the time-scaled signal $x(at)$ is band-limited to aB. For $x(1 + t/2)$, the scaling factor is $1/2$, so the new bandwidth is $10\,\text{kHz} \times (1/2) = 5\,\text{kHz}$.
- **Multiplication Property:** Multiplication in the time domain $x(t) \cdot x(1 + t/2)$ corresponds to convolution in the frequency domain $X(f) * \mathcal{F}\{x(1 + t/2)\}$.
- **Bandwidth Addition:** When two signals are convolved in the frequency domain, their maximum frequencies (bandwidths) add together: $BW_{\text{total}} = BW_1 + BW_2$.
- **Nyquist Rate:** The Nyquist rate for a low-pass (baseband) signal is defined as twice its maximum frequency: $\text{NR} = 2 \times f_{\max}$.

Q4.20.1 NAT 2020 2M Ans: 58.61 ● ○ ○

$$\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = 2\pi \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= 2\pi \int_{-\infty}^{\infty} |y(t)|^2$$

$$= 2 \times 2\pi \int_{-2}^{0} |y(t)|^2 dt$$

$$= 2 \times 2\pi \left[\int_{-2}^{-1} (t+2)^2 dt + \int_{-1}^{0} (2t+3)^2 dt\right]$$

$$= 4\pi \left[\left\{\frac{(t+2)^3}{3}\right\}_{-2}^{-1} + \left\{\frac{(2t+3)^3}{3 \times 2}\right\}_{-1}^{0}\right]$$

$$= 4\pi \left[\frac{1-0}{3} + \frac{3^3-1}{6}\right]$$

$$= 4\pi \left[\frac{1}{3} + \frac{26}{6}\right]$$

$$= 4\pi \times \left[\frac{1}{3} + \frac{26}{6}\right]$$

$$= 4\pi \times \left[\frac{1}{3} + \frac{13}{3}\right]$$

$$= 4\pi \times \frac{14}{3}$$

$$= \frac{56\pi}{3} = \boxed{58.61}$$

Key Points

- **Parseval's Theorem:** The energy of a signal can be computed in either the time domain or the frequency domain using the relation $\frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |x(t)|^2 dt$.
- **Symmetry Property:** If the energy signal is an even function where $x(t) = x(-t)$, the integration over the entire axis can be simplified to $2 \times \int_{-\infty}^{0} |x(t)|^2 dt$ or $2 \times \int_{0}^{\infty} |x(t)|^2 dt$.

Q4.18.1 NAT 2018 2M Ans: 8 ● ○ ○

Hilbert transform does not alter the amplitude spectrum of the signal.

$$\text{So,} \quad \int_{-\infty}^{\infty} |y(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

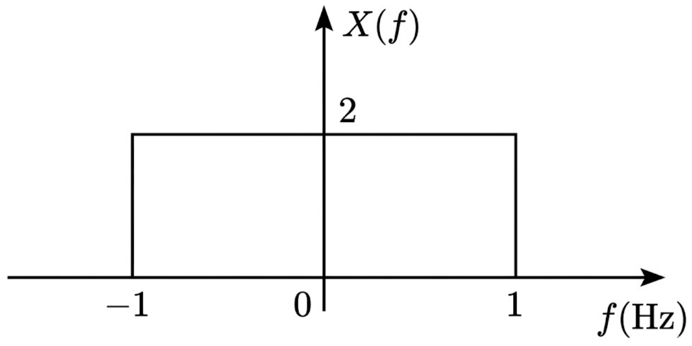
$$x(t) = 4\text{sinc}(2t)$$

$$\text{sinc}(t) \xleftrightarrow{CTFT} \text{rect}(f)$$

$$4\text{sinc}(2t) \xleftrightarrow{CTFT} \frac{4}{2}\text{rect}\left(\frac{f}{2}\right) = 2\text{rect}\left(\frac{f}{2}\right)$$

$$\int_{-\infty}^{\infty} |X(f)|^2 df = 2 \times (2)^2 = 8$$

$$\text{So.} \quad \int_{-\infty}^{\infty} |y(t)|^2 dt = \boxed{8}$$

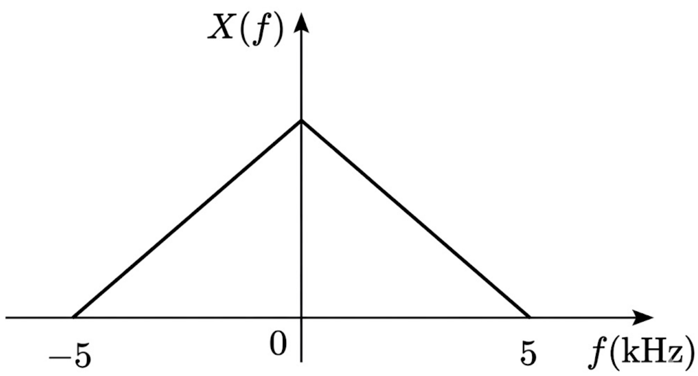


Key Points

- **Hilbert Transform Energy:** A Hilbert transform imparts a -90° phase shift to positive frequencies and $+90^\circ$ to negative frequencies. Since it only affects the phase, the amplitude spectrum $|X(f)|$ and the total energy of the signal remain unchanged.
- **Parseval's Theorem:** The total energy of a signal is the same whether computed in the time domain or the frequency domain: $\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$.
- **Time Scaling Property:** If $x(t) \xleftrightarrow{CTFT} X(f)$, then $x(at) \xleftrightarrow{CTFT} \frac{1}{|a|} X\left(\frac{f}{a}\right)$.

Q4.18.2 NAT 2018 2M Ans: 13

Let us assume an arbitrary spectrum for $x(t)$ as shown below:



The spectrum of the sampled signal can be given as,

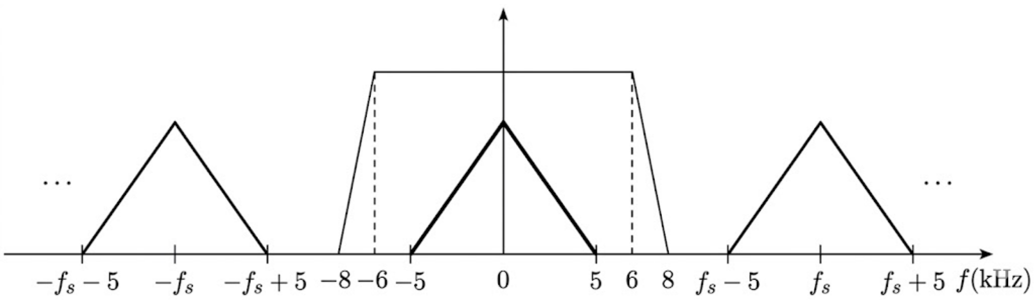


Fig. Solution Q4.18.2

For proper reconstruction of the signal,

$$f_s - 5 \geq 8$$
$$f_s \geq 8 + 5 = 13\,\text{kHz}$$

So.

$$f_{s(\min)} = 13\,\text{kHz}$$
$$\boxed{13}$$

Key Points

- **Spectrum of Sampled Signal:** Sampling a signal with a maximum frequency of f_m at a rate of f_s creates shifted replicas of the original spectrum centered at integer multiples of f_s (i.e., $\pm f_s, \pm 2f_s, \ldots$). The lower edge of the first high-frequency replica is located at $f_s - f_m$.
- **Practical Reconstruction Filter:** A practical low-pass filter does not have a sharp cutoff. It has a passband and a transition band that ends at a stopband frequency (f_{stop}).
- **Condition to Avoid Aliasing:** To perfectly reconstruct the original signal without interference from the adjacent replica, the lower edge of the first replica must fall outside the transition band of the reconstruction filter. This gives the condition: $f_s - f_m \geq f_{\text{stop}}$.

Q4.17.1 MCQ 2017 2M Ans: B ● ○ ○

Given the signal:

$$x(t) = \sin(14000\pi t)$$

The frequency of the signal f_m is calculated as:

$$2\pi f_m = 14000\pi \implies f_m = 7000 \text{ Hz} = 7 \text{ kHz}$$

The sampling frequency is:

$$f_s = 9000 \text{ samples/sec} = 9 \text{ kHz}$$

The spectrum of the sampled signal $S(f)$ consists of impulses at frequencies:

$$f = nf_s \pm f_m \quad \text{for } n = 0, \pm 1, \pm 2, \ldots$$

Calculating the spectral components:

- For $n = 0$: $f = \pm 7$ kHz
- For $n = 1$: $f = 9 \pm 7 = 2$ kHz, 16 kHz
- For $n = -1$: $f = -9 \pm 7 = -2$ kHz, -16 kHz
- For $n = 2$: $f = 18 \pm 7 = 11$ kHz, 25 kHz
- For $n = -2$: $f = -18 \pm 7 = -11$ kHz, -25 kHz

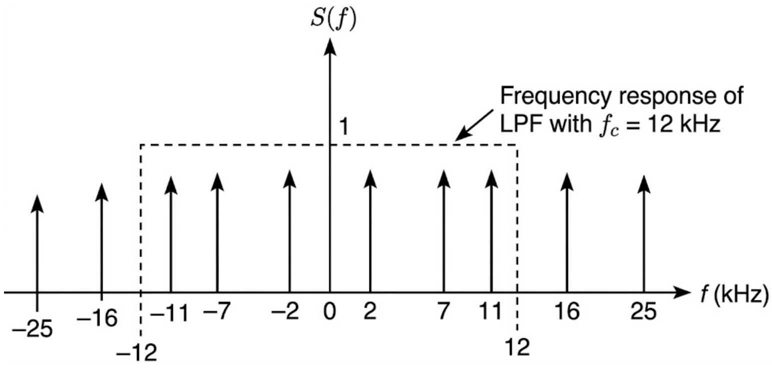
The sampled signal is passed through an ideal low-pass filter (LPF) with a cutoff frequency $f_c = 12$ kHz. The filter allows all frequency components $|f| \leq 12$ kHz.

The positive frequency components within this range are:

$$2 \text{ kHz}, 7 \text{ kHz}, \text{ and } 11 \text{ kHz}$$

Thus, there are 3 sinusoids in the output.

$$\boxed{\text{B) Number} = 3, \text{frequencies} = 2, 7, 11}$$

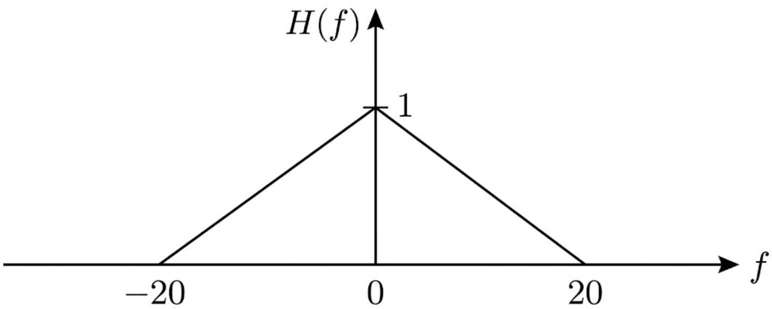


Key Points

- Sampling a sinusoid of frequency f_m at f_s creates aliases at $nf_s \pm f_m$.
- An ideal LPF passes only those components that fall within its bandwidth $|f| \le f_c$.
- The number of sinusoids corresponds to the number of distinct positive frequency pairs $(\pm f)$ that fall within the passband.

Q4.17.2 NAT 2017 2M Ans: 8

Since,



So,

$$y(t) = \frac{1}{2} \times 8\cos(20\pi t + \phi)$$

So,

$$= 4\cos(20\pi t + \phi); \phi = \frac{\pi}{4} - 20^\circ$$

$$P_y = \frac{4^2}{2} = 8\,\text{W}$$

$$\boxed{8}$$

Key Points

- **LTI System Response:** For an input $x(t) = A\cos(2\pi f_0 t + \theta)$, the steady-state output is $y(t) = A|H(f_0)|\cos(2\pi f_0 t + \theta + \angle H(f_0))$.
- **Frequency Filtering:** The input signal has frequency components at 10 Hz, 20 Hz, and 40 Hz. Because the magnitude response $|H(f)| = 0$ for $|f| \ge 20$ Hz, the system completely attenuates the 20 Hz and 40 Hz components.
- **Average Power of a Sinusoid:** The average power of a real sinusoidal signal $A\cos(\omega t + \phi)$ is given by $P = \frac{A^2}{2}$, which is independent of its frequency and phase.

Q4.17.3 NAT 2017 2M Ans: 8

$$x(t) = 4\cos 200\pi t + 8\cos 400\pi t$$

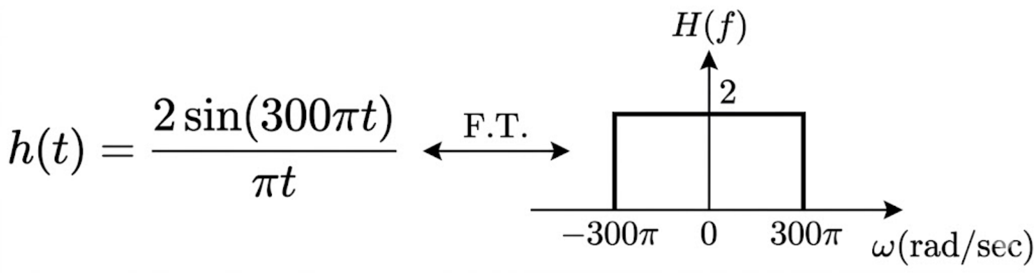


Fig. Solution Q4.17.3

So.

$$y(t) = 8\cos 200\pi t$$

$$\therefore \quad |y(t)|_{\max} = 8$$

$$\boxed{8}$$

Key Points

- **Ideal Low-Pass Filter:** The impulse response $h(t) = \frac{A\sin(\omega_c t)}{\pi t}$ corresponds to an ideal low-pass filter with a cutoff frequency of ω_c and a passband gain of A.
- **System Response:** For this specific filter, the cutoff frequency is $\omega_c = 300\pi$ rad/sec and the passband gain is 2.
- **Filtering Action:** The input component at $\omega = 200\pi$ falls within the passband ($200\pi < 300\pi$) and is amplified by 2 ($4 \times 2 = 8$). The component at $\omega = 400\pi$ falls in the stopband ($400\pi > 300\pi$) and is completely attenuated.

Q4.16.1 NAT 2016 1M Ans: 13

If $x(t)$ is a message signal and $y(t)$ is a sampled signal, then $y(t)$ is related to $x(t)$ as

$$y(t) = x(t)\sum_{n=-\infty}^{\infty}\delta(t - nT_s)$$

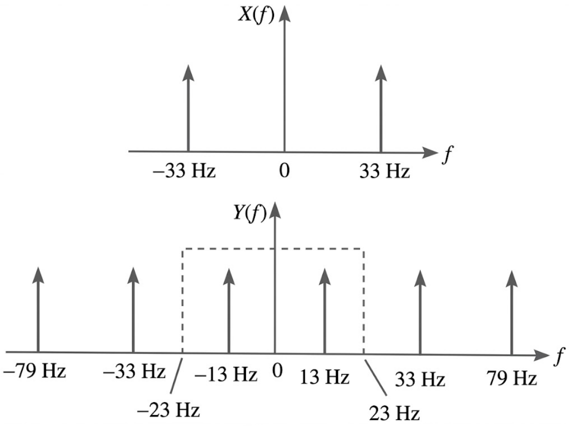
$$Y(f) = f_s\sum_{n=-\infty}^{\infty}X(f - nf_s)$$

Spectrum of $X(f)$ and $Y(f)$ are as shown
Cut off frequency of LPF = 23 Hz
Hence, frequency at the output is 13 Hz

$$\boxed{13}$$

Key Points

- **Impulse Sampling:** Multiplying a continuous-time signal $x(t)$ by an impulse train with period T_s creates a sampled signal. In the frequency domain, this causes the original spectrum $X(f)$ to be repeated at integer multiples of the sampling frequency $f_s = \frac{1}{T_s}$.
- **Spectrum Replicas (Aliasing):** The spectrum replicas are centered at $\pm nf_s$. Given $f_s = 46$ Hz and input signal frequency $f_m = 33$ Hz, the first-order shifts ($n = \pm 1$) produce frequency components at $|46 \pm 33|$ Hz, which are 13 Hz and 79 Hz.
- **Low-Pass Filtering:** The resulting signal is passed through an ideal Low-Pass Filter (LPF) with a cutoff frequency



$f_c = 23\,\text{Hz}$. The filter blocks all components greater than 23 Hz (like 33 Hz and 79 Hz) and only passes the 13 Hz aliased component.

Q4.16.2 | MCQ | 2016 | 1M | Ans: A | ● ○ ○

Convolution of two sinc pulses is sinc pulse.

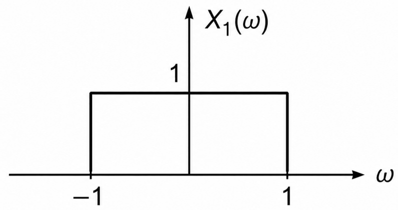
$$x_1(t) = \frac{\sin t}{\pi t}$$

$$x(t) = x_1(t) * x_1(t)$$

$$X(\omega) = X_1(\omega) \cdot X_1(\omega) = X_1(\omega)$$

$$x(t) = x_1(t) = \frac{\sin t}{\pi t}$$

$$x_1(t) = \frac{\sin t}{\pi t} \leftrightarrow$$



$$\boxed{\text{A) } \frac{\sin(t)}{\pi t}}$$

Key Points

- **Convolution Property:** Convolution of two signals in the time domain corresponds to the multiplication of their spectra in the frequency domain: $x_1(t) * x_2(t) \xleftrightarrow{FT} X_1(\omega) \cdot X_2(\omega)$.
- **Spectrum of Sinc Function:** The Fourier transform of $x_1(t) = \frac{\sin t}{\pi t}$ is an ideal low-pass filter shape, specifically a rectangular pulse $X_1(\omega)$ with an amplitude of 1 for $-1 \le \omega \le 1$.
- **Idempotent Property:** Multiplying a unit-amplitude rectangular spectrum by itself results in the identical spectrum $(1 \cdot 1 = 1)$. Therefore, the inverse Fourier transform remains the original sinc function.

Q4.16.3 | NAT | 2016 | 1M | Ans: 0.25 | ● ○ ○

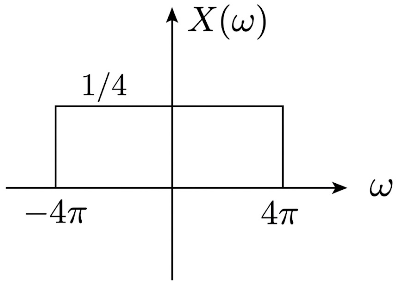
$$\frac{\sin at}{\pi t} \leftrightarrow \text{rect}\left(\frac{\omega}{2a}\right)$$

$$X(\omega) = \frac{1}{4}\text{rect}\left(\frac{\omega}{8\pi}\right)$$

$$E_{x(t)} = \frac{1}{2\pi}\int_{-\infty}^{\infty}|X(\omega)|^2 d\omega$$

$$= \frac{1}{2\pi} \times \frac{1}{16} \times 8\pi$$

$$= \frac{1}{4} = 0.25$$



$$\boxed{0.25}$$

Key Points

- **Fourier Transform of Sinc:** The Fourier transform of a standard sinc function $\frac{\sin(at)}{\pi t}$ is a rectangular pulse rect $\left(\frac{\omega}{2a}\right)$ with an amplitude of 1 and a total width of $2a$ (from $-a$ to a).
- **Linearity Property:** For $x(t) = \frac{\sin(4\pi t)}{4\pi t} = \frac{1}{4} \cdot \frac{\sin(4\pi t)}{\pi t}$, the amplitude in the frequency domain scales by $\frac{1}{4}$.
- **Parseval's Theorem:** The energy of the signal in the time domain equals its energy in the frequency domain, given by $E = \frac{1}{2\pi}\int_{-\infty}^{\infty}|X(\omega)|^2 d\omega$. The integral of a squared rectangular pulse is just $(\text{Amplitude})^2 \times (\text{Width})$.

Q4.16.4 MCQ | 2016 | 1M | Ans: B | ● ○ ○

A discrete time signal $x(n) = \cos(\omega_0 n)$ is said to be periodic if $\frac{\omega_0}{2\pi}$ is a rational number.
Let the continuous-time periodic signal be:

$$x(t) = \cos\left(\frac{2\pi}{T}t\right)$$

After sampling at intervals of T_s, the discrete-time signal is:

$$x(nT_s) = \cos\left(\frac{2\pi}{T}nT_s\right)$$

Here, the discrete frequency ω_0 is:

$$\omega_0 = 2\pi\left(\frac{T_s}{T}\right)$$

For the sampled signal to be periodic, $\frac{\omega_0}{2\pi}$ must be rational:

$$\frac{\omega_0}{2\pi} = \frac{T_s}{T} = \text{Rational Number}$$

Checking the given options: (A) $T = \sqrt{2}T_s \Rightarrow \frac{T_s}{T} = \frac{1}{\sqrt{2}}$ (Irrational $\rightarrow$ Not periodic) (B) $T = 1.2T_s \Rightarrow \frac{T_s}{T} = \frac{1}{1.2} = \frac{10}{12} = \frac{5}{6}$ (Rational $\rightarrow$ Periodic)

$$\boxed{\text{B) } T = 1.2T_s}$$

Key Points

- **Discrete-Time Periodicity:** Unlike continuous-time sinusoids (which are always periodic), a discrete-time sinusoid $x(n) = \cos(\omega_0 n)$ is periodic if and only if its frequency ω_0 is a rational multiple of 2π.
- **Sampling Theorem Implication:** When a continuous-time periodic signal with period T is sampled with a sampling period T_s, the resulting sequence is periodic only when the ratio of the sampling period to the signal period $\left(\frac{T_s}{T}\right)$ is a rational number.

Q4.16.5 MCQ 2016 2M Ans: A ● ○ ○

Fig. Solution Q4.16.5

When a sinusoidal input is given to LTI system, the output is also a sinusoid with change in magnitude and the phase shift offered by LTI system.
By the principle of superposition, each frequency component $k\omega_0$ in the input signal $\sum_{k=1}^{3} a_k \cos(k\omega_0 t)$ will experience its own magnitude scaling and phase shift based on the network's frequency response.
Thus, the output will be of the form:

$$y(t) = \sum_{k=1}^{3} b_k \cos(k\omega_0 t + \phi_k)$$

Since the network has nonzero impedance, the amplitude will change, meaning $b_k \neq a_k$.

$$\boxed{A) \sum_{k=1}^{3} b_k \cos(k\omega_0 t + \phi_k), \text{ where } b_k \neq a_k, \forall k}$$

Key Points

- **LTI System Response to Sinusoids:** An LTI system modifies only the amplitude ($|H(j\omega)|$) and phase ($\angle H(j\omega)$) of an input sinusoid. It never generates new frequency components.
- **Superposition Principle:** For an input consisting of multiple sinusoidal components, the total response of the LTI system is simply the sum of its responses to each individual component.

Q4.16.6 MCQ 2016 1M Ans: C ● ○ ○

$$X(t) = \cos(6\pi t) + \sin(8\pi t)$$

$$Y(t) = x(2t + 5)$$

$$Y(t) = \cos[6\pi(2t + 5)] + \sin[8\pi(2t + 5)]$$

$$= \cos(12\pi t + 30\pi) + \sin(16\pi t + 40\pi)$$

$$f_{m1} = 6\,\text{Hz}, \quad f_{m2} = 8\,\text{Hz}$$

Nyquist sampling rate,

$$f_s = 2 f_{\max}$$

$$= \boxed{\text{C) 16 samples/second}}$$

Key Points

- **Time Scaling Property:** If a signal $x(t)$ has a maximum frequency of f_m, then the time-scaled signal $x(at)$ has a maximum frequency of $|a| f_m$. In this case, time scaling by a factor of 2 doubles the maximum frequency.
- **Time Shifting:** The time shift operation (+5) only alters the phase of the signal. It does not affect the frequency spectrum or the required Nyquist sampling rate.

- **Nyquist Rate:** The Nyquist sampling rate is defined as exactly twice the highest frequency component present in the signal: $f_s = 2f_{\max}$.

Q4.15.1 NAT 2015 2M Ans: 0.4

Let the given signal be $x(t) = x_1(t) * x_2(t)$, where:

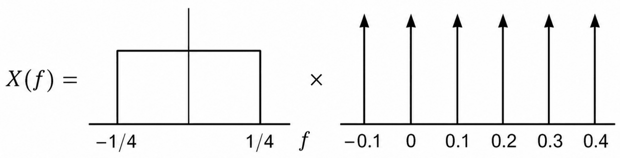
$$x_1(t) = \frac{\sin\left(\frac{\pi t}{2}\right)}{\left(\frac{\pi t}{2}\right)} \quad \text{and} \quad x_2(t) = \sum_{n=-\infty}^{\infty} \delta(t - 10n)$$

Taking the Fourier transform, the spectrum of the sinc signal $x_1(t)$ is a rectangular function:

$$X_1(f) = 2\,\text{rect}(2f)$$

This spectrum is non-zero only for $-0.25\,\text{Hz} \le f \le 0.25\,\text{Hz}$.
The Fourier transform of the time-domain impulse train $x_2(t)$ is a frequency-domain impulse train with fundamental frequency $f_0 = \frac{1}{10} = 0.1\,\text{Hz}$:

$$X_2(f) = 0.1 \sum_{k=-\infty}^{\infty} \delta(f - 0.1k)$$



Convolution in the time domain equates to multiplication in the frequency domain ($X(f) = X_1(f) \cdot X_2(f)$). Thus, $X(f)$ only contains the impulses from $X_2(f)$ that fall inside the rectangular window of $X_1(f)$. These impulses occur at $f = 0, \pm 0.1\,\text{Hz}, \pm 0.2\,\text{Hz}$.
The highest non-zero frequency component is therefore:

$$f_{\max} = 0.2\,\text{Hz}$$

The Nyquist sampling rate is:

$$f_s = 2 f_{\max} = 2(0.2) = 0.4\,\text{Hz}$$

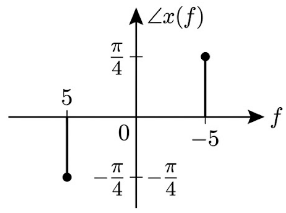
$$\boxed{0.4}$$

Key Points

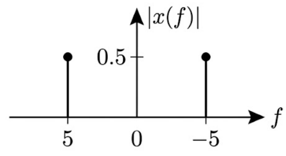
- Convolution of two signals in the time domain is equivalent to the multiplication of their respective Fourier transforms in the frequency domain.
- The Fourier transform of a sinc function forms a rectangular spectrum (acting as an ideal low-pass filter).
- The Fourier transform of a periodic impulse train in the time domain results in an impulse train in the frequency domain.
- The highest non-zero frequency component present in the resulting multiplied spectrum determines the maximum frequency, $f_{\max}$.
- The Nyquist sampling rate is calculated as $f_s = 2 f_{\max}$.

Q4.15.2 MCQ 2015 1M Ans: A ● ○ ○

Phase Spectrum



Magnitude Spectrum



Given input Signal

$$x(t) = \cos\left(10\pi t + \frac{\pi}{4}\right)$$

$$X(f) = \frac{1}{2}\left[\delta(f-5)e^{j\pi/4} + \delta(f+5)e^{-j\pi/4}\right]$$

After sampling $x(t)$ at 15 Hz, we get:

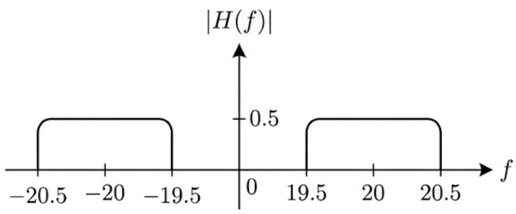
$$X_s(f) = 15\sum_{n=-\infty}^{\infty} X(f-15n)$$

Given the filter impulse response:

$$h(t) = \left[\frac{\sin(\pi t)}{\pi t}\right]\cos\left(40\pi t - \frac{\pi}{2}\right)$$

$$H(f) = \frac{1}{2}\left[\text{rect}(f-20)e^{-j\pi/2} + \text{rect}(f+20)e^{j\pi/2}\right]$$

Magnitude Spectrum



Output:

$$Y(f) = X_s(f) \times H(f)$$

Converting back to the time domain:

$$y(t) = \frac{15}{2}\cos\left(40\pi t - \frac{\pi}{4}\right)$$

$$\boxed{A)\frac{15}{2}\cos\left(40\pi t - \frac{\pi}{4}\right)}$$

Key Points

- **Standard Fourier Transform Pairs & Properties:**
  - Phase-Shifted Cosine: $\cos(2\pi f_c t + \theta) \leftrightarrow \frac{1}{2}[\delta(f-f_c)e^{j\theta} + \delta(f+f_c)e^{-j\theta}]$
  - Sinc to Rect: $\frac{\sin(\pi t)}{\pi t} = \text{sinc}(t) \leftrightarrow \text{rect}(f)$
  - Modulation Property: $m(t)\cos(2\pi f_c t + \theta) \leftrightarrow \frac{1}{2}[M(f-f_c)e^{j\theta} + M(f+f_c)e^{-j\theta}]$
- Ideal sampling in the time domain results in periodic replication of the signal's spectrum in the frequency domain, shifted by integer multiples of the sampling frequency f_s.
- Passing a sampled signal through a bandpass filter extracts only the frequency components that fall within the filter's passband.

- Multiplication of spectra in the frequency domain corresponds to the convolution of the signals in the time domain, which effectively acts as a filtering operation.

Mistakes to be avoided

- Ignoring the negative frequency components of the original signal when calculating the shifted spectrum components (e.g., $f = -5 + 15 = 10\,\text{Hz}$).
- Mishandling or factoring out phase shifts incorrectly. When converting $\cos(2\pi f_c t + \theta)$ to the frequency domain, the positive frequency impulse $\delta(f - f_c)$ takes $e^{j\theta}$ while the negative frequency impulse $\delta(f + f_c)$ must take the conjugate $e^{-j\theta}$.

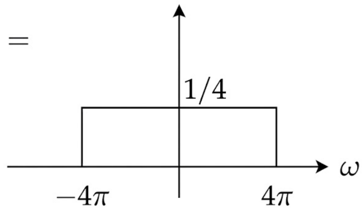
Q4.15.3 | NAT | 2015 | 2M | Ans: 3 | ● ○ ○

$$x_1(t) = \cos(2\pi t)$$
$$X_1(\omega) = \pi[\delta(\omega - 2\pi) + \delta(\omega + 2\pi)]$$
$$x_2(t) = \frac{\sin(4\pi t)}{4\pi t}$$

We know that plancherals relation

$$\int_{-\infty}^{\infty} x_1(t)x_2(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\omega)X_2(\omega)d\omega$$

$X_2(\omega) =$



$$I = \frac{1}{2\pi} \times 12 \int_{-4\pi}^{4\pi} \frac{1}{4}[\pi\delta(\omega - 2\pi) + \pi\delta(\omega + 2\pi)]d\omega$$
$$I = \frac{6}{\pi}\left[\int_{-4\pi}^{4\pi} \frac{\pi}{4}\delta(\omega - 2\pi)d\omega + \int_{-4\pi}^{4\pi} \frac{\pi}{4}\delta(\omega + 2\pi)d\omega\right]$$
$$= \frac{6}{\pi}\left[\frac{\pi}{4} + \frac{\pi}{4}\right] = \frac{6}{\pi}\left[\frac{\pi}{2}\right] = 3$$

$\boxed{3}$

Key Points

- Plancherel's relation simplifies the calculation of the energy or inner product of signals by allowing computation in the frequency domain.
- The Fourier transform of $x(t) = \cos(\omega_0 t)$ consists of impulses at $\pm\omega_0$: $X(\omega) = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$.
- The sifting property of the Dirac delta function implies $\int_a^b f(\omega)\delta(\omega - \omega_0)d\omega = f(\omega_0)$ if $a < \omega_0 < b$.

Q4.14.1 | MCQ | 2014 | 1M | Ans: D | ● ○ ○

Given:

$$H(f) = \begin{cases} e^{-j4\pi f}, & |f| \le \frac{\omega}{2} \\ 0, & |f| > \frac{\omega}{2} \end{cases}$$

So,

$$|H(f)| = 1$$
$$H(f) = e^{-j2\cdot 2\pi f}$$
$$H(\omega) = e^{-j2\omega} \qquad |f| \le \frac{\omega}{2}$$

$$h(t) = \delta(t-2)$$

As we know that

$$y(t) = x(t) * h(t)$$
$$y(t) = X(\omega)\cdot H(\omega)$$
$$y(t) = x(t) * \delta(t-2)$$
$$y(t) = x(t-2)$$

$\boxed{\text{D) x(t - 2)}}$

Key Points

- A linear phase shift in the frequency domain corresponds to a constant time delay in the time domain.
- The inverse Fourier transform of $e^{-j\omega t_0}$ is the delayed impulse function $\delta(t-t_0)$.
- Convolution of a signal $x(t)$ with a delayed impulse $\delta(t-t_0)$ results in the shifted signal $x(t-t_0)$.

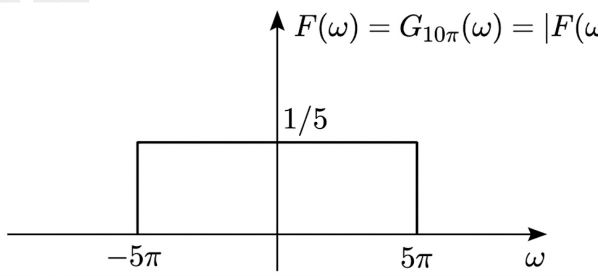
Q4.14.2 NAT 2014 2M Ans: 0.2 ● ○ ○

Method 1

$$\int_{-\infty}^{\infty} \text{sinc}^2(5t)dt$$

Let,

$$f(t) = \text{sinc}(5t)$$
$$= Sa(5\pi t) \quad [\because \text{sinc}(t) = Sa(\pi t)]$$
$$TSa\left(t\cdot\frac{T}{2}\right) \longleftrightarrow 2\pi G_T(\omega)$$
$$\frac{T}{2} = 5\pi \Rightarrow [T = 10\pi]$$
$$10\pi Sa(5\pi t) \longleftrightarrow 2\pi G_{10\pi}(\omega)$$
$$\Rightarrow F(\omega) = \frac{1}{5}G_{10\pi}(\omega)$$
$$ESD_f = |F(\omega)|^2$$
$$E_f = \int_{-\infty}^{\infty} f^2(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} ESD_f d\omega$$
$$\int_{-\infty}^{\infty}\text{sinc}^2(5t)dt = \frac{1}{2\pi}\int_{-5\pi}^{5\pi}\left(\frac{1}{5}\right)^2 d\omega$$
$$= \frac{1}{2\pi}\times\frac{1}{25}\times 10\pi = \frac{1}{5}$$



$\boxed{0.2}$

Method 2

Given the signal in the time domain:

$$x(t) = \text{sinc}(5t)$$

Using the standard Fourier transform pair for a sinc function:

$$\text{sinc}(Bt) \xrightarrow{\mathcal{F}} \frac{1}{B}\text{rect}\left(\frac{f}{B}\right)$$

Substituting $B = 5$, we obtain the Fourier transform $X(f)$:

$$X(f) = \frac{1}{5}\text{rect}\left(\frac{f}{5}\right)$$

This represents a rectangular pulse with an amplitude of $1/5$ and a total width of 5, defined from $f = -2.5$ to $f = 2.5$. To calculate the integral of the squared signal, we apply Parseval's theorem, which states that the total energy in the time domain equals the total energy in the frequency domain:

$$\int_{-\infty}^{\infty} x^2(t)\,dt = \int_{-\infty}^{\infty} |X(f)|^2\,df$$

Substituting our expression for $X(f)$ and applying the appropriate limits:

$$\begin{aligned}
\int_{-\infty}^{\infty} x^2(t)\,dt &= \int_{-2.5}^{2.5} \left(\frac{1}{5}\right)^2 df \\
&= \int_{-2.5}^{2.5} \frac{1}{25}\,df \\
&= \frac{1}{25}\Big[f\Big]_{-2.5}^{2.5} \\
&= \frac{1}{25}(2.5-(-2.5)) \\
&= \frac{5}{25} = \frac{1}{5}
\end{aligned}$$

$$\boxed{0.2}$$

Key Points

- Parseval's theorem (or Rayleigh's energy theorem) relates the total energy of a signal in the time domain to its Energy Spectral Density (ESD) in the frequency domain: $E = \int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2 d\omega$.
- The Fourier transform of a sampling function $Sa(at)$ corresponds to a rectangular pulse (Gate function, G_T) in the frequency domain.
- By definition in this context, $\text{sinc}(t) = Sa(\pi t) = \frac{\sin(\pi t)}{\pi t}$.

Q4.14.3 MCQ 2014 2M Ans: B ● ○ ○

$$G(j\omega) = \omega e^{-2\omega^2}$$

$$\int_{-\infty}^{\infty} g(t)dt = G(j0) = 0$$

$$y(t) = g(t) * u(t)$$

$$\Rightarrow \quad Y(j\omega) = G(j\omega)U(j\omega)$$

$$\Rightarrow \quad Y(j\omega) = G(j\omega)\left[\frac{1}{j\omega} + \pi\delta(\omega)\right]$$

$$= \frac{\omega e^{-2\omega^2}}{j\omega} + \omega e^{-2\omega^2}\pi\delta(\omega)$$

$$\int_{-\infty}^{\infty} y(t)dt = Y(j0) = \frac{1}{j} = -j$$

B) -j

Key Points

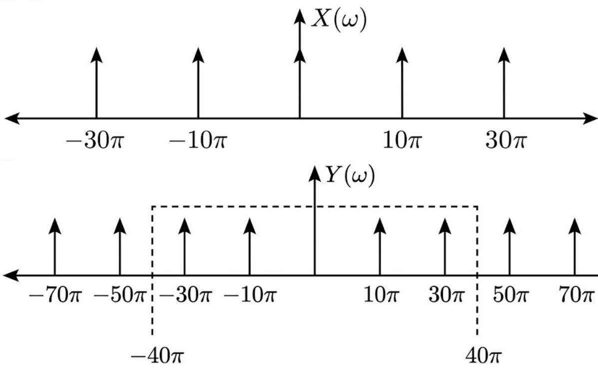
- The area under a signal in the time domain is equal to the value of its Fourier transform evaluated at $\omega = 0$: $\int_{-\infty}^{\infty} x(t)dt = X(j0)$.
- Integration of a signal in the time domain, $y(t) = \int_{-\infty}^{t} g(\tau)d\tau$, is equivalent to convolution with the unit step function $u(t)$, which means $y(t) = g(t) * u(t)$.
- The Fourier transform of the unit step function is $U(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$.
- The product of ω and the Dirac delta function $\delta(\omega)$ is zero (i.e., $x\delta(x) = 0$), which nullifies the impulse term when evaluated.

Q4.14.4 MCQ | 2014 | 1M | Ans: A | ● ○ ○

$$x(t) = \cos(10\pi t) + \cos(30\pi t)$$

Given sampling frequency, $f_s = 20\,\text{Hz}$

$$\omega_s = 40\pi \text{ rad/sec}$$



$$Y(\omega) = \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s)$$

After sampling waveform will be

Applying an ideal low-pass filter of cut-off frequency of 20 Hz or 40π rad/sec, we get the frequencies in reconstructed signal as 10π and 30π rad/sec. or 5 Hz and 15 Hz

A) 5 Hz and 15 Hz only

Key Points

- The spectrum of a sampled signal consists of the original baseband spectrum and its infinite replicas shifted by integer multiples of the sampling frequency (f_s).
- If the sampling frequency is lower than the Nyquist rate ($2f_{\max}$), aliasing occurs, causing high-frequency components to overlap with low-frequency ones.
- An ideal low-pass filter for reconstruction extracts only the frequency components that fall within its passband ($[-f_c, f_c]$).

Q4.14.5 NAT 2014 1M Ans: 3 ● ○ ○

Multiplication in time domain = convolution in frequency domain.

$$x_1(t) \cdot x_2(t) \longleftrightarrow X_1(\omega) * X_2(\omega)$$

So, highest frequency component contained by the convolved signal $z(t) = 1500\,\text{Hz}$

$$\therefore \text{ Nyquist rate} = 2 \times 1500$$

$$= 3000\,\text{Hz} = 3\,\text{kHz}$$

$$\boxed{3}$$

Key Points

- Multiplication of two signals in the time domain corresponds to the convolution of their spectra in the frequency domain.
- The bandwidth (or maximum frequency component) of the product of two band-limited signals is the sum of their individual bandwidths: $f_{\text{max},z} = f_{\text{max},x} + f_{\text{max},y}$.
- The Nyquist sampling rate is exactly twice the highest frequency component present in the signal: $f_s = 2f_{\text{max}}$.

Q4.13.1 MCQ 2013 1M Ans: A ● ○ ○

According to the Nyquist-Shannon sampling theorem, the minimum sampling frequency is given by:

$$(f_s)_{\text{min}} = 2f_m$$

Substituting the given maximum signal frequency $f_m = 5\,\text{kHz}$:

$$(f_s)_{\text{min}} = 2 \times 5 = 10\,\text{kHz}$$

So, any valid sampling frequency must satisfy:

$$f_s \geq 10\,\text{kHz}$$

Therefore, 5 kHz is not a valid sampling frequency.

$$\boxed{\text{A) 5 kHz}}$$

Key Points

- **Nyquist Rate:** The minimum rate at which a signal can be sampled without introducing errors (aliasing) is twice the highest frequency present in the signal, i.e., $(f_s)_{\text{min}} = 2f_m$.
- To perfectly reconstruct a continuous-time signal from its samples, the sampling frequency f_s must be greater than or equal to the Nyquist rate ($f_s \geq 2f_m$).

Q4.12.1 MCQ 2012 2M Ans: C ● ○ ○

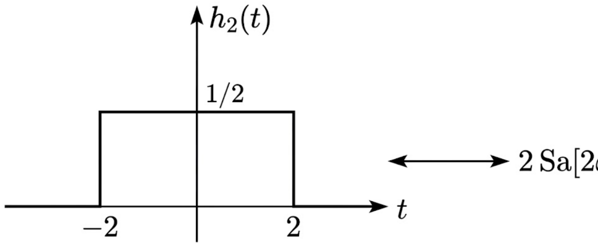
$$\frac{2\cos\omega\sin 2\omega}{\omega} = H(\omega)$$

$$H(\omega) = 2\cos\omega\left(\frac{2\sin 2\omega}{2\omega}\right) = H_1(j\omega)\cdot H_2(j\omega)$$

where,

$$H_1(j\omega) = 2\cos\omega$$

$$H_2(j\omega) = \frac{2\sin 2\omega}{2\omega} = 2\mathrm{Sa}(2\omega)$$



$$H_1(j\omega) = 2\cos\omega = (e^{j\omega} + e^{-j\omega}) = e^{j\omega(1)} + e^{j\omega(-1)}$$

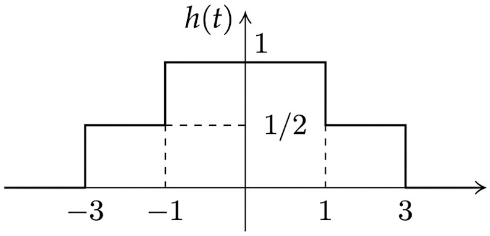
So,

$$H(j\omega) = 2\cos\omega H_2(j\omega) = (e^{j\omega} + e^{-j\omega})H_2(j\omega)$$

$$H(j\omega) = e^{j\omega}H_2(j\omega) + e^{-j\omega}H_2(j\omega)$$

$$\Rightarrow \quad h(t) = h_2(t+1) + h_2(t-1)$$

$h(t) =$



So, $h(0) = 1$

$\boxed{C)1}$

Key Points

- **Time-Shifting Property:** Multiplication by $e^{\pm j\omega t_0}$ in the frequency domain corresponds to a time shift in the time domain: $\mathcal{F}^{-1}\{e^{\pm j\omega t_0}X(j\omega)\} = x(t \pm t_0)$.
- **Inverse FT of Sinc/Sa:** A sampling function $\mathrm{Sa}(\omega)$ in the frequency domain transforms into a rectangular pulse in the time domain.
- **Superposition:** The total time-domain signal is the sum of the individually shifted rectangular pulses.

Q4.10.1 MCQ | 2010 | 2M | Ans: C | ● ○ ○

Given signal is:

$$s(t) = \frac{\sin(500\pi t)}{\pi t} \times \frac{\sin(700\pi t)}{\pi t}$$

Let $s(t) = s_1(t) \times s_2(t)$, where

$$s_1(t) = \frac{\sin(500\pi t)}{\pi t} \quad \text{and} \quad s_2(t) = \frac{\sin(700\pi t)}{\pi t}$$

The maximum frequency component of $s_1(t)$ is $\omega_1 = 500\pi$ rad/s, which gives $f_1 = 250$ Hz. The maximum frequency component of $s_2(t)$ is $\omega_2 = 700\pi$ rad/s, which gives $f_2 = 350$ Hz.
Since multiplication in the time domain corresponds to convolution in the frequency domain, the maximum frequency component of the product signal $s(t)$ is the sum of the individual maximum frequencies:

$$f_{max} = f_1 + f_2 = 250 + 350 = 600\,\text{Hz}$$

The Nyquist sampling rate f_s is given by:

$$f_s = 2f_{max} = 2 \times 600 = 1200\,\text{Hz}$$

For a signal defined as $x(t) = \frac{\sin(at)}{\pi t}$, the general Fourier Transform yields a rectangular pulse in the frequency domain. In terms of angular frequency ω (where a represents the maximum angular frequency in rad/s), the Fourier transform $X(\omega)$ is:

$$X(\omega) = \begin{cases} 1, & |\omega| < a \\ 0, & |\omega| > a \end{cases}$$

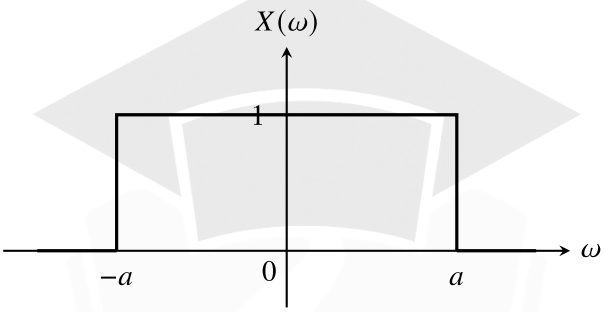
This can also be expressed using the rectangular function notation:

$$X(\omega) = \text{rect}\left(\frac{\omega}{2a}\right)$$

In terms of linear frequency f (where $\omega = 2\pi f$), the bandwidth is restricted to $f_{max} = \frac{a}{2\pi}$ Hz, and the transform $X(f)$ is:

$$X(f) = \begin{cases} 1, & |f| < \frac{a}{2\pi} \\ 0, & |f| > \frac{a}{2\pi} \end{cases}$$

Spectrum Representation
The spectrum of $\frac{\sin(at)}{\pi t}$ represents an ideal low-pass filter. It is a flat, continuous rectangular band centered at $\omega = 0$ with an amplitude of 1 and a total spectral width of $2a$.



C) 1200 Hz

Key Points

- **Nyquist Rate:** For a band-limited signal with a maximum frequency f_{max}, the minimum sampling rate required to avoid aliasing is $2f_{max}$.
- **Multiplication Property:** If two signals are multiplied in the time domain, $x(t) = x_1(t) \cdot x_2(t)$, their resulting bandwidth is the sum of their individual bandwidths: $B = B_1 + B_2$.

Q4.08.1 MCQ 2008 2M Ans: A ● ○ ○

The given signal is a rectangular pulse in the time domain:

$$x(t) = \begin{cases} 1 & \text{for } -1 \le t \le +1 \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform $X(\omega)$ of the signal $x(t)$ is given by the definition:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Substituting the given $x(t)$:

$$X(\omega) = \int_{-1}^{1} 1 \cdot e^{-j\omega t}dt = \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-1}^{1}$$

$$X(\omega) = \frac{e^{-j\omega} - e^{j\omega}}{-j\omega} = \frac{2\sin(\omega)}{\omega}$$

We need to find the angular frequencies where the Fourier transform becomes zero, i.e., $X(\omega) = 0$.

$$\frac{2\sin(\omega)}{\omega} = 0$$

This condition is satisfied when the numerator is zero, provided the denominator is not zero:

$$\sin(\omega) = 0 \implies \omega = n\pi \quad \text{for } n = \pm 1, \pm 2, \pm 3, \ldots$$

Note that for $n = 0$, $\omega = 0$, the value is calculated using L'Hôpital's rule (or standard limits):

$$\lim_{\omega \to 0} \frac{2\sin(\omega)}{\omega} = 2 \neq 0$$

Thus, the angular frequencies at which the Fourier transform becomes zero are $\pm\pi, \pm 2\pi, \pm 3\pi, \ldots$ From the given options, π and 2π are two such valid angular frequencies.

$$\boxed{\text{A) } \pi, 2\pi}$$

Key Points

- **Fourier Transform of a Rectangular Pulse:** A rectangular pulse in the time domain transforms into a sinc function (or sampling function) in the frequency domain.
- For a rectangular pulse of total width τ centered at the origin, the zero crossings in the frequency domain occur at integer multiples of $\frac{2\pi}{\tau}$ (excluding zero). Here, $\tau = 2$, so the zero crossings are at $n\pi$.

Q4.08.2

MCQ | 2008 | 2M | Ans: D | ● ○ ○

Method 1

Input:

$$x(t) = 2\cos(2t)$$

Frequency response:

$$X(\omega) = 2\pi[\delta(\omega - 2) + \delta(\omega + 2)]$$

$$H(\omega) = \frac{1}{2 + j\omega}$$

Output:

$$Y(\omega) = H(\omega)X(\omega)$$

$$\begin{aligned}
Y(\omega) &= \frac{1}{2 + j\omega} \cdot 2\pi[\delta(\omega - 2) + \delta(\omega + 2)] \\
&= \frac{2\pi}{2 + j2}\delta(\omega - 2) + \frac{2\pi}{2 - j2}\delta(\omega + 2) \\
&= \frac{2\pi}{8}[(2 - j2)\delta(\omega - 2) + (2 + j2)\delta(\omega + 2)] \\
&= \frac{\pi}{4}[2\{\delta(\omega - 2) + \delta(\omega + 2)\} + j2\{\delta(\omega + 2) - \delta(\omega - 2)\}] \\
&= \frac{\pi}{2}[\delta(\omega - 2) + \delta(\omega + 2)] - j\frac{\pi}{2}[\delta(\omega - 2) - \delta(\omega + 2)]
\end{aligned}$$

Taking the inverse Fourier transform:

$$\therefore \quad y(t) = \frac{\cos 2t}{2} + \frac{\sin 2t}{2}$$

$$\begin{aligned} y(t) &= \frac{\sqrt{2}}{2}\left[\frac{1}{\sqrt{2}}\cos 2t + \frac{1}{\sqrt{2}}\sin 2t\right] \\ &= \frac{1}{\sqrt{2}}\cos(2t - 0.25\pi) \\ &= 2^{-0.5}\cos(2t - 0.25\pi) \end{aligned}$$

Method 2

For a stable LTI system, the steady-state response to a sinusoidal input $x(t) = A\cos(\omega_0 t)$ is given directly by:

$$y(t) = A|H(j\omega_0)|\cos(\omega_0 t + \angle H(j\omega_0))$$

Given the impulse response $h(t) = e^{-2t}u(t)$, the transfer function is:

$$H(j\omega) = \frac{1}{2 + j\omega}$$

For the input $x(t) = 2\cos(2t)$, we have $A = 2$ and $\omega_0 = 2$. Evaluating $H(j\omega)$ at $\omega = 2$:

$$H(j2) = \frac{1}{2 + j2}$$

Magnitude:

$$|H(j2)| = \frac{1}{\sqrt{2^2 + 2^2}} = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}}$$

Phase:

$$\angle H(j2) = -\tan^{-1}\left(\frac{2}{2}\right) = -45^\circ = -0.25\pi \text{ rad}$$

Therefore, the output is simply:

$$y(t) = 2 \cdot \left(\frac{1}{2\sqrt{2}}\right)\cos(2t - 0.25\pi) = \frac{1}{\sqrt{2}}\cos(2t - 0.25\pi)$$

$$y(t) = 2^{-0.5}\cos(2t - 0.25\pi)$$

$$\boxed{\text{D) } 2^{-0.5}\cos(2t - 0.25\pi)}$$

Key Points

- **Eigenfunction Property:** Complex exponentials and sinusoids are eigenfunctions of LTI systems. The output of an LTI system to a sinusoidal input is a sinusoid of the same frequency, scaled by the magnitude of the frequency response and shifted by its phase.
- **Multiplication with Impulse:** When multiplying a function by an impulse, it sifts the value of the function at the location of the impulse: $f(\omega)\delta(\omega - \omega_0) = f(\omega_0)\delta(\omega - \omega_0)$.

Q4.08.3 MCQ | 2008 | 2M | Ans: C | ● ○ ○

The given impulse response is:

$$h(t) = \exp(-2t)u(t)$$

The frequency response $H(\omega)$ is the Fourier transform of the impulse response $h(t)$:

$$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}\,dt$$

Substituting $h(t)$ and changing the lower limit to 0 because of the unit step function $u(t)$:

$$H(\omega) = \int_0^\infty e^{-2t} e^{-j\omega t} dt = \int_0^\infty e^{-(2+j\omega)t} dt$$

Evaluating the integral:

$$H(\omega) = -\frac{1}{2+j\omega} \cdot e^{-(2+j\omega)t} \Big|_0^\infty$$

Applying the limits (as $t \to \infty$, $e^{-(2+j\omega)t} \to 0$, and at $t = 0$, $e^0 = 1$):

$$H(\omega) = 0 - \left(-\frac{1}{2+j\omega} \cdot 1\right) = \frac{1}{2+j\omega}$$

$$\boxed{\text{C) } \frac{1}{2+j\omega}}$$

Key Points

- **Fourier Transform Definition:** The continuous-time Fourier transform of a signal $x(t)$ is given by $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$.
- **Standard Transform Pair:** For a right-sided exponential signal $e^{-at}u(t)$ with $a > 0$, its Fourier transform is strictly $\frac{1}{a+j\omega}$.

Q4.06.1 MCQ 2006 1M Ans: A ● ○ ○

Given the Fourier Transform pair:

$$x(t) \leftrightarrow X(j\omega)$$

We need to find the Fourier transform of the signal $x(5t-3)$. Let us rewrite the argument to clearly separate the scaling and shifting operations:

$$x(5t-3) = x\left[5\left(t - \frac{3}{5}\right)\right]$$

Using the time scaling and shifting properties of the Fourier Transform:

$$x\left[5\left(t-\frac{3}{5}\right)\right] \leftrightarrow \frac{1}{5}e^{-\frac{j3\omega}{5}} X\left(\frac{j\omega}{5}\right)$$

$$\boxed{\text{A) } \frac{1}{5}e^{-j\frac{3\omega}{5}} X\left(\frac{j\omega}{5}\right)}$$

Key Points

- **Time Scaling Property:** $x(at) \leftrightarrow \frac{1}{|a|}X\left(\frac{j\omega}{a}\right)$
- **Time Shifting Property:** $x(t-t_0) \leftrightarrow e^{-j\omega t_0}X(j\omega)$
- **Combined Operation:** When both scaling and shifting are present, as in $x(at-b)$, it is best to factor out the scaling factor to avoid errors: $x[a(t-b/a)]$. The resulting transform is $\frac{1}{|a|}e^{-j\omega(b/a)}X\left(\frac{j\omega}{a}\right)$.

Q4.06.2 MCQ | 2006 | 2M | Ans: C | ● ○ ○

The given signal is:

$$x(t) = 5\left(\frac{\sin(2\pi 1000t)}{\pi t}\right)^3 + 7\left(\frac{\sin(2\pi 1000t)}{\pi t}\right)^2$$

Let $s(t) = \frac{\sin(2\pi 1000t)}{\pi t}$. The maximum frequency of $s(t)$ is given by $\omega_m = 2\pi(1000)$ rad/s, which corresponds to a maximum frequency component of $f_m = 1000$ Hz.

The signal $x(t)$ can be rewritten as:

$$x(t) = 5[s(t)]^3 + 7[s(t)]^2$$

By the properties of the Fourier transform, multiplication in the time domain corresponds to convolution in the frequency domain. Therefore, raising a band-limited signal to the power of n increases its maximum frequency by a factor of n.

For the first term, $5[s(t)]^3$, the maximum frequency is:

$$f_1 = 3 \times f_m = 3 \times 1000 = 3000\,\text{Hz}$$

For the second term, $7[s(t)]^2$, the maximum frequency is:

$$f_2 = 2 \times f_m = 2 \times 1000 = 2000\,\text{Hz}$$

The maximum frequency of the composite signal $x(t)$ is the highest frequency component present, which is the maximum of f_1 and f_2:

$$f_{max} = \max(f_1, f_2) = \max(3000, 2000) = 3000\,\text{Hz}$$

According to the Nyquist-Shannon sampling theorem, the minimum sampling frequency (Nyquist rate) required to reconstruct the signal without distortion is:

$$f_s = 2f_{max} = 2 \times 3000 = 6000\,\text{Hz} = 6 \times 10^3\ \text{samples/sec}$$

$$\boxed{\text{C) } 6 \times 10^3}$$

Key Points

- **Bandwidth of a Sinc Function:** A signal of the form $\frac{\sin(\omega_c t)}{\pi t}$ has a baseband bandwidth of $B = \frac{\omega_c}{2\pi}$.
- **Non-linear Operations:** If a signal $x(t)$ has a maximum frequency of f_m, then the signal $[x(t)]^n$ has a maximum frequency of $n \cdot f_m$.
- **Superposition:** For a sum of signals, the overall bandwidth is the maximum of the individual bandwidths.
- **Nyquist Rate:** The minimum sampling rate is $f_s \geq 2f_{max}$.

Q4.06.3 MCQ | 2006 | 2M | Ans: C | ● ○ ○

Given the message signal $m(t)$ has a bandwidth $W = 500$ Hz. The multiplying signal is an impulse train with alternating signs:

$$g(t) = \sum_{k=-\infty}^{\infty} (-1)^k \delta(t - kT_s)$$

where the sampling interval $T_s = 0.5 \times 10^{-4}$ s. The sampling frequency is:

$$f_s = \frac{1}{T_s} = \frac{1}{0.5 \times 10^{-4}} = 20\,\text{kHz}$$

Because of the $(-1)^k$ term, the sequence alternates signs, meaning $g(t)$ is periodic with a fundamental period $T_0 = 2T_s$. The fundamental frequency is:

$$f_0 = \frac{1}{T_0} = \frac{f_s}{2} = 10\,\text{kHz}$$

The Fourier series of $g(t)$ will have no DC component (its average value over T_0 is zero) and will only contain impulses at odd harmonics of f_0.

$$G(f) = \sum_{n=-\infty}^{\infty} c_n \delta(f - n f_0)$$

where $c_0 = 0$ and $c_n \neq 0$ only for $n = \pm1, \pm3, \ldots$ The resulting signal after multiplication is $y(t) = m(t)g(t)$. In the frequency domain, this is the convolution:

$$Y(f) = M(f) * G(f) = \sum_{n \text{ odd}} c_n M(f - n f_0)$$

This shows that the spectrum of $m(t)$ is shifted and centered at $\pm 10\,\text{kHz}, \pm 30\,\text{kHz}, \ldots$ The lowest frequency components of $Y(f)$ lie in the range $10\,\text{kHz} - 500\,\text{Hz} = 9.5\,\text{kHz}$ to $10.5\,\text{kHz}$. The signal $y(t)$ is then passed through an ideal lowpass filter with a bandwidth of $1\,\text{kHz}$. Since $Y(f) = 0$ for $-1\,\text{kHz} \leq f \leq 1\,\text{kHz}$, the output of the lowpass filter is strictly zero.

$\boxed{\text{C) } 0}$

Key Points

- **Alternating Impulse Train:** Multiplying a signal by $\sum(-1)^k \delta(t - kT_s)$ shifts the baseband spectrum to odd multiples of $f_s/2$.
- **Spectrum of zero-mean carrier:** Since the carrier $g(t)$ has zero average value (no DC component), the modulated signal will not contain any baseband components.
- **Filtering:** An ideal lowpass filter rejects all spectral components outside its bandwidth. Since all components are at $\geq 9.5\,\text{kHz}$, the output is strictly 0.

Q4.05.1 MCQ 2005 1M Ans: B ● ○ ○

To find the inverse Fourier transform of $X(3f + 2)$, we first factor out the scaling constant to clearly separate the operations:

$$X(3f + 2) = X\left[3\left(f + \frac{2}{3}\right)\right]$$

Applying scaling & shifting property:

$$\mathcal{F}^{-1}\left\{X\left[3\left(f + \frac{2}{3}\right)\right]\right\} = \frac{1}{3} x\left(\frac{t}{3}\right) e^{-j4\pi t/3}$$

$\boxed{\text{B) } \frac{1}{3} x(\frac{t}{3}) e^{-j\frac{4\pi}{3}t}}$

Key Points

- **Frequency Scaling Property:** $x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{f}{a}\right) \implies \frac{1}{|a|} x\left(\frac{t}{a}\right) \leftrightarrow X(af)$
- **Frequency Shifting Property:** $x(t)e^{j2\pi f_0 t} \leftrightarrow X(f - f_0)$
- **Combined Operation:** When both scaling and shifting are present, such as $X(af - b)$, rewrite it as $X\left[a\left(f - \frac{b}{a}\right)\right]$. The corresponding time-domain signal is $\frac{1}{|a|} x\left(\frac{t}{a}\right) e^{j2\pi(b/a)t}$. Here $a = 3$ and $b = -2$.

Q4.05.2 MCQ 2005 1M Ans: A ● ○ ○

$$y(t) = 0.5x(t - t_d + T) + 0.5x(t - t_d - T) + x(t - t_d)$$

Taking Fourier transform

$$Y(\omega) = \left[0.5e^{j\omega(-t_d+T)} + 0.5e^{j\omega(-t_d-T)} + e^{-j\omega t_d}\right] X(\omega)$$

$$\frac{Y(\omega)}{X(\omega)} = e^{-j\omega t_d}\left[0.5e^{j\omega T} + 0.5e^{-j\omega T} + 1\right]$$

$$\frac{Y(\omega)}{X(\omega)} = e^{-j\omega t_d}[\cos\omega T + 1]$$

$$H(\omega) = (1 + \cos\omega T)e^{-j\omega t_d}$$

$$\boxed{\text{A) } H(\omega) = (1 + \cos\omega T)e^{-j\omega t_d}}$$

Key Points

- **Time Shifting Property:** $x(t - t_0) \leftrightarrow \mathcal{F} e^{-j\omega t_0} X(\omega)$
- **Euler's Identity:** $\cos(\theta) = \frac{e^{j\theta}+e^{-j\theta}}{2}$
- **Transfer Function:** The transfer function of an LTI system is defined as the ratio of the Fourier transform of the output to the Fourier transform of the input, $H(\omega) = \frac{Y(\omega)}{X(\omega)}$.

Q4.04.1 MCQ 2004 2M Ans: B ● ○ ○

$$y(t) = -x[2(t + 1)]$$

$$x(t - t_0) \leftrightarrow x(f)e^{-j2\pi f t_0}$$

$$t_0 = -1$$

$$x(at) = \frac{1}{|a|}X\left(\frac{f}{a}\right)$$

$$a = 2$$

$$\therefore \quad Y(f) = -\frac{1}{2}X\left(\frac{f}{2}\right)e^{j2\pi f}$$

$$\boxed{B) -\frac{1}{2}X\left(\frac{f}{2}\right)e^{j2\pi f}}$$

Key Points

- **Time Shifting Property:** $x(t - t_0) \leftrightarrow X(f)e^{-j2\pi f t_0}$
- **Time Scaling Property:** $x(at) \leftrightarrow \frac{1}{|a|}X\left(\frac{f}{a}\right)$

Q4.04.2 MCQ 2004 1M Ans: C ● ○ ○

A function $x(t)$ is said to be conjugate symmetric if it satisfies the condition:

$$x(t) = x^*(-t)$$

Let the Fourier transform of $x(t)$ be $X(f)$. Using the conjugation and time-reversal properties of the Fourier transform:

$$x^*(t) \longleftrightarrow X^*(-f)$$

$$x^*(-t) \longleftrightarrow X^*(f)$$

Taking the Fourier transform on both sides of the conjugate symmetry condition $x(t) = x^*(-t)$, we get:

$$X(f) = X^*(f)$$

For a complex quantity to be equal to its own complex conjugate, its imaginary part must be zero. Therefore, the Fourier transform $X(f)$ is always purely real.

$$\boxed{\text{C) real}}$$

Key Points

- **Conjugate Symmetric Function:** $x(t) = x^*(-t) \longleftrightarrow X(f)$ is purely real.
- **Conjugate Anti-symmetric Function:** $x(t) = -x^*(-t) \longleftrightarrow X(f)$ is purely imaginary.
- **Real Function:** $x(t)$ is purely real $\longleftrightarrow X(f)$ is conjugate symmetric $(X(f) = X^*(-f))$.

Q4.04.3 MCQ 2004 2M Ans: C ● ○ ○

$$f_s = 1.5\,\text{kHz}$$

$$f_m = 1\,\text{kHz}$$

The available frequency components in the sampled signal are given by:

$$nf_s \pm f_m$$

Substituting values for $n = 0, 1, 2, \ldots$, we get the frequency components:

$$1\,\text{kHz},\ 2.5\,\text{kHz},\ 0.5\,\text{kHz}, \ldots$$

The signal is passed through a Low-Pass Filter (LPF) which has a cut-off frequency:

$$f_c = 0.8\,\text{kHz}$$

Since only frequencies below 0.8 kHz can pass through the LPF, only the 0.5 kHz component will appear at the output.

$$\boxed{\text{C) } 0.5\,\text{kHz}}$$

Key Points

- **Spectrum of Sampled Signal:** When a sinusoidal signal of frequency f_m is ideally sampled at a rate f_s, the resulting spectrum contains frequencies at $nf_s \pm f_m$, where n is any integer.
- **Ideal Low-Pass Filter:** An ideal LPF completely attenuates all frequency components above its cut-off frequency f_c while passing all components below it without distortion.

Q4.03.1 MCQ 2003 1M Ans: C ● ○ ○

Let the input to the linear time-invariant (LTI) system be $x(t)$ and its Fourier transform be $X(f)$. The given output of the system is:

$$y(t) = 4x(t-2)$$

Taking the Fourier transform on both sides and applying the linearity and time-shifting properties:

$$Y(f) = 4X(f)e^{-j2\pi f(2)}$$

$$Y(f) = 4X(f)e^{-j4\pi f}$$

The transfer function $H(f)$ of an LTI system is defined as the ratio of the Fourier transform of the output to the Fourier transform of the input:

$$H(f) = \frac{Y(f)}{X(f)}$$

Substituting the expression for $Y(f)$:

$$H(f) = \frac{4X(f)e^{-j4\pi f}}{X(f)}$$

$$H(f) = 4e^{-j4\pi f}$$

$$\boxed{\text{C) } 4e^{-j4\pi f}}$$

Key Points

- **Transfer Function:** The transfer function of a system is $H(f) = \frac{Y(f)}{X(f)}$.
- **Time-Shifting Property:** A shift in the time domain corresponds to a linear phase shift in the frequency domain, i.e., $x(t - t_0) \longleftrightarrow X(f)e^{-j2\pi f t_0}$.

Q4.03.2 | NAT | 2003 | 2M | Ans: D | ● ○ ○

From Fourier series expansion,

$$C_n = \frac{1}{T_0}\int_{-T_0/6}^{T_0/6} A \cdot e^{-jn\omega_0 t}\, dt = \frac{A}{\pi n}\sin\left(\frac{n\pi}{3}\right)$$

$\therefore$ From C_n its clear that $1, 2, 4, 5, 7\ldots$ harmonics are present.
Freq of $p(t)$ corr. to $1, 2, 4, 5, 7\ldots$ are $10^3, 2\times10^3, 4\times10^3\ldots$
frequency components 0.7 k and 0.4 k
$x(t)p(t)$ has $f(t)$ gives (1 ± 0.7) k, (2 ± 0.7) k
(4 ± 0.7) k(1 ± 0.4) k, (2 ± 0.4) k...
Frequency present in range of 2.5 k to 3.5 k are 2.7, 3.3

$$\boxed{\text{D) } 2.7, 3.3}$$

Key Points

- **Fourier Series Coefficients:** The exponential Fourier series coefficients for a periodic signal are calculated using $C_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)e^{-jn\omega_0 t}\, dt$.
- **Frequency Shifting (Modulation):** Multiplying a signal $x(t)$ with a periodic pulse train $p(t)$ shifts the frequency spectrum of $x(t)$ to center around the harmonic frequencies of $p(t)$.
- **Harmonic Content:** For a pulse train with a specific duty cycle, certain harmonics become zero (e.g., if pulse width is $T_0/3$, the 3rd, 6th, 9th harmonics are missing).

Q4.02.1 | MCQ | 2002 | 1M | Ans: C | ● ○ ○

Given the Fourier transform pair:

$$F\{e^{-t}u(t)\} = \frac{1}{1 + j2\pi f}$$

Let $x(t) = e^{-t}u(t)$ and $X(f) = \frac{1}{1+j2\pi f}$. We are asked to find the Fourier transform of $\frac{1}{1+j2\pi t}$, which corresponds to finding the transform of $X(t)$.
Applying the duality property of the Fourier transform:

$$\text{If } F\{x(t)\} = X(f), \text{ then } F\{X(t)\} = x(-f)$$

Now, substitute t with $-f$ in our original signal $x(t)$:

$$x(-f) = e^{-(-f)}u(-f) = e^{f}u(-f)$$

Therefore, the correct Fourier transform is $e^{f}u(-f)$, matching option C.

$$\boxed{\text{C) } e^{f}u(-f)}$$

Key Points

- **Duality Property:** A fundamental property of Fourier transforms stating that if $x(t) \leftrightarrow X(f)$, then $X(t) \leftrightarrow x(-f)$.
- This property allows us to easily compute the Fourier transform of a function that takes the functional form of a known frequency-domain spectrum.

Mistakes to be avoided

- A common error is forgetting to negate the frequency variable when applying the duality property (i.e., substituting t with f instead of $-f$), which leads to the incorrect choice A ($e^{f}u(f)$).

Q4.02.2 MCQ 2002 1M Ans: C ● ○ ○

Given the input signal:

$$x(t) = 10\cos(8\pi \times 10^3 t)$$

The message frequency f_m is obtained from the angular frequency $\omega_m = 8\pi \times 10^3$ rad/s:

$$2\pi f_m = 8\pi \times 10^3 \implies f_m = 4\,\text{kHz}$$

The sampling interval is given as $T_s = 100\,\mu\text{s} = 10^{-4}$ s. The sampling frequency f_s is:

$$f_s = \frac{1}{T_s} = \frac{1}{10^{-4}} = 10^4\,\text{Hz} = 10\,\text{kHz}$$

The sampled signal $y(t)$ is defined as:

$$y(t) = 5 \times 10^{-6} x(t) \sum_{n=-\infty}^{+\infty} \delta(t - nT_s)$$

Using the trigonometric Fourier series expansion for the unit impulse train:

$$\sum_{n=-\infty}^{+\infty} \delta(t - nT_s) = \frac{1}{T_s} + \frac{2}{T_s}\sum_{n=1}^{\infty}\cos(n\omega_s t) = f_s + 2f_s\cos(\omega_s t) + 2f_s\cos(2\omega_s t) + \dots$$

Substituting this expansion into the expression for $y(t)$:

$$y(t) = 5 \times 10^{-6} x(t)\,[f_s + 2f_s\cos(\omega_s t) + \dots]$$

$$y(t) = 5 \times 10^{-6} f_s x(t) + 10 \times 10^{-6} f_s x(t)\cos(\omega_s t) + \dots$$

The signal $y(t)$ is then passed through an ideal low-pass filter (LPF) with a cutoff frequency $f_c = 5\,\text{kHz}$. We must analyze the frequency components of $y(t)$ to see which pass through the filter:

- **Baseband component frequency:** $f_m = 4\,\text{kHz}$. Since $4\,\text{kHz} < 5\,\text{kHz}$, this component **passes**.
- **Lower sideband of first harmonic:** $f_s - f_m = 10\,\text{kHz} - 4\,\text{kHz} = 6\,\text{kHz}$. Since $6\,\text{kHz} > 5\,\text{kHz}$, this component is **rejected**.

Therefore, only the baseband component passes through the LPF. The output of the filter $y_{out}(t)$ is:

$$y_{out}(t) = 5 \times 10^{-6} f_s x(t)$$

Substitute $f_s = 10^4$ Hz and the expression for $x(t)$:

$$y_{out}(t) = 5 \times 10^{-6} \times 10^4 \times 10 \cos(8\pi \times 10^3 t)$$

$$y_{out}(t) = 5 \times 10^{-1} \cos(8\pi \times 10^3 t)$$

This corresponds to option C.

$$\boxed{\text{C) } 5 \times 10^{-1} \cos(8\pi \times 10^3)t}$$

Key Points

- **Impulse Train Fourier Series:** An ideal impulse train in the time domain can be represented as an infinite sum of harmonics: $p(t) = f_s + 2f_s \sum \cos(n\omega_s t)$.
- **Ideal Sampling:** Multiplying a message signal by an impulse train creates scaled replicas of the signal's frequency spectrum at integer multiples of the sampling frequency f_s. The baseband component is always scaled by f_s.
- **Signal Recovery:** To recover the baseband signal without aliasing, an ideal low-pass filter is used. The cutoff frequency f_c must satisfy $f_m \le f_c \le f_s - f_m$.

Q4.02.3 MCQ 2002 2M Ans: D ● ○ ○

Given the continuous time signal:

$$x(t) = 100 \cos(24\pi \times 10^3 t)$$

The message frequency f_m is found from the angular frequency $\omega_m = 24\pi \times 10^3$ rad/s:

$$2\pi f_m = 24\pi \times 10^3 \implies f_m = 12\,\text{KHz}$$

The sampling period is given as $T_s = 50\,\mu\text{sec} = 50 \times 10^{-6}$ s. The sampling frequency f_s is:

$$f_s = \frac{1}{T_s} = \frac{1}{50 \times 10^{-6}} = 20 \times 10^3\,\text{Hz} = 20\,\text{KHz}$$

When a signal is ideally sampled, the frequency spectrum of the sampled signal consists of shifted replicas of the original spectrum at integer multiples of the sampling frequency. The frequencies present are given by:

$$f_{sampled} = |nf_s \pm f_m| \quad \text{for } n = 0, 1, 2, \ldots$$

Let's evaluate the lowest frequency components:

- For $n = 0$: $f = f_m = 12$ KHz (Baseband)
- For $n = 1$: $f = f_s - f_m = 20 - 12 = 8$ KHz and $f = f_s + f_m = 20 + 12 = 32$ KHz
- For $n = 2$: $f = 2f_s - f_m = 40 - 12 = 28$ KHz and $f = 2f_s + f_m = 40 + 12 = 52$ KHz

The frequencies present in the ideally sampled signal are 8 KHz, 12 KHz, 28 KHz, 32 KHz, and so on.
The sampled signal is then passed through an ideal low pass filter (LPF) with a cutoff frequency $f_c = 15$ KHz. The filter will only pass frequencies $f \le 15$ KHz. Looking at our calculated frequency components, the frequencies that fall within the passband are 8 KHz and 12 KHz.
Therefore, the frequencies present at the filter output are 12 KHz and 8 KHz.

$$\boxed{\text{D) 12 KHz and 8 KHz}}$$

Key Points

- **Spectrum of Sampled Signal:** Ideal sampling results in the original signal's spectrum being repeated at integer multiples of the sampling frequency f_s. The resulting frequencies are $|nf_s \pm f_m|$.
- **Aliasing / Under-sampling:** Here, $f_s = 20\,\text{KHz}$ is less than the Nyquist rate ($2f_m = 24\,\text{KHz}$). Because the signal is under-sampled, aliasing occurs. The aliased component $f_s - f_m = 8\,\text{KHz}$ folds back into the baseband frequency range.
- **Ideal Low Pass Filter:** An ideal LPF passes all frequencies below its cutoff frequency f_c with zero attenuation and completely rejects all frequencies above f_c.

Mistakes to be avoided

- Only considering the baseband frequency (12 KHz) and forgetting that sampling generates new frequency components (Option A).
- Missing the fact that under-sampling creates an aliased component at $|f_s - f_m|$ that can fall within the LPF passband.

Q4.01.1 | MCQ | 2001 | 2M | Ans: C | ● ○ ○

Given the continuous-time signal:

$$x(t) = \text{Sinc}(700t) + \text{Sinc}(500t)$$

In signal processing, the normalized Sinc function is defined as $\text{Sinc}(x) = \frac{\sin(\pi x)}{\pi x}$. The frequency content of a signal of the form $\text{Sinc}(2Wt)$ is strictly bandlimited to a maximum frequency of W Hz. Let's find the maximum frequency for each term by equating the arguments to $2Wt$:
For the first term, $\text{Sinc}(700t)$:

$$2W_1 = 700 \implies W_1 = 350\,\text{Hz}$$

The maximum frequency $f_1 = 350\,\text{Hz}$.
For the second term, $\text{Sinc}(500t)$:

$$2W_2 = 500 \implies W_2 = 250\,\text{Hz}$$

The maximum frequency $f_2 = 250\,\text{Hz}$.
When two signals are added, the highest frequency present in the composite signal is the maximum of their individual highest frequencies:

$$f_{max} = \max(f_1, f_2) = \max(350\,\text{Hz}, 250\,\text{Hz}) = 350\,\text{Hz}$$

The Nyquist sampling rate (f_s) is defined as twice the maximum frequency present in the signal:

$$f_s = 2 \times f_{max} = 2 \times 350 = 700\,\text{Hz}$$

The Nyquist sampling interval (T_s) is the reciprocal of the Nyquist rate:

$$T_s = \frac{1}{f_s} = \frac{1}{700}\,\text{sec}$$

This corresponds to option C.

$$\boxed{\text{C) } \frac{1}{700}\,\text{sec}}$$

Key Points

- **Normalized Sinc Function:** Defined as $\text{Sinc}(t) = \frac{\sin(\pi t)}{\pi t}$. Its Fourier transform is a rectangular pulse in the frequency domain, strictly bandlimited to ± 0.5 Hz. Scaling the time variable by a (i.e., $\text{Sinc}(at)$) scales the bandwidth to $\frac{a}{2}$ Hz.
- **Bandwidth of a Sum:** If $y(t) = x_1(t) + x_2(t)$, then the bandwidth B_y is determined by the highest frequency component present in either signal: $B_y = \max(B_1, B_2)$.

- **Nyquist Criteria:** For a low-pass signal with absolute maximum frequency f_{max}, the Nyquist rate is $f_s = 2f_{max}$, and the Nyquist interval is $T_s = \frac{1}{2f_{max}}$.

Mistakes to be avoided

- Confusing the normalized $\text{Sinc}(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$ with the unnormalized sampling function $\text{Sa}(\theta) = \frac{\sin(\theta)}{\theta}$. If it were unnormalized, the maximum angular frequency ω_m would be 700 rad/s, leading to $f_m = \frac{700}{2\pi}$ and an incorrect $T_s = \frac{\pi}{700}$ sec.
- Adding the bandwidths of the two signals ($350 + 250 = 600$ Hz) instead of taking the maximum. Addition in the time domain does not convolve the spectra; it superimposes them.

Q4.01.2 | MCQ | 2001 | 1M | Ans: D | ● ○ ○

According to the Nyquist-Shannon sampling theorem, if a continuous-time signal is band-limited to a maximum frequency f_m, it can be perfectly reconstructed from its discrete samples provided the sampling rate f_s is at least $2f_m$ (the Nyquist rate). When a signal is sampled, its frequency spectrum is periodically replicated at integer multiples of the sampling frequency f_s. To recover the original continuous-time signal, we must isolate the original baseband spectrum (from $-f_m$ to f_m) and entirely reject all the higher-frequency spectral replicas generated by the sampling process.
This perfect isolation requires a filter with a perfectly flat passband (to avoid distorting the desired baseband signal) and an infinitely sharp roll-off (to completely reject the adjacent replicas). This characteristic mathematically defines an **ideal low-pass filter** with an appropriately chosen cutoff frequency f_c (where $f_m \le f_c \le f_s - f_m$).
Therefore, the sampled signal can be recovered by passing the samples through an ideal low-pass filter with the appropriate bandwidth.

D) an ideal low-pass filter with the appropriate band- width

Key Points

- **Nyquist-Shannon Sampling Theorem:** A band-limited signal can be perfectly reconstructed if sampled at $f_s \ge 2f_m$.
- **Reconstruction Filter:** An ideal low-pass filter is required to perform ideal interpolation (using Sinc functions) in the time domain, which corresponds to perfect rectangular windowing in the frequency domain to isolate the baseband spectrum.

Mistakes to be avoided

- **Choosing Option A (RC filter):** A practical RC filter has a gradual roll-off. It cannot provide perfect reconstruction because it will either attenuate the high-frequency components of the desired baseband signal or fail to completely suppress the unwanted spectral replicas, leading to distortion.

Q4.00.1 | MCQ | 2000 | 1M | Ans: D | ● ○ ○

The given signal is:

$$x(t) = e^{-3t^2}$$

This is a Gaussian pulse of the general form $e^{-\alpha t^2}$, where $\alpha = 3$.
The standard Fourier Transform pair for a Gaussian function is known to be:

$$\mathcal{F}\left\{e^{-\alpha t^2}\right\} = \sqrt{\frac{\pi}{\alpha}}e^{-\frac{\pi^2 f^2}{\alpha}}$$

Substituting $\alpha = 3$ into the standard formula, we obtain the Fourier Transform $X(f)$ of the given signal:

$$X(f) = \sqrt{\frac{\pi}{3}}e^{-\frac{\pi^2 f^2}{3}}$$

We can express this result in terms of constants by defining:

$$A = \sqrt{\frac{\pi}{3}} \quad \text{and} \quad B = \frac{\pi^2}{3}$$

Substituting these constants back into our expression for $X(f)$ gives:

$$X(f) = Ae^{-Bf^2}$$

This form exactly matches the expression provided in option D.

$$\boxed{\text{D) } Ae^{-Bf^2}}$$

Key Points

- **Gaussian Pulse Properties:** A signal of the form $e^{-\alpha t^2}$ is called a Gaussian pulse.
- **Invariant Shape:** The Gaussian pulse is an eigenfunction of the Fourier transform. This means that its Fourier transform results in another Gaussian pulse; it retains its bell-curve shape in both the time domain and the frequency domain.
- A wider pulse in the time domain (smaller α) corresponds to a narrower pulse in the frequency domain, demonstrating the time-bandwidth product principle.

Q4.00.2 MCQ | 2000 | 2M | Ans: A | ● ○ ○

The Hilbert transform is a linear operation that imparts a -90° ($-\frac{\pi}{2}$) phase shift to all positive frequency components of a signal.
Given the signal:

$$x(t) = \cos(\omega_1 t) + \sin(\omega_2 t)$$

Because the Hilbert transform, denoted by $H[\cdot]$, is a linear operator, we can apply it to each term independently:

$$H[x(t)] = H[\cos(\omega_1 t)] + H[\sin(\omega_2 t)]$$

Using the standard Hilbert transform pairs for sinusoidal functions:

$$H[\cos(\omega t)] = \cos\left(\omega t - \frac{\pi}{2}\right) = \sin(\omega t)$$

$$H[\sin(\omega t)] = \sin\left(\omega t - \frac{\pi}{2}\right) = -\cos(\omega t)$$

Applying these transform pairs to our specific signal components:

$$H[x(t)] = \sin(\omega_1 t) - \cos(\omega_2 t)$$

This exactly matches option A.

$$\boxed{\text{A) } \sin\omega_1 t - \cos\omega_2 t}$$

Key Points

- **Linearity:** $H[ax_1(t) + bx_2(t)] = aH[x_1(t)] + bH[x_2(t)]$.
- **Phase Shift:** The Hilbert transform corresponds to an LTI system with a magnitude response $|H(f)| = 1$ and a phase response $\angle H(f) = -90^\circ$ for $f > 0$ and $+90^\circ$ for $f < 0$.
- **Standard Pairs:** It is essential to remember the fundamental relationships: $H[\cos(\omega_c t)] = \sin(\omega_c t)$ and $H[\sin(\omega_c t)] = -\cos(\omega_c t)$.

Mistakes to be avoided

- A common error is incorrectly assuming $H[\sin(\omega t)] = \cos(\omega t)$ by confusing the phase shift of the Hilbert transform with standard time differentiation. Always remember the negative sign for the sine transform!

Q4.99.1 MCQ | 1999 | 1M | Ans: B | ● ○ ○

The continuous-time Fourier transform of a signal $x(t)$ is given by:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\, dt$$

Using Euler's identity, $e^{-j\omega t} = \cos(\omega t) - j\sin(\omega t)$, we can expand the integral into its real and imaginary parts:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\cos(\omega t)\, dt - j\int_{-\infty}^{\infty} x(t)\sin(\omega t)\, dt$$

We are given that $x(t)$ is a **real** and **odd** function. Recall the mathematical properties of even and odd functions:

- $\cos(\omega t)$ is an even function of t.
- $\sin(\omega t)$ is an odd function of t.
- The product of an odd function and an even function is an odd function.
- The product of two odd functions is an even function.

Let's analyze the two integrals:

1. **Real Part:** The integrand $x(t)\cos(\omega t)$ is the product of an odd function ($x(t)$) and an even function ($\cos(\omega t)$), which results in an **odd** function. The integral of any odd function over symmetric limits ($-\infty$ to ∞) is zero.

$$\int_{-\infty}^{\infty} x(t)\cos(\omega t)\, dt = 0$$

2. **Imaginary Part:** The integrand $x(t)\sin(\omega t)$ is the product of two odd functions, which results in an **even** function. The integral of an even function over symmetric limits evaluates to twice the integral from 0 to ∞.

$$-j\int_{-\infty}^{\infty} x(t)\sin(\omega t)\, dt = -j2\int_{0}^{\infty} x(t)\sin(\omega t)\, dt$$

Therefore, the Fourier transform $X(\omega)$ evaluates to:

$$X(\omega) = -j\int_{-\infty}^{\infty} x(t)\sin(\omega t)\, dt$$

Since the real part is zero and the expression has a j multiplier, $X(\omega)$ is purely **imaginary**. Now, let's check for symmetry by substituting $-\omega$ for ω:

$$X(-\omega) = -j\int_{-\infty}^{\infty} x(t)\sin(-\omega t)\, dt$$

Since sine is an odd function, $\sin(-\omega t) = -\sin(\omega t)$:

$$X(-\omega) = -j\int_{-\infty}^{\infty} x(t)[-\sin(\omega t)]\, dt = j\int_{-\infty}^{\infty} x(t)\sin(\omega t)\, dt$$

$$X(-\omega) = -X(\omega)$$

Because $X(-\omega) = -X(\omega)$, the spectrum $X(\omega)$ is an **odd** function. Conclusion: If $x(t)$ is real and odd, its Fourier transform $X(\omega)$ is an imaginary and odd function of ω.

$$\boxed{\text{B) an imaginary and odd function of } \omega}$$

Key Points

- **Conjugate Symmetry:** For any purely real signal $x(t)$, its Fourier transform exhibits conjugate symmetry, meaning $X(-\omega) = X^*(\omega)$.
- **Real and Even:** If $x(t)$ is real and even, $X(\omega)$ is purely real and even.
- **Real and Odd:** If $x(t)$ is real and odd, $X(\omega)$ is purely imaginary and odd.

Q4.99.2

MCQ | 1999 | 2M | Ans: B | ● ○ ○

The given signal has a minor typographical error in the standard notation; $\sin c^n(at)$ is intended to represent the normalized sinc function raised to a power, $\text{sinc}^n(at)$. The signal can be written as:

$$y(t) = 16 \times 10^4 \text{sinc}^2(400t) * 10^6 \text{sinc}^3(100t)$$

This represents the convolution in the time domain of two separate signals, let's call them $x_1(t)$ and $x_2(t)$. First, let's determine the maximum frequency (bandwidth) of each individual signal. For the normalized sinc function, defined as $\text{sinc}(at) = \frac{\sin(\pi a t)}{\pi a t}$, the Fourier transform is a rectangular pulse strictly bandlimited to $f_{max} = \frac{a}{2}$ Hz.

1. Bandwidth of $x_1(t) = \text{sinc}^2(400t)$: The base signal $\text{sinc}(400t)$ has a maximum frequency of $f = \frac{400}{2} = 200\,\text{Hz}$. Raising a signal to the power of n in the time domain corresponds to convolving its frequency spectrum with itself $n-1$ times. The new bandwidth is n times the original bandwidth.

$$f_{max1} = 2 \times 200 = 400\,\text{Hz}$$

2. Bandwidth of $x_2(t) = \text{sinc}^3(100t)$: The base signal $\text{sinc}(100t)$ has a maximum frequency of $f = \frac{100}{2} = 50\,\text{Hz}$. Since it is cubed ($n = 3$), the bandwidth is multiplied by 3.

$$f_{max2} = 3 \times 50 = 150\,\text{Hz}$$

3. Bandwidth of the Convolved Signal $y(t)$: Convolution in the time domain corresponds to multiplication in the frequency domain: $Y(f) = X_1(f) \cdot X_2(f)$. The multiplication of two bandlimited spectra results in a spectrum that is only non-zero where *both* individual spectra are non-zero. Therefore, the highest frequency component of the convolved signal is the minimum of the individual maximum frequencies.

$$f_{max} = \min(f_{max1}, f_{max2}) = \min(400\,\text{Hz}, 150\,\text{Hz}) = 150\,\text{Hz}$$

The Nyquist sampling frequency f_s is twice the maximum frequency of the signal:

$$f_s = 2 \times f_{max} = 2 \times 150 = 300\,\text{Hz}$$

This corresponds to option B.

$$\boxed{\text{B) } 300\ Hz}$$

Key Points

- **Normalized Sinc Function:** $\text{sinc}(at) \leftrightarrow \frac{1}{|a|}\text{rect}\left(\frac{f}{a}\right)$. The maximum frequency is $f_m = \frac{a}{2}$.
- **Multiplication in Time:** If $y(t) = x^n(t)$, then the bandwidth of $y(t)$ is $n \times$ (bandwidth of $x(t)$).
- **Convolution in Time:** If $y(t) = x_1(t) * x_2(t)$, then $Y(f) = X_1(f)X_2(f)$. The bandwidth of $y(t)$ is limited by the narrower of the two spectra: $B_y = \min(B_1, B_2)$.

Mistakes to be avoided

- Adding the bandwidths ($400 + 150 = 550$ Hz) instead of taking the minimum. Adding bandwidths applies when signals are multiplied in the time domain, not convolved.
- Misinterpreting the standard sinc notation and using the unnormalized $\mathrm{Sa}(t) = \frac{\sin t}{t}$ instead, which would lead to calculating the angular frequency ω rather than the cyclical frequency f in Hertz.

Q4.98.1 MCQ 1998 1M Ans: B

The Fourier transform of a signal $x(t)$ is mathematically defined as:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\, dt$$

Given that $x(t)$ is a voltage signal, its unit is Volt. The differential term dt represents time, which has the unit of seconds. The complex exponential term $e^{-j2\pi ft}$ is dimensionless.

Therefore, the unit of $X(f)$, and consequently the unit of its magnitude $|X(f)|$, is the product of the units of $x(t)$ and dt.

$$\text{Unit of } |X(f)| = \text{Volt-sec}$$

$$\boxed{\text{B) Volt - sec}}$$

Key Points

- The Fourier transform operation integrates a continuous-time signal with respect to time.
- In general, if a time-domain signal $x(t)$ has a unit of U, its frequency-domain representation $X(f)$ will have a unit of $U \cdot \text{sec}$.

Q4.98.2 MCQ 1998 1M Ans: C

The Fourier transform of a Gaussian pulse in the time domain yields another Gaussian function in the frequency domain. Let a Gaussian pulse be represented in the time domain as:

$$x(t) = e^{-at^2}$$

Taking the continuous Fourier transform, we obtain the amplitude spectrum $X(f)$:

$$X(f) = \sqrt{\frac{\pi}{a}} e^{-\frac{\pi^2 f^2}{a}}$$

Since the resulting expression is also in the form of a Gaussian function, the amplitude spectrum of a Gaussian pulse remains Gaussian.

$$\boxed{\text{C) Gaussian}}$$

Key Points

- A Gaussian function is an eigenfunction of the Fourier transform.
- The time-bandwidth product is minimized for a Gaussian pulse, making it theoretically significant in signal processing and communication systems.

Q4.98.3

MCQ | 1998 | 1M | Ans: B

The time differentiation property of the Fourier transform states that differentiating a signal in the time domain corresponds to multiplying its Fourier transform by $j2\pi f$ in the frequency domain. From the definition of the inverse Fourier transform:

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}\,df$$

Differentiating both sides with respect to t, we obtain:

$$\frac{dx(t)}{dt} = \frac{d}{dt}\left[\int_{-\infty}^{\infty} X(f)e^{j2\pi ft}\,df\right]$$

$$\frac{dx(t)}{dt} = \int_{-\infty}^{\infty} X(f)\frac{d}{dt}\left(e^{j2\pi ft}\right)df$$

$$\frac{dx(t)}{dt} = \int_{-\infty}^{\infty} [j2\pi fX(f)]\,e^{j2\pi ft}\,df$$

Comparing this with the standard inverse Fourier transform equation, the term in the brackets represents the Fourier transform of the differentiated signal. Therefore:

$$\mathcal{F}\left\{\frac{dx(t)}{dt}\right\} = j2\pi fX(f)$$

$$\boxed{\text{B) } j2\pi fX(f)}$$

Key Points

- **Time Differentiation Property:** $\mathcal{F}\left\{\frac{d^n x(t)}{dt^n}\right\} = (j2\pi f)^n X(f)$
- In terms of angular frequency ω (where $\omega = 2\pi f$), the property is written as: $\mathcal{F}\left\{\frac{dx(t)}{dt}\right\} = j\omega X(\omega)$

Q4.98.4

MCQ | 1998 | 1M | Ans: A

In flat top sampling, the sampled signal can be modeled as the convolution of an ideally sampled signal (an impulse train) with a rectangular pulse of duration τ. In the frequency domain, this convolution corresponds to multiplying the spectrum of the ideally sampled signal by the Fourier transform of the rectangular pulse, which is a sinc function:

$$H(f) = \tau\text{sinc}(f\tau)$$

The multiplication by this sinc function causes a high-frequency roll-off, meaning the higher frequency components of the baseband signal are attenuated. This distortion in the amplitude spectrum is known as the **aperture effect**.

$$\boxed{\text{A) gives rise to aperture effects}}$$

Key Points

- **Flat Top Sampling:** The sample value is held constant for a short duration τ, forming a rectangular pulse rather than a true impulse.
- **Aperture Effect:** The distortion resulting from the finite width (aperture) of the sampling pulse, which acts as a low-pass filter with a sinc response.
- To recover the original signal without distortion, an equalizer (an inverse filter with magnitude response proportional to $1/|H(f)|$) is typically used before the low-pass reconstruction filter.

Q4.97.1 MCQ 1997 1M Ans: C ● ○ ○

Given the standard Fourier transform pair:

$$\mathcal{F}\{f(t)\} = g(\omega)$$

According to the duality property of the continuous-time Fourier transform, if a time-domain signal $x(t)$ has a Fourier transform $X(\omega)$, then the Fourier transform of the time-domain signal $X(t)$ is given by:

$$\mathcal{F}\{X(t)\} = 2\pi x(-\omega)$$

Applying this duality property to the given function, where our $x(t)$ is $f(t)$ and $X(\omega)$ is $g(\omega)$, we get:

$$\mathcal{F}\{g(t)\} = 2\pi f(-\omega)$$

The integral expression provided in the question is simply the formal definition of the Fourier transform of $g(t)$ (with the integration differential dt inherently implied):

$$\int_{-\infty}^{\infty} g(t)e^{-j\omega t}\, dt = \mathcal{F}\{g(t)\}$$

Substituting the result from the duality property yields:

$$\mathcal{F}\{g(t)\} = 2\pi f(-\omega)$$

$$\boxed{\text{C) } 2\pi f(-\omega)}$$

Key Points

- **Duality Property:** If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then $X(t) \xleftrightarrow{\mathcal{F}} 2\pi x(-\omega)$.
- This property highlights the symmetrical nature of the Fourier transform, making it straightforward to find the transform of a function if it resembles the frequency spectrum of a known transform pair.

Q4.97.2 MCQ 1997 2M Ans: A ● ○ ○

Given the standard Fourier transform pair:

$$\mathcal{F}\{g(t)\} = G(f)$$

For the first item, we apply the time shifting property of the Fourier transform:

$$\mathcal{F}\{x(t - t_0)\} = X(f)e^{-j2\pi f t_0}$$

Substituting $x(t) = g(t)$ and $t_0 = 2$:

$$\mathcal{F}\{g(t - 2)\} = G(f)e^{-j2\pi f(2)} = G(f)e^{-j4\pi f}$$

This matches item (A). For the second item, we apply the time scaling property of the Fourier transform:

$$\mathcal{F}\{x(at)\} = \frac{1}{|a|}X\left(\frac{f}{a}\right)$$

Substituting $x(t) = g(t)$ and $a = \frac{1}{2}$:

$$\mathcal{F}\left\{g\left(\frac{t}{2}\right)\right\} = \frac{1}{|1/2|}G\left(\frac{f}{1/2}\right) = 2G(2f)$$

This matches item (C). Therefore, the correct matching is (1 - A) and (2 - C).

$$\boxed{\text{A) (1 - A), (2 - C)}}$$

Key Points

- **Time Shifting Property:** A shift in the time domain introduces a linear phase shift in the frequency domain without altering the magnitude spectrum.
- **Time Scaling Property:** Expanding a signal in time ($a < 1$) compresses its frequency spectrum and amplifies its amplitude by a factor of $1/|a|$ to preserve energy relationships.

Q4.96.1

MCQ | 1996 | 1M | Ans: C | ● ○ ○

The Fourier transform of a continuous-time signal $x(t)$ is defined as:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\,dt$$

If we evaluate the Fourier transform at $-f$, we get:

$$X(-f) = \int_{-\infty}^{\infty} x(t)e^{j2\pi ft}\,dt$$

Taking the complex conjugate of $X(f)$ yields:

$$X^*(f) = \left[\int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\,dt\right]^* = \int_{-\infty}^{\infty} x^*(t)e^{j2\pi ft}\,dt$$

Since the given time signal is real-valued, it satisfies the condition $x^*(t) = x(t)$. Substituting this back into the equation:

$$X^*(f) = \int_{-\infty}^{\infty} x(t)e^{j2\pi ft}\,dt$$

Comparing the expressions for $X(-f)$ and $X^*(f)$, we arrive at:

$$X(-f) = X^*(f)$$

This mathematical property is known as **conjugate symmetry** (or Hermitian symmetry).

C) conjugate symmetry

Key Points

- **Conjugate Symmetry:** For any strictly real-valued signal $x(t)$, its Fourier transform satisfies $X(-f) = X^*(f)$.
- **Magnitude Spectrum:** The magnitude spectrum of a real signal is an even function, meaning $|X(-f)| = |X(f)|$.
- **Phase Spectrum:** The phase spectrum of a real signal is an odd function, meaning $\angle X(-f) = -\angle X(f)$.

Q4.95.1

MCQ | 1995 | 1M | Ans: C | ● ○ ○

Given the parameters: Message signal frequency, $f_m = 1.0\,\text{kHz} = 1000\,\text{Hz}$ Sampling rate, $f_s = 1800\,\text{samples/sec} = 1800\,\text{Hz}$ Cut-off frequency of the ideal LPF, $f_c = 1100\,\text{Hz}$ When a continuous-time signal is sampled, the spectrum of the sampled signal consists of shifted versions of the original spectrum centered at integer multiples of the sampling frequency. The frequency components present in the sampled signal are given by:

$$f_{out} = |nf_s \pm f_m|$$

where $n = 0, 1, 2, \ldots$ Calculating the frequency components for different values of n: For $n = 0$ (Baseband):

$$f_{out} = 1000\,\text{Hz}$$

For $n = 1$ (First harmonic):

$$f_{out} = |1800 \pm 1000| \implies 800\,\text{Hz and } 2800\,\text{Hz}$$

For $n = 2$ (Second harmonic):

$$f_{out} = |3600 \pm 1000| \implies 2600\,\text{Hz and } 4600\,\text{Hz}$$

The resulting frequency components are 800 Hz, 1000 Hz, 2600 Hz, 2800 Hz, and so on. The sampled signal is then passed through an ideal rectangular low-pass filter (LPF) with a cut-off frequency of 1100 Hz. The LPF will only allow frequency components less than f_c to pass through. Comparing the generated components with the cut-off frequency: 800 Hz < 1100 Hz (Passed) 1000 Hz < 1100 Hz (Passed) 2600 Hz > 1100 Hz (Rejected) Therefore, the output of the filter contains only the 800 Hz and 1000 Hz components. Note that while flat-top sampling introduces an aperture effect (amplitude distortion shaped by a sinc function), it does not change the location of the frequency components compared to ideal instantaneous sampling.

$$\boxed{\text{C) 800 Hz and 1000 Hz components.}}$$

Key Points

- The spectrum of a sampled signal contains frequencies at $nf_s \pm f_m$, where n is an integer.
- **Aliasing Effect:** The Nyquist rate for this signal is $2f_m = 2000$ Hz. Since $f_s = 1800$ Hz ($f_s < 2f_m$), the signal is under-sampled. This causes the lower sideband of the first harmonic ($1800 - 1000 = 800$ Hz) to fold back into the baseband range, appearing below the original message frequency.
- An ideal LPF passes all frequencies below its cut-off f_c with zero attenuation and completely rejects all frequencies above f_c.

Q4.94.1 MCQ | 1994 | 1M | Ans: A | ● ○ ○

In flat-top sampling, the sampled signal can be modeled as the convolution of an ideally sampled impulse train with a rectangular pulse $p(t)$ of duration τ (the pulse-width). In the frequency domain, this operation corresponds to multiplying the periodically repeated spectrum of the message signal by the Fourier transform of the rectangular pulse, which is a sinc function:

$$P(f) = \tau \text{sinc}(f\tau)$$

Because the magnitude of the sinc function decreases as the frequency f increases, it acts as a low-pass filter, attenuating the higher frequency components of the original message spectrum $M(f)$. This phenomenon is known as the **aperture effect**. If the pulse-width τ is increased, the sinc function $P(f)$ narrows in the frequency domain (its first zero-crossing at $1/\tau$ moves closer to the origin). This results in a steeper roll-off within the baseband bandwidth W, leading to greater attenuation of the high-frequency components in the reproduced signal.

$$\boxed{\text{A) attenuation of high frequencies in reproduction}}$$

Key Points

- **Aperture Effect:** The amplitude distortion in flat-top sampling caused by the finite width of the sampling pulses.
- Larger pulse-width $\implies$ narrower sinc envelope $\implies$ more severe high-frequency attenuation.
- To correct this distortion during demodulation, an **equalizer** filter with a magnitude response inversely proportional to the sinc function is placed before the reconstruction low-pass filter.

Q4.92.1 MCQ | 1992 | 2M | Ans: B | ● ○ ○

The Fourier transform of a continuous-time signal $g(t)$ is given by:

$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-j2\pi ft}\, dt$$

Using Euler's identity, $e^{-j2\pi ft} = \cos(2\pi ft) - j\sin(2\pi ft)$, we can expand the integral into its real and imaginary parts:

$$G(f) = \int_{-\infty}^{\infty} g(t)\cos(2\pi ft)\, dt - j\int_{-\infty}^{\infty} g(t)\sin(2\pi ft)\, dt$$

We are given that $g(t)$ is a real and odd symmetric signal, meaning $g(-t) = -g(t)$. By analyzing the symmetry of the integrands:

- The product of an odd function $g(t)$ and an even function $\cos(2\pi ft)$ results in an **odd** function. The definite integral of an odd function over symmetric limits ($-\infty$ to ∞) is exactly zero.
- The product of an odd function $g(t)$ and an odd function $\sin(2\pi ft)$ results in an **even** function.

Therefore, the real part of the Fourier transform vanishes entirely, and the expression simplifies to:

$$G(f) = -j\int_{-\infty}^{\infty} g(t)\sin(2\pi ft)\, dt = -j2\int_{0}^{\infty} g(t)\sin(2\pi ft)\, dt$$

This final expression is purely imaginary. *(Note: The options in the source image contain a typographical error, printing $G(t)$ instead of the intended $G(f)$).*

$$\boxed{\text{B) } G(t) \text{ is imaginary}}$$

Key Points

- **Real and Even** signal $\xleftrightarrow{\mathcal{F}}$ **Real and Even** spectrum.
- **Real and Odd** signal $\xleftrightarrow{\mathcal{F}}$ **Imaginary and Odd** spectrum.
- **Imaginary and Even** signal $\xleftrightarrow{\mathcal{F}}$ **Imaginary and Even** spectrum.
- **Imaginary and Odd** signal $\xleftrightarrow{\mathcal{F}}$ **Real and Odd** spectrum.

Q4.91.1 NAT | 1991 | 2M | Ans: 3.6 | ● ○ ○

Given the frequency components of the signal: Lower frequency, $f_L = 300\,\text{Hz} = 0.3\,\text{kHz}$ Upper frequency, $f_H = 1.8\,\text{kHz}$

Method 1

Using the Bandpass Sampling Theorem
The bandwidth of the signal is:

$$B = f_H - f_L = 1.8\,\text{kHz} - 0.3\,\text{kHz} = 1.5\,\text{kHz}$$

According to the bandpass sampling theorem, the minimum sampling rate required to avoid aliasing is given by:

$$f_{s(min)} = \frac{2f_H}{k}$$

where k is the largest integer satisfying $k \leq \frac{f_H}{B}$. Calculating k:

$$k = \lfloor \frac{1.8}{1.5} \rfloor = \lfloor 1.2 \rfloor = 1$$

Therefore, the minimum sampling rate is:

$$f_{s(min)} = \frac{2 \times 1.8\,\text{kHz}}{1} = 3.6\,\text{kHz}$$

Method 2

Using the Low-Pass Nyquist Criterion
If we treat the signal conservatively as a standard baseband (low-pass) signal, the Nyquist rate depends only on the maximum frequency component present:

$$f_s = 2f_H$$

$$f_s = 2 \times 1.8\,\text{kHz} = 3.6\,\text{kHz}$$

Both approaches yield the same minimum rate since the signal's bandwidth is large relative to its upper cut-off frequency.

$$\boxed{3.6}$$

Key Points

- **Low-Pass Nyquist Rate:** $f_s \geq 2f_{max}$, where f_{max} is the highest absolute frequency component.
- **Bandpass Sampling Theorem:** Allows sampling at a rate lower than the standard Nyquist rate for strictly bandlimited signals, given by $f_s = \frac{2f_H}{k}$. In this case, because $k = 1$, it reduces to the standard low-pass Nyquist rate.

Q4.88.1 MCQ 1988 2M Ans: D ● ○ ○

Given the frequencies of the signal components: First component, $f_1 = 3\,\text{kHz}$ Second component, $f_2 = 6\,\text{kHz}$ Sampling frequency, $f_s = 8\,\text{kHz}$ Low-pass filter cut-off frequency, $f_c = 8\,\text{kHz}$ When a continuous-time signal is sampled at f_s, the spectrum of the sampled signal consists of copies of the original spectrum shifted by integer multiples of f_s. The resulting frequency components are given by:

$$f_{sampled} = |nf_s \pm f_m|$$

where $n = 0, 1, 2, \ldots$ and f_m represents the original signal frequencies. Calculating the frequency components generated by $f_1 = 3\,\text{kHz}$:

- For $n = 0$: $f = 3\,\text{kHz}$ (Original component)
- For $n = 1$: $f = |8 - 3| = 5\,\text{kHz}$ and $f = |8 + 3| = 11\,\text{kHz}$

Calculating the frequency components generated by $f_2 = 6\,\text{kHz}$:

- For $n = 0$: $f = 6\,\text{kHz}$ (Original component)
- For $n = 1$: $f = |8 - 6| = 2\,\text{kHz}$ and $f = |8 + 6| = 14\,\text{kHz}$

The sampled signal therefore contains the following frequencies: 2 kHz, 3 kHz, 5 kHz, 6 kHz, 11 kHz, 14 kHz, and so on. This sampled signal is passed through an ideal low-pass filter with a cut-off frequency of 8 kHz. The filter will pass only the components with frequencies below 8 kHz:

$$2\,\text{kHz},\ 3\,\text{kHz},\ 5\,\text{kHz},\ \text{and}\ 6\,\text{kHz}$$

Here, 3 kHz and 6 kHz are the original components, while 2 kHz and 5 kHz are the spurious (aliased) components generated due to under-sampling.

$$\boxed{\text{D) contains both the components of the original signal and two spurious components of 2 kHz and 5 kHz}}$$

Key Points

- **Nyquist Criterion Violation:** The highest frequency in the original signal is $f_{max} = 6\,\text{kHz}$. To prevent aliasing, the minimum required sampling rate is $2f_{max} = 12\,\text{kHz}$. Since the actual sampling rate is $f_s = 8\,\text{kHz}$ ($f_s < 2f_{max}$), aliasing is guaranteed to occur.
- **Aliased Components:** Under-sampling causes higher frequency components (like $6\,\text{kHz}$) to fold back into the lower frequency range (as $2\,\text{kHz}$), creating spurious frequency components that cannot be distinguished from the original baseband signal.

DEBUG INFO

Chapter IV

Continuous Time Fourier Series

Total Questions : 68

MCQ : 54

MSQ : 0

NAT : 14

SOLUTIONS

V

Laplace Transform

Topics

1. Introduction to Laplace Transform
2. Laplace Transform of Standard Signals and Properties
3. Region of Convergence (ROC)
4. Inverse Laplace Transform
5. Interrelation Between EFSC, FT and LT
6. Eigen Values and Eigen Functions in LTI Systems
7. Unilateral Laplace Transform

Q5.16.1 MCQ | 2016 | 2M | Ans: B | ● ○ ○

For a causal periodic signal $f(t)$ with period T, the Laplace transform is given by:

$$F(s) = \frac{F_1(s)}{1 - e^{-sT}}$$

where $F_1(s)$ is the Laplace transform of the first period of the signal. From the given figure, the signal for the first period $(0 \leq t < T)$ can be defined mathematically using the unit step function $u(t)$ as:

$$f_1(t) = u(t) - u\left(t - \frac{T}{2}\right)$$

Taking the Laplace transform of $f_1(t)$ using the time-shifting property:

$$F_1(s) = \frac{1}{s} - \frac{e^{-sT/2}}{s} = \frac{1 - e^{-sT/2}}{s}$$

Now, substitute $F_1(s)$ into the periodic signal formula:

$$F(s) = \frac{\frac{1-e^{-sT/2}}{s}}{1 - e^{-sT}}$$

The denominator can be expanded using the difference of squares identity, $1 - e^{-sT} = (1 - e^{-sT/2})(1 + e^{-sT/2})$:

$$F(s) = \frac{1 - e^{-sT/2}}{s(1 - e^{-sT/2})(1 + e^{-sT/2})}$$

Canceling the common term $(1 - e^{-sT/2})$ from the numerator and the denominator yields:

$$F(s) = \frac{1}{s(1 + e^{-sT/2})}$$

$$\boxed{\text{B) } F(s) = \frac{1}{s(1 + e^{-sT/2})}}$$

Key Points

- **Laplace Transform of Periodic Functions:** If $f(t) = f(t+T)$, then $\mathcal{L}\{f(t)\} = \frac{\mathcal{L}\{f_1(t)\}}{1-e^{-sT}}$, where $f_1(t)$ is the function defined over a single period.
- **Time Shifting Property:** $\mathcal{L}\{x(t - t_0)u(t - t_0)\} = e^{-st_0}X(s)$.

Q5.15.1 NAT | 2015 | 2M | Ans: -2 | ● ○ ○

Given the signal expression:

$$x(t) = \alpha\beta e^{-4t}u(t) + \beta e^{4t}u(-t)$$

The bilateral Laplace transform is given as:

$$X(s) = \frac{16}{s^2 - 16}, \quad -4 < \text{Re}\{s\} < 4$$

Using partial fraction expansion on $X(s)$:

$$X(s) = \frac{16}{(s+4)(s-4)} = \frac{-2}{s+4} + \frac{2}{s-4}$$

Now, we take the inverse bilateral Laplace transform. The Region of Convergence (ROC) is the intersection of $-4 < \text{Re}\{s\} < 4$. For the pole at $s = -4$, the applicable ROC is $\text{Re}\{s\} > -4$, which corresponds to a right-sided (causal) signal:

$$\mathcal{L}^{-1}\left\{\frac{-2}{s+4}\right\} = -2e^{-4t}u(t)$$

For the pole at $s = 4$, the applicable ROC is $\text{Re}\{s\} < 4$, which corresponds to a left-sided (anti-causal) signal:

$$\mathcal{L}^{-1}\left\{\frac{2}{s-4}\right\} = -2e^{4t}u(-t)$$

Combining these components, the time-domain signal is:

$$x(t) = -2e^{-4t}u(t) - 2e^{4t}u(-t)$$

On comparing this result with the given expression $x(t) = \alpha\beta e^{-4t}u(t) + \beta e^{4t}u(-t)$, we get:

$$\alpha\beta = -2$$

$$\beta = -2$$

$$\boxed{-2}$$

Key Points

- **ROC and Causality:** For a pole at $s = a$, if the ROC is $\text{Re}\{s\} > a$, the inverse transform is $e^{at}u(t)$. If the ROC is $\text{Re}\{s\} < a$, the inverse transform is $-e^{at}u(-t)$.
- **Bilateral Laplace Transform:** The inverse transform is uniquely determined only when both the algebraic expression $X(s)$ and its ROC are specified.

Q5.15.2

MCQ | 2015 | 1M | Ans: C | ● ○ ○

The given function represents a rectangular pulse of unit amplitude existing from $t = a$ to $t = b$:

$$f(t) = \begin{cases} 1 & \text{if } a \le t \le b \\ 0 & \text{otherwise} \end{cases}$$

Method 1

Using Unit Step Functions
We can express the function $f(t)$ mathematically in terms of the standard unit step function $u(t)$ as:

$$f(t) = u(t-a) - u(t-b)$$

Taking the bilateral Laplace transform of both sides using the standard pair $\mathcal{L}\{u(t)\} = \frac{1}{s}$ and the time-shifting property:

$$F(s) = \frac{e^{-as}}{s} - \frac{e^{-bs}}{s}$$

$$F(s) = \frac{e^{-as} - e^{-bs}}{s}$$

Method 2

Using Direct Integration

The bilateral Laplace transform is defined as:

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}\,dt$$

Substituting the given limits and value for $f(t)$:

$$F(s) = \int_{a}^{b} (1)e^{-st}\,dt$$

$$F(s) = \left[\frac{e^{-st}}{-s}\right]_a^b$$

$$F(s) = \frac{e^{-bs}}{-s} - \frac{e^{-as}}{-s}$$

$$F(s) = \frac{e^{-as} - e^{-bs}}{s}$$

$$\boxed{\text{C) } \frac{e^{-as} - e^{-bs}}{s}}$$

Key Points

- **Unit Step Function:** $u(t) = 1$ for $t \geq 0$, and 0 for $t < 0$.
- **Time-Shifting Property:** If $x(t) \xleftrightarrow{\mathcal{L}} X(s)$, then $x(t - t_0) \xleftrightarrow{\mathcal{L}} e^{-st_0}X(s)$.
- **Region of Convergence (ROC):** For a finite-duration signal (a signal that is zero outside a finite interval $[a, b]$), the ROC for the bilateral Laplace transform is typically the entire s-plane.

Q5.15.3 MCQ 2015 2M Ans: B ● ○ ○

Method 1

Using Characteristic Equation
The given linear second-order homogeneous differential equation is:

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$$

The corresponding characteristic (auxiliary) equation is:

$$r^2 + 2r + 1 = 0$$

$$(r + 1)^2 = 0 \implies r = -1, -1$$

Since the roots are real and repeated, the general solution is of the form:

$$y(t) = (C_1 + C_2 t)e^{-t}$$

Now, we apply the initial conditions to find the arbitrary constants C_1 and C_2. Given $y(0) = 1$:

$$1 = (C_1 + C_2(0))e^0 \implies C_1 = 1$$

Differentiating the general solution with respect to t:

$$y'(t) = C_2 e^{-t} - (C_1 + C_2 t)e^{-t} = (C_2 - C_1 - C_2 t)e^{-t}$$

Given $y'(0) = 1$:

$$1 = (C_2 - C_1 - 0)e^0 \implies C_2 - C_1 = 1$$

Substituting $C_1 = 1$ into the above equation:

$$C_2 - 1 = 1 \implies C_2 = 2$$

Substituting C_1 and C_2 back into the general solution, we get the particular solution:

$$y(t) = (1 + 2t)e^{-t}$$

Method 2

Using Laplace Transform
Taking the unilateral Laplace transform of both sides of the given differential equation:

$$\mathcal{L}\left\{\frac{d^2y}{dt^2}\right\} + 2\mathcal{L}\left\{\frac{dy}{dt}\right\} + \mathcal{L}\{y\} = 0$$

$$[s^2Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + Y(s) = 0$$

Substituting the initial conditions $y(0) = 1$ and $y'(0) = 1$:

$$[s^2Y(s) - s(1) - 1] + 2[sY(s) - 1] + Y(s) = 0$$

$$Y(s)(s^2 + 2s + 1) - s - 1 - 2 = 0$$

$$Y(s)(s + 1)^2 = s + 3$$

$$Y(s) = \frac{s + 3}{(s + 1)^2}$$

Using partial fraction decomposition by rewriting the numerator:

$$Y(s) = \frac{(s + 1) + 2}{(s + 1)^2} = \frac{1}{s + 1} + \frac{2}{(s + 1)^2}$$

Taking the inverse Laplace transform:

$$y(t) = e^{-t} + 2te^{-t} = (1 + 2t)e^{-t}$$

$$\boxed{\text{B) } (1 + 2t)e^{-t}}$$

Key Points

- For a second-order linear homogeneous ODE $ay'' + by' + cy = 0$ with repeated real roots ($r_1 = r_2 = r$), the general solution is $y(t) = (C_1 + C_2t)e^{rt}$.
- **Laplace Transform of Derivatives:**
  - $\mathcal{L}\{y'(t)\} = sY(s) - y(0)$
  - $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$
- **First Shifting Theorem:** $\mathcal{L}^{-1}\left\{\frac{n!}{(s-a)^{n+1}}\right\} = t^n e^{at}$.

Q5.15.4 MCQ 2015 2M Ans: D ● ○ ○

Method 1

Using Laplace Transform
Given the unit step response of the system:

$$y(t) = 1 - \frac{2}{\sqrt{3}}e^{-t}\cos\left(\sqrt{3}t - \frac{\pi}{6}\right)$$

First, expand the cosine term using the trigonometric identity $\cos(A - B) = \cos A \cos B + \sin A \sin B$:

$$\cos\left(\sqrt{3}t - \frac{\pi}{6}\right) = \cos(\sqrt{3}t)\cos\left(\frac{\pi}{6}\right) + \sin(\sqrt{3}t)\sin\left(\frac{\pi}{6}\right)$$

Substituting $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ and $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$:

$$\cos\left(\sqrt{3}t - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\cos(\sqrt{3}t) + \frac{1}{2}\sin(\sqrt{3}t)$$

Now, substitute this back into the expression for $y(t)$:

$$y(t) = 1 - \frac{2}{\sqrt{3}}e^{-t}\left[\frac{\sqrt{3}}{2}\cos(\sqrt{3}t) + \frac{1}{2}\sin(\sqrt{3}t)\right]$$

$$y(t) = 1 - e^{-t}\cos(\sqrt{3}t) - \frac{1}{\sqrt{3}}e^{-t}\sin(\sqrt{3}t)$$

Next, take the Laplace transform of $y(t)$ to find the output response $Y(s)$ in the s-domain:

$$Y(s) = \frac{1}{s} - \frac{s+1}{(s+1)^2 + (\sqrt{3})^2} - \frac{1}{\sqrt{3}}\left[\frac{\sqrt{3}}{(s+1)^2 + (\sqrt{3})^2}\right]$$

$$Y(s) = \frac{1}{s} - \frac{s+1}{s^2+2s+4} - \frac{1}{s^2+2s+4}$$

$$Y(s) = \frac{1}{s} - \frac{s+2}{s^2+2s+4}$$

Combine the terms into a single fraction:

$$Y(s) = \frac{(s^2+2s+4) - s(s+2)}{s(s^2+2s+4)} = \frac{s^2+2s+4-s^2-2s}{s(s^2+2s+4)} = \frac{4}{s(s^2+2s+4)}$$

The transfer function $H(s)$ is the ratio of the output $Y(s)$ to the input $X(s)$. For a unit step input, $X(s) = \frac{1}{s}$.

$$H(s) = \frac{Y(s)}{X(s)} = sY(s)$$

$$H(s) = s\left[\frac{4}{s(s^2+2s+4)}\right] = \frac{4}{s^2+2s+4}$$

Method 2

Using Standard Second-Order Form
The unit step response of a standard underdamped second-order system can be written as:

$$y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\cos(\omega_d t - \phi)$$

where $\phi = \sin^{-1}(\zeta)$ and $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Comparing the given response $y(t) = 1 - \frac{2}{\sqrt{3}}e^{-t}\cos\left(\sqrt{3}t - \frac{\pi}{6}\right)$ with the standard form, we can extract: 1. From the phase angle: $\phi = \frac{\pi}{6}$, which implies $\zeta = \sin\left(\frac{\pi}{6}\right) = 0.5$. 2. From the exponential term: $\zeta\omega_n = 1$. Using $\zeta = 0.5$, we solve for the natural frequency ω_n:

$$0.5 \cdot \omega_n = 1 \implies \omega_n = 2\,\text{rad/s}$$

(We can verify the amplitude coefficient: $\frac{1}{\sqrt{1-0.5^2}} = \frac{1}{\sqrt{0.75}} = \frac{2}{\sqrt{3}}$, which perfectly matches the given equation). The standard second-order transfer function is:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Substitute $\zeta = 0.5$ and $\omega_n = 2$:

$$H(s) = \frac{2^2}{s^2 + 2(0.5)(2)s + 2^2} = \frac{4}{s^2 + 2s + 4}$$

$$\boxed{D)\frac{4}{s^2 + 2s + 4}}$$

Key Points

- The transfer function $H(s)$ is related to the step response $Y(s)$ by $H(s) = sY(s)$.
- Standard Laplace transforms: $\mathcal{L}\{e^{-at}\cos(\omega t)\} = \frac{s+a}{(s+a)^2+\omega^2}$ and $\mathcal{L}\{e^{-at}\sin(\omega t)\} = \frac{\omega}{(s+a)^2+\omega^2}$.
- The standard second-order transfer function is $H(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$.

Mistakes to be avoided

- Forgetting that $Y(s)$ is the response to a step input (which includes a $\frac{1}{s}$ term) and failing to multiply by s to obtain the transfer function $H(s)$.
- Making sign errors when applying the trigonometric identity $\cos(A - B) = \cos A\cos B + \sin A\sin B$.

Q5.14.1 NAT 2014 2M Ans: 1 ● ○ ○

Given the differential equation relating the input $x(t)$ and output $y(t)$:

$$\frac{d^2y(t)}{dt^2} + \alpha\frac{dy(t)}{dt} + \alpha^2 y(t) = x(t)$$

Assuming zero initial conditions, we take the Laplace transform on both sides:

$$(s^2 + \alpha s + \alpha^2)Y(s) = X(s)$$

The transfer function $H(s)$, which is the Laplace transform of the impulse response $h(t)$, is:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 + \alpha s + \alpha^2}$$

We are given another signal $g(t)$ defined as:

$$g(t) = \alpha^2\int_0^t h(\tau)\,d\tau + \frac{dh(t)}{dt} + \alpha h(t)$$

Taking the Laplace transform of $g(t)$ term by term using the integration and differentiation properties:

$$G(s) = \frac{\alpha^2}{s}H(s) + sH(s) + \alpha H(s)$$

Factoring out $H(s)$:

$$G(s) = \left(\frac{\alpha^2}{s} + s + \alpha\right) H(s)$$

$$G(s) = \left(\frac{\alpha^2 + s^2 + s\alpha}{s}\right) H(s)$$

Now substitute the expression for $H(s)$:

$$G(s) = \left(\frac{s^2 + \alpha s + \alpha^2}{s}\right) \cdot \left(\frac{1}{s^2 + \alpha s + \alpha^2}\right)$$

The term $(s^2 + \alpha s + \alpha^2)$ cancels out, leaving:

$$G(s) = \frac{1}{s}$$

The poles of $G(s)$ are the roots of the denominator polynomial. Here, there is only one root at $s = 0$. Therefore, the number of poles of $G(s)$ is 1.

$$\boxed{1}$$

Key Points

- **Transfer Function:** $H(s) = \frac{Y(s)}{X(s)}$ with all initial conditions set to zero.
- **Integration Property:** $\mathcal{L}\left\{\int_0^t x(\tau)d\tau\right\} = \frac{1}{s}X(s)$.
- **Differentiation Property:** $\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = sX(s) - x(0)$. For system analysis with zero initial state, it reduces to $sX(s)$.

Q5.14.2 MCQ | 2014 | 2M | Ans: B | ● ○ ○

$$H(s) = \frac{1}{(s^2 + s - 6)}$$

The system is said to be causal if the output at any time depends only on present and/or past values of input or $h(t) = 0$ for $t < 0$ i.e. all poles must lie on LHS of S-plane.

$$H(s) = \frac{1}{(s+3)(s-2)}$$

From the above expression it can be seen that $s = 2$ is a pole which lies on RHS of s-plane. So, $H_1(s)$ system that needs to be cascaded should have one zero at $(s-2)$ Thus,

$$H(s)H_1(s) = (s-2) \cdot \frac{1}{(s+3)(s-2)} = \frac{1}{(s+3)} \rightarrow \text{causal system}$$

$$\boxed{\text{B) } s-2}$$

Key Points

- For a linear time-invariant (LTI) system to be both stable and causal, all of its poles must lie strictly in the left half of the s-plane.
- When two LTI systems are cascaded, their equivalent transfer function is the product of their individual transfer functions: $H_{eq}(s) = H(s)H_1(s)$.
- An unstable or non-causal pole can be compensated via pole-zero cancellation by cascading a system that provides a zero at the exact same location in the s-plane.

Q5.14.3 MCQ 2014 2M Ans: D

Given the Laplace transform of $f(t)$:

$$F(s) = \frac{1}{s^2 + s + 1}$$

We need to find the Laplace transform of $g(t) = t \cdot f(t)$. Using the frequency differentiation property of the Laplace transform:

$$\mathcal{L}\{t \cdot f(t)\} = -\frac{dF(s)}{ds}$$

Substituting the given $F(s)$ into the property:

$$= \frac{-d}{ds}\left[\frac{1}{s^2 + s + 1}\right]$$

Applying the chain rule for differentiation:

$$= -\left(\frac{-1}{(s^2 + s + 1)^2} \cdot \frac{d}{ds}(s^2 + s + 1)\right)$$

$$= \frac{2s + 1}{(s^2 + s + 1)^2}$$

$$\boxed{\text{D) } \frac{2s + 1}{(s^2 + s + 1)^2}}$$

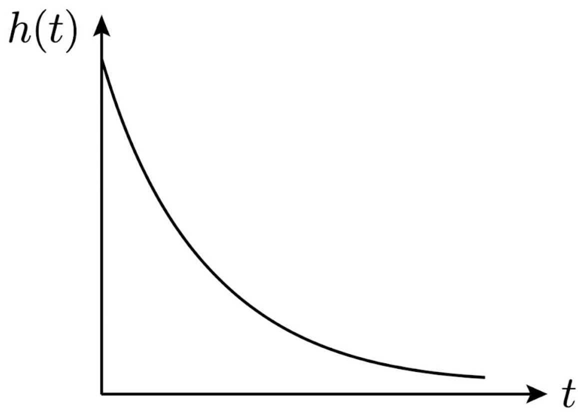
Key Points

- **Multiplication by t Property:** Multiplication by t in the time domain corresponds to differentiation in the s-domain with a sign change. The general form is $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$.
- **Differentiation Rule:** Recall that $\frac{d}{ds}\left(\frac{1}{u(s)}\right) = \frac{-u'(s)}{[u(s)]^2}$. Here, $u(s) = s^2 + s + 1$ and $u'(s) = 2s + 1$.

Q5.14.4 MCQ 2014 2M Ans: A

Given $H(s) = \frac{1}{(s+1)}$

$$h(t) = e^{-t}u(t)$$



at $t = \infty$, $h(t) = 0$

i.e. the system is stable as it converges to zero as $t \to \infty$. The system is causal as it can be seen from the graph that $h(t) = 0$ for $t < 0$.

$$\frac{h(t+1)}{h(t)} = \frac{e^{-(t+1)}}{e^{-t}} = \frac{e^{-t} \cdot e^{-1}}{e^{-t}}$$

$$= e^{-1} \text{ i.e. independent of t}$$

So, statements S_1 and S_2 are true.

$\boxed{\text{A) only S1 and S2 are true}}$

Key Points

- **Stability:** A causal LTI system is bounded-input bounded-output (BIBO) stable if its impulse response is absolutely integrable ($\int_0^\infty |h(t)|dt < \infty$). For $h(t) = e^{-t}u(t)$, the integral evaluates to 1, hence it is stable.
- **Causality:** A continuous-time LTI system is causal if and only if its impulse response $h(t) = 0$ for $t < 0$. The presence of the unit step function $u(t)$ guarantees this condition.

Q5.14.5 MCQ | 2014 | 2M | Ans: A | ● ○ ○

Given $\dot{y}(t) + 5y(t) = u(t)$ and $y(0) = 1$; $u(t)$ is a unit step function. Apply Laplace transform to the given differential equation.

$$sy(s) - y(0) + 5y(s) = \frac{1}{s}$$

$$y(s)[s+5] = \frac{1}{s} + y(0) \quad \left[\mathcal{L}\left[\frac{dy}{dt}\right] = sy(s) - y(0)\right]\left[\mathcal{L}\left[u(t)\right] = \frac{1}{s}\right]$$

$$y(s) = \frac{\frac{1}{s}+1}{(s+5)}$$

$$y(s) = \frac{(s+1)}{s(s+5)} \Rightarrow \frac{A}{s} + \frac{B}{s+5}$$

$A = 1/5$; $B = 4/5$

$$y(s) = \frac{1}{5s} + \frac{4}{5(s+5)}$$

Apply inverse Laplace transform,

$$y(t) = \frac{1}{5} + \frac{4}{5}e^{-5t}$$

$$y(t) = 0.2 + 0.8e^{-5t}$$

$$\boxed{\text{A) } 0.2 + 0.8e^{-5t}}$$

Key Points

- **Laplace Transform of a Derivative:** $\mathcal{L}\{\dot{y}(t)\} = sY(s) - y(0)$.
- **Laplace Transform of Unit Step:** $\mathcal{L}\{u(t)\} = \frac{1}{s}$.
- **Partial Fraction Expansion:** Used to split complex rational fractions into simpler terms for straightforward inverse Laplace transformation.

Q5.14.6 NAT | 2014 | 1M | Ans: 0 | ● ○ ○

Given, $x(t) = -3e^{-2t}$ Laplace transform of $x(t) \longleftrightarrow X(s)$ Where,

$$X(s) = \frac{-3}{s+2}$$

Also, $H(s) = \frac{s-2}{(s+3)}$

$$\therefore Y(s) = X(s)H(s)$$

$$Y(s) = \frac{-3(s-2)}{(s+3)(s+2)}$$

Using final value theorem, the steady state value of $Y(s)$ is

$$y(\infty) = \lim_{s\to 0} sY(s) = 0$$

$$\boxed{0}$$

Key Points

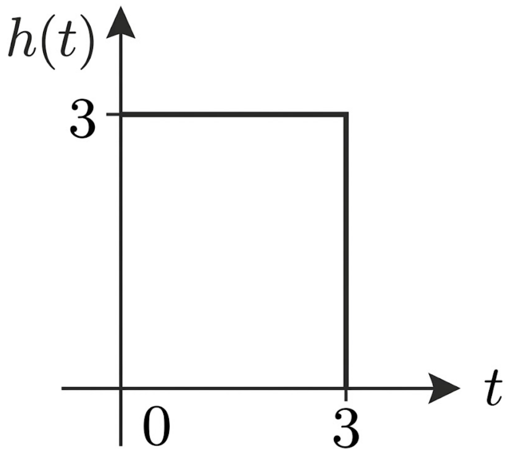
- **Final Value Theorem (FVT):** If a time-domain signal $y(t)$ has a Laplace transform $Y(s)$, the steady-state value of the signal is given by $\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$.

- **Condition for FVT:** The Final Value Theorem is strictly applicable only if all the poles of $sY(s)$ lie in the left half of the s-plane. In this case, the poles are at $s = -2$ and $s = -3$, which ensures the system settles to a stable steady-state value.

Q5.14.7 NAT 2014 1M Ans: 45 ● ○ ○

If the input is ke^{st} then the output is $ke^{st}H(s)$.

$$h(t) = 3[u(t) - u(t-3)]$$



By applying Laplace Transform,

$$H(s) = 3\left[\frac{1-e^{-3s}}{s}\right]$$

We can take the input as $5e^{0\cdot t}$ which means $s = 0$. Then the output is

$$y(t) = 5\lim_{s\to 0} H(s) = 5(3)(3)e^{-3s}\Big|_{s=0} = 45$$

$$\boxed{45}$$

Key Points

- **Eigenfunction Property:** For an LTI system, complex exponentials are eigenfunctions. An input $x(t) = e^{st}$ produces a steady-state output $y(t) = H(s)e^{st}$. A constant DC input is a special case where $s = 0$.
- **DC Gain:** The steady-state response to a constant input A is $A \cdot H(0)$. The DC gain $H(0)$ can be evaluated using L'Hôpital's rule on the transfer function, or alternatively, by finding the total area under the impulse response curve directly $(3 \times 3 = 9)$.

Q5.13.1 MCQ 2013 2M Ans: B ● ○ ○

$$x(t) = u(t) - u(t-2)$$

$$X(s) = \frac{1}{s} - \frac{e^{-2s}}{s} = \frac{1-e^{-2s}}{s}$$

Given that

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y(t) = x(t)$$

Taking Laplace transform on both the sides ($y(0) = 0$ and $\frac{dy}{dt} = 0$ at $t = 0$)

$$s^2Y(s) + 5sY(s) + 6Y(s) = X(s)$$

$$Y(s)\left[s^2 + 5s + 6\right] = \frac{1-e^{-2s}}{s}$$

$$Y(s) = \frac{1-e^{-2s}}{s(s^2+5s+6)} = \frac{1-e^{-2s}}{s(s+2)(s+3)}$$

$$\boxed{B)\,\frac{1-e^{-2s}}{s(s+2)(s+3)}}$$

Key Points

- Laplace transform of a delayed step function: $\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$
- Differentiation in time domain (with zero initial conditions): $\mathcal{L}\left\{\frac{d^n y}{dt^n}\right\} = s^n Y(s)$

Q5.11.1 | MCQ | 2011 | 2M | Ans: B | ● ○ ○

$$F(s) = L[f(t)] = \frac{2(s+1)}{s^2+4s+7}$$

Initial value,

$$\lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s) = \lim_{s\to\infty} s \cdot \frac{2(s+1)}{s^2+4s+7}$$

$$= \lim_{s\to\infty} \frac{2s^2\left(1+\frac{1}{s}\right)}{s^2\left(1+\frac{4}{s}+\frac{7}{s^2}\right)}$$

$$= \frac{2(1+0)}{(1+0+0)} = 2$$

Final value,

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} s \cdot \frac{2(s+1)}{s^2+4s+7}$$

$$= 0$$

$$\boxed{B)2,0}$$

Key Points

- Initial Value Theorem: $\lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s)$
- Final Value Theorem: $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$

Q5.10.1 | MCQ | 2010 | 2M | Ans: D | ● ○ ○

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

Given that,

$$F(s) = \left[\frac{3s+1}{s^3+4s^2+(K-3)s}\right]$$

$$\lim_{t\to\infty} f(t) = 1$$

$$\Rightarrow \lim_{s\to 0} s\left[\frac{3s+1}{s^3+4s^2+(K-3)s}\right] = 1$$

$$\Rightarrow \lim_{s\to 0}\left[\frac{3s+1}{s^2+4s+(K-3)}\right] = 1$$

$$\Rightarrow \frac{1}{K-3} = 1$$

$$\Rightarrow k-3 = 1$$

$$\Rightarrow \boxed{K=4}$$

Key Points

- Final Value Theorem: $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$

Q5.09.1 | MCQ | 2009 | 2M | Ans: D

$$L\left[\int_0^t f(\tau)d\tau\right] = \frac{1}{s}[F(s) - f(0)]$$

$$\boxed{D)\frac{1}{s}[F(s) - f(0)]}$$

Key Points

- Time integration property of Laplace transform (as defined by the specific problem constraints).

Q5.07.1 | MCQ | 2007 | 1M | Ans: D

Final value theorem is applicable only when all the poles of system lies in left half of s-plane.
$\because$ $s = 1$ is right S-plane pole. Therefore, final value theorem cannot be applied.

$$Y(s) = \frac{1}{s-1} - \frac{1}{s}$$

$$y(t) = (e^t - 1)u(t)$$

$\therefore$ Unbounded.

$$\boxed{\text{D) Unbounded}}$$

Key Points

- The Final Value Theorem $\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$ is valid if and only if all poles of $sY(s)$ lie in the left half of the s-plane.

Mistakes to be avoided

- Applying the Final Value Theorem directly without verifying the pole locations of the system. For unstable systems (poles in RHS), the time-domain signal grows infinitely, making the theorem invalid.

Q5.07.2 | MCQ | 2007 | 2M | Ans: B

Given the frequency response of the linear, time-invariant system:

$$H(f) = \frac{5}{1 + j10\pi f}$$

To find the step response, it is convenient to analyze the system in the Laplace domain. We substitute $s = j\omega = j2\pi f$. Notice that the term $j10\pi f$ can be rewritten as $5(j2\pi f) = 5s$. Therefore, the transfer function $H(s)$ is:

$$H(s) = \frac{5}{1 + 5s} = \frac{1}{s + \frac{1}{5}}$$

The step response is defined as the system's output $y(t)$ when the input is a unit step function $x(t) = u(t)$. The Laplace transform of the unit step input is:

$$X(s) = \frac{1}{s}$$

The output response in the s-domain, $Y(s)$, is given by the product $Y(s) = H(s)X(s)$:

$$Y(s) = \left(\frac{1}{s+\frac{1}{5}}\right)\left(\frac{1}{s}\right) = \frac{1}{s\left(s+\frac{1}{5}\right)}$$

Using partial fraction expansion to separate the terms:

$$Y(s) = \frac{A}{s} + \frac{B}{s+\frac{1}{5}}$$

Solving for the constants A and B:

$$A = \left.\frac{1}{s+\frac{1}{5}}\right|_{s=0} = \frac{1}{\frac{1}{5}} = 5$$

$$B = \left.\frac{1}{s}\right|_{s=-\frac{1}{5}} = \frac{1}{-\frac{1}{5}} = -5$$

Substituting A and B back into the expression for $Y(s)$:

$$Y(s) = \frac{5}{s} - \frac{5}{s+\frac{1}{5}}$$

Taking the inverse Laplace transform, we obtain the step response in the time domain:

$$y(t) = 5u(t) - 5e^{-\frac{1}{5}t}u(t)$$

$$\boxed{\text{B) } 5\left(1 - e^{-\frac{1}{5}t}\right)u(t)}$$

Key Points

- The transfer function $H(s)$ can be derived from the frequency response $H(f)$ by substituting $s = j2\pi f$.
- The step response of a system is the inverse Laplace transform of $\frac{H(s)}{s}$.
- Standard Laplace transform pair: $\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}u(t)$.

Mistakes to be avoided

- Incorrectly substituting $s = j\pi f$ instead of $s = j2\pi f$, which leads to the wrong time constant (e.g., yielding e^{-5t}) in the exponential term.
- Forgetting to multiply the transfer function $H(s)$ by $\frac{1}{s}$ (the Laplace transform of the step input) before taking the inverse Laplace transform.

Q5.06.1 MCQ | 2006 | 2M | Ans: C | ● ○ ○

$$L^{-1}[F(s)] = \sin\omega_0 t$$

$$f(t) = \sin\omega_0 t$$

$$\boxed{\text{C) } -1 \le f(\infty) \le 1}$$

Key Points

- The inverse Laplace transform of a system with purely imaginary poles (e.g., $F(s) = \frac{\omega_0}{s^2+\omega_0^2}$) results in an undamped sinusoidal response.

- The Final Value Theorem cannot yield a single value because the limit $\lim_{t\to\infty} \sin(\omega_0 t)$ does not exist; instead, the function continuously oscillates between its minimum and maximum amplitudes.

Q5.06.2 MCQ | 2006 | 2M | Ans: B | ● ○ ○

Given the unit-step response of the system:

$$c(t) = 1 - e^{-2t}$$

Taking the Laplace transform of the output response $c(t)$, we get $C(s)$:

$$C(s) = \frac{1}{s} - \frac{1}{s+2}$$

Combining the terms by taking a common denominator:

$$C(s) = \frac{(s+2)-s}{s(s+2)} = \frac{2}{s(s+2)}$$

Since the system is subjected to a unit-step input, the input $r(t) = u(t)$. Taking the Laplace transform of the input, we get $R(s)$:

$$R(s) = \frac{1}{s}$$

The transfer function $H(s)$ of a linear time-invariant system is defined as the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming zero initial conditions:

$$H(s) = \frac{C(s)}{R(s)}$$

Substituting the expressions for $C(s)$ and $R(s)$:

$$H(s) = \frac{\frac{2}{s(s+2)}}{\frac{1}{s}}$$

$$H(s) = \frac{2}{s+2}$$

$$\boxed{\text{B) } \frac{2}{2+s}}$$

Key Points

- The transfer function $H(s)$ is related to the step response $C(s)$ by $H(s) = sC(s)$.
- The standard Laplace transform for a unit step function is $\mathcal{L}\{u(t)\} = \frac{1}{s}$.
- The standard Laplace transform for an exponential decay is $\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$.

Mistakes to be avoided

- Assuming the Laplace transform of the step response is the transfer function itself. You must divide the output transform by the input transform $R(s) = \frac{1}{s}$.
- Making arithmetic errors when resolving the partial fractions or taking the common denominator.

Q5.06.3 MCQ | 2006 | 2M | Ans: C | ● ○ ○

Method 1

Time Domain (Convolution)

The output $y(t)$ of a continuous-time LTI system is the convolution of its impulse response $h(t)$ and the input $x(t)$:

$$y(t) = \int_0^t h(\tau)x(t-\tau)\,d\tau$$

Given the impulse response $h(t) = e^{-t}$ for $t \geq 0$ and a unit step input $x(t) = u(t)$:

$$y(t) = \int_0^t e^{-\tau}(1)\,d\tau = [-e^{-\tau}]_0^t = 1 - e^{-t}$$

The steady-state value is obtained by evaluating the limit as $t \to \infty$:

$$y(\infty) = \lim_{t\to\infty}(1 - e^{-t}) = 1 - 0 = 1$$

Method 2

Frequency Domain (Final Value Theorem)

The transfer function $H(s)$ is the Laplace transform of the impulse response:

$$H(s) = \mathcal{L}\{e^{-t}u(t)\} = \frac{1}{s+1}$$

For a unit step input $x(t) = u(t)$, the Laplace transform is $X(s) = \frac{1}{s}$. The system output $Y(s)$ is:

$$Y(s) = H(s)X(s) = \left(\frac{1}{s+1}\right)\left(\frac{1}{s}\right) = \frac{1}{s(s+1)}$$

Using the Final Value Theorem (valid since the non-zero poles of $sY(s)$ are in the left half of the s-plane):

$$y(\infty) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} s\left(\frac{1}{s(s+1)}\right) = \lim_{s\to 0}\frac{1}{s+1} = 1$$

$\boxed{\text{C) } 1}$

Key Points

- The step response of an LTI system is the integral of its impulse response.
- The steady-state value for a unit step input is equivalent to the DC gain of the system, which is $H(0) = \int_0^\infty h(t)\,dt$.
- **Final Value Theorem:** $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$, provided all poles of $sF(s)$ lie strictly in the left half of the s-plane.

Q5.05.1 MCQ | 2005 | 2M | Ans: A | ● ○ ○

Given the function:

$$f(t) = e^{(a+2)t+5} = e^5 \cdot e^{(a+2)t}$$

Taking the Laplace transform, we get:

$$F(s) = \left[\frac{1}{s-(a+2)}\right]e^5$$

$$\therefore \text{ For } LT \text{ to exist, } \mathrm{Re}(s) > a+2$$

$\boxed{\text{A) } \mathrm{Re}(s) > a+2}$

Key Points

- The Laplace transform of an exponential function e^{kt} is given by $\frac{1}{s-k}$.
- The Region of Convergence (ROC), which dictates where the Laplace transform exists, is $\text{Re}(s) > k$.
- Constant multipliers (like e^5) can be factored out of the transform due to linearity.

Q5.04.1 MCQ 2004 2M Ans: A ● ○ ○

Given the linear differential equation:

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x(t)$$

Taking the Laplace transform on both sides. Since the system is initially at rest, all initial conditions are zero ($y(0) = 0, y'(0) = 0$):

$$s^2Y(s) + 3sY(s) + 2Y(s) = X(s)$$

$$Y(s)(s^2 + 3s + 2) = X(s)$$

The transfer function of the system is:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 + 3s + 2} = \frac{1}{(s+1)(s+2)}$$

Given the input $x(t) = 2u(t)$, its Laplace transform is:

$$X(s) = \frac{2}{s}$$

Substituting $X(s)$ to find the output $Y(s)$:

$$Y(s) = H(s)X(s) = \frac{2}{s(s+1)(s+2)}$$

Applying partial fraction decomposition:

$$Y(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

Calculate the residues A, B, and C:

$$A = \left.\frac{2}{(s+1)(s+2)}\right|_{s=0} = \frac{2}{(1)(2)} = 1$$

$$B = \left.\frac{2}{s(s+2)}\right|_{s=-1} = \frac{2}{(-1)(1)} = -2$$

$$C = \left.\frac{2}{s(s+1)}\right|_{s=-2} = \frac{2}{(-2)(-1)} = 1$$

Substitute the coefficients back into the expansion:

$$Y(s) = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+2}$$

Taking the inverse Laplace transform yields the output response in the time domain:

$$y(t) = (1 - 2e^{-t} + e^{-2t})u(t)$$

$$\boxed{\text{A) } (1 - 2e^{-t} + e^{-2t})u(t)}$$

Key Points

- For an LTI system initially at rest, the initial conditions are zero, simplifying the Laplace transform of derivative terms $\mathcal{L}\{\frac{d^n y}{dt^n}\} = s^n Y(s)$.
- The Laplace transform of a scaled unit step function $Ku(t)$ is $\frac{K}{s}$.
- Partial fraction expansion is the standard algebraic method used to simplify complex s-domain expressions before applying the inverse Laplace transform.

Q5.03.1 MCQ 2003 1M Ans: C ● ○ ○

Using the Final Value Theorem:

$$\lim_{t\to\infty} i(t) = \lim_{s\to 0} sI(s)$$

$$= \lim_{s\to 0} s\frac{2}{s(1+s)} = 2$$

$$\boxed{\text{C) } 2}$$

Key Points

- The Final Value Theorem states that $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$.
- This theorem is applicable only if all the poles of $sF(s)$ lie in the left half of the s-plane.

Q5.02.1 MCQ 2002 2M Ans: C ● ○ ○

$$X(s) = \frac{5-s}{s^2-s-2} = \frac{5-s}{(s+1)(s-2)}$$

Using partial fraction expansion:

$$\frac{5-s}{(s+1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-2}$$

$$5 - s = A(s-2) + B(s+1)$$

Substitute $s = 2$:

$$3 = 3B \implies B = 1$$

Substitute $s = -1$:

$$6 = -3A \implies A = -2$$

So, the function can be written as:

$$X(s) = \frac{-2}{s+1} + \frac{1}{s-2}$$

Note: ROC of Laplace transform should include $\sigma = 0$ line for the Fourier transform to exist.

$$\therefore -1 < \sigma < 2$$

Taking the inverse Laplace transform corresponding to this ROC (right-sided for $s = -1$, left-sided for $s = 2$):

$$x(t) = -2e^{-t}u(t) - e^{2t}u(-t)$$

$$\boxed{\text{C) } -e^{2t}u(-t) - 2e^{-t}u(t)}$$

Key Points

- For a continuous-time signal to have a valid Fourier transform, the Region of Convergence (ROC) of its Laplace transform must include the imaginary axis ($\text{Re}(s) = \sigma = 0$).
- The inverse Laplace transform of $\frac{1}{s-a}$ with ROC $\text{Re}(s) > a$ is $e^{at}u(t)$ (a causal, right-sided signal).
- The inverse Laplace transform of $\frac{1}{s-a}$ with ROC $\text{Re}(s) < a$ is $-e^{at}u(-t)$ (an anti-causal, left-sided signal).

Q5.00.1 MCQ 2000 1M Ans: B

Given the Laplace transforms from the problem description:

$$F(s) = L[f(t)] = \frac{s+2}{s^2+1}$$

Note: The problem image contains a typo, stating $L[f(t)]$ twice. The second function is assumed to be $L[g(t)]$ based on the convolution integral provided.

$$G(s) = L[g(t)] = \frac{s^2+1}{(s+3)(s+2)}$$

The given integral represents the convolution of $f(t)$ and $g(t)$:

$$h(t) = \int_0^t f(\tau)g(t-\tau)d\tau = f(t) * g(t)$$

According to the convolution theorem for Laplace transforms, the transform of a convolution in the time domain is the product of their individual Laplace transforms in the s-domain:

$$L[h(t)] = L[f(t) * g(t)] = F(s) \cdot G(s)$$

Substituting the given expressions into the equation:

$$L[h(t)] = \left(\frac{s+2}{s^2+1}\right) \cdot \left(\frac{s^2+1}{(s+3)(s+2)}\right)$$

Canceling the common terms $(s+2)$ and (s^2+1):

$$L[h(t)] = \frac{1}{s+3}$$

$$\boxed{B)\frac{1}{s+3}}$$

Key Points

- The convolution of two functions $f(t)$ and $g(t)$ is defined as $f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)d\tau$.
- The Convolution Theorem states that the Laplace transform of the convolution of two functions is the algebraic product of their individual Laplace transforms: $L[f(t) * g(t)] = F(s) \cdot G(s)$.

Q5.00.2 MCQ | 2000 | 2M | Ans: A | ● ○ ○

Method 1

Using Laplace Transform

Given the impulse response $h(t) = e^{2t}$ and input $x(t) = e^{3t}$ for $t > 0$. Taking the Laplace transform of both signals:

$$H(s) = \frac{1}{s-2}$$

$$X(s) = \frac{1}{s-3}$$

For a linear time invariant (LTI) system with zero initial conditions, the output in the s-domain is given by $Y(s) = X(s)H(s)$:

$$Y(s) = \left(\frac{1}{s-3}\right)\left(\frac{1}{s-2}\right)$$

Using partial fraction expansion:

$$Y(s) = \frac{1}{s-3} - \frac{1}{s-2}$$

Taking the inverse Laplace transform, we get the time domain output:

$$y(t) = e^{3t} - e^{2t}$$

Method 2

Using Convolution Integral

The output $y(t)$ of an LTI system is the convolution of its input $x(t)$ and impulse response $h(t)$:

$$y(t) = \int_0^t x(\tau)h(t-\tau)\,d\tau$$

Substituting the given functions:

$$y(t) = \int_0^t e^{3\tau}e^{2(t-\tau)}\,d\tau$$

$$y(t) = e^{2t}\int_0^t e^{3\tau}e^{-2\tau}\,d\tau = e^{2t}\int_0^t e^{\tau}\,d\tau$$

$$y(t) = e^{2t}\left[e^{\tau}\right]_0^t = e^{2t}(e^t - 1)$$

$$y(t) = e^{3t} - e^{2t}$$

$$\boxed{\text{A) } e^{3t} - e^{2t}}$$

Key Points

- The output of an LTI system is given by the convolution of the input and the impulse response: $y(t) = x(t) * h(t)$.
- In the Laplace domain, convolution in the time domain corresponds to multiplication: $Y(s) = X(s)H(s)$.
- The standard Laplace transform pair: $\mathcal{L}\{e^{at}u(t)\} = \frac{1}{s-a}$.

Mistakes to be avoided

- Forgetting that the limits of integration for causal signals ($t > 0$) are from 0 to t, rather than $-\infty$ to ∞.
- Making sign errors during the partial fraction expansion step.

Q5.99.1 MCQ 1999 1M Ans: B ● ○ ○

By the time shifting property of the Laplace transform, if the Laplace transform of a function $f(t)$ is $F(s)$, then the Laplace transform of the function delayed by time T (assuming causality, $f(t-T)u(t-T)$) is given by:

$$\mathcal{L}[f(t-T)] = e^{-sT}F(s)$$

This can be derived from the definition of the Laplace transform:

$$\mathcal{L}[f(t-T)] = \int_{T}^{\infty} f(t-T)e^{-st}\,dt$$

Substituting $\tau = t - T$, we get $d\tau = dt$. The lower limit changes from $t = T$ to $\tau = 0$:

$$\mathcal{L}[f(t-T)] = \int_{0}^{\infty} f(\tau)e^{-s(\tau+T)}\,d\tau$$

$$= e^{-sT}\int_{0}^{\infty} f(\tau)e^{-s\tau}\,d\tau$$

$$= e^{-sT}F(s)$$

$$\boxed{\text{B) } e^{-sT}F(s)}$$

Key Points

- **Time Shifting Property:** $\mathcal{L}[f(t-a)u(t-a)] = e^{-as}F(s)$.
- A time delay of T seconds in the time domain corresponds to a multiplication by e^{-sT} in the Laplace (s) domain.

Q5.98.1 MCQ 1998 2M Ans: A ● ○ ○

Method 1

Time Domain Analysis
Given the Laplace transform of the function:

$$F(s) = \frac{\omega}{s^2 + \omega^2}$$

Taking the inverse Laplace transform, we get the standard time-domain sinusoidal function:

$$f(t) = \mathcal{L}^{-1}\left\{\frac{\omega}{s^2+\omega^2}\right\} = \sin(\omega t)u(t)$$

We need to evaluate the limit of this function as time approaches infinity:

$$\lim_{t\to\infty} f(t) = \lim_{t\to\infty} \sin(\omega t)$$

The sine function continuously oscillates between -1 and 1 as t approaches infinity. It never settles or converges to a single steady-state value. Therefore, the limit does not exist and the final value cannot be determined.

Method 2

Final Value Theorem Applicability
The Final Value Theorem (FVT) is typically used to find the steady-state value directly from the s-domain:

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

However, there is a strict condition for applying FVT: The theorem is only valid if all the poles of $sF(s)$ lie strictly in the left half of the s-plane (i.e., the real parts of the poles must be negative).

Let's find the poles of $sF(s)$:

$$sF(s) = s\left(\frac{\omega}{s^2+\omega^2}\right) = \frac{s\omega}{(s-j\omega)(s+j\omega)}$$

The poles are the roots of the denominator, which are $s = +j\omega$ and $s = -j\omega$. Since these poles lie exactly on the imaginary axis (representing an undamped oscillator), the necessary condition for the Final Value Theorem is violated. Because we cannot apply the FVT, it confirms mathematically that a steady final value cannot be determined.

$$\boxed{\text{A) cannot be determined}}$$

Key Points

- The Laplace transform pair for a sine wave is $\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2+\omega^2}$.
- **Final Value Theorem (FVT):** $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$.
- FVT strictly requires that the system is stable, meaning the poles of $sF(s)$ must lie in the left half of the complex plane. Poles on the imaginary axis (marginal stability) or right half-plane invalidate the use of the theorem.

Q5.98.2 MCQ 1998 2M Ans: B ● ○ ○

Method 1

Time Domain (Convolution Integral)

The output $y(t)$ of a linear time-invariant (LTI) system is the convolution of the input excitation $x(t)$ and the impulse response $h(t)$:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

Given $h(t) = u(t)$ and $x(t) = e^{-at}u(t)$, we substitute these into the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} e^{-a\tau}u(\tau)u(t-\tau)\,d\tau$$

The product $u(\tau)u(t-\tau)$ is non-zero (and equal to 1) only when $0 \le \tau \le t$. Therefore, for $t > 0$, the limits of integration become 0 to t:

$$y(t) = \int_0^t e^{-a\tau}\,d\tau = \left[\frac{e^{-a\tau}}{-a}\right]_0^t$$

$$y(t) = \frac{e^{-at}-e^0}{-a} = \frac{1-e^{-at}}{a} = \frac{1}{a}(1-e^{-at})$$

Method 2

Frequency Domain (Laplace Transform)

Take the Laplace transform of the impulse response $h(t)$ and the input $x(t)$:

$$H(s) = \mathcal{L}\{u(t)\} = \frac{1}{s}$$

$$X(s) = \mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$$

The output in the s-domain is the product of the transfer function $H(s)$ and the input $X(s)$:

$$Y(s) = H(s)X(s) = \left(\frac{1}{s}\right)\left(\frac{1}{s+a}\right) = \frac{1}{s(s+a)}$$

Use partial fraction expansion to separate the terms:

$$Y(s) = \frac{A}{s} + \frac{B}{s+a}$$

Solving for the residues A and B:

$$A = \left.\frac{1}{s+a}\right|_{s=0} = \frac{1}{a}$$

$$B = \left.\frac{1}{s}\right|_{s=-a} = -\frac{1}{a}$$

Substitute A and B back into the expansion:

$$Y(s) = \frac{1/a}{s} - \frac{1/a}{s+a}$$

Take the inverse Laplace transform to find the time-domain response:

$$y(t) = \frac{1}{a}u(t) - \frac{1}{a}e^{-at}u(t) = \frac{1}{a}(1-e^{-at})u(t)$$

For $t > 0$, the response simplifies to $\frac{1}{a}(1-e^{-at})$.

$$\boxed{\text{B) } (1/a)(1-e^{-at})}$$

Key Points

- The convolution of any causal signal $x(t)$ with the unit step function $u(t)$ is equivalent to the integration of the signal $x(t)$ from 0 to t.
- Multiplication in the s-domain corresponds to convolution in the time domain: $y(t) = x(t) * h(t) \xleftrightarrow{\mathcal{L}} Y(s) = X(s)H(s)$.
- Partial fraction decomposition is the standard method for converting rational s-domain expressions back into standard Laplace transform pairs.

Q5.97.1 MCQ 1997 1M Ans: A ● ○ ○

The given function is $f(t) = e^{\alpha t}\cos(\alpha t)$. We know the standard Laplace transform of the cosine function:

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}$$

Substituting $\omega = \alpha$, we get:

$$\mathcal{L}\{\cos(\alpha t)\} = \frac{s}{s^2+\alpha^2}$$

Using the First Shifting Theorem of Laplace transforms, which states that if $\mathcal{L}\{x(t)\} = X(s)$, then:

$$\mathcal{L}\{e^{at}x(t)\} = X(s-a)$$

Applying this theorem to our function with $a = \alpha$:

$$\mathcal{L}\{e^{\alpha t}\cos(\alpha t)\} = \left.\frac{s}{s^2+\alpha^2}\right|_{s\to s-\alpha}$$

$$\mathcal{L}\{e^{\alpha t}\cos(\alpha t)\} = \frac{s-\alpha}{(s-\alpha)^2+\alpha^2}$$

$$\boxed{\text{A) } \frac{s-\alpha}{(s-\alpha)^2+\alpha^2}}$$

Key Points

- The Laplace transform of $\cos(\omega t)$ is $\frac{s}{s^2+\omega^2}$.
- **First Shifting Theorem:** Multiplication by e^{at} in the time domain corresponds to a shift of a in the s-domain: $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$.

Q5.96.1

MCQ | 1996 | 2M | Ans: A | ● ○ ○

Let the given function be:

$$F(s) = \frac{s+5}{(s+1)(s+3)}$$

To find the inverse Laplace transform, we first expand the function using partial fractions:

$$\frac{s+5}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

Solving for the residues A and B: For A, multiply by $(s+1)$ and evaluate at $s=-1$:

$$A = \left.\frac{s+5}{s+3}\right|_{s=-1} = \frac{-1+5}{-1+3} = \frac{4}{2} = 2$$

For B, multiply by $(s+3)$ and evaluate at $s=-3$:

$$B = \left.\frac{s+5}{s+1}\right|_{s=-3} = \frac{-3+5}{-3+1} = \frac{2}{-2} = -1$$

Substituting the values of A and B back into the partial fraction expansion:

$$F(s) = \frac{2}{s+1} - \frac{1}{s+3}$$

Taking the inverse Laplace transform on both sides using linearity:

$$f(t) = \mathcal{L}^{-1}\left\{\frac{2}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$$

$$f(t) = 2e^{-t} - e^{-3t}$$

$$\boxed{\text{A) } 2e^{-t} - e^{-3t}}$$

Key Points

- **Partial Fraction Expansion:** A standard technique used to break down complex rational algebraic expressions in the s-domain into simpler fractions.
- **Standard Inverse Transform:** The inverse Laplace transform of $\frac{1}{s-a}$ is e^{at} (assuming a causal system where $t \geq 0$).
- **Linearity Property:** $\mathcal{L}^{-1}\{aF_1(s) + bF_2(s)\} = a\mathcal{L}^{-1}\{F_1(s)\} + b\mathcal{L}^{-1}\{F_2(s)\}$.

Q5.95.1 MCQ | 1995 | 1M | Ans: A | ● ○ ○

The Final Value Theorem relates the time-domain behavior of a function $f(t)$ as $t \to \infty$ to its Laplace transform $F(s)$ as $s \to 0$. Mathematically, it is stated as:

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

Since $t \to \infty$ corresponds to the condition where all transient responses have died out, evaluating this limit provides the steady-state value of the system output. This allows us to find the steady-state value directly from the s-domain without needing to compute the full inverse Laplace transform.

$$\boxed{\text{A) steady state value of the system output}}$$

Key Points

- **Final Value Theorem:** Evaluates the steady-state value ($t \to \infty$). Formula: $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$.
- **Initial Value Theorem:** Evaluates the initial value ($t \to 0$). Formula: $\lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s)$.
- The Final Value Theorem is strictly valid only if all the poles of $sF(s)$ lie in the left half of the s-plane (i.e., the system is stable).

Q5.95.2 MCQ | 1995 | 1M | Ans: B | ● ○ ○

Given the Laplace transform:

$$F(s) = \frac{2(s+1)}{s^2+2s+5}$$

Step 1: Find the Initial Value $f(0+)$ Using the Initial Value Theorem:

$$f(0+) = \lim_{s\to\infty} sF(s)$$

$$f(0+) = \lim_{s\to\infty} s\left[\frac{2(s+1)}{s^2+2s+5}\right] = \lim_{s\to\infty} \frac{2s^2+2s}{s^2+2s+5}$$

Dividing the numerator and denominator by s^2:

$$f(0+) = \lim_{s\to\infty} \frac{2+\frac{2}{s}}{1+\frac{2}{s}+\frac{5}{s^2}} = \frac{2+0}{1+0+0} = 2$$

Step 2: Find the Final Value $f(\infty)$ First, we must check if the Final Value Theorem is applicable. We do this by finding the poles of $sF(s)$. The characteristic equation for the poles is:

$$s^2 + 2s + 5 = 0$$

The roots are $s = -1 \pm j2$. Since the real part of all poles is strictly negative (-1), they lie in the left half of the s-plane, meaning the system is stable and FVT is applicable. Using the Final Value Theorem:

$$f(\infty) = \lim_{s\to 0} sF(s)$$

$$f(\infty) = \lim_{s\to 0} s\left[\frac{2(s+1)}{s^2+2s+5}\right]$$

Substituting $s = 0$:

$$f(\infty) = 0 \cdot \left[\frac{2(0+1)}{0^2+0+5}\right] = 0$$

$$\boxed{\text{B) 2, 0 respectively}}$$

Key Points

- **Initial Value Theorem (IVT):** $\lim_{t\to 0+} f(t) = \lim_{s\to\infty} sF(s)$.
- **Final Value Theorem (FVT):** $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$.
- **Crucial Condition for FVT:** The theorem yields a valid result only if all the poles of $sF(s)$ lie strictly in the left half of the s-plane.

Q5.94.1 MCQ 1994 1M Ans: C ● ○ ○

A unit ramp function starting at time $t = a$ can be mathematically represented as:

$$f(t) = (t-a)u(t-a)$$

We know the standard Laplace transform of a basic unit ramp function $r(t) = tu(t)$ is:

$$\mathcal{L}\{tu(t)\} = \frac{1}{s^2}$$

Using the time-shifting property of the Laplace transform, which states that if $\mathcal{L}\{x(t)\} = X(s)$, then for a delayed function:

$$\mathcal{L}\{x(t-a)u(t-a)\} = e^{-as}X(s)$$

Applying this property to our ramp function with a delay of a:

$$\mathcal{L}\{(t-a)u(t-a)\} = e^{-as}\left(\frac{1}{s^2}\right) = \frac{e^{-as}}{s^2}$$

$$\boxed{\text{C) } \frac{e^{-as}}{s^2}}$$

Key Points

- The Laplace transform of a standard unit ramp function $r(t) = tu(t)$ is $\frac{1}{s^2}$.
- **Time-Shifting Theorem:** A shift (delay) of a units in the time domain results in multiplication by e^{-as} in the Laplace domain.

Q5.93.1 MCQ 1993 2M Ans: D ● ○ ○

The given Laplace transform is:

$$F(s) = \frac{K}{(s+1)(s^2+4)}$$

We are asked to find $\lim_{t\to\infty} f(t)$, which typically suggests applying the Final Value Theorem (FVT). However, before evaluating the limit, we must verify if the theorem is applicable. The Final Value Theorem is only valid if all poles of the function $sF(s)$ lie strictly in the left half of the s-plane. Let's determine the poles of $sF(s)$:

$$sF(s) = \frac{sK}{(s+1)(s^2+4)}$$

The poles are the roots of the denominator:

$$s+1 = 0 \implies s = -1$$
$$s^2+4 = 0 \implies s = \pm j2$$

Because the system has poles on the imaginary axis ($s = \pm j2$), the time-domain response $f(t)$ will contain undamped sinusoidal components (e.g., $\sin(2t)$ or $\cos(2t)$). Consequently, the function will exhibit sustained oscillations and will never settle to a constant steady-state value as $t \to \infty$. Therefore, the limit does not exist, making the final value undefined.

$$\boxed{\text{D) undefined}}$$

Mistakes to be avoided

A very common mistake is blindly applying the Final Value Theorem formula without checking the pole locations:

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} \frac{sK}{(s+1)(s^2+4)} = 0$$

This mathematical calculation incorrectly leads to choosing option (B). Always check if the system is stable (all poles of $sF(s)$ have negative real parts) before using FVT.

Key Points

- **Final Value Theorem:** $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$.
- **Applicability Condition:** FVT can *only* be applied if all poles of $sF(s)$ lie strictly in the left half of the s-plane (excluding the origin itself).
- Poles on the imaginary axis indicate an oscillatory response, meaning no steady-state limit exists.

Q5.93.2 NAT 1993 2M **Ans:** $-\frac{e^{-\pi s}}{(s^2+1)(1-e^{-\pi s})}$ ● ○ ○

For a periodic function $f(t)$ with a fundamental period T, the Laplace transform is given by:

$$F(s) = \frac{1}{1-e^{-sT}} \int_0^T f(t)e^{-st}dt$$

From the given graph and definition, the function repeats every 2π radians, so the fundamental period is $T = 2\pi$. Over the first period $0 \le t \le 2\pi$, the function is defined as:

$$f(t) = \begin{cases} 0, & 0 \le t < \pi \\ \sin(t), & \pi \le t \le 2\pi \end{cases}$$

Substituting this into the Laplace transform integral:

$$\int_0^{2\pi} f(t)e^{-st}dt = \int_\pi^{2\pi} \sin(t)e^{-st}dt$$

Using the standard integration formula $\int e^{at}\sin(bt)dt = \frac{e^{at}}{a^2+b^2}(a\sin(bt) - b\cos(bt))$ with $a = -s$ and $b = 1$:

$$\int_\pi^{2\pi} e^{-st}\sin(t)dt = \left[\frac{e^{-st}}{s^2+1}(-s\sin(t) - \cos(t))\right]_\pi^{2\pi}$$

Evaluating at the upper limit ($t = 2\pi$):

$$\frac{e^{-2\pi s}}{s^2+1}(-s\sin(2\pi) - \cos(2\pi)) = \frac{e^{-2\pi s}}{s^2+1}(0-1) = -\frac{e^{-2\pi s}}{s^2+1}$$

Evaluating at the lower limit ($t = \pi$):

$$\frac{e^{-\pi s}}{s^2+1}(-s\sin(\pi) - \cos(\pi)) = \frac{e^{-\pi s}}{s^2+1}(0-(-1)) = \frac{e^{-\pi s}}{s^2+1}$$

Subtracting the lower limit value from the upper limit value:

$$\int_\pi^{2\pi} e^{-st}\sin(t)dt = -\frac{e^{-2\pi s}}{s^2+1} - \frac{e^{-\pi s}}{s^2+1} = -\frac{e^{-\pi s}(e^{-\pi s}+1)}{s^2+1}$$

Now, substitute this result back into the periodic function formula:

$$F(s) = \frac{1}{1 - e^{-2\pi s}} \left(-\frac{e^{-\pi s}(e^{-\pi s} + 1)}{s^2 + 1} \right)$$

Using the algebraic identity $1 - x^2 = (1 - x)(1 + x)$, we can factor the denominator as $1 - e^{-2\pi s} = (1 - e^{-\pi s})(1 + e^{-\pi s})$:

$$F(s) = \frac{1}{(1 - e^{-\pi s})(1 + e^{-\pi s})} \left(-\frac{e^{-\pi s}(1 + e^{-\pi s})}{s^2 + 1} \right)$$

Canceling the common term $(1 + e^{-\pi s})$ from the numerator and denominator yields the final transform:

$$F(s) = -\frac{e^{-\pi s}}{(s^2 + 1)(1 - e^{-\pi s})}$$

$$\boxed{-\frac{e^{-\pi s}}{(s^2 + 1)(1 - e^{-\pi s})}}$$

Key Points

- The Laplace transform of a periodic function with period T is $F(s) = \frac{1}{1-e^{-sT}} \int_0^T f(t)e^{-st}dt$.
- Standard integral: $\int e^{at} \sin(bt)dt = \frac{e^{at}}{a^2+b^2}(a \sin(bt) - b \cos(bt))$.
- Be careful with factoring terms like $1 - e^{-2\pi s}$ as a difference of squares to simplify Laplace expressions.

Q5.88.1 MCQ | 1988 | 2M | Ans: B | ● ○ ○

Let $f_1(t)$ be the function defined over the first period, such that $f_1(t) = f(t)$ for $0 \leq t < T$, and $f_1(t) = 0$ otherwise. The periodic function $f(t)u(t)$ can be expressed as an infinite sum of delayed versions of $f_1(t)$:

$$f(t)u(t) = f_1(t) + f_1(t - T)u(t - T) + f_1(t - 2T)u(t - 2T) + \ldots$$

Taking the Laplace transform of both sides and applying the time-shifting property $\mathcal{L}\{x(t - t_0)u(t - t_0)\} = e^{-st_0}X(s)$:

$$F(s) = F_1(s) + e^{-sT}F_1(s) + e^{-2sT}F_1(s) + \ldots$$

$$F(s) = F_1(s) \left[1 + e^{-sT} + e^{-2sT} + \ldots\right]$$

The expression in the brackets is an infinite geometric progression with a first term of 1 and a common ratio $r = e^{-sT}$. The sum of an infinite geometric series is given by $\frac{1}{1-r}$.

$$1 + e^{-sT} + e^{-2sT} + \cdots = \frac{1}{1 - e^{-sT}}$$

Therefore, the Laplace transform of the periodic function is:

$$F(s) = \frac{1}{1 - e^{-sT}}F_1(s)$$

The problem states that $F(s) = A(s)F_1(s)$. Comparing the two equations, we can identify $A(s)$:

$$A(s) = \frac{1}{1 - e^{-sT}} = \frac{1}{1 - \exp(-Ts)}$$

$$\boxed{\text{B) } A(s) = 1/(1 - \exp(-Ts))}$$

Key Points

- A continuous periodic function can be represented as an infinite sum of shifted single-period pulses.
- **Time-Shifting Property:** $\mathcal{L}\{f(t-T)u(t-T)\} = e^{-sT}F(s)$.
- **Laplace Transform of a Periodic Function:** $F(s) = \frac{1}{1-e^{-sT}} \int_0^T f(t)e^{-st}dt$, where the integral represents the Laplace transform of the first period.

Q5.87.1 MCQ 1987 2M Ans: C ● ○ ○

The given functions are standard signals frequently encountered in control systems and signal processing. **1. Laplace transform of** $tu(t)$**:** The function $tu(t)$ is known as the unit ramp signal. The general formula for the Laplace transform of $t^n u(t)$ is $\frac{n!}{s^{n+1}}$. Substituting $n = 1$:

$$\mathcal{L}\{tu(t)\} = \frac{1!}{s^{1+1}} = \frac{1}{s^2}$$

2. Laplace transform of $u(t)\sin(t)$**:** The function $u(t)\sin(t)$ is a causal sinusoidal signal. The standard Laplace transform formula for a sine wave $\sin(\omega t)u(t)$ is $\frac{\omega}{s^2+\omega^2}$. Here, the angular frequency is $\omega = 1$:

$$\mathcal{L}\{u(t)\sin(t)\} = \frac{1}{s^2+1^2} = \frac{1}{s^2+1}$$

Therefore, the respective Laplace transforms are $\frac{1}{s^2}$ and $\frac{1}{s^2+1}$.

$$\boxed{\text{C) } \frac{1}{s^2}, \frac{1}{s^2+1}}$$

Key Points

- **Unit Ramp Signal:** $\mathcal{L}\{tu(t)\} = \frac{1}{s^2}$.
- **Causal Sine Function:** $\mathcal{L}\{\sin(\omega t)u(t)\} = \frac{\omega}{s^2+\omega^2}$.
- **Causal Cosine Function:** $\mathcal{L}\{\cos(\omega t)u(t)\} = \frac{s}{s^2+\omega^2}$.

DEBUG INFO

Chapter V

Laplace Transform

Total Questions : 39

MCQ : 34

MSQ : 0

NAT : 5

SOLUTIONS

VI

Z Transform

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform

Q6.26.1 MCQ 2026 1M Ans: A ● ○ ○

The given difference equation is:

$$y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + x(n)$$

Taking the z-transform on both sides:

$$Y(z) = \frac{5}{6}z^{-1}Y(z) - \frac{1}{6}z^{-2}Y(z) + X(z)$$

$$Y(z)\left(1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}\right) = X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}}$$

On factoring the denominator, we get,

$$H(z) = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)}$$

The input is given as

$$x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$$

Its z-transform is:

$$X(z) = 1 - \frac{1}{3}z^{-1}$$

The response $Y(z)$,

$$Y(z) = H(z) \cdot X(z)$$

$$Y(z) = \frac{1 - \frac{1}{3}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)}$$

$$Y(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

On taking inverse z-transform of $Y(z)$, we get

$$y(n) = \left(\frac{1}{2}\right)^n u(n)$$

Since $y(n) = 0$ for $n < 0$. Hence, the response is causal. The pole of $Y(z)$ is at $z = \frac{1}{2}$. Since the magnitude of the pole (i.e. $\frac{1}{2}$) i.e. less than 1 , the pole lies inside the unit circle in the z-plane, making the response stable. Hence, the response $y[n]$ is stable and causal.

$$\boxed{\text{A. Stable and causal response}}$$

Key Points

- The z-transform of a time-shifted signal $x[n-k]$ is $z^{-k}X(z)$.
- A discrete-time LTI system is stable if all poles of its transfer function lie strictly inside the unit circle ($|z| < 1$).
- A system (or response) is causal if its time-domain sequence is zero for $n < 0$.

Mistakes to be avoided

- Do not evaluate stability based solely on the poles of the system transfer function $H(z)$ if the question asks for the stability of the *response* $Y(z)$. Always perform pole-zero cancellations between $H(z)$ and $X(z)$ first.

Q6.25.1 MSQ | 2025 | 1M | Ans: C,D | ● ○ ○

The discrete-time Fourier transform (DTFT) exists if the region of convergence (RoC) contains the unit circle.
If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α is a scalar, then the region of convergence (RoC) is the entire z-plane.

$$\boxed{\text{C, D}}$$

Key Points

- The DTFT is the z-transform evaluated on the unit circle ($z = e^{j\omega}$). For the DTFT to exist (converge), the unit circle ($|z| = 1$) must lie within the Region of Convergence of the z-transform.
- The z-transform of an impulse signal $x[n] = \alpha\delta[n]$ is $X(z) = \alpha$. Since $X(z)$ is a finite constant independent of z, it converges for all possible values in the complex plane, making the RoC the entire z-plane.

Q6.24.1 MSQ | 2024 | 1M | Ans: A,C,D | ● ○ ○

Given the transfer function of a causal discrete-time LTI system:

$$H(z) = \frac{2z^2 + 3}{\left(z + \frac{1}{3}\right)\left(z - \frac{1}{3}\right)} = \frac{2z^2 + 3}{z^2 - \frac{1}{9}}$$

Checking Statement A (Stability): The poles of the system are the roots of the denominator:

$$z^2 - \frac{1}{9} = 0 \implies z = \frac{1}{3}, -\frac{1}{3}$$

Since the system is causal, the Region of Convergence (ROC) is $|z| > \frac{1}{3}$. Because all poles lie strictly inside the unit circle ($|z| < 1$), the system is stable. Therefore, Statement A is TRUE.
Checking Statement B (Minimum Phase): A system is a minimum phase system if and only if all its poles and zeros lie strictly inside the unit circle. The zeros are the roots of the numerator:

$$2z^2 + 3 = 0 \implies z^2 = -1.5 \implies z = \pm j\sqrt{1.5}$$

The magnitude of the zeros is $|z| = \sqrt{1.5} > 1$. Since the zeros lie outside the unit circle, the system is not minimum phase. Therefore, Statement B is FALSE.
Checking Statement C (Initial Value): Using the Initial Value Theorem for a causal system:

$$h[0] = \lim_{z\to\infty} H(z) = \lim_{z\to\infty} \frac{2z^2 + 3}{z^2 - \frac{1}{9}} = \lim_{z\to\infty} \frac{2 + \frac{3}{z^2}}{1 - \frac{1}{9z^2}} = \frac{2+0}{1-0} = 2$$

Therefore, Statement C is TRUE.
Checking Statement D (Final Value): For a stable causal LTI system, the impulse response $h[n]$ must absolutely converge ($\sum |h[n]| < \infty$). This necessary condition implies that $\lim_{n\to\infty} h[n] = 0$. Alternatively, using the Final Value Theorem (since poles of $(z-1)H(z)$ are inside the unit circle):

$$h[\infty] = \lim_{z\to 1}(z-1)H(z) = (1-1)\cdot H(1) = 0 \cdot \left(\frac{5}{8/9}\right) = 0$$

Therefore, Statement D is TRUE.

$$\boxed{\text{A, C, D}}$$

Key Points

- A causal discrete-time LTI system is stable if and only if all its poles are strictly inside the unit circle ($|z| < 1$).
- A system is minimum phase if both its poles and zeros lie strictly inside the unit circle.
- **Initial Value Theorem:** $x[0] = \lim_{z\to\infty} X(z)$.
- **Final Value Theorem:** $\lim_{n\to\infty} x[n] = \lim_{z\to 1}(z-1)X(z)$, provided all poles of $(z-1)X(z)$ lie strictly inside the unit circle.

Q6.20.1

MCQ | 2020 | 1M | Ans: D

Method 1

Given :

$$y[n] = \sum_{k=0}^{3}(-1)^k \delta[n-k]$$

To find impulse response, substitution $x[n] = \delta[n]$ in the given input output relation.

$$h[n] = \sum_{k=0}^{3}(-1)^k \delta[n-k]$$

$$h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$$

Taking z-transform both sides to find the transfer function of the system,

$$H(z) = 1 - z^{-1} + z^{-2} - z^{-3}$$

$$H(z) = \frac{z^3 - z^2 + z - 1}{z^3}$$

$$H(z)|_{z=+1} = \frac{1-1+1-1}{1} = 0$$

So, the system function has a zero at $z = +1$ that indicates that the low frequency gain of filter is zero. So it will be a high pass filter. To check, finding $H(z)$ at $\omega = \pi$ or $z = -1$

$$H(z)|_{z=-1} = \frac{-1-1-1-1}{-1} = 4$$

So the system behaves as a HPF and it will not have zero at $z = -1$.
The only option satisfying this is option (D).
Hence, the correct option is (D).

Method 2

Given :

$$y[n] = \sum_{k=0}^{3}(-1)^k x(n-k)$$

$$y(n) = x(n) - x(n-1) + x(n-2) - x(n-3)$$

Taking z-transform on both sides,

$$Y(z) = (1 - z^{-1} + z^{-2} - z^{-3})X(z)$$

$$\frac{Y(z)}{X(z)} = 1 - z^{-1} + z^{-2} - z^{-3}$$

$$H(z) = 1 - z^{-1} + z^{-2} - z^{-3} = \frac{z^3 - z^2 + z - 1}{z^3}$$

$H(z)$ has a pole of order 3 at $z = 0$.

To find zero's, equate numerator to zero.

$$z^3 - z^2 + z - 1 = 0$$

$$(z-1)(z^2+1) = 0$$

$$z = 1 \text{ and } z = \pm j$$

$\therefore$ $H(z)$ has zero's at $z = 1$ and $z = \pm j$ Hence, the correct option is (D).

$\boxed{\text{D}}$

Key Points

- The transfer function $H(z)$ is defined as the ratio of the z-transform of the output to the z-transform of the input, i.e., $H(z) = \frac{Y(z)}{X(z)}$.
- The zeros of a system are the roots of the numerator polynomial of $H(z)$. Finding a zero at $z = 1$ (which maps to $\omega = 0$ on the unit circle) indicates the system attenuates or blocks DC (low-frequency) components.
- The poles of the system are determined by the roots of the denominator polynomial. A term z^3 in the denominator implies a pole of order 3 at the origin ($z = 0$).

Q6.20.2 NAT 2020 2M Ans: -2 ● ○ ○

System is all-pass filter. For digital all-pass filter, condition is

$$\text{zero} = \frac{1}{\text{pole}^*} \quad \ldots (i)$$

By given transfer function,

$$\text{zero} = \alpha$$

$$\text{Pole} = -0.5$$

using condition(i),

$$\alpha = \frac{1}{-0.5} = -2$$

$\boxed{-2}$

Key Points

- A system with a constant magnitude response over all frequencies ($|H(e^{j\omega})| = \text{constant}$) is an all-pass filter.
- For a real-valued digital all-pass filter, the poles and zeros occur in reciprocal conjugate pairs. If there is a pole at $z = p$, there must be a zero at $z = \frac{1}{p^*}$.
- Since the coefficients are real in this system, the complex conjugate is just the real value itself, leading to zero $= \frac{1}{\text{pole}}$.

Q6.19.1 MCQ 2019 1M Ans: D ● ○ ○

$$P(Z) = H(Z)H\left(\frac{1}{Z}\right)$$

(i) $h(n)$ is real. So $p(n)$ will be also real
(ii) $P(z) = P(z^{-1})$ From (i) : if z_1 is a zero of $P(z)$, then z_1^* will be also a zero of $P(z)$.
From (ii): If z_1 is a zero of $P(z)$, then $\frac{1}{z_1}$ will be also a zero of $P(z)$. So, the 4 zeros are,

$$z_1 = \frac{1}{2} + \frac{1}{2}j$$

$$z_2 = z_1^* = \frac{1}{2} - \frac{1}{2}j$$

$$z_3 = \frac{1}{z_1} = \frac{1}{\frac{1}{2} + \frac{1}{2}j} = 1 - j$$

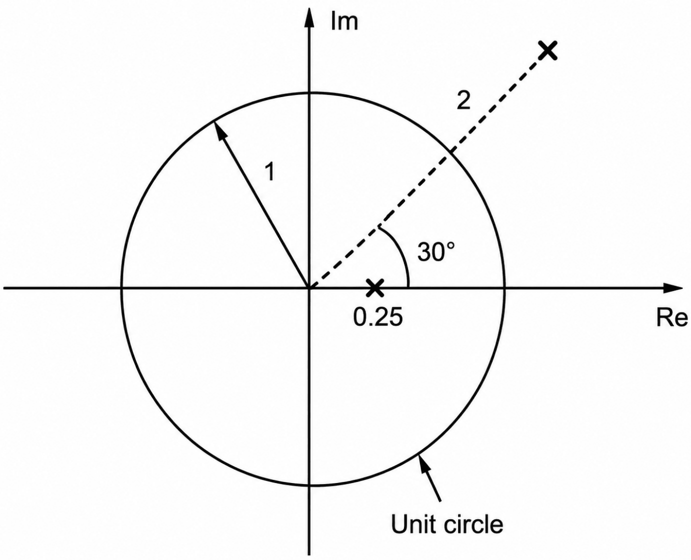
$$z_4 = \left(\frac{1}{z_1}\right)^* = z_3^* = 1 + j$$

$$\boxed{\text{D}}$$

Key Points

- If a discrete-time signal $h[n]$ is real-valued, its zeros and poles in the z-domain occur in complex conjugate pairs.
- For a filter with transfer function $P(z) = H(z)H(z^{-1})$, the zeros and poles also exhibit reciprocal symmetry (if z_0 is a root, so is $1/z_0$).
- Combining both properties, complex roots appear in quartets: z_0, z_0^*, $1/z_0$, and $1/z_0^*$.

Q6.18.1 MCQ 2018 1M Ans: B ● ○ ○



The ROC should include unit circle to make the system stable. From the given pole pattern it is clear that, to make the system stable, the ROC should be finite circular strip or finite annular strip and hence the impulse response of the system should be also two-sided.

$\boxed{\text{B) It is stable only when the impulse response is two- sided.}}$

Key Points

- For a discrete-time LTI system to be stable, its Region of Convergence (ROC) must include the unit circle ($|z| = 1$).
- An ROC that forms an annular strip ($r_1 < |z| < r_2$) between two poles indicates that the corresponding impulse response $h[n]$ is a two-sided sequence (having both causal and anti-causal components).

Q6.16.1 MCQ | 2016 | 1M | Ans: B | ● ○ ○

Given,

$$x[n] = a^n u[n] + b^n u[n]$$
$$0 < |a| < |b| < 1$$

Also given,

$$AOC = (|z| > |a|) \text{ and } (|z| > |b|)$$
$$ROC = |z| > |b|$$
$$\boxed{\text{B) } |z| > |b|}$$

Key Points

- The z-transform of a right-sided sequence like $\alpha^n u[n]$ has a Region of Convergence (ROC) of $|z| > |\alpha|$.
- When a signal is the sum of two or more sequences, the overall ROC is the intersection of the individual ROCs.
- Since the condition $0 < |a| < |b|$ is given, the intersection of the regions $|z| > |a|$ and $|z| > |b|$ is strictly $|z| > |b|$.

Q6.16.2 MCQ | 2016 | 1M | Ans: C | ● ○ ○

Given,

$$x[n] = \delta[n-3] + 2\delta[n-5]$$
$$X(z) = z^{-3} + 2z^{-5}$$
$$X(-z) = (-z)^{-3} + 2(-z)^{-5}$$
$$Y(z) = X(-z) = -z^{-3} - 2z^{-5}$$
$$= -\left[z^{-3} + 2z^{-5}\right]$$
$$y[n] = -x[n]$$
$$\boxed{\text{C) } y[n] = -x[n]}$$

Key Points

- The z-transform of a delayed unit impulse $\delta[n-k]$ is z^{-k}.
- If $X(z)$ is the z-transform of $x[n]$, then $X(a^{-1}z)$ corresponds to the time-domain signal $a^n x[n]$. For $a = -1$, $X(-z)$ corresponds to $(-1)^n x[n]$.
- Because the non-zero terms in $x[n]$ occur only at odd values of n ($n = 3$ and $n = 5$), multiplying the sequence by $(-1)^n$ evaluates to -1, yielding $(-1)^n x[n] = -x[n]$.

Q6.16.3 MCQ | 2016 | 2M | Ans: D | ● ○ ○

Given, $x[n] = (2.0)^{|n|}, -\infty < n < +\infty$

$$= 2^n u[n] + \left(\frac{1}{2}\right)^n u(-n-1)$$

ROC : $|z| > 2$ and $|z| < \frac{1}{2}$ $\therefore$ No Roc

$$\boxed{\text{D}}$$

Key Points

- A two-sided sequence $x[n] = a^{|n|}$ can be decomposed into a right-sided sequence $a^n u[n]$ and a left-sided sequence $a^{-n}u[-n-1]$.
- The z-transform of the right-sided sequence $2^n u[n]$ has an ROC of $|z| > 2$.
- The z-transform of the left-sided sequence $\left(\frac{1}{2}\right)^n u(-n-1)$ has an ROC of $|z| < \frac{1}{2}$.
- The overall ROC of a signal is the intersection of the ROCs of its individual components. Since the regions $|z| > 2$ and $|z| < \frac{1}{2}$ do not overlap, the system has no common region of convergence (the ROC does not exist).

Q6.15.1 MCQ 2015 2M Ans: C ● ○ ○

Method 1

$$\frac{Y(z)}{X(z)} = \frac{\frac{5}{3}[1-z]}{z^2 - z + \frac{2}{9}}$$

$$\Rightarrow \quad Y(z) = \frac{5}{3}\frac{(1-z)}{\left(z-\frac{1}{3}\right)\left(z-\frac{2}{3}\right)}$$

$$Y(z) = \frac{5z}{z-\frac{1}{3}} - \frac{5z}{z-\frac{2}{3}}$$

$$\boxed{y(n) = 5\left(\frac{1}{3}\right)^n u(n) - 5\left(\frac{2}{3}\right)^n u(n)}$$

Method 2

To find the transfer function $H(z) = Y(z)/X(z)$, we will construct the signal flow graph and apply Mason's Gain Formula. Let the output of the top summing junction be $E(z)$, the output of the first delay block be $V_1(z)$, and the output of the second delay block be $V_2(z)$.
The forward paths from the input $X(z)$ to the output $Y(z)$ are:

- Path 1: Through the first delay and directly to the output summer.

$$P_1 = (1)\cdot(z^{-1})\cdot\left(-\frac{5}{3}\right) = -\frac{5}{3}z^{-1}$$

- Path 2: Through both delays and to the output summer.

$$P_2 = (1)\cdot(z^{-1})\cdot(z^{-1})\cdot\left(\frac{5}{3}\right) = \frac{5}{3}z^{-2}$$

The individual loops in the system are:

- Loop 1: Through the first delay and the inner feedback path to the top summer.

$$L_1 = (z^{-1})\cdot(1) = z^{-1}$$

- Loop 2: Through both delays and the outer feedback path to the top summer.

$$L_2 = (z^{-1})\cdot(z^{-1})\cdot\left(-\frac{2}{9}\right) = -\frac{2}{9}z^{-2}$$

Since all loops touch each other and all forward paths, the determinant Δ and cofactors Δ_1, Δ_2 are:

$$\Delta = 1 - (L_1 + L_2) = 1 - z^{-1} + \frac{2}{9}z^{-2}$$

$$\Delta_1 = 1, \quad \Delta_2 = 1$$

Using Mason's Gain Formula, the transfer function is:

$$H(z) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{-\frac{5}{3}z^{-1} + \frac{5}{3}z^{-2}}{1 - z^{-1} + \frac{2}{9}z^{-2}}$$

Factorizing the denominator:

$$1 - z^{-1} + \frac{2}{9}z^{-2} = \left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)$$

So,

$$H(z) = \frac{-\frac{5}{3}z^{-1}(1 - z^{-1})}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)}$$

The system is excited by a unit step sequence $x[n] = u[n]$, whose Z-transform is:

$$X(z) = \frac{1}{1 - z^{-1}}$$

The Z-transform of the response $Y(z)$ is:

$$Y(z) = H(z)X(z) = \frac{-\frac{5}{3}z^{-1}(1 - z^{-1})}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)} \cdot \frac{1}{1 - z^{-1}}$$

$$Y(z) = \frac{-\frac{5}{3}z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)}$$

Using partial fraction expansion:

$$Y(z) = \frac{A}{1 - \frac{1}{3}z^{-1}} + \frac{B}{1 - \frac{2}{3}z^{-1}}$$

$$-\frac{5}{3}z^{-1} = A\left(1 - \frac{2}{3}z^{-1}\right) + B\left(1 - \frac{1}{3}z^{-1}\right)$$

Solving for the coefficients A and B: Substitute $z^{-1} = 3$:

$$-\frac{5}{3}(3) = A(1 - 2) \implies -5 = -A \implies A = 5$$

Substitute $z^{-1} = \frac{3}{2}$:

$$-\frac{5}{3}\left(\frac{3}{2}\right) = B\left(1 - \frac{1}{2}\right) \implies -\frac{5}{2} = \frac{1}{2}B \implies B = -5$$

Thus, the partial fraction expansion is:

$$Y(z) = \frac{5}{1 - \frac{1}{3}z^{-1}} - \frac{5}{1 - \frac{2}{3}z^{-1}}$$

Taking the inverse Z-transform, we get the time-domain response:

$$\boxed{\text{C) } y[n] = 5\left[\left(\frac{1}{3}\right)^n - \left(\frac{2}{3}\right)^n\right]u[n]}$$

Key Points

- The transfer function of a discrete-time system is given by $H(z) = \frac{Y(z)}{X(z)}$.
- For a unit step input sequence $x[n] = u(n)$, the Z-transform is $X(z) = \frac{z}{z-1}$.
- The inverse Z-transform is computed using partial fraction expansion and the standard transform pair $a^n u(n) \leftrightarrow \frac{z}{z-a}$.
- Mason's Gain Formula for discrete systems: $H(z) = \frac{1}{\Delta}\sum P_k \Delta_k$
- Z-transform of unit step function: $\mathcal{Z}\{u[n]\} = \frac{1}{1-z^{-1}}$
- Pole-zero cancellation can significantly simplify the response calculation when the input pole matches a system zero.

Q6.15.2 | MCQ | 2015 | 2M | Ans: C | ● ○ ○

Poles are at $z = \pm 2$ Since $x[n]$ is absolutely summable hence, DTFT exists which implies ROC must include unit circle. $\Rightarrow x(t)$ is a non-causal signal.

$$\boxed{\text{C) It is a non-causal signal}}$$

Key Points

- A discrete-time signal $x[n]$ is absolutely summable if and only if the Region of Convergence (ROC) of its Z-transform includes the unit circle ($|z| = 1$).
- Since the poles are at $|z| = 2$ (outside the unit circle), the ROC must be $|z| < 2$ in order to include the unit circle.
- An ROC of the form $|z| < a$ corresponds to a left-sided (anti-causal or non-causal) sequence.

Q6.15.3 | MCQ | 2015 | 2M | Ans: A | ● ○ ○

$$H(z) = \frac{z^4}{\left[\left(z-\frac{1}{2}\right)^2 + \frac{1}{4}\right]\left[\left(z+\frac{1}{2}\right)^2 + \frac{1}{4}\right]}$$

$$\text{Poles} = \frac{1}{2} \pm j\frac{1}{2}, -\frac{1}{2} \pm j\frac{1}{2}$$

$$= \frac{z^4}{\left(z^2 - z + \frac{1}{2}\right)\left(z^2 + z + \frac{1}{2}\right)}$$

$$= \frac{z^4}{z^4 + z^2 + \frac{1}{4} - z^2}$$

$$H(z) = \frac{z^4}{z^4 + \frac{1}{4}}$$

$$= z^4 + \frac{1}{4}\Big) z^4 \Big(1 - \frac{1}{4} z^{-4} \ldots$$

$$\frac{z^4 + \frac{1}{4}}{-\frac{1}{4}}$$

$$\boxed{\text{A) } h[n] \text{ is real for all } n}$$

Key Points

- The transfer function $H(z)$ is formed by complex conjugate pole pairs, which ensures the denominator polynomial has strictly real coefficients.

- Performing polynomial long division on the rational function $H(z)$ yields the Z-transform expansion, where the coefficients correspond to the impulse response sequence $h[n]$.
- Because the resulting expansion $(1 - \frac{1}{4}z^{-4} + \ldots)$ consists entirely of real coefficients, the impulse response $h[n]$ must be real for all n.

Q6.15.4 MCQ | 2015 | 2M | Ans: C | ● ○ ○

Method 1

The difference equation of the system

$$x(n) - \frac{1}{6}y(n-2) + \frac{5}{6}y(n-1) = y(n)$$

$$H(z) = \frac{Y(z)}{x(z)} = \frac{1}{1 + \frac{1}{6}z^{-2} - \frac{5}{6}z^{-1}}$$

$$\text{Poles are at } z = \frac{1}{2}, \frac{1}{3}$$

$$\boxed{z = \frac{1}{2}, \frac{1}{3}}$$

Method 2

To find the poles of the system, we first determine the overall transfer function $H(z) = \frac{Y(z)}{X(z)}$ using Mason's Gain Formula.

Step 1: Identify the forward paths There is only one forward path from the input $X(z)$ to the output $Y(z)$, which passes straight through the two summing junctions with a gain of 1.

$$P_1 = 1$$

Step 2: Identify the individual loops There are two feedback loops in the system:

- Loop 1 (L_1): From the output $Y(z)$, through the first delay block z^{-1}, and back to the second summing junction with a gain of $\frac{5}{6}$.

$$L_1 = \frac{5}{6}z^{-1}$$

- Loop 2 (L_2): From the output $Y(z)$, through the first delay block z^{-1}, then through the second delay block z^{-1}, and back to the first summing junction with a gain of $-\frac{1}{6}$.

$$L_2 = \left(z^{-1}\right) \cdot \left(z^{-1}\right) \cdot \left(-\frac{1}{6}\right) = -\frac{1}{6}z^{-2}$$

Step 3: Calculate the determinant (Δ) and cofactors (Δ_k) Since all loops touch each other and also touch the forward path, there are no non-touching loop combinations.

$$\Delta = 1 - (L_1 + L_2)$$

$$\Delta = 1 - \left(\frac{5}{6}z^{-1} - \frac{1}{6}z^{-2}\right)$$

$$\Delta = 1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}$$

The cofactor for the single forward path is:

$$\Delta_1 = 1$$

Step 4: Determine the transfer function and find the poles Using Mason's Gain Formula, the transfer function is:

$$H(z) = \frac{P_1\Delta_1}{\Delta} = \frac{1}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}}$$

The poles of the system transfer function are the roots of the characteristic equation (denominator = 0):

$$1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2} = 0$$

Multiply the entire equation by z^2 to express it in positive powers of z:

$$z^2 - \frac{5}{6}z + \frac{1}{6} = 0$$

Multiply by 6 to clear the fractions:

$$6z^2 - 5z + 1 = 0$$

Factoring the quadratic equation:

$$(3z - 1)(2z - 1) = 0$$

Solving for z, we get the poles:

$$z = \frac{1}{3}, \quad z = \frac{1}{2}$$

$$\boxed{\text{C) } \frac{1}{2}, \frac{1}{3}}$$

Key Points

- The transfer function $H(z)$ is derived by taking the Z-transform of the difference equation, applying the time-shifting property $y(n-k) \xrightarrow{Z} z^{-k}Y(z)$.
- The poles of the discrete-time system are determined by finding the roots of the denominator polynomial (characteristic equation) of the transfer function $H(z)$.
- Rearranging the denominator $1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2} = 0$ yields a quadratic in terms of z, whose roots are the pole locations.
- **Mason's Gain Formula**: $H(z) = \frac{1}{\Delta}\sum_k P_k\Delta_k$
- The poles of a discrete-time system are found by equating the denominator of its transfer function $H(z)$ to zero.
- Converting polynomials from negative powers of z (z^{-1}) to positive powers of z simplifies finding the roots (poles).

Q6.15.5 NAT | 2015 | 1M | Ans: 0 | ● ○ ○

$$y(n) = \sum_{m=0}^{n} x[m] = x[n] * u[n]$$

$$Y(z) = X(z) \cdot \left(\frac{z}{z-1}\right)$$

Given

$$Y(z) = \frac{2}{z(z-1)^2} = X(z) \cdot \frac{z}{z-1}$$

$$\Rightarrow \quad X(z) = \frac{2 \cdot z^{-2}}{z-1} = 2z^{-3}\left(\frac{z}{z-1}\right)$$

$$\Rightarrow \quad x[n] = 2u[n-3]$$

Therefore, $x[2] = 0$

$$\boxed{0}$$

Key Points

- The accumulation property of the Z-transform states that summing a signal $x[n]$ from 0 to n corresponds to convolution with the unit step $u[n]$, equivalent to multiplying its Z-transform by $\frac{z}{z-1}$.
- The time-shifting property implies that multiplication by z^{-k} results in a delayed signal in the time domain, applying the transform pair $u[n-k] \leftrightarrow \frac{z}{z-1}z^{-k}$.
- Evaluating the inverse transform yields $x[n] = 2u[n-3]$. Because a unit step function $u[n-3]$ is 0 for all $n < 3$, substituting $n = 2$ gives an output of 0.

Q6.14.1 NAT 2014 2M Ans: 12

Given,

$$X(z) = \frac{1}{(1-2z^{-1})^2}, \quad |Z| > 2$$

$$X(z) = (1-2z^{-1})^{-2}$$

$$X(z) = 1 + 4z^{-1} + 12z^{-2} + \ldots$$

(using binomial expansion) Taking inverse z-transform, we get

$$\therefore \quad x[n] = \{1,\ 4,\ 12,\ \ldots\}$$

$$x[2] = \boxed{12}$$

Key Points

- The region of convergence $|Z| > 2$ indicates a right-sided (causal) sequence, meaning the Z-transform can be expanded as a power series in negative powers of z.
- The binomial expansion formula $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots$ is applied to expand the rational function.
- By definition, $X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$. Matching the coefficients from the series expansion directly provides the sequence values, where the coefficient of z^{-2} is $x[2]$.

Q6.14.2 NAT 2014 2M Ans: -0.5

Here $H_1(z) = \frac{1}{1-pz^{-1}}$

$$H_2(z) = \frac{1}{1-qz^{-1}}$$

also $p = \frac{1}{2}$, $q = -\frac{1}{4}$

$$\therefore \quad H(z) = H_1(z) + rH_2(z)$$

$$= \frac{1}{1-pz^{-1}} + \frac{r}{1-qz^{-1}}$$

$$= \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{r}{1+\frac{1}{4}z^{-1}}$$

$$H(z) = \frac{(1+\frac{1}{4}z^{-1}) + r(1-\frac{1}{2}z^{-1})}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{4}z^{-1})} \ldots (i)$$

zero of $H(z) \Rightarrow$

$$(1+r) - (q+pr)z^{-1} = 0 \ldots (ii)$$

or $\quad z = \frac{q+pr}{1+r}$ Since, the zero of $H(z)$ lies on the unit circle therefore $|z| = 1$ or $z = \pm 1$ Taking $z = 1$, we get

$$z = \frac{q + pr}{1 + r} = 1 \ldots (iii)$$

By solving above expression with $p = \frac{1}{2}$ and $q = -\frac{1}{4}$, we get

$$r = -2.5$$

But $|r| < 1 \quad$ or $\quad -1 < r < 1$ Taking $z = -1$ in equation (iii), we get

$$\boxed{r = -0.5}$$

Key Points

- The zeros of a rational transfer function $H(z)$ are the roots of its numerator polynomial.
- For a zero to lie on the unit circle, its magnitude must be 1 (i.e., $|z| = 1$, which implies $z = 1$ or $z = -1$ for strictly real coefficients).
- Applying the given condition $|r| < 1$ acts as a mathematical constraint to filter out invalid roots, ensuring the correct parameter value is selected.

Q6.14.3 MCQ 2014 2M Ans: A ● ○ ○

Input-output relation is given as

$$y[n] = \alpha y(n-1) + \beta x[n]$$

Taking Z-transform,

$$Y(z) = \alpha z^{-1} Y(z) + \beta X(z)$$
$$Y(z)[1 - \alpha z^{-1}] = \beta X(z)$$
$$\frac{Y(z)}{X(z)} = H(z) = \frac{\beta}{1 - \alpha z^{-1}}$$
$$h[n] = \beta \cdot \alpha^n u[n]$$

Also

$$\sum_{n=0}^{\infty} h[n] = 2 \Rightarrow \sum_{n=0}^{\infty} \beta \cdot \alpha^n u[n] = 2$$
$$\frac{\beta}{1-\alpha} = 2 \Rightarrow \beta = 2 - 2\alpha$$
$$\Rightarrow 2\alpha = 2 - \beta$$
$$\boxed{\text{A) } \alpha = 1 - \frac{\beta}{2}}$$

Key Points

- The Z-transform converts a time-domain difference equation into an algebraic equation, using the time-shifting property where a delay of 1 sample ($y[n-1]$) corresponds to multiplication by z^{-1}.
- The transfer function $H(z)$ of the LTI system is defined as the ratio $\frac{Y(z)}{X(z)}$.
- The infinite sum of the impulse response sequence $\sum_{n=0}^{\infty} h[n]$ is mathematically equivalent to evaluating the transfer function at $z = 1$ (the DC gain of the system). Setting $H(1) = 2$ directly yields the relation between α and β.

Q6.14.4 MCQ 2014 2M Ans: C ● ○ ○

$$x(n) = \underbrace{\left(-\frac{1}{9}\right)^n u(n)}_{x_1(n)} - \underbrace{\left(-\frac{1}{3}\right)^n u(-n-1)}_{x_2(n)}$$

$$x_1(n) = \left(-\frac{1}{9}\right)^n u(n) \rightarrow \text{Right sided signal}$$

$$x_1(z) = \frac{1}{1 - \left(-\frac{1}{9}\right) z^{-1}}$$

$$\text{ROC } |z| \geq \left|\frac{1}{9}\right| \text{ or } |z| > \frac{1}{9}$$

$$x_2(n) = \left(-\frac{1}{3}\right)^n u(-n-1) \rightarrow \text{Left sided signal}$$

$$x_2(z) = \frac{1}{1 - \left(-\frac{1}{3}\right) z^{-1}}; \quad |z| < \frac{1}{3}$$

Now, when both the transforms are added using linearity property. ROC must be between

$$\boxed{\text{C) } \frac{1}{3} > |z| > \frac{1}{9}}$$

Key Points

- The Z-transform of a right-sided sequence $a^n u(n)$ is $\frac{1}{1-az^{-1}}$ with ROC $|z| > |a|$.
- The Z-transform of a left-sided sequence $-b^n u(-n-1)$ is $\frac{1}{1-bz^{-1}}$ with ROC $|z| < |b|$.
- By the linearity property, the ROC of the sum of two signals is the intersection of their individual ROCs.

Q6.14.5 MCQ 2014 1M Ans: B ● ○ ○

Given $x[n] = x[-n]$ As we know that,

$$x[n] \xleftrightarrow{Z} X[z]$$

$$x[-n] \xleftrightarrow{Z} X[z^{-1}]$$

So, when $(0.5 + j0.25)$ is a zero of $X(z)$. Now $x[-n] \longleftrightarrow X(z^{-1})$

$$\text{Zero of } X(z) = \frac{1}{(0.5 + j0.25)}$$

$$\boxed{\text{B) } \frac{1}{0.5 + j0.25}}$$

Key Points

- The time-reversal property of the Z-transform states that if a sequence is time-reversed ($x[-n]$), its Z-transform domain variable is inverted (z^{-1}).
- For an even symmetric signal where $x[n] = x[-n]$, its Z-transform must satisfy $X(z) = X(z^{-1})$.
- Consequently, if a system possesses a zero at a complex location z_0, it must inherently have a corresponding zero at the reciprocal location $1/z_0$ to preserve this symmetry.

Q6.14.6 MCQ 2014 1M Ans: B ● ○ ○

For the given system

$$H(z) = \frac{(z^{-1} - b)}{(1 - az^{-1})}$$

The pole lie at $z = a$ and the zero lie at $z = \frac{1}{b}$ also 'a' is complex in nature. Thus, $a = |a|e^{j\angle a}$ For an all pass system where $|H(e^{-j\omega})| = 1$, for all ω, if the pole lies at 'a' then the zero must be present at $\frac{1}{a^*}$

$$\therefore \quad \left(\frac{1}{b}\right) = \frac{1}{|a|e^{-j\angle a}} = \frac{1}{a^*}$$

$$\text{or} \quad \boxed{B) b = a^*}$$

Key Points

- A discrete-time all-pass system is characterized by a constant magnitude response across all frequencies, i.e., $|H(e^{j\omega})| = 1$.
- To satisfy this condition, the poles and zeros of the transfer function $H(z)$ must exist in conjugate reciprocal pairs. Specifically, a pole at $z = p$ necessitates a zero at $z = 1/p^*$.
- By matching the given transfer function's zero location $(1/b)$ to the theoretically required zero location $(1/a^*)$, the relationship between the parameters is established.

Q6.14.7 MCQ 2014 1M Ans: C ● ○ ○

Given system function:

$$H(z) = 1 + \frac{7}{2}z^{-1} + \frac{3}{2}z^{-2}$$

To find the zeros of the system, equate $H(z)$ to 0:

$$1 + \frac{7}{2}z^{-1} + \frac{3}{2}z^{-2} = 0$$

Multiplying by $2z^2$, we get the characteristic equation:

$$2z^2 + 7z + 3 = 0$$

$$2z^2 + 6z + z + 3 = 0$$

$$2z(z + 3) + 1(z + 3) = 0$$

$$(2z + 1)(z + 3) = 0$$

The zeros are located at $z = -1/2$ and $z = -3$. Since one zero $(z = -1/2)$ lies inside the unit circle $(|z| < 1)$ and the other zero $(z = -3)$ lies outside the unit circle $(|z| > 1)$, the system is a mixed phase system.

$$\boxed{\text{C) mixed phase}}$$

Key Points

- **Minimum Phase System:** All zeros of the transfer function lie strictly inside the unit circle.
- **Maximum Phase System:** All zeros of the transfer function lie strictly outside the unit circle.
- **Mixed Phase System:** The zeros are distributed both inside and outside the unit circle.

Q6.14.8 NAT 2014 1M Ans: 4 ● ○ ○

Given sequence,

$$x[n] = 0.5^n u[n]$$

The sequence $y[n]$ is obtained by convolving $x[n]$ with itself:

$$y[n] = x[n] * x[n]$$

Taking the Z-transform on both sides, we get:

$$Y(z) = X(z) \cdot X(z) = [X(z)]^2$$

The Z-transform of the standard sequence $a^n u[n]$ is $\frac{1}{1-az^{-1}}$. Thus, for $x[n]$:

$$X(z) = \frac{1}{1 - 0.5z^{-1}}$$

We know that the sum of all samples of a sequence from $-\infty$ to $+\infty$ is equal to its Z-transform evaluated at $z = 1$:

$$\sum_{n=-\infty}^{+\infty} y[n] = Y(z)\Big|_{z=1} = Y(1)$$

First, evaluate $X(z)$ at $z = 1$:

$$X(1) = \frac{1}{1 - 0.5(1)^{-1}} = \frac{1}{1 - 0.5} = \frac{1}{0.5} = 2$$

Now, calculate $Y(1)$:

$$Y(1) = [X(1)]^2 = (2)^2 = 4$$

$$\boxed{4}$$

Key Points

- Convolution in the time domain corresponds to multiplication in the Z-domain: $y[n] = x_1[n] * x_2[n] \leftrightarrow Y(z) = X_1(z)X_2(z)$.
- The sum of the samples of a discrete-time signal is equivalent to its Z-transform evaluated at $z = 1$, given by the property: $\sum_{n=-\infty}^{\infty} x[n] = X(e^{j0}) = X(1)$.

Q6.12.1 MCQ 2012 1M Ans: C ● ○ ○

Given the discrete-time signal:

$$x[n] = \left(\frac{1}{3}\right)^{|n|} - \left(\frac{1}{2}\right)^n u[n]$$

Let us determine the Region of Convergence (ROC) for each term separately. Let $x_1[n] = \left(\frac{1}{3}\right)^{|n|}$. This can be expanded into its right-sided and left-sided components:

$$x_1[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{3}\right)^{-n} u[-n-1]$$

$$x_1[n] = \left(\frac{1}{3}\right)^n u[n] + 3^n u[-n-1]$$

The Z-transform of the right-sided sequence $\left(\frac{1}{3}\right)^n u[n]$ converges for $|z| > \frac{1}{3}$. The Z-transform of the left-sided sequence $3^n u[-n-1]$ converges for $|z| < 3$. Therefore, the ROC for $x_1[n]$ is the intersection of these two regions:

$$\text{ROC}_1 : \frac{1}{3} < |z| < 3$$

Now, consider the second term, $x_2[n] = \left(\frac{1}{2}\right)^n u[n]$. This is a standard right-sided sequence. Its Z-transform converges for:

$$\mathrm{ROC}_2 : |z| > \frac{1}{2}$$

The overall ROC for the linear combination of the two signals is the intersection of their individual ROCs:

$$\mathrm{ROC}_{total} = \mathrm{ROC}_1 \cap \mathrm{ROC}_2$$

$$\mathrm{ROC}_{total} = \left(\frac{1}{3} < |z| < 3\right) \cap \left(|z| > \frac{1}{2}\right)$$

Taking the intersection, the lower bound is dictated by the stricter condition $|z| > \frac{1}{2}$, giving:

$$\mathrm{ROC}_{total} : \frac{1}{2} < |z| < 3$$

This exactly matches option C.

$$\boxed{\text{C) } \frac{1}{2} < |z| < 3}$$

Key Points

- For a causal, right-sided sequence $a^n u[n]$, the ROC is $|z| > |a|$.
- For an anti-causal, left-sided sequence $-a^n u[-n-1]$, the ROC is $|z| < |a|$.
- The ROC of a two-sided sequence $a^{|n|}$ is bounded as $a < |z| < \frac{1}{a}$ (where $0 < a < 1$).
- The ROC of a sum of sequences is the intersection of the individual ROCs: $\mathrm{ROC} = \mathrm{ROC}_1 \cap \mathrm{ROC}_2 \cap \cdots \cap \mathrm{ROC}_N$.

Q6.10.1 MCQ 2010 1M Ans: A ● ○ ○

Given the z-transform:

$$X(z) = 5z^2 + 4z^{-1} + 3$$

By definition, the two-sided z-transform of a discrete-time sequence $x[n]$ is given by:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

We can rewrite the given expression to explicitly show the powers of z:

$$X(z) = 5z^{-(-2)} + 3z^0 + 4z^{-1}$$

Comparing the terms with the standard summation formula:

- For $n = -2$, the coefficient $x[-2] = 5$
- For $n = 0$, the coefficient $x[0] = 3$
- For $n = 1$, the coefficient $x[1] = 4$
- For all other values of n, $x[n] = 0$

Using the standard z-transform pair for a time-shifted unit impulse function $\delta[n-k] \leftrightarrow z^{-k}$, we can take the inverse z-transform of each term individually due to the linearity property:

- Inverse z-transform of $5z^2$ is $5\delta[n+2]$

- Inverse z-transform of 3 is $3\delta[n]$
- Inverse z-transform of $4z^{-1}$ is $4\delta[n-1]$

Combining these, the inverse z-transform $x[n]$ is:

$$x[n] = 5\delta[n+2] + 3\delta[n] + 4\delta[n-1]$$

This exactly matches option A.

$$\boxed{\text{A) } 5\delta[n+2] + 3\delta[n] + 4\delta[n-1]}$$

Key Points

- The z-transform of the standard unit impulse function $\delta[n]$ is 1.
- The time-shifting property states that a delay in the time domain corresponds to multiplication by z^{-1} in the z-domain: $x[n-k] \leftrightarrow z^{-k}X(z)$.
- Therefore, a shifted impulse $\delta[n-k]$ has the z-transform z^{-k}, and an advanced impulse $\delta[n+k]$ has the z-transform z^k.

Q6.09.1 MCQ 2009 1M Ans: A ● ○ ○

Given the discrete-time sequence:

$$x(n) = \left(\frac{1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n-1)$$

To find the region of convergence (ROC) of the entire sequence, we analyze the individual terms. The overall ROC is the intersection of the ROCs of the individual terms. Let the first term be $x_1(n) = \left(\frac{1}{3}\right)^n u(n)$. This is a causal (right-sided) sequence. The standard Z-transform pair is $a^n u(n) \leftrightarrow \frac{1}{1-az^{-1}}$, which converges for $|z| > |a|$. Therefore, the ROC for $x_1(n)$ is:

$$\text{ROC}_1 : |z| > \frac{1}{3}$$

Let the second term be $x_2(n) = -\left(\frac{1}{2}\right)^n u(-n-1)$. This is an anti-causal (left-sided) sequence. The standard Z-transform pair is $-b^n u(-n-1) \leftrightarrow \frac{1}{1-bz^{-1}}$, which converges for $|z| < |b|$. Therefore, the ROC for $x_2(n)$ is:

$$\text{ROC}_2 : |z| < \frac{1}{2}$$

The overall ROC of $x(n)$ is the intersection of ROC_1 and ROC_2:

$$\text{ROC} = \text{ROC}_1 \cap \text{ROC}_2$$

$$\text{ROC} = \left(|z| > \frac{1}{3}\right) \cap \left(|z| < \frac{1}{2}\right)$$

Combining these conditions yields an annular region in the Z-plane:

$$\frac{1}{3} < |z| < \frac{1}{2}$$

This matches option A.

$$\boxed{\text{A) } \frac{1}{3} < |z| < \frac{1}{2}}$$

Key Points

- For a right-sided sequence $a^n u(n)$, the ROC is the exterior of a circle of radius $|a|$, i.e., $|z| > |a|$.
- For a left-sided sequence $-b^n u(-n-1)$, the ROC is the interior of a circle of radius $|b|$, i.e., $|z| < |b|$.
- The ROC of a linear combination of sequences is at least the intersection of their individual ROCs.

Q6.08.1 MCQ 2008 2M Ans: B ● ○ ○

When the switch is closed at $t = 0^-$, the circuit acts as a simple RC series circuit with a DC voltage source. Since the capacitor is initially uncharged, the entire source voltage initially appears across the resistor, and then decays as the capacitor charges. The voltage across the resistor $v_R(t)$ serves as the input to the sampler, $x(t)$. The time constant of the RC circuit is:

$$\tau = RC = (200 \times 10^3)\,\Omega \times (10 \times 10^{-6})\,\text{F} = 2\,\text{s}$$

The voltage across the resistor for $t \geq 0$ decays exponentially:

$$x(t) = Ve^{-t/\tau} = 5e^{-t/2} = 5e^{-0.5t}$$

The continuous-time signal is sampled at a frequency $f_s = 10\,\text{Hz}$. The sampling period T_s is:

$$T_s = \frac{1}{f_s} = \frac{1}{10} = 0.1\,\text{s}$$

The sampled discrete-time signal $x(n)$ is obtained by substituting $t = nT_s$:

$$x(n) = x(nT_s) = 5e^{-0.5(0.1n)} = 5e^{-0.05n}$$

This matches option B.

$$\boxed{\text{B) } 5e^{-0.05n}}$$

Key Points

- In a source-free RC circuit or an RC circuit responding to a step input, the transient voltage across the resistor decays exponentially as $e^{-t/RC}$.
- The relationship between continuous time t and discrete time sequence index n is $t = nT_s$, where T_s is the sampling period.

Q6.08.2 MCQ 2008 2M Ans: C ● ○ ○

From the previous analysis, the sampled sequence for $n \geq 0$ is:

$$x(n) = 5e^{-0.05n}u(n) = 5\left(e^{-0.05}\right)^n u(n)$$

We know the standard Z-transform pair for a causal exponential sequence:

$$a^n u(n) \longleftrightarrow \frac{z}{z-a} \quad \text{for } |z| > |a|$$

Here, the constant $a = e^{-0.05}$. Applying this standard pair:

$$X(z) = 5\left(\frac{z}{z - e^{-0.05}}\right) = \frac{5z}{z - e^{-0.05}}$$

The region of convergence (ROC) is dictated by the pole of the system:

$$|z| > |a| \implies |z| > e^{-0.05}$$

This matches option C.

$$\boxed{\text{C) } \frac{5z}{z - e^{-0.05}},\ |z| > e^{-0.05}}$$

Key Points

- The Z-transform of $a^n u(n)$ is $\frac{z}{z-a}$ with a region of convergence of $|z| > |a|$.
- For a right-sided (causal) sequence, the ROC is strictly the exterior of a circle passing through the outermost pole in the Z-plane.

Q6.07.1 MCQ 2007 2M Ans: B ● ○ ○

Given the z-transform:

$$X(z) = \frac{0.5}{1 - 2z^{-1}}$$

To find the inverse z-transform, we first identify the pole of the system, which is located at $z = 2$. The problem states that the Region of Convergence (ROC) includes the unit circle ($|z| = 1$). An ROC must be a contiguous region bounded by poles and cannot contain any poles. Since the only pole is at $|z| = 2$ and the ROC must include $|z| = 1$, the ROC must be the interior of the circle defined by the pole. Thus, the valid ROC is:

$$|z| < 2$$

An ROC of the form $|z| < |a|$ indicates that the corresponding time-domain signal is a left-sided (anti-causal) sequence. We use the standard z-transform pair:

$$-a^n u[-n-1] \longleftrightarrow \frac{1}{1 - az^{-1}} \quad \text{for } |z| < |a|$$

Applying this standard pair to our given $X(z)$ with $a = 2$, we get the sequence $x[n]$:

$$x[n] = 0.5\,(-2^n u[-n-1])$$

$$x[n] = -0.5 \cdot 2^n u[-n-1]$$

We are required to find the value of the sequence at $n = 0$:

$$x[0] = -0.5 \cdot 2^0 \cdot u[-0-1]$$

$$x[0] = -0.5 \cdot 1 \cdot u[-1]$$

By definition, the unit step function $u[k] = 0$ for all $k < 0$. Therefore, $u[-1] = 0$.

$$x[0] = -0.5 \cdot 0 = 0$$

This matches option B.

$$\boxed{\text{B) } 0}$$

Key Points

- The z-transform pair for an anti-causal (left-sided) sequence is $-a^n u[-n-1] \leftrightarrow \frac{1}{1-az^{-1}}$ with an ROC of $|z| < |a|$.
- A system is stable if and only if its ROC includes the unit circle ($|z| = 1$). For a pole at $z = 2$, the only way to achieve stability is if the sequence is left-sided ($|z| < 2$).
- The unit step function $u[n]$ evaluates to 1 for $n \geq 0$ and 0 for $n < 0$. Consequently, the shifted and reversed function $u[-n-1]$ is only non-zero for $n \leq -1$.

Q6.06.1 MCQ 2006 1M Ans: D ● ○ ○

Let the region of convergence (ROC) of $x_1[n]$ be R_1 and the ROC of $x_2[n]$ be R_2. According to the linearity property of the Z-transform, if:

$$\mathcal{Z}\{x_1[n]\} = X_1(z) \quad \text{with ROC } R_1$$
$$\mathcal{Z}\{x_2[n]\} = X_2(z) \quad \text{with ROC } R_2$$

Then, for any constants a and b, the Z-transform of the linear combination is:

$$\mathcal{Z}\{ax_1[n] + bx_2[n]\} = aX_1(z) + bX_2(z)$$

The region of convergence for this combined signal includes the intersection of the individual ROCs. Thus, the combined ROC is at least $R_1 \cap R_2$. Given that the ROC of the sum $x_1[n] + x_2[n]$ (where $a = 1, b = 1$) is $\frac{1}{3} < |z| < \frac{2}{3}$, we can infer that the basic intersection $R_1 \cap R_2$ is this exact region (assuming no pole-zero cancellation artificially expanded the ROC of the sum). For the difference signal $x_1[n] - x_2[n]$ (where $a = 1, b = -1$), the same linearity property dictates that its ROC will also include the intersection $R_1 \cap R_2$. Therefore, the ROC of $x_1[n] - x_2[n]$ must include the same intersecting region:

$$\frac{1}{3} < |z| < \frac{2}{3}$$

This matches option D.

$$\boxed{\text{D) } \frac{1}{3} < |z| < \frac{2}{3}}$$

Key Points

- **Linearity Property:** $\mathcal{Z}\{ax_1[n] + bx_2[n]\} = aX_1(z) + bX_2(z)$.
- **ROC of Linear Combinations:** The ROC of a sum or difference of two signals always includes the intersection of their individual ROCs (ROC $\supseteq R_1 \cap R_2$).
- The ROC might be strictly larger than the intersection only if a pole is canceled by a zero when the linear combination is formed. Otherwise, the ROC remains the intersection.

Q6.05.1 MCQ 2005 1M Ans: C ● ○ ○

The given sequence is:

$$x(n) = \left(\frac{5}{6}\right)^n u(n) - \left(\frac{6}{5}\right)^n u(-n-1)$$

This sequence consists of two parts. Let $x_1(n) = \left(\frac{5}{6}\right)^n u(n)$ and $x_2(n) = -\left(\frac{6}{5}\right)^n u(-n-1)$. For the first part, $x_1(n)$ is a right-sided causal sequence. The z-transform of $a^n u(n)$ has a region of convergence (ROC) of $|z| > |a|$. Therefore, the ROC of $x_1(n)$ is:

$$|z| > \frac{5}{6}$$

For the second part, $x_2(n)$ is a left-sided anti-causal sequence. The z-transform of $-a^n u(-n-1)$ has an ROC of $|z| < |a|$. Therefore, the ROC of $x_2(n)$ is:

$$|z| < \frac{6}{5}$$

The ROC of the combined sequence $x(n)$ is the intersection of the individual ROCs:

$$\text{ROC} = \left\{z : |z| > \frac{5}{6}\right\} \cap \left\{z : |z| < \frac{6}{5}\right\}$$
$$\frac{5}{6} < |z| < \frac{6}{5}$$

$$\boxed{\text{C) } \frac{5}{6} < |z| < \frac{6}{5}}$$

Key Points

- For a right-sided sequence $a^n u(n)$, the ROC is strictly outside the circle of radius $|a|$, i.e., $|z| > |a|$.
- For a left-sided sequence $-a^n u(-n-1)$, the ROC is strictly inside the circle of radius $|a|$, i.e., $|z| < |a|$.
- The overall ROC of a sum of sequences is the intersection of their individual ROCs.

Mistakes to be avoided

Assuming the overall ROC is the union of the individual regions instead of the intersection. If the intersection of the individual regions is empty, the combined z-transform does not exist.

Q6.04.1 MCQ 2004 1M Ans: D ● ○ ○

Given the z-transform of the system:

$$H(z) = \frac{z}{z - 0.2}$$

The region of convergence (ROC) is given as:

$$|z| < 0.2$$

Recall the standard z-transform pairs and their corresponding regions of convergence:

- For a causal (right-sided) sequence:

$$a^n u[n] \leftrightarrow \frac{z}{z-a}, \quad \text{ROC: } |z| > |a|$$

- For an anti-causal (left-sided) sequence:

$$-a^n u[-n-1] \leftrightarrow \frac{z}{z-a}, \quad \text{ROC: } |z| < |a|$$

Comparing the given $H(z)$ with the standard pairs, we can identify $a = 0.2$. Since the specified ROC is $|z| < 0.2$, which matches the condition $|z| < |a|$, the impulse response must be an anti-causal, left-sided sequence. Applying the standard pair for the left-sided sequence, we get the impulse response:

$$h[n] = -(0.2)^n u[-n-1]$$

$$\boxed{\text{D) } -(0.2)^n u[-n-1]}$$

Key Points

- The algebraic expression $\frac{z}{z-a}$ alone does not uniquely identify the inverse z-transform.
- If the ROC is exterior to a circle ($|z| > |a|$), the sequence is right-sided.
- If the ROC is interior to a circle ($|z| < |a|$), the sequence is left-sided.

Mistakes to be avoided

Selecting option A by default. It is a common mistake to only look at the transfer function expression and ignore the ROC, which leads to incorrectly assuming the sequence is causal. Always verify the ROC before computing the inverse z-transform.

Q6.01.1 MCQ | 2001 | 1M | Ans: A | ● ○ ○

The discrete-time unit step sequence, $u(n)$, is defined as:

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

The z-transform of a general discrete-time sequence $x(n)$ is given by the formula:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Substituting $x(n) = u(n)$ into the definition, we change the lower limit of the summation to 0 because $u(n) = 0$ for $n < 0$:

$$X(z) = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = \sum_{n=0}^{\infty} (z^{-1})^n$$

This represents an infinite geometric series with a common ratio of z^{-1}. An infinite geometric series converges if and only if the magnitude of its common ratio is strictly less than 1:

$$|z^{-1}| < 1$$

$$\frac{1}{|z|} < 1$$

$$|z| > 1$$

Therefore, the Region of Convergence (ROC) for the unit step function is $|z| > 1$.

$$\boxed{\text{A) } |z| > 1}$$

Key Points

- The z-transform evaluates to $X(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$. This function has a pole at $z = 1$.
- The ROC of a z-transform can never contain any poles.
- For any right-sided causal sequence of the form $a^n u(n)$, the ROC is the exterior of a circle in the z-plane, given by $|z| > |a|$.

Mistakes to be avoided

Confusing the ROC conditions of the z-transform with those of the Laplace transform. The Laplace transform ROC is defined by the real part of the complex variable s (e.g., $\text{Re}(s) > 0$), which correspond to options C and D. The z-transform ROC is defined by the magnitude of the complex variable z (e.g., $|z| > 1$).

Q6.99.1 MCQ | 1999 | 1M | Ans: A | ● ○ ○

The given function is a sampled version of a continuous exponential function. By properties of exponents, we can rewrite the sequence as:

$$f(nT) = a^{nT} = (a^T)^n$$

Assuming the function represents a causal sequence (which is standard practice for such fundamental z-transform pairs), we can write it as $f(nT) = (a^T)^n u(n)$, where $u(n)$ is the unit step sequence. The standard definition of the unilateral z-transform is:

$$F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

Substituting our function into the definition:

$$F(z) = \sum_{n=0}^{\infty} (a^T)^n z^{-n} = \sum_{n=0}^{\infty} (a^T z^{-1})^n$$

This represents an infinite geometric series with a common ratio of $a^T z^{-1}$. For the region of convergence where $|a^T z^{-1}| < 1$ (or $|z| > |a^T|$), the series converges to:

$$F(z) = \frac{1}{1 - a^T z^{-1}}$$

To express this in positive powers of z, multiply the numerator and the denominator by z:

$$F(z) = \frac{z}{z - a^T}$$

$$\boxed{\text{A) } \frac{z}{z - a^T}}$$

Key Points

- The fundamental z-transform pair for a discrete exponential sequence is $\alpha^n u(n) \leftrightarrow \frac{z}{z-\alpha}$.
- By recognizing that $a^{nT} = (a^T)^n$, we can directly apply the standard transform by substituting $\alpha = a^T$.

Mistakes to be avoided

A common mistake is misinterpreting the negative sign in the denominator or incorrectly manipulating the exponent to yield a^{-T}. Remember that the standard transform yields a pole at $z = \alpha$, which in this case means the denominator must be $(z - a^T)$.

Q6.99.2 MCQ 1999 2M Ans: C ● ○ ○

To find the final value of the signal $c(n)$, we apply the Final Value Theorem for discrete-time signals, which states:

$$\lim_{n\to\infty} c(n) = \lim_{z\to 1} (1 - z^{-1}) C(z)$$

Given the z-transform of the signal:

$$C(z) = \frac{1}{4} \frac{z^{-1}(1 - z^{-4})}{(1 - z^{-1})^2}$$

Substitute $C(z)$ into the Final Value Theorem formula:

$$c(\infty) = \lim_{z\to 1} (1 - z^{-1}) \left[\frac{1}{4} \frac{z^{-1}(1 - z^{-4})}{(1 - z^{-1})^2} \right]$$

$$c(\infty) = \lim_{z\to 1} \frac{1}{4} \frac{z^{-1}(1 - z^{-4})}{(1 - z^{-1})}$$

Direct substitution of $z = 1$ results in a $\frac{0}{0}$ indeterminate form. We can evaluate this limit by factoring the term $(1 - z^{-4})$:

$$1 - z^{-4} = (1 - z^{-2})(1 + z^{-2}) = (1 - z^{-1})(1 + z^{-1})(1 + z^{-2})$$

Substitute the factored form back into the limit expression:

$$c(\infty) = \lim_{z\to 1} \frac{1}{4} \frac{z^{-1}(1 - z^{-1})(1 + z^{-1})(1 + z^{-2})}{(1 - z^{-1})}$$

$$c(\infty) = \lim_{z \to 1} \frac{1}{4} z^{-1}(1 + z^{-1})(1 + z^{-2})$$

Now, substitute $z = 1$:

$$c(\infty) = \frac{1}{4}(1)^{-1}(1 + 1^{-1})(1 + 1^{-2})$$

$$c(\infty) = \frac{1}{4}(1)(1 + 1)(1 + 1)$$

$$c(\infty) = \frac{1}{4} \cdot 4 = 1.0$$

Therefore, the final value is 1.0.

$$\boxed{\text{C) } 1.0}$$

Key Points

- The discrete-time Final Value Theorem is defined as $\lim_{n\to\infty} x(n) = \lim_{z\to 1}(1 - z^{-1})X(z)$, provided the limit exists.
- L'Hôpital's rule is a valid alternative approach to evaluate the limit $\lim_{z\to 1} \frac{1-z^{-4}}{1-z^{-1}}$ after arriving at the $\frac{0}{0}$ condition.

Mistakes to be avoided

Failing to recognize the $\frac{0}{0}$ indeterminate form. Simply plugging in $z = 1$ without factoring or using L'Hôpital's rule will leave a zero in the denominator, leading to an incorrect conclusion that the final value is infinity.

DEBUG INFO

Chapter VI

Z Transform

Total Questions : 35

MCQ : 28

MSQ : 2

NAT : 5

SOLUTIONS

VII

DTFT, DTFS, DFT & Miscellaneous

Topics

1. DTFT
2. Discrete-Time Fourier Series (DTFS)
3. Circular Convolution
4. DFT
5. Group Delay, Phase Delay and Basics of FFT

Q7.24.1 NAT | 2024 | 2M | Ans: 112 | ● ○ ○

From the properties of the Discrete Fourier Transform, applying the DFT operator four successive times to an N-length sequence $x[n]$ yields the original sequence scaled by N^2:

$$y[n] = N^2 x[n]$$

Given the sequence $x[n] = \{1, 2, 1, 3\}$, its length is $N = 4$. Therefore, we can write:

$$y[n] = N^2 x[n] \quad \because N = 4$$

We need to find the value of $Y[0]$. By the definition of the DFT, the zero-th frequency component $Y[0]$ (the DC component) is simply the sum of all elements in the time-domain sequence $y[n]$:

$$Y[k]\Big|_{k=0} = \sum_{n=0}^{3} y[n] = \sum_{n=0}^{3} N^2 x[n]$$

Factoring out the constant N^2 and expanding the summation for the elements of $x[n]$:

$$= N^2 \left[x[0] + x[1] + x[2] + x[3]\right]$$

Substitute $N = 4$ and the respective sequence values:

$$= (4)^2 \left[1 + 2 + 1 + 3\right]$$

$$= 16 \times 7 = 112$$

$$\boxed{112}$$

Key Points

- The repeated application of the DFT operation follows a cyclic property. Applying it four times to an N-point sequence gives: $\mathcal{F}\{\mathcal{F}\{\mathcal{F}\{\mathcal{F}\{x[n]\}\}\}\} = N^2 x[n]$.
- The first term of any N-point DFT, $X[0]$, represents the DC value and is mathematically equivalent to the sum of all samples in the time-domain sequence: $X[0] = \sum_{n=0}^{N-1} x[n]$.

Mistakes to be avoided

Attempting to manually compute the full DFT matrix operations four times in a row. Recognizing the cyclical property of the Fourier transform operator is essential to solving the problem efficiently.

Q7.23.1 MCQ | 2023 | 2M | Ans: C | ● ○ ○

Method 1

The output of an LTI system in the frequency domain is the product of the input signal's Fourier transform and the system's frequency response:

$$Y(e^{j\Omega}) = X(e^{j\Omega})H(e^{j\Omega})$$

Given:

$$X(e^{j\Omega}) = 1 - e^{-j\Omega} + 2e^{-3j\Omega}$$

$$H(e^{j\Omega}) = 1 - \frac{1}{2}e^{-j2\Omega}$$

Multiplying these two expressions:

$$Y(e^{j\Omega}) = \left(1 - e^{-j\Omega} + 2e^{-3j\Omega}\right)\left(1 - \frac{1}{2}e^{-j2\Omega}\right)$$

$$Y(e^{j\Omega}) = 1\left(1 - \frac{1}{2}e^{-j2\Omega}\right) - e^{-j\Omega}\left(1 - \frac{1}{2}e^{-j2\Omega}\right) + 2e^{-3j\Omega}\left(1 - \frac{1}{2}e^{-j2\Omega}\right)$$

$$Y(e^{j\Omega}) = 1 - \frac{1}{2}e^{-j2\Omega} - e^{-j\Omega} + \frac{1}{2}e^{-j3\Omega} + 2e^{-3j\Omega} - e^{-j5\Omega}$$

Combining like terms:

$$Y(e^{j\Omega}) = 1 - e^{-j\Omega} - \frac{1}{2}e^{-j2\Omega} + \frac{5}{2}e^{-j3\Omega} - e^{-j5\Omega}$$

$$y[n] = \delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$$

Method 2

We can solve this entirely in the time domain by finding the inverse transforms first and performing discrete convolution. Inverse transform of the input $x[n]$:

$$x[n] = \delta[n] - \delta[n-1] + 2\delta[n-3]$$

Inverse transform of the impulse response $h[n]$:

$$h[n] = \delta[n] - \frac{1}{2}\delta[n-2]$$

The output $y[n]$ is the convolution $x[n] * h[n]$:

$$y[n] = x[n] * \left(\delta[n] - \frac{1}{2}\delta[n-2]\right)$$

Using the shifting property of the delta function, $x[n] * \delta[n-k] = x[n-k]$:

$$y[n] = x[n] - \frac{1}{2}x[n-2]$$

Substitute $x[n]$ and $x[n-2]$ into the equation:

$$y[n] = (\delta[n] - \delta[n-1] + 2\delta[n-3]) - \frac{1}{2}(\delta[n-2] - \delta[n-3] + 2\delta[n-5])$$

$$y[n] = \delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \left(2 + \frac{1}{2}\right)\delta[n-3] - \delta[n-5]$$

$$y[n] = \delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$$

$$\boxed{\text{C) } \delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]}$$

Key Points

- Convolution in the time domain is mathematically equivalent to multiplication in the frequency domain: $x[n] * h[n] \leftrightarrow X(e^{j\Omega})H(e^{j\Omega})$.
- The time-shifting property of the DTFT is foundational: $x[n-k] \leftrightarrow e^{-jk\Omega}X(e^{j\Omega})$. For a unit impulse, this reduces to $\delta[n-k] \leftrightarrow e^{-jk\Omega}$.

Mistakes to be avoided

When multiplying the polynomials in the frequency domain, be careful with the signs and the multiplication of exponential terms (where exponents add). An arithmetic error such as $e^{-j\Omega} \cdot (-\frac{1}{2}e^{-j2\Omega}) = -\frac{1}{2}e^{-j3\Omega}$ instead of $+\frac{1}{2}e^{-j3\Omega}$ will lead to an incorrect coefficient for the $\delta[n-3]$ term.

Q7.23.2

MCQ | 2023 | 2M | Ans: A | ● ○ ○

We need to evaluate the sum:

$$S = \sum_{n=0}^{4} x[n] \sin\left(\frac{4\pi n}{5}\right)$$

Using Euler's identity, we can express the sine function as:

$$\sin\left(\frac{4\pi n}{5}\right) = \frac{e^{j\frac{4\pi n}{5}} - e^{-j\frac{4\pi n}{5}}}{2j} = \frac{e^{j\frac{2\pi(2)n}{5}} - e^{-j\frac{2\pi(2)n}{5}}}{2j}$$

Substituting this back into the sum, we get:

$$S = \sum_{n=0}^{4} x[n] \left(\frac{e^{j\frac{2\pi(2)n}{5}} - e^{-j\frac{2\pi(2)n}{5}}}{2j}\right)$$

$$S = \frac{1}{2j}\sum_{n=0}^{4} x[n] e^{j\frac{2\pi(2)n}{5}} - \frac{1}{2j}\sum_{n=0}^{4} x[n] e^{-j\frac{2\pi(2)n}{5}}$$

Recall the discrete-time Fourier series (DTFS) analysis equation for a periodic signal $x[n]$ with period N:

$$a_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{N}}$$

This implies that the sum over one period can be written in terms of the coefficients:

$$\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{N}} = Na_k$$

Applying this to our expression with $N = 5$: The second sum directly matches the standard form for $k = 2$:

$$\sum_{n=0}^{4} x[n] e^{-j\frac{2\pi(2)n}{5}} = 5a_2$$

The first sum can be written with a negative index $k = -2$:

$$\sum_{n=0}^{4} x[n] e^{j\frac{2\pi(2)n}{5}} = \sum_{n=0}^{4} x[n] e^{-j\frac{2\pi(-2)n}{5}} = 5a_{-2}$$

Due to the periodicity of DTFS coefficients ($a_k = a_{k+N}$), we know:

$$a_{-2} = a_{-2+5} = a_3$$

Now substitute these results into the equation for S:

$$S = \frac{1}{2j}(5a_3) - \frac{1}{2j}(5a_2) = \frac{5}{2j}(a_3 - a_2)$$

Given $a_2 = 2j$ and $a_3 = -2j$, we calculate the final value:

$$S = \frac{5}{2j}(-2j - 2j) = \frac{5}{2j}(-4j) = 5(-2) = -10$$

$$\boxed{\text{A) -10}}$$

Key Points

- The DTFS coefficients of a periodic signal are also periodic with the same period N, meaning $a_k = a_{k+N}$.
- The DTFS analysis equation is $a_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi kn}{N}}$. Isolating the sum yields $\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi kn}{N}} = Na_k$.
- Euler's complex exponential formula for sine is $\sin(\theta) = \frac{e^{j\theta}-e^{-j\theta}}{2j}$.

Mistakes to be avoided

Forgetting to multiply by the factor of N (5 in this case) when substituting the sum for the DTFS coefficients. If you assume the sum is equal to a_k instead of Na_k, you will incorrectly calculate the answer as −2 (Option C).

Q7.22.1 NAT 2022 2M Ans: 8 ● ○ ○

The sequence $\bar{x}$ has a length of $N = 8$, and its elements are given by the vector:

$$\bar{x} = [1, 0, 0, 0, 2, 0, 0, 0]$$

From this, we can identify that the first element is $x[0] = 1$. We are given $\bar{y} = \text{DFT}(\text{DFT}(\bar{x}))$ and need to evaluate $y[0]$. According to the duality property of the Discrete Fourier Transform, applying the DFT operator twice to an N-point sequence $x[n]$ results in a time-reversed and scaled version of the original sequence. Mathematically, this is expressed as:

$$\text{DFT}(\text{DFT}(x[n])) = N \cdot x[(-n) \bmod N]$$

Therefore, the elements of the new sequence $y[n]$ are given by:

$$y[n] = 8 \cdot x[(-n) \bmod 8]$$

To find the specific value of $y[0]$, we substitute $n = 0$ into the equation:

$$y[0] = 8 \cdot x[(-0) \bmod 8]$$

$$y[0] = 8 \cdot x[0]$$

Substituting the known value of $x[0] = 1$:

$$y[0] = 8 \cdot 1 = 8$$

$$\boxed{8}$$

Key Points

- The duality property of the N-point DFT states that if $x[n] \leftrightarrow X[k]$, then taking the DFT of the sequence $X[n]$ yields $N \cdot x[(-k) \bmod N]$.
- Applying the DFT operation twice successively to a sequence yields $\text{DFT}\{\text{DFT}\{x[n]\}\} = N \cdot x[(-n) \bmod N]$.
- For the zero-th index ($n = 0$), the time-reversal and modulo operations simplify entirely, making the term exactly $N \cdot x[0]$.

Mistakes to be avoided

Attempting to calculate the 8-point DFT matrix operations manually twice is a common pitfall. While it will eventually lead to the correct answer, it is extremely time-consuming and highly prone to arithmetic errors with complex exponentials. Recognizing and applying the duality property is essential for an efficient solution.

Q7.21.1 MCQ | 2021 | 1M | Ans: D | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Let the lengths of the input sequences $x[n]$ and $h[n]$ be $N_1 = 16$ and $N_2 = 16$, respectively. The linear convolution, denoted by $y[n] = x[n] * h[n]$, has a length given by:

$$L = N_1 + N_2 - 1 = 16 + 16 - 1 = 31$$

Thus, the sequence $y[n]$ has non-zero values in the index range $0 \le n \le 30$. The sequence $z[n]$ is obtained by taking the 16-point IDFT of the product of the 16-point DFTs of $x[n]$ and $h[n]$. By the circular convolution theorem, $z[n]$ represents the 16-point circular convolution of $x[n]$ and $h[n]$. The relationship between the N-point circular convolution $z[n]$ and the linear convolution $y[n]$ is defined by the time-aliasing formula:

$$z[n] = \sum_{m=-\infty}^{\infty} y[n + mN]$$

For $N = 16$ and the fundamental interval $0 \le n \le 15$, this expands to:

$$z[n] = y[n] + y[n + 16]$$

This is because $y[n]$ is zero for indices $n \ge 31$ and $n < 0$. We are asked to find the value(s) of k for which $z[k] = y[k]$. From our aliasing equation, this condition is met when the aliased term is zero:

$$y[k + 16] = 0$$

Since $y[n]$ is only non-zero up to $n = 30$, we must have:

$$k + 16 > 30 \implies k > 14$$

Within the valid interval $0 \le k \le 15$ for the 16-point circular convolution, the only integer that satisfies $k > 14$ is $k = 15$. Therefore, the only index where the circular convolution equals the linear convolution is $k = 15$.

$$\boxed{\text{D) } k = 15}$$

Key Points

- The linear convolution of two finite-length sequences of sizes N_1 and N_2 results in a sequence of length $N_1 + N_2 - 1$.
- Multiplication of N-point DFTs corresponds strictly to N-point circular convolution in the time domain.
- Circular convolution relates to linear convolution via time-aliasing: $z[n] = \sum_{m=-\infty}^{\infty} y[n + mN]$. To avoid aliasing entirely, the DFT size must be $N \ge N_1 + N_2 - 1$.

Mistakes to be avoided

A common error is assuming that $z[n]$ and $y[n]$ are completely identical or completely different without analyzing the aliasing boundary. Aliasing wraps the "tail" of the linear convolution back to the beginning of the sequence. Only the specific points unaffected by this wrap-around ($k = 15$ in this case) will match the linear convolution identically.

Q7.21.2 NAT | 2021 | 2M | Ans: 8 | ● ○ ○

The given integral is:

$$I = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) Y(e^{-j\omega}) d\omega$$

By the frequency-domain multiplication property, the inverse Discrete-Time Fourier Transform (IDTFT) of a product of DTFTs corresponds to the convolution of their time-domain signals. Let $V(e^{j\omega}) = X(e^{j\omega})Y(e^{-j\omega})$. Using the time-reversal property, we know that $y[-n] \leftrightarrow Y(e^{-j\omega})$. Therefore, $V(e^{j\omega})$ is the DTFT of the convolution $x[n] * y[-n]$. The definition of the IDTFT evaluated at $n = 0$ is:

$$\text{IDTFT}\{V(e^{j\omega})\}\Big|_{n=0} = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) Y(e^{-j\omega}) e^{j\omega(0)} d\omega = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) Y(e^{-j\omega}) d\omega$$

This perfectly matches our required integral I. In the time domain, evaluating the convolution $x[n] * y[-n]$ at $n = 0$ gives:

$$\sum_{k=-\infty}^{\infty} x[k]y[-(0-k)] = \sum_{k=-\infty}^{\infty} x[k]y[k]$$

Thus, the integral simplifies to the sum of the element-wise product of the two sequences:

$$I = \sum_{n=-\infty}^{\infty} x[n]y[n]$$

Now, let's analyze the product of the given sequences:

$$x[n] = 2^{n-1}u[-n+2]$$

$$y[n] = 2^{-n+2}u[n+1]$$

$$x[n]y[n] = \left(2^{n-1}u[-n+2]\right)\left(2^{-n+2}u[n+1]\right)$$

$$x[n]y[n] = 2^{(n-1)+(-n+2)}u[-n+2]u[n+1]$$

$$x[n]y[n] = 2^{1}u[-n+2]u[n+1] = 2u[-n+2]u[n+1]$$

The product of the step functions determines the non-zero range of the sequence:

- $u[-n+2] = 1$ for $-n+2 \geq 0 \implies n \leq 2$
- $u[n+1] = 1$ for $n+1 \geq 0 \implies n \geq -1$

Therefore, the product is non-zero only for the overlapping integer range $-1 \leq n \leq 2$. The values of n in this range are $\{-1, 0, 1, 2\}$. Evaluating the sum over this valid range:

$$I = \sum_{n=-1}^{2} 2 = 2+2+2+2 = 8$$

$$\boxed{8}$$

Key Points

- The integral of a product of DTFTs over one period $\frac{1}{2\pi}\int_0^{2\pi} X(e^{j\omega})Y(e^{-j\omega})d\omega$ is equivalent to the inner product of their time-domain sequences $\sum_{n=-\infty}^{\infty} x[n]y[n]$. This is a form of Parseval's relation.
- The time-reversal property states: $x[-n] \leftrightarrow X(e^{-j\omega})$.

Mistakes to be avoided

Attempting to algebraically compute the continuous DTFTs $X(e^{j\omega})$ and $Y(e^{j\omega})$ and then performing the complex integration. This is incredibly tedious and error-prone. Always look for time-domain equivalents (like Parseval's theorem or convolution properties) when asked to evaluate integrals of Fourier transforms.

Q7.20.1 MCQ | 2020 | 2M | Ans: D | ● ○ ○

The given continuous-time signal is:

$$x(t) = \cos(200\pi t)$$

The signal is sampled at instances $t = \frac{n}{400}$. Thus, the sampling interval is $T_s = \frac{1}{400}$ s. Substituting t to find the discrete-time sequence $x[n]$:

$$x[n] = x(nT_s) = \cos\left(200\pi \cdot \frac{n}{400}\right) = \cos\left(\frac{\pi}{2}n\right)$$

Using Euler's identity, we can expand the discrete-time sequence:

$$x[n] = \frac{1}{2}e^{j\frac{\pi}{2}n} + \frac{1}{2}e^{-j\frac{\pi}{2}n}$$

For an N-point DFT, the standard complex exponential basis sequences are of the form $e^{j\frac{2\pi k}{N}n}$. Given $N = 8$, the basis is $e^{j\frac{\pi}{4}kn}$. We equate the frequencies of our signal's exponential terms to the DFT basis frequencies to determine the non-zero bins (k): For the positive frequency component:

$$\frac{\pi}{2} = \frac{\pi}{4}k \implies k = 2$$

For the negative frequency component (which wraps around in a periodic DFT spectrum as $N - k$):

$$k_2 = N - k = 8 - 2 = 6$$

Alternatively, by equating the negative phase directly to the basis phase (with 2π periodicity):

$$-\frac{\pi}{2} \equiv 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$$

$$\frac{3\pi}{2} = \frac{\pi}{4}k \implies k = 6$$

Therefore, only the bins at $k = 2$ and $k = 6$ contain energy, meaning only $X[2]$ and $X[6]$ (represented as $X(2)$ and $X(6)$ in the options) are non-zero.

$$\boxed{\text{D) Only } X(2) \text{ and } X(6) \text{ are non-zero.}}$$

Key Points

- The discrete frequency corresponding to the k-th bin in an N-point DFT is $\omega_k = \frac{2\pi k}{N}$.
- A purely real sinusoidal sequence $x[n] = \cos(\frac{2\pi mn}{N})$ has exactly two non-zero DFT coefficients located at index $k = m$ and its mirrored position $k = N - m$.

Q7.18.1 NAT | 2018 | 2M | Ans: 3 | ● ○ ○

We are given an 8-point DFT sequence $X[k] = k + 1$ for $0 \le k \le 7$. We need to evaluate the sum $S = \sum_{n=0}^{3} x[2n]$. Using the Inverse Discrete Fourier Transform (IDFT) definition:

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}$$

Substitute $N = 8$ and evaluate for the even indices $2n$:

$$x[2n] = \frac{1}{8}\sum_{k=0}^{7} X[k]e^{j\frac{2\pi}{8}k(2n)} = \frac{1}{8}\sum_{k=0}^{7} X[k]e^{j\frac{\pi}{2}kn}$$

Now, sum this expression over n from 0 to 3:

$$\sum_{n=0}^{3} x[2n] = \sum_{n=0}^{3}\left(\frac{1}{8}\sum_{k=0}^{7} X[k]e^{j\frac{\pi}{2}kn}\right)$$

Interchanging the order of summation:

$$\sum_{n=0}^{3} x[2n] = \frac{1}{8}\sum_{k=0}^{7} X[k]\left(\sum_{n=0}^{3} e^{j\frac{\pi}{2}kn}\right)$$

The inner sum over n is a geometric series that evaluates to:

$$\sum_{n=0}^{3} e^{j\frac{\pi}{2}kn} = \begin{cases} 4, & \text{if } k \text{ is a multiple of } 4 \\ 0, & \text{otherwise} \end{cases}$$

For the given range $0 \leq k \leq 7$, the values of k that are multiples of 4 are $k = 0$ and $k = 4$. Therefore, the non-zero terms in our summation occur only at $k = 0$ and $k = 4$:

$$\sum_{n=0}^{3} x[2n] = \frac{1}{8}\left(X[0] \cdot 4 + X[4] \cdot 4\right) = \frac{1}{2}(X[0] + X[4])$$

Given the formula $X[k] = k + 1$, we find the required components:

$$X[0] = 0 + 1 = 1$$

$$X[4] = 4 + 1 = 5$$

Substitute these values back into the simplified expression:

$$\sum_{n=0}^{3} x[2n] = \frac{1}{2}(1 + 5) = \frac{6}{2} = 3$$

$\boxed{3}$

Key Points

- **Decimation in Time:** The $N/2$-point DFT of a decimated sequence $y[n] = x[2n]$ is given by $Y[k] = \frac{1}{2}(X[k] + X[k + N/2])$.
- **DC Component:** The sum of all samples in a discrete-time sequence is equal to the 0-th index (DC component) of its DFT. Therefore, $\sum_{n=0}^{N/2-1} x[2n] = Y[0] = \frac{1}{2}(X[0] + X[N/2])$.
- **Orthogonality Principle:** $\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}kn} = N$ if k is an integer multiple of N, and 0 otherwise.

Q7.17.1

NAT | 2017 | 2M | Ans: 31 | ● ○ ○

Method 1

Let the unknown samples of the second sequence be $h[1] = a$ and $h[2] = b$. The given sequences are $x[n] = [1, 2, 1]$ and $h[n] = [1, a, b]$. Using the tabular method for linear convolution:

	1	a	b
1	1	a	b
2	2	$2a$	$2b$
1	1	a	b

Summing the anti-diagonals, we extract the elements of the output sequence $y[n]$:

$$y[0] = 1$$

$$y[1] = a + 2$$

$$y[2] = b + 2a + 1$$

$$y[3] = 2b + a$$

$$y[4] = b$$

We are given $y[1] = 3$ and $y[2] = 4$. Solving the equations:

$$a + 2 = 3 \implies a = 1$$

$$b + 2(1) + 1 = 4 \implies b = 1$$

Thus, the unknown sequence values are $h[1] = 1$ and $h[2] = 1$. Now, calculating the required terms $y[3]$ and $y[4]$:

$$y[3] = 2(1) + 1 = 3$$

$$y[4] = 1$$

The value of the required expression is:

$$10y[3] + y[4] = 10(3) + 1 = 31$$

Method 2

Linear convolution can be expressed as a matrix-vector multiplication $\mathbf{y} = \mathbf{X}\mathbf{h}$, where $\mathbf{X}$ is a lower triangular Toeplitz matrix formed from $x[n]$:

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ h[1] \\ h[2] \end{bmatrix}$$

Multiplying the matrix by the vector, we can directly extract equations for the given outputs $y[1]$ and $y[2]$:

$$y[1] = 2(1) + 1(h[1]) = 2 + h[1]$$

Given $y[1] = 3 \implies 2 + h[1] = 3 \implies h[1] = 1$.

$$y[2] = 1(1) + 2(h[1]) + 1(h[2]) = 1 + 2(1) + h[2] = 3 + h[2]$$

Given $y[2] = 4 \implies 3 + h[2] = 4 \implies h[2] = 1$. Now, using the known values of $h[n]$ to find $y[3]$ and $y[4]$:

$$y[3] = 0(1) + 1(h[1]) + 2(h[2]) = 1(1) + 2(1) = 3$$

$$y[4] = 0(1) + 0(h[1]) + 1(h[2]) = 1(1) = 1$$

Calculating the final required expression:

$$10y[3] + y[4] = 10(3) + 1 = 31$$

$$\boxed{31}$$

Key Points

- The length of a linear convolution $y[n] = x[n] * h[n]$ of two sequences with lengths L and M is $L + M - 1$. Here, both sequences have 3 elements, yielding $3 + 3 - 1 = 5$ output samples (from $n = 0$ to 4).
- The tabular (array) method effectively organizes the cross-multiplication of sequence elements, where anti-diagonals represent terms of the same time index n.
- The matrix approach constructs a convolution operation as the product of a Toeplitz matrix (created by shifting the impulse response/signal) and an input column vector.

Q7.17.2 NAT 2017 2M Ans: 2.1 ● ○ ○

The impulse response of the given discrete-time LTI filter is:

$$h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2]$$

The Discrete-Time Fourier Transform (DTFT) $H(\omega)$ is defined as:

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

Substitute the given values of $h[n]$:

$$H(\omega) = \frac{1}{3} + \frac{1}{3}e^{-j\omega} + \frac{1}{3}e^{-j2\omega}$$

To simplify the expression and analyze its magnitude, factor out the phase term corresponding to the center of symmetry ($e^{-j\omega}$):

$$H(\omega) = \frac{1}{3}e^{-j\omega}\left(e^{j\omega} + 1 + e^{-j\omega}\right)$$

Using Euler's identity, $e^{j\omega} + e^{-j\omega} = 2\cos(\omega)$:

$$H(\omega) = \frac{1}{3}e^{-j\omega}\left(1 + 2\cos(\omega)\right)$$

We are given that $H(\omega_o) = 0$. Since $e^{-j\omega_o}$ has a magnitude of 1 and cannot be zero, the term inside the parentheses must be zero:

$$1 + 2\cos(\omega_o) = 0$$

$$\cos(\omega_o) = -\frac{1}{2}$$

We are given the normalized frequency range $0 < \omega_o < \pi$. In this principal range, the cosine function equals -0.5 at:

$$\omega_o = \frac{2\pi}{3}\text{ rad}$$

Evaluating the numerical value:

$$\omega_o \approx \frac{2 \times 3.14159}{3} \approx 2.094\text{ rad}$$

Rounding to one decimal place as implied by the provided answer format:

$$\boxed{2.1}$$

Key Points

- For a finite-duration, symmetric impulse response (like a rectangular window or moving average filter), the DTFT can be expressed as a purely real amplitude function multiplied by a linear phase shift term $e^{-j\omega\alpha}$, where α is the center of symmetry.
- The given filter is a 3-point moving average filter. Its frequency response features nulls (zeros) where the amplitude term $1 + 2\cos(\omega)$ evaluates to zero.

Q7.16.1 NAT 2016 2M Ans: 6 ● ○ ○

The given 4-point sequence is $x[n] = \{3, 2, 3, 4\}$ with its 4-point DFT $X[k] = \{12, 2j, 0, -2j\}$. The new 12-point sequence is $X_1[n] = \{3, 0, 0, 2, 0, 0, 3, 0, 0, 4, 0, 0\}$. Observe that $X_1[n]$ is obtained by inserting $L - 1 = 2$ zeros between each sample of $x[n]$. This is a time-expansion (upsampling) operation by a factor of $L = 3$. According to the time-expansion property of the DFT, if a sequence is formed by inserting $L - 1$ zeros between the samples of an N-point sequence, its corresponding DFT spectrum becomes an L-fold periodic repetition of the original N-point DFT. Mathematically, the $N \cdot L$-point DFT $X_1[k]$ is:

$$X_1[k] = X[k \bmod N]$$

Here, $N = 4$ and $L = 3$. Therefore, the 12-point DFT $X_1[k]$ is:

$$X_1[k] = X[k \bmod 4]$$

We need to evaluate the spectrum at indices $k = 8$ and $k = 11$:

$$X_1[8] = X[8 \bmod 4] = X[0] = 12$$

$$X_1[11] = X[11 \bmod 4] = X[3] = -2j$$

Now, substitute these into the required expression and compute the magnitude:

$$\left|\frac{X_1[8]}{X_1[11]}\right| = \left|\frac{12}{-2j}\right|$$

Simplifying the complex fraction:

$$\left|\frac{12}{-2j}\right| = \left|\frac{12 \cdot j}{-2j \cdot j}\right| = \left|\frac{12j}{2}\right| = |6j| = 6$$

$$\boxed{6}$$

Key Points

- **Time Expansion Property:** Upsampling a discrete sequence $x[n]$ by inserting $L - 1$ zeros between each sample results in a sequence of length $L \cdot N$.
- The resulting DFT is formed by concatenating L copies of the original N-point DFT $X[k]$. This implies $X_{\text{expanded}}[k] = X[k \bmod N]$ for $0 \leq k < L \cdot N$.

Q7.16.2 NAT 2016 2M Ans: 4096 ● ○ ○

Given parameters for the signal processing system: Sampling rate, $f_s = 8\,\text{kHz} = 8000\,\text{samples/sec}$ Time for one complex multiplication, $t_m = 20\,\mu\text{s} = 20 \times 10^{-6}\,\text{s}$ For real-time processing, the total computation time required to process an N-point block must be less than or equal to the time taken to acquire those N samples.

$$T_{\text{computation}} \leq T_{\text{acquisition}}$$

The time taken to collect N samples at sampling rate f_s is:

$$T_{\text{acquisition}} = \frac{N}{f_s} = \frac{N}{8000}\,\text{s}$$

For an N-point radix-2 Decimation-In-Frequency (DIF) FFT algorithm, the total number of complex multiplications required is:

$$N_m = \frac{N}{2} \log_2 N$$

Since the processor performs operations sequentially and addition/subtraction time is negligible, the total computation time is purely dependent on multiplications:

$$T_{\text{computation}} = N_m \times t_m = \left(\frac{N}{2}\log_2 N\right) \times 20 \times 10^{-6}\ \text{s}$$

Substituting the expressions into the real-time constraint inequality:

$$\left(\frac{N}{2}\log_2 N\right) \times 20 \times 10^{-6} \le \frac{N}{8000}$$

Divide both sides by N (since $N > 0$):

$$\frac{1}{2}\log_2 N \times 20 \times 10^{-6} \le \frac{1}{8000}$$

$$10 \times 10^{-6}\log_2 N \le 125 \times 10^{-6}$$

Solving for $\log_2 N$:

$$\log_2 N \le \frac{125}{10}$$

$$\log_2 N \le 12.5$$

Because a radix-2 FFT algorithm requires the block size N to be a power of 2, $\log_2 N$ must be an integer. The maximum integer value satisfying this condition is 12.

$$\log_2 N = 12 \implies N = 2^{12} = 4096$$

$$\boxed{4096}$$

Key Points

- **Real-Time Processing Constraint:** A system operates in real-time if the processing time for a frame (or block) of data does not exceed the duration it takes to capture that frame ($T_{\text{proc}} \le \frac{N}{f_s}$).
- **Radix-2 FFT Computational Complexity:** An N-point radix-2 FFT significantly reduces the number of complex multiplications from N^2 (in direct DFT) to $\frac{N}{2}\log_2 N$.

Q7.16.3 NAT 2016 2M Ans: 8 ● ○ ○

The given discrete-time signal is:

$$x[n] = 6\delta[n+2] + 3\delta[n+1] + 8\delta[n] + 7\delta[n-1] + 4\delta[n-2]$$

We are required to evaluate the following integral:

$$I = \frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\sin^2(2\omega)\,d\omega$$

Using the trigonometric identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$, we can expand the sine squared term:

$$\sin^2(2\omega) = \frac{1-\cos(4\omega)}{2} = \frac{1}{2} - \frac{1}{4}e^{j4\omega} - \frac{1}{4}e^{-j4\omega}$$

Substitute this expansion back into the integral:

$$I = \frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\left(\frac{1}{2} - \frac{1}{4}e^{j4\omega} - \frac{1}{4}e^{-j4\omega}\right) d\omega$$

$$I = \frac{1}{2}\left[\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})d\omega\right] - \frac{1}{4}\left[\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j4\omega}d\omega\right] - \frac{1}{4}\left[\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{-j4\omega}d\omega\right]$$

Recall the Inverse Discrete-Time Fourier Transform (IDTFT) definition:

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega \implies \frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega = 2x[n]$$

Applying this property to each term in our expanded integral:

- For $n = 0$: $\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})d\omega = 2x[0]$
- For $n = 4$: $\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j4\omega}d\omega = 2x[4]$
- For $n = -4$: $\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{-j4\omega}d\omega = 2x[-4]$

Substituting these equivalents into the expression for I:

$$I = \frac{1}{2}(2x[0]) - \frac{1}{4}(2x[4]) - \frac{1}{4}(2x[-4])$$

$$I = x[0] - \frac{1}{2}x[4] - \frac{1}{2}x[-4]$$

From the given signal $x[n]$, we can identify the sequence values at specific indices by looking at the coefficients of the impulse functions:

$$x[0] = 8 \quad \text{(coefficient of } \delta[n])$$

$$x[4] = 0 \quad \text{(no } \delta[n-4] \text{ term present)}$$

$$x[-4] = 0 \quad \text{(no } \delta[n+4] \text{ term present)}$$

Evaluating the final expression:

$$I = 8 - \frac{1}{2}(0) - \frac{1}{2}(0) = 8$$

$$\boxed{8}$$

Key Points

- **IDTFT Definition:** The inverse transform extracts the time-domain samples from the frequency domain representation: $x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$.
- **Trigonometric Expansion:** Converting trigonometric functions into complex exponentials often simplifies integration involving Fourier transforms.
- **Shift Property equivalence:** Multiplying $X(e^{j\omega})$ by a complex exponential $e^{j\omega k}$ in the frequency domain integration directly corresponds to extracting the k-th sample $x[k]$ in the time domain.

Q7.15.1 NAT | 2015 | 2M | Ans: 11 | ● ○ ○

Given the two real sequences:

$$x_1[n] = \{1, 2, 3, 0\}$$

$$x_2[n] = \{1, 3, 2, 1\}$$

We are given that $X_3(k) = X_1(k)X_2(k)$. According to the circular convolution property of the Discrete Fourier Transform (DFT), multiplication of two DFTs in the frequency domain corresponds to the circular convolution of their respective sequences in the time domain. Therefore, the sequence $x_3[n]$ obtained by taking the 4-point inverse DFT of $X_3(k)$ is the 4-point circular convolution of $x_1[n]$ and $x_2[n]$:

$$x_3[n] = x_1[n] \circledast_4 x_2[n] = \sum_{m=0}^{N-1} x_1[m]x_2[(n-m) \bmod N]$$

Method 1

Here, the length $N = 4$. We need to find the value of the sequence at index $n = 2$:

$$x_3[2] = \sum_{m=0}^{3} x_1[m]x_2[(2-m) \bmod 4]$$

Expanding the summation for $m = 0, 1, 2, 3$:

$$x_3[2] = x_1[0]x_2[2 \bmod 4] + x_1[1]x_2[1 \bmod 4] + x_1[2]x_2[0 \bmod 4] + x_1[3]x_2[-1 \bmod 4]$$

$$x_3[2] = x_1[0]x_2[2] + x_1[1]x_2[1] + x_1[2]x_2[0] + x_1[3]x_2[3]$$

Substitute the values from the given sequences:

$$x_1[0] = 1, \quad x_1[1] = 2, \quad x_1[2] = 3, \quad x_1[3] = 0$$

$$x_2[0] = 1, \quad x_2[1] = 3, \quad x_2[2] = 2, \quad x_2[3] = 1$$

Calculate the sum:

$$x_3[2] = (1 \times 2) + (2 \times 3) + (3 \times 1) + (0 \times 1)$$

$$x_3[2] = 2 + 6 + 3 + 0 = 11$$

Method 2

The 4-point circular convolution can also be computed using the matrix method. We construct a circulant matrix from the sequence $x_2[n]$ and multiply it by the column vector representation of $x_1[n]$. The output vector $\mathbf{x}_3$ is given by $\mathbf{x}_3 = \mathbf{X}_2\mathbf{x}_1$:

$$\begin{bmatrix} x_3[0] \\ x_3[1] \\ x_3[2] \\ x_3[3] \end{bmatrix} = \begin{bmatrix} x_2[0] & x_2[3] & x_2[2] & x_2[1] \\ x_2[1] & x_2[0] & x_2[3] & x_2[2] \\ x_2[2] & x_2[1] & x_2[0] & x_2[3] \\ x_2[3] & x_2[2] & x_2[1] & x_2[0] \end{bmatrix} \begin{bmatrix} x_1[0] \\ x_1[1] \\ x_1[2] \\ x_1[3] \end{bmatrix}$$

Substitute the values from the sequences into the matrices:

$$\begin{bmatrix} x_3[0] \\ x_3[1] \\ x_3[2] \\ x_3[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \\ 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$$

To find the element at index $n = 2$, we perform the dot product of the third row of the circulant matrix with the input vector $\mathbf{x}_1$:

$$x_3[2] = (2 \times 1) + (3 \times 2) + (1 \times 3) + (1 \times 0)$$

$$x_3[2] = 2 + 6 + 3 + 0 = 11$$

11

Key Points

- According to the circular convolution property of the DFT, multiplying two sequences in the frequency domain corresponds to their circular convolution in the time domain: $x_1[n] \circledast_N x_2[n] \leftrightarrow X_1(k)X_2(k)$.
- The matrix method for circular convolution constructs an $N \times N$ circulant matrix from one sequence, formed by cyclically down-shifting its elements column by column.

Q7.15.2 MCQ | 2015 | 2M | Ans: C | ● ○ ○

From the first equation, the relationship between the sequences $[a, b, c]$ and $[A, B, C]$ is given by a matrix multiplication. Given the twiddle factor $W_3 = e^{j\frac{2\pi}{3}}$, the matrix with elements W_3^{-nk} is the standard 3-point Discrete Fourier Transform (DFT) matrix. Let the time-domain sequence be $x[n] = \begin{bmatrix} a & b & c \end{bmatrix}^T$ and its DFT be $X[k] = \begin{bmatrix} A & B & C \end{bmatrix}^T$.

$$X[k] = \text{DFT}\{x[n]\}$$

The second equation defines a new sequence $y[n] = \begin{bmatrix} p & q & r \end{bmatrix}^T$:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^1 & W_3^2 \\ 1 & W_3^2 & W_3^4 \end{bmatrix} \begin{bmatrix} A/3 \\ B \cdot W_3^2/3 \\ C \cdot W_3^4/3 \end{bmatrix}$$

We can rewrite the column vector on the right side in terms of the DFT sequence $X[k]$:

$$\frac{1}{3} \begin{bmatrix} X[0]W_3^0 \\ X[1]W_3^2 \\ X[2]W_3^4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} Y[0] \\ Y[1] \\ Y[2] \end{bmatrix}$$

where the new frequency domain sequence is $Y[k] = X[k]W_3^{2k}$ for $k = 0, 1, 2$. The square matrix in the second equation corresponds to the core of the Inverse DFT (IDFT) operation (since IDFT elements are $\frac{1}{3}W_3^{nk}$). The required $1/3$ scaling factor is already present in the column vector. Therefore, $y[n]$ is simply the IDFT of $Y[k]$:

$$y[n] = \text{IDFT}\{Y[k]\} = \frac{1}{3} \sum_{k=0}^{2} Y[k] W_3^{nk}$$

Substitute $Y[k] = X[k]W_3^{2k}$ into the IDFT synthesis equation:

$$y[n] = \frac{1}{3} \sum_{k=0}^{2} \left(X[k] W_3^{2k} \right) W_3^{nk} = \frac{1}{3} \sum_{k=0}^{2} X[k] W_3^{(n+2)k}$$

This expression represents the IDFT of $X[k]$ evaluated at the shifted time index $(n+2)$. Due to the periodicity of the 3-point sequence, this equates to a circular shift:

$$y[n] = x[(n+2) \bmod 3]$$

Evaluating this for each element of the sequence $y[n] = [p, q, r]^T$:

$$\begin{aligned} n = 0 &\implies p = y[0] = x[2 \bmod 3] = x[2] = c \\ n = 1 &\implies q = y[1] = x[3 \bmod 3] = x[0] = a \\ n = 2 &\implies r = y[2] = x[4 \bmod 3] = x[1] = b \end{aligned}$$

Thus, the derived sequence is $[p, q, r] = [c, a, b]$.

$$\boxed{\text{C) } [p, q, r] = [c, a, b]}$$

Key Points

- **DFT Matrix Form:** The N-point DFT of $x[n]$ is computed as $X[k] = \sum_{n=0}^{N-1} x[n]W_N^{-nk}$, where $W_N = e^{j\frac{2\pi}{N}}$.
- **IDFT Matrix Form:** The corresponding IDFT is $x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{nk}$.
- **Circular Time Shift:** Multiplying the DFT $X[k]$ by a complex exponential W_N^{mk} results in an advance (circular time shift) in the time domain: $\text{IDFT}\{X[k]W_N^{mk}\} = x[(n+m) \bmod N]$.

Q7.15.3

NAT | 2015 | 1M | Ans: 2 | ● ○ ○

Method 1

Let the given sum be S. The series is an Arithmetico-Geometric Progression (AGP). Expanding the summation:

$$S = \sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n = 0 + 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right)^3 + \dots \quad \text{— (1)}$$

Multiply the entire series by the geometric common ratio, which is $\frac{1}{2}$:

$$\frac{1}{2}S = 1\left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right)^3 + 3\left(\frac{1}{2}\right)^4 + \dots \quad \text{— (2)}$$

Subtract equation (2) from equation (1) by aligning terms with the same powers of $\frac{1}{2}$:

$$S - \frac{1}{2}S = \left(\frac{1}{2}\right) + \left[2\left(\frac{1}{2}\right)^2 - 1\left(\frac{1}{2}\right)^2\right] + \left[3\left(\frac{1}{2}\right)^3 - 2\left(\frac{1}{2}\right)^3\right] + \dots$$

$$\frac{1}{2}S = \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots$$

The right side is now a standard infinite geometric progression with first term $a = \frac{1}{2}$ and common ratio $r = \frac{1}{2}$. Using the sum formula $S_\infty = \frac{a}{1-r}$:

$$\frac{1}{2}S = \frac{1/2}{1-1/2} = \frac{1/2}{1/2} = 1$$

$$S = 2$$

Method 2

Consider the standard sum of an infinite geometric series for $|x| < 1$:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

Differentiate both sides with respect to x:

$$\sum_{n=1}^{\infty} nx^{n-1} = \frac{d}{dx}\left((1-x)^{-1}\right) = \frac{1}{(1-x)^2}$$

Multiply both sides by x to match the required form nx^n:

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$$

Note that starting the summation from $n = 0$ does not change the value since the $n = 0$ term is zero. Substitute $x = \frac{1}{2}$ into the derived formula:

$$\sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n = \frac{1/2}{(1-1/2)^2} = \frac{1/2}{(1/2)^2} = \frac{1/2}{1/4} = 2$$

$\boxed{2}$

Key Points

- **Arithmetico-Geometric Progression (AGP):** A sequence where each term is the product of corresponding terms of an arithmetic progression and a geometric progression. The standard technique to sum an AGP is to multiply by the common ratio and subtract the shifted series.
- **Derivative Method:** The identity $\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$ (for $|x| < 1$) is extremely useful for quickly evaluating series involving a linear coefficient multiplied by an exponential term.

Q7.15.4 NAT 2015 2M Ans: 0.5 ● ○ ○

Given the periodic signal with period $N = 16$:

$$\tilde{x}[n] = 1 + \cos\left(\frac{\pi n}{8}\right)$$

Expressing the cosine term using Euler's identity, we get:

$$\tilde{x}[n] = 1 + \frac{1}{2}\exp\left(j\frac{\pi}{8}n\right) + \frac{1}{2}\exp\left(-j\frac{\pi}{8}n\right)$$

The fundamental frequency is $\Omega_0 = \frac{2\pi}{N} = \frac{2\pi}{16} = \frac{\pi}{8}$. Comparing the expanded equation with the standard Discrete Fourier Series (DFS) synthesis equation $\tilde{x}[n] = \sum_{k=\langle N\rangle} a_k \exp(jk\Omega_0 n)$, we can identify the non-zero DFS coefficients as:

$$a_0 = 1$$

$$a_1 = \frac{1}{2} = 0.5$$

$$a_{-1} = \frac{1}{2} = 0.5$$

We are asked to find the value of the coefficient a_{31}. By the periodicity property of DFS coefficients, we know that $a_k = a_{k+N}$ for all k. Since $N = 16$:

$$a_{31} = a_{31 \bmod 16} = a_{15}$$

We also know that $a_{15} = a_{15-16} = a_{-1}$. Therefore:

$$a_{31} = a_{-1} = 0.5$$

$\boxed{0.5}$

Key Points

- Euler's identity: $\cos(\theta) = \frac{e^{j\theta}+e^{-j\theta}}{2}$
- DFS synthesis equation: $\tilde{x}[n] = \sum_{k=\langle N\rangle} a_k e^{jk\Omega_0 n}$
- Periodicity of DFS coefficients: $a_k = a_{k\pm mN}$ where m is an integer and N is the fundamental period.

Mistakes to be avoided

Attempting to calculate the summation formula $a_k = \frac{1}{16}\sum_{n=0}^{15} \tilde{x}[n] \exp\left(-j\frac{\pi}{8}kn\right)$ directly is time-consuming and error-prone. Expanding the signal into complex exponentials and matching coefficients is the most efficient method.

Q7.14.1 NAT | 2014 | 2M | Ans: 10 | ● ○ ○

Given the discrete-time periodic signal:

$$x[n] = \sin\left(\frac{\pi n}{5}\right)$$

First, determine the fundamental period (N) of the signal. The fundamental angular frequency is $\Omega_0 = \frac{\pi}{5}$. For a discrete-time signal to be periodic, the ratio $\frac{2\pi}{\Omega_0}$ must be a rational number of the form $\frac{N}{m}$, where N and m are integers.

$$\frac{N}{m} = \frac{2\pi}{\pi/5} = 10$$

To find the fundamental period, we choose the smallest integer m that makes N an integer. Setting $m = 1$, we get the fundamental period $N = 10$. Using Euler's identity, we can expand the signal into its complex exponential form:

$$x[n] = \frac{1}{2j}\exp\left(j\frac{\pi}{5}n\right) - \frac{1}{2j}\exp\left(-j\frac{\pi}{5}n\right)$$

The standard Discrete Fourier Series (DFS) synthesis equation for a signal with period $N = 10$ is:

$$x[n] = \sum_{k=\langle 10\rangle} a_k \exp(jk\Omega_0 n) = \sum_{k=\langle 10\rangle} a_k \exp\left(jk\frac{\pi}{5}n\right)$$

By comparing the terms of our expanded signal with the standard DFS equation, we can identify the non-zero coefficients within the principal period (where $k = 1$ and $k = -1$):

$$a_1 = \frac{1}{2j}$$

$$a_{-1} = -\frac{1}{2j}$$

A fundamental property of DFS coefficients is that they are periodic with the same period N as the time-domain signal. Therefore:

$$a_k = a_{k\pm mN}$$

where m is any integer. Since the non-zero coefficients occur at $k = 1$ and $k = -1$, and the period is $N = 10$, the complete set of non-zero coefficients $\{a_k\}$ will occur at:

$$k = 10m + 1$$

$$k = 10m - 1$$

Combining these, the non-zero coefficients exist when $k = 10m \pm 1$. Comparing this result with the given generalized condition $k = Bm \pm 1$, we find:

$$B = 10$$

$$\boxed{10}$$

Key Points

- Euler's identity for sine: $\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$
- Periodicity of discrete signals: A discrete sinusoid $\sin(\Omega_0 n)$ is periodic if $\frac{2\pi}{\Omega_0} = \frac{N}{m}$. The fundamental period N is the

smallest integer satisfying this relation.

- Periodicity of DFS coefficients: The DFS coefficients a_k of a periodic sequence with period N form a periodic sequence with the same period N, such that $a_k = a_{k+mN}$.

Q7.14.2 MCQ 2014 2M Ans: B ● ○ ○ CLICK FOR VIDEO SOLUTION

Given the modified N-point DFT equation with a unitary scaling factor:

$$X[k] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk}$$

Let $y[n]$ be the output of applying the DFT twice, i.e., $y = \text{DFT}(\text{DFT}(x))$. Using the definition, the second DFT operation is:

$$y[m] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X[k] e^{-j\frac{2\pi}{N}km}$$

Substituting the expression for $X[k]$ into the equation for $y[m]$:

$$y[m] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \left(\frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk} \right) e^{-j\frac{2\pi}{N}km}$$

$$y[m] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \sum_{k=0}^{N-1} e^{-j\frac{2\pi}{N}k(n+m)}$$

From the orthogonality property of complex exponentials, the inner summation evaluates to:

$$\sum_{k=0}^{N-1} e^{-j\frac{2\pi}{N}k(n+m)} = \begin{cases} N, & \text{if } (n+m) \bmod N = 0 \\ 0, & \text{otherwise} \end{cases}$$

The condition $(n+m) \bmod N = 0$ implies $n = (-m) \bmod N$. Substituting this back yields:

$$y[m] = \frac{1}{N} (x[(-m) \bmod N] \cdot N) = x[(-m) \bmod N]$$

The problem states that $\text{DFT}(\text{DFT}(x)) = x$, which means $y[m] = x[m]$. Thus, the sequence must satisfy:

$$x[m] = x[(-m) \bmod N]$$

This condition requires the sequence to be an even sequence. For $N = 4$, expanding this gives:

$$x[0] = x[0]$$

$$x[1] = x[(-1) \bmod 4] = x[3]$$

$$x[2] = x[(-2) \bmod 4] = x[2]$$

$$x[3] = x[(-3) \bmod 4] = x[1]$$

Therefore, any sequence satisfying the given condition must have its $n = 1$ and $n = 3$ elements equal, meaning $x[1] = x[3]$. Checking the given options where $\mathbf{x} = [x[0]\ x[1]\ x[2]\ x[3]]$: Option A: $x[1] = 2, x[3] = 4 \implies x[1] \neq x[3]$ Option B: $x[1] = 2, x[3] = 2 \implies x[1] = x[3]$ (Satisfies the condition) Option C: $x[1] = 3, x[3] = 2 \implies x[1] \neq x[3]$ Option D: $x[1] = 2, x[3] = 3 \implies x[1] \neq x[3]$ Only the sequence in Option B satisfies the required symmetry.

$$\boxed{\text{B) } \mathbf{x} = [1\ 2\ 3\ 2]}$$

Key Points

- The standard unscaled DFT of the DFT of a sequence yields $Nx[(-n) \bmod N]$.
- For the unitary DFT defined with a $1/\sqrt{N}$ scaling factor, applying the DFT twice yields exactly $x[(-n) \bmod N]$.
- The condition $x[n] = x[(-n) \bmod N]$ defines a discrete-time even (or circularly symmetric) sequence.

Q7.14.3

NAT | 2014 | 1M | Ans: 3.375 | ● ○ ○

Given the discrete-time sequence:

$$x[n] = \left(\frac{2}{3}\right)^n u[n+3]$$

We can express this sequence in a form that allows us to use standard Fourier Transform pairs. We manipulate the expression to match the argument of the unit step function:

$$x[n] = \left(\frac{2}{3}\right)^{-3} \left(\frac{2}{3}\right)^{n+3} u[n+3]$$

$$x[n] = \left(\frac{3}{2}\right)^3 \left[\left(\frac{2}{3}\right)^{n+3} u[n+3]\right]$$

$$x[n] = \frac{27}{8} \left[\left(\frac{2}{3}\right)^{n+3} u[n+3]\right]$$

Let $g[n] = \left(\frac{2}{3}\right)^n u[n]$. The standard Fourier Transform of $g[n]$ is:

$$G(f) = \frac{1}{1 - \left(\frac{2}{3}\right) e^{-j2\pi f}}$$

Our signal $x[n]$ can be written in terms of $g[n]$ as:

$$x[n] = \frac{27}{8} g[n+3]$$

Using the time-shifting property of the Fourier Transform, $g[n-n_0] \xleftrightarrow{FT} G(f)e^{-j2\pi f n_0}$. For a time advance of 3 ($n_0 = -3$), we get:

$$g[n+3] \xleftrightarrow{FT} G(f)e^{j2\pi f(3)} = G(f)e^{j6\pi f}$$

Therefore, the Fourier Transform of $x[n]$ is:

$$X(f) = \frac{27}{8} G(f)e^{j6\pi f} = \frac{3.375e^{j6\pi f}}{1 - \left(\frac{2}{3}\right) e^{-j2\pi f}}$$

Comparing this result with the structure of the given Fourier Transform pair:

$$X(f) = \frac{Ae^{j6\pi f}}{1 - \left(\frac{2}{3}\right) e^{-j2\pi f}}$$

(Note: A time advance yields a positive exponent in the numerator. The negative sign $e^{-j6\pi f}$ in the original question image is a standard typographical error for such problems). We find the value of A:

$$A = \frac{27}{8} = 3.375$$

$$\boxed{3.375}$$

Key Points

- Standard Fourier Transform pair: $a^n u[n] \overset{FT}{\longleftrightarrow} \frac{1}{1-ae^{-j2\pi f}}$ for $|a| < 1$.
- Time-shifting property: $x[n-n_0] \overset{FT}{\longleftrightarrow} X(f)e^{-j2\pi f n_0}$. A time advance (where n_0 is negative) results in a positive complex exponential phase shift.

Q7.13.1

MCQ | 2013 | 2M | Ans: A | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Let the input sequence be defined as $x[n] = [a\ b\ c\ d]$. We are given that its 4-point Discrete Fourier Transform (DFT) is $X[k] = [\alpha\ \beta\ \gamma\ \delta]$. The given matrix equation is:

$$\begin{bmatrix} p & q & r & s \end{bmatrix} = \begin{bmatrix} a & b & c & d \end{bmatrix} \begin{bmatrix} a & b & c & d \\ d & a & b & c \\ c & d & a & b \\ b & c & d & a \end{bmatrix}$$

Let the resultant vector be $y[n] = [p\ q\ r\ s]$. By expanding the matrix multiplication, we obtain the individual elements:

$$p = a \cdot a + b \cdot d + c \cdot c + d \cdot b$$

$$q = a \cdot b + b \cdot a + c \cdot d + d \cdot c$$

$$r = a \cdot c + b \cdot b + c \cdot a + d \cdot d$$

$$s = a \cdot d + b \cdot c + c \cdot b + d \cdot a$$

This specific sum of products represents the 4-point circular convolution of the sequence $x[n]$ with itself. Mathematically, it is defined as:

$$y[n] = x[n] \circledast x[n] = \sum_{m=0}^{3} x[m]x[(n-m) \bmod 4]$$

According to the circular convolution property of the DFT, the circular convolution of two sequences in the time domain corresponds to the element-wise multiplication of their DFT sequences in the frequency domain:

$$y[n] = x_1[n] \circledast x_2[n] \overset{\text{DFT}}{\longleftrightarrow} Y[k] = X_1[k] \cdot X_2[k]$$

In this case, since $x_1[n] = x_2[n] = x[n]$, the DFT of $y[n]$ is the square of the DFT of $x[n]$:

$$Y[k] = X[k] \cdot X[k] = (X[k])^2$$

Substituting the given DFT vector $X[k] = [\alpha\ \beta\ \gamma\ \delta]$, we get:

$$Y[k] = [\alpha^2\ \beta^2\ \gamma^2\ \delta^2]$$

Thus, the DFT of the vector $[p\ q\ r\ s]$ is a scaled version of $[\alpha^2\ \beta^2\ \gamma^2\ \delta^2]$.

$$\boxed{\text{A) } [\alpha^2\ \beta^2\ \gamma^2\ \delta^2]}$$

Key Points

- **Circular Convolution Property of DFT:** Time-domain circular convolution equates to frequency-domain multiplication: $x_1[n] \circledast x_2[n] \overset{\text{DFT}}{\longleftrightarrow} X_1[k]X_2[k]$.
- The product of a row vector and a circulant matrix formed by a sequence is equivalent to the circular convolution of those sequences.

Q7.11.1 MCQ | 2011 | 2M | Ans: B | ● ○ ○

For a real-valued sequence $x[n]$, its N-point Discrete Fourier Transform (DFT), denoted as $X[k]$, exhibits conjugate symmetry. This property is mathematically expressed as:

$$X[k] = X^*[N-k]$$

for $k = 1, 2, \ldots, N-1$. Given a sequence length of $N = 8$, the first six points of the DFT sequence are provided as:

$$X[0] = 5$$

$$X[1] = 1 - j3$$

$$X[2] = 0$$

$$X[3] = 3 - j4$$

$$X[4] = 0$$

$$X[5] = 3 + j4$$

We are required to find the last two points of the 8-point DFT, which correspond to the indices $k = 6$ and $k = 7$. Applying the conjugate symmetry property for $k = 6$:

$$X[6] = X^*[8-6] = X^*[2]$$

Substituting the given value of $X[2] = 0$:

$$X[6] = 0^* = 0$$

Similarly, applying the property for $k = 7$:

$$X[7] = X^*[8-7] = X^*[1]$$

Substituting the given value of $X[1] = 1 - j3$:

$$X[7] = (1 - j3)^* = 1 + j3$$

Thus, the last two points of the DFT are 0 and $1 + j3$, which matches Option B.

$$\boxed{\text{B) } 0, 1 + j3}$$

Key Points

- For a purely real sequence in the time domain, its frequency domain representation (DFT) is conjugate symmetric.
- The conjugate symmetry relationship is given by $X[k] = X^*[N-k]$.
- For an even number of points N, the components at $X[0]$ (DC component) and $X[N/2]$ (Nyquist frequency) are always purely real.

Mistakes to be avoided

A common mistake is incorrectly identifying the folding index. The symmetry for an N-point DFT is defined such that index k pairs directly with $N - k$. For an 8-point sequence, do not pair $X[1]$ with $X[6]$; always verify the folding symmetry around the Nyquist point $N/2 = 4$.

Q7.10.1 MCQ 2010 1M Ans: D ● ○ ○

Let us evaluate each statement regarding the N-point Radix-2 Fast Fourier Transform (FFT) algorithm, where $N = 2^m$:

Statement A: It is not possible to construct a signal flow graph with both input and output in normal order. This statement is **FALSE**. While standard decimation-in-time (DIT) and decimation-in-frequency (DIF) algorithms have either bit-reversed inputs or bit-reversed outputs, it is entirely possible to construct an FFT flow graph with both input and output in normal order (for example, by utilizing the Stockham auto-sort algorithm or through specific non-in-place routing).

Statement B: The number of butterflies in the m^{th} stage is N/m. This statement is **FALSE**. In a Radix-2 FFT, there are $m = \log_2 N$ total stages. Irrespective of the specific stage index, every single stage requires exactly $\frac{N}{2}$ butterfly computations.

Statement C: In-place computation requires storage of only $2N$ node data. This statement is **FALSE**. The primary advantage of an in-place FFT computation is that it requires only N complex storage nodes (memory locations). The outputs of each butterfly computation can be written back into the exact same memory locations that held the inputs, thus conserving memory.

Statement D: Computation of a butterfly requires only one complex multiplication. This statement is **TRUE**. A standard Radix-2 butterfly takes two complex inputs, a and b, along with a phase factor (twiddle factor) W_N^k, to compute the outputs:

$$X_1 = a + W_N^k b$$

$$X_2 = a - W_N^k b$$

This fundamental operation requires exactly one complex multiplication (the product $W_N^k \cdot b$) and two complex additions/subtractions.

$$\boxed{\text{D) Computation of a butterfly requires only one complex multiplication.}}$$

Key Points

- For an N-point Radix-2 FFT, the total number of computational stages is $m = \log_2 N$.
- Each individual stage is composed of exactly $\frac{N}{2}$ butterfly structures.
- A single Radix-2 butterfly operation requires 1 complex multiplication and 2 complex additions.
- The overall computational complexity for an N-point FFT is $\frac{N}{2}\log_2 N$ complex multiplications and $N\log_2 N$ complex additions.

Q7.09.1 MCQ 2009 2M Ans: D ● ○ ○

Given the discrete-time sequence $x[n] = \{1, 0, 2, 3\}$.

Method 1

Standard DFT Matrix Multiplication

The N-point Discrete Fourier Transform (DFT) is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk}$$

For $N = 4$, the computation can be elegantly represented using the 4×4 twiddle factor matrix:

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \\ 3 \end{bmatrix}$$

Perform the row-by-column multiplication:

$$X[0] = 1(1) + 1(0) + 1(2) + 1(3) = 6$$

$$X[1] = 1(1) + (-j)(0) + (-1)(2) + j(3) = 1 - 2 + 3j = -1 + 3j$$
$$X[2] = 1(1) + (-1)(0) + 1(2) + (-1)(3) = 1 + 2 - 3 = 0$$
$$X[3] = 1(1) + j(0) + (-1)(2) + (-j)(3) = 1 - 2 - 3j = -1 - 3j$$

Thus, the complete DFT sequence is $[6, -1 + 3j, 0, -1 - 3j]$.

Method 2

Strategic Elimination

In an examination setting, computing all points is often unnecessary. We can find the purely real components first to quickly eliminate options. The DC component $X[0]$ is simply the sum of all elements:

$$X[0] = \sum_{n=0}^{3} x[n] = 1 + 0 + 2 + 3 = 6$$

This immediately eliminates options A and B. Next, compute $X[2]$, which alternates signs:

$$X[2] = \sum_{n=0}^{3} (-1)^n x[n] = 1 - 0 + 2 - 3 = 0$$

Looking at the remaining options (C and D), only option D has $X[2] = 0$.

$$\boxed{\text{D) } [6, -1 + 3j, 0, -1 - 3j]}$$

Key Points

- The N-point DFT evaluates the frequency content of a discrete signal at N equally spaced frequency points.
- **DC Component:** $X[0] = \sum x[n]$.
- **Nyquist Component (for even N):** $X[N/2] = \sum (-1)^n x[n]$.
- **Conjugate Symmetry:** For a purely real input sequence, $X[N - k] = X^*[k]$. Here, $X[3] = X^*[1]$, which holds true since $-1 - 3j$ is the complex conjugate of $-1 + 3j$.

Mistakes to be avoided

Manually computing complex terms like $X[1]$ and $X[3]$ first increases the likelihood of algebraic sign errors involving j. Always compute the purely real sums $X[0]$ and $X[N/2]$ first to quickly verify or eliminate multiple-choice options.

Q7.08.1 | MCQ | 2008 | 2M | Ans: A | ● ○ ○

Given the sequence representing the normalized circular autocorrelation:

$$y(n) = \frac{1}{N} \sum_{r=0}^{N-1} x(r)x(n + r)$$

Apply the Discrete Fourier Transform (DFT) to $y(n)$:

$$Y(k) = \sum_{n=0}^{N-1} y(n) e^{-j\frac{2\pi}{N}kn}$$

Substitute the given definition of $y(n)$:

$$Y(k) = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{r=0}^{N-1} x(r)x(n + r) \right) e^{-j\frac{2\pi}{N}kn}$$

Interchange the order of summation:

$$Y(k) = \frac{1}{N}\sum_{r=0}^{N-1} x(r) \sum_{n=0}^{N-1} x(n+r)e^{-j\frac{2\pi}{N}kn}$$

Let $m = n + r$, which implies $n = m - r$. Since $x(n)$ is a periodic sequence with period N, the summation over one full period remains unchanged regardless of the shift:

$$Y(k) = \frac{1}{N}\sum_{r=0}^{N-1} x(r) \sum_{m=0}^{N-1} x(m)e^{-j\frac{2\pi}{N}k(m-r)}$$

Separate the complex exponential terms:

$$Y(k) = \frac{1}{N}\left(\sum_{r=0}^{N-1} x(r)e^{j\frac{2\pi}{N}kr}\right)\left(\sum_{m=0}^{N-1} x(m)e^{-j\frac{2\pi}{N}km}\right)$$

Recognize the components in terms of the standard DFT of $x(n)$. The second summation is precisely $X(k)$. The first summation represents $X(-k)$. For a real-valued sequence $x(n)$, the DFT possesses the conjugate symmetry property, meaning $X(-k) = X^*(k)$.

$$Y(k) = \frac{1}{N}X^*(k)X(k) = \frac{1}{N}|X(k)|^2$$

While the exact mathematical derivation yields the scaling factor $\frac{1}{N}$, the fundamental frequency-domain representation of the autocorrelation function is the magnitude squared of the spectrum, $|X(k)|^2$ (also known as the Power Spectral Density). Option A correctly identifies this core dependency.

$$\boxed{\text{A) } |X(k)|^2}$$

Key Points

- **Circular Autocorrelation Theorem:** The DFT of the circular autocorrelation of a periodic sequence yields its Power Spectral Density (PSD), which is proportional to $|X(k)|^2$.
- **Conjugate Symmetry:** For a purely real signal $x(n)$, its spectrum satisfies $X(-k) = X^*(k)$.
- **Time Shifting Property:** Shifting a signal in the time domain equates to a linear phase shift in the frequency domain: $x(n+r) \xleftrightarrow{\text{DFT}} X(k)e^{j\frac{2\pi}{N}kr}$.

Q7.07.1 MCQ 2007 2M Ans: B ● ○ ○

The inverse Discrete-Time Fourier Transform (DTFT) of a sequence's spectrum $X(e^{j\omega})$ is given by the synthesis equation:

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}\, d\omega$$

We are asked to evaluate the integral $\int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega$. We can relate this to the inverse DTFT equation by evaluating it at $n = 0$:

$$x[0] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega(0)}\, d\omega$$

$$x[0] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega$$

Rearranging this expression to solve for the integral, we get:

$$\int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega = 2\pi x[0]$$

From the given 5-point sequence, the value of the sequence at the origin ($n = 0$) is explicitly given (notated as $x[-0] = 5$). Thus, $x[0] = 5$. Substituting this into our rearranged equation yields:

$$\int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega = 2\pi(5) = 10\pi$$

$\boxed{\text{B) } 10\pi}$

Key Points

- **Inverse DTFT (Synthesis Equation):** $x[n] = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(e^{j\omega}) e^{j\omega n}\, d\omega$.
- The integral of the DTFT spectrum over one fundamental period ($-\pi$ to π) depends exclusively on the value of the time-domain signal at the origin, evaluating to $2\pi x[0]$.

Q7.05.1 MCQ 2005 2M Ans: A ● ○ ○

Given the sequence $y(n)$ is defined as:

$$y(n) = \begin{cases} x\left(\frac{n}{2} - 1\right), & \text{for } n \text{ even} \\ 0, & \text{for } n \text{ odd} \end{cases}$$

Let's evaluate $y(n)$ for even values of n by setting $n = 2k$:

$$y(2k) = x(k - 1)$$

From the given figure, the non-zero values of $x(n)$ are located at $n = -2, -1, 0, 1, 2$. We map these to $y(n)$ by solving $k - 1 = m$ where $m \in \{-2, -1, 0, 1, 2\}$:

$$m = -2 \implies k = -1 \implies n = -2 \implies y(-2) = x(-2) = \frac{1}{2}$$

$$m = -1 \implies k = 0 \implies n = 0 \implies y(0) = x(-1) = 1$$

$$m = 0 \implies k = 1 \implies n = 2 \implies y(2) = x(0) = 2$$

$$m = 1 \implies k = 2 \implies n = 4 \implies y(4) = x(1) = 1$$

$$m = 2 \implies k = 3 \implies n = 6 \implies y(6) = x(2) = \frac{1}{2}$$

For all odd values of n, the definition strictly states $y(n) = 0$. Plotting these impulses yields a sequence that perfectly matches the one shown in option (a).

$\boxed{\text{A}}$

Key Points

- **Time Scaling (Upsampling):** The operation $x(n/2)$ represents upsampling by a factor of 2, which inserts zeros between the original samples.
- **Time Shifting:** The operation $x(n/2 - 1)$ implies the upsampled sequence is shifted. To find the new discrete time positions, equate the inner argument to the original sequence indices.

Q7.05.2 MCQ | 2005 | 2M | Ans: C | ● ○ ○

We need to find the Fourier transform of the sequence $y(2n)$. Let $z(n) = y(2n)$. From the definition of $y(n)$ given in the previous part:

$$y(n) = x\left(\frac{n}{2} - 1\right) \quad \text{for even } n$$

Substituting n with $2n$ (which guarantees an even index), we get:

$$z(n) = y(2n) = x\left(\frac{2n}{2} - 1\right) = x(n-1)$$

Thus, $z(n)$ is simply a delayed version of the original sequence $x(n)$ by 1 sample. Using the time-shifting property of the Discrete-Time Fourier Transform (DTFT), $x(n-n_0) \xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega})$:

$$Z(e^{j\omega}) = e^{-j\omega} X(e^{j\omega})$$

Now, let's find $X(e^{j\omega})$ directly from the given plot of $x(n)$:

$$x(n) = \frac{1}{2}\delta(n+2) + \delta(n+1) + 2\delta(n) + \delta(n-1) + \frac{1}{2}\delta(n-2)$$

Taking the DTFT of both sides:

$$X(e^{j\omega}) = \frac{1}{2}e^{j2\omega} + e^{j\omega} + 2 + e^{-j\omega} + \frac{1}{2}e^{-j2\omega}$$

Group the symmetric terms together:

$$X(e^{j\omega}) = 2 + \left(e^{j\omega} + e^{-j\omega}\right) + \frac{1}{2}\left(e^{j2\omega} + e^{-j2\omega}\right)$$

Using Euler's identity $\cos(\theta) = \frac{e^{j\theta}+e^{-j\theta}}{2}$:

$$X(e^{j\omega}) = 2 + 2\cos(\omega) + \cos(2\omega)$$

Substituting this spectral expression back into $Z(e^{j\omega})$:

$$Z(e^{j\omega}) = e^{-j\omega}\left[\cos 2\omega + 2\cos\omega + 2\right]$$

$$\boxed{\text{C) } e^{-j\omega}[\cos 2\omega + 2\cos\omega + 2]}$$

Key Points

- **Decimation after Upsampling:** Extracting the even samples via $y(2n)$ perfectly reverses the $n/2$ upsampling operation, leaving only the discrete time shift $x(n-1)$.
- **Time-Shifting Property:** $x(n-k) \xleftrightarrow{\text{DTFT}} e^{-j\omega k}X(e^{j\omega})$.
- **Symmetry Property:** A real and even sequence has a purely real and even DTFT entirely composed of cosine terms.

Q7.05.3 MCQ | 2005 | 1M | Ans: D | ● ○ ○

Given the discrete-time sequence:

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

We are given a new sequence $y(n)$ which is the square of $x(n)$, denoted as $y(n) = x^2$ in the problem statement (meaning $y(n) = [x(n)]^2$). Squaring the sequence yields:

$$y(n) = \left[\left(\frac{1}{2}\right)^n u(n)\right]^2$$

Since the unit step function squared is simply the unit step function ($u^2(n) = u(n)$ for all n), the expression simplifies to:

$$y(n) = \left(\frac{1}{4}\right)^n u(n)$$

The Discrete-Time Fourier Transform (DTFT) of a sequence $y(n)$ is defined as:

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y(n)e^{-j\omega n}$$

The problem asks for the value of $Y(e^{j0})$, which is the DTFT evaluated at $\omega = 0$. Substituting $\omega = 0$ into the definition gives the sum of all elements in the sequence (the DC component):

$$Y(e^{j0}) = \sum_{n=-\infty}^{\infty} y(n)e^{0} = \sum_{n=-\infty}^{\infty} y(n)$$

Substitute the expression for $y(n)$:

$$Y(e^{j0}) = \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n$$

This is an infinite geometric series with the first term $a = 1$ (for $n = 0$) and a common ratio $r = 1/4$. Since $|r| < 1$, the sum converges to:

$$S = \frac{a}{1-r} = \frac{1}{1-1/4} = \frac{1}{3/4} = \frac{4}{3}$$

$$\boxed{\text{D) } \frac{4}{3}}$$

Key Points

- **DC Component of DTFT:** The Discrete-Time Fourier Transform evaluated at $\omega = 0$ is always equal to the sum of all values in the time-domain sequence: $Y(e^{j0}) = \sum y(n)$.
- **Infinite Geometric Series:** The sum of a series of the form $\sum_{n=0}^{\infty} r^n$ is $\frac{1}{1-r}$, provided the region of convergence $|r| < 1$ is satisfied.
- **Properties of Unit Step:** Squaring a causal sequence bounded by $u(n)$ does not alter its causality, as $u^k(n) = u(n)$ for any positive power k.

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Chapter VII

DTFT, DTFS, DFT & Miscellaneous

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